Natalyx MVP Product Flow Reshape — consolidated programme plan

> Planning-only. Do not implement, deploy, migrate, change cloud resources, alter existing PRs or clean protected worktrees from this document. Alan should review/sign off this consolidated plan before proposed assignments become expected work.

Inspection snapshot: 2026-07-12 19:30 America/New_York. Planning branch: `plan/mvp-product-flow-reshape`. GitHub Project: MVP Product Flow Reshape. Proposed PRs: 52.

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Executive summary

The programme is a controlled rebuild around free Node 0 workspace → explicit entitlement/commitment/payment → typed promotion authorization → enrollment-pending paid case, not a patch stack on the current case-shaped Node 0 branch. PR #122 is the named paid-case invitation/case-presentation behavior baseline; PR #124 is extraction evidence only; preserved Wave 2 at 5b63e3a is adapted test/semantic evidence and not a merge unit. Alan's completed review is independently dispositioned in the frozen matrix.

The canonical package proposes 52 focused PRs across app, infra, intelligence, Learn, participant design system and organisation planning. It separates identity/participation/access, metadata coordination/document content, free workspace/paid case, payment/truth state; uses one processor-neutral promotion authorization; and requires forced RLS/non-owner runtime before B1 while PHI remains prohibited until B2.

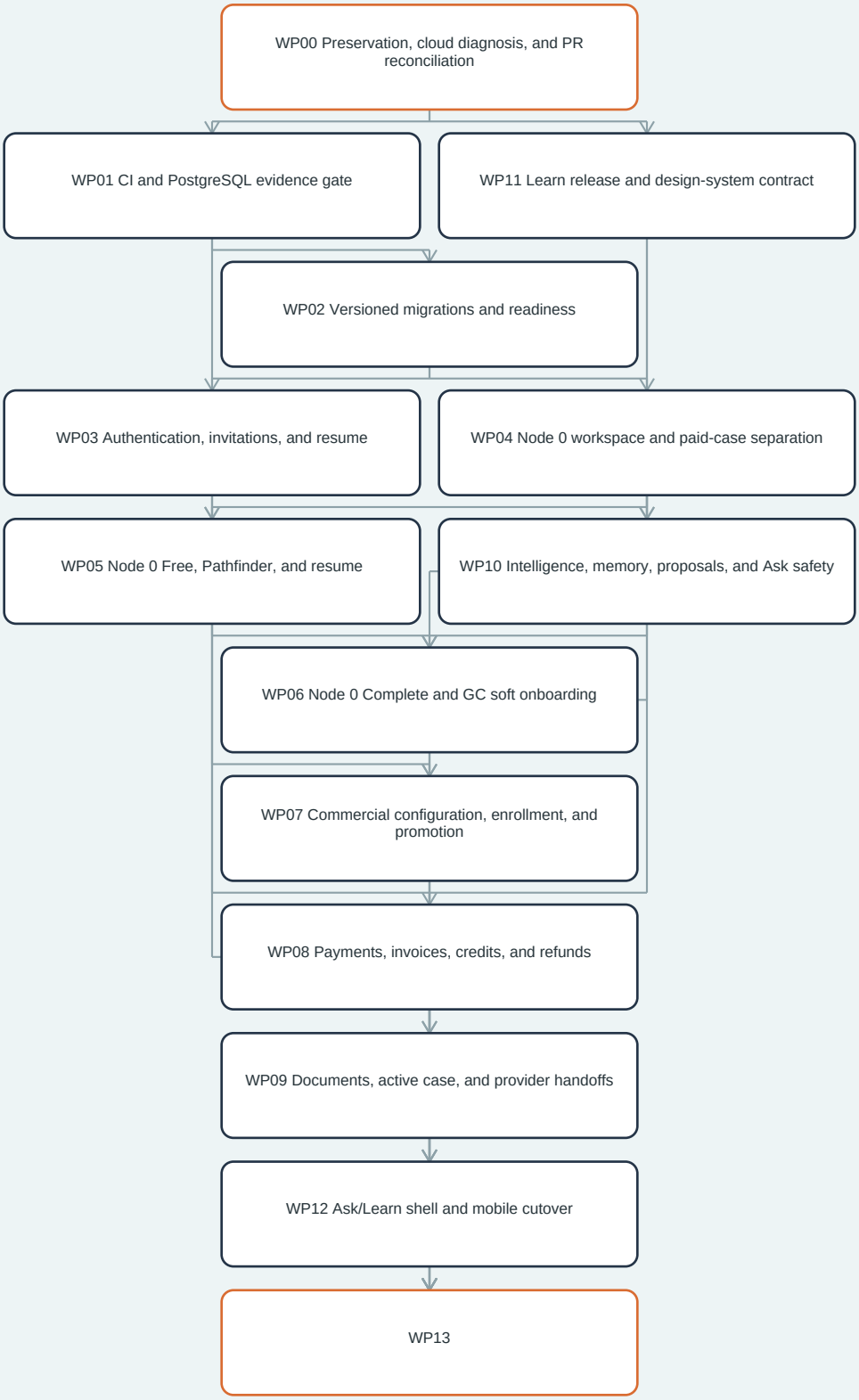
Current verified verdict

Area	Verdict	Evidence / consequence
Live cloud	Not usable now	dev.app.natalyx.health returned ALB 503 at the inspection snapshot; A0 is open.
AWS detail	Partially unverified	Local natalyx-dev SSO expired; service counts/task revision must be captured during A0.
CI/Postgres	Insufficient on main	Container build + contract smoke exist; no main Python/PostgreSQL evidence job.
PR #122	Materially improved, not final architecture	Latest e853821 fixes return_to/TTL/CAS/lens/CI and should be selectively retained.
PR #124	Do not merge unchanged	Free Node 0 creates a canonical case; unaddressed link can lead to case membership; no commercial promotion/provenance contract.
Wave 2	Preserve and adapt, do not merge	535 tests passed locally, but restart loss, stale approval, metadata/content conflation, caller-derived roles and absent RLS block direct reuse.
CI governance	Workflow accepted; enforcement decision open	D-CI-01 records the private-repository plan limitation and required ruleset readback/negative bypass evidence.
Payments/documents/PHI	Not implemented/ready	No money ledger; metadata is not content; forced RLS precedes B1 and PHI remains blocked until B2.

Architecture principles

- natalyx-app owns routing, validation, permissions, participant/workspace/case/commercial truth and PHI policy.
- Workspace, entitlement, commitment, payment, case, invitation/claim, document and journey-stage state machines remain separate.
- Identity, participation and access grants remain distinct; application services and RLS consume one subject-derived authorization context plus ownership and purpose.
- PostgreSQL is production authority; established migrations are preferred pending ADR; runtime identities lose DDL authority.
- Only `byos.workspace_promotion_authorized.v1` authorizes workspace-to-case promotion; redirects and raw webhooks never do.
- Commercial/money events are append-only; balances/statuses are projections.
- IP observations/provisional assertions remain attributed; GC/provider confirmation appends revisions and never silently erases/elevates source.
- Provider/intelligence/Learn projections are purpose-, role-, consent-, provenance- and version-bound.
- Security/accessibility/responsive interaction can block release; aesthetic/token purity alone generally cannot.
- Metadata requirements are not documents; content versions follow a separate upload/quarantine/scan lifecycle.
- Synthetic only through A2; forced RLS/non-owner runtime and approved PII scope at B1; PHI only after formal B2.

Dependency graph



The graph is package-level; the PR table below is the executable dependency authority.

Global PR table and execution order

P00-01

ID	P00-01
Title	Approve programme baseline and extraction ledger
Repository	.github
Work package	WP00 / Preservation, cloud diagnosis, and PR reconciliation
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	None
Gate	Pre-A0
Local preparation	Ready for Alan review and joint acceptance; no new implementation
Merge readiness	Blocked pending Alan review and joint programme acceptance
Blocking decision	None
Project item	P00-01

P01-01

ID	P01-01
Title	Establish current-test and PostgreSQL CI evidence
Repository	app
Work package	WP01 / CI and PostgreSQL evidence gate
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-CI-01: Alan - immediate control configuration/evidence
Reviewer	Luke
Final approver	Luke
Dependencies	P00-01
Gate	A0
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P00-01; technical decisions/controls D-CI-01; Luke review/final founder approval
Blocking decision	D-CI-01
Project item	P01-01

P11-01

ID	P11-01
Title	Publish validated Learn artifact and provenance contract
Repository	learn
Work package	WP11 / Learn release and design-system contract
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P00-01
Gate	A2
Local preparation	Eligible for isolated, uncommitted, synthetic/no-invented-content preparation only
Merge readiness	Blocked pending PR dependencies P00-01; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P11-01

P11-03

ID	P11-03
Title	Release participant design-system accessibility and interaction contract
Repository	design-system
Work package	WP11 / Learn release and design-system contract
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P00-01
Gate	A1
Local preparation	Eligible for isolated, uncommitted, synthetic/no-invented-content preparation only
Merge readiness	Blocked pending PR dependencies P00-01; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P11-03

P02-01

ID	P02-01
Title	Decide migration architecture and compatibility contract
Repository	app
Work package	WP02 / Versioned migrations and readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01
Gate	A0
Local preparation	Technical-decision preparation only; implementation waits on named dependencies
Merge readiness	Blocked pending PR dependencies P01-01; Luke review/final founder approval
Blocking decision	None
Project item	P02-01

P03-01

ID	P03-01
Title	Harden auth return-to and participant resume
Repository	app
Work package	WP03 / Authentication, invitations, and resume
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P03-01

P13-01

ID	P13-01
Title	Expose known-revision health, readiness, and dependency diagnostics
Repository	app
Work package	WP13 / Deployment, cloud gates, and PHI readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01
Gate	A0
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01; Luke review/final founder approval
Blocking decision	None
Project item	P13-01

P11-02

ID	P11-02
Title	Serve approved Learn catalog versions from intelligence
Repository	intelligence
Work package	WP11 / Learn release and design-system contract
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P11-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P11-01; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P11-02

P02-02

ID	P02-02
Title	Introduce versioned migration runner and readiness checks
Repository	app
Work package	WP02 / Versioned migrations and readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P02-01
Gate	A0
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-01; technical decisions/controls D-MIG-01; Luke review/final founder approval
Blocking decision	D-MIG-01
Project item	P02-02

P09-01

ID	P09-01
Title	Define document lifecycle, authorization, and retention contract
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Dependencies	P00-01, P02-01
Gate	A2
Local preparation	Technical-decision preparation only; implementation waits on named dependencies
Merge readiness	Blocked pending PR dependencies P00-01, P02-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P09-01

P13-02

ID	P13-02
Title	Harden guarded deployment, rollback, and evidence capture
Repository	infra
Work package	WP13 / Deployment, cloud gates, and PHI readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-OPS-01: Alan - immediate recovery/rollback/forward-repair mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01, P13-01
Gate	A0
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01, P13-01; technical decisions/controls D-OPS-01; Luke review/final founder approval
Blocking decision	D-OPS-01
Project item	P13-02

P11-04

ID	P11-04
Title	Adopt approved Learn and source-card recipes in app
Repository	app
Work package	WP11 / Learn release and design-system contract
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P11-02, P11-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P11-02, P11-03; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P11-04

P02-03

ID	P02-03
Title	Baseline canonical schema and prove expand-contract repair
Repository	app
Work package	WP02 / Versioned migrations and readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P02-02
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-02; technical decisions/controls D-MIG-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	D-MIG-01
Project item	P02-03

P02-04

ID	P02-04
Title	Move intelligence persistence to a revisioned migration namespace
Repository	intelligence
Work package	WP02 / Versioned migrations and readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P02-01, P02-02
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-01, P02-02; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P02-04

P09-02

ID	P09-02
Title	Provision private encrypted document and quarantine storage
Repository	infra
Work package	WP09 / Documents, active case, and provider handoffs
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Dependencies	P09-01, P13-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-01, P13-02; technical decisions/controls D-DOC-02; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	D-DOC-02
Project item	P09-02

P03-05

ID	P03-05
Title	Implement canonical identity, participation, access, and authorization contracts
Repository	app
Work package	WP03 / Authentication, invitations, and resume
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P02-03, P03-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P03-05

P07-01

ID	P07-01
Title	Add versioned commercial configuration and terms acceptance
Repository	app
Work package	WP07 / Commercial configuration, enrollment, and promotion
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P02-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P07-01

P09-03

ID	P09-03
Title	Implement resilient document upload, scan, and download
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P09-02, P02-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-02, P02-03; technical decisions/controls D-DOC-01, D-DOC-02; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	D-DOC-01, D-DOC-02
Project item	P09-03

P03-03

ID	P03-03
Title	Bind paid-case invitations to identity and consent
Repository	app
Work package	WP03 / Authentication, invitations, and resume
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P03-01, P03-05
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P03-05; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P03-03

P04-01

ID	P04-01
Title	Separate Node 0 workspace storage from paid cases
Repository	app
Work package	WP04 / Node 0 workspace and paid-case separation
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P02-03, P03-05
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-05; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P04-01

P08-01

ID	P08-01
Title	Add append-only commercial and money ledgers
Repository	app
Work package	WP08 / Payments, invoices, credits, and refunds
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P02-03, P07-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P07-01; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P08-01

P03-02

ID	P03-02
Title	Add low-risk workspace claim lifecycle
Repository	app
Work package	WP03 / Authentication, invitations, and resume
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P03-01, P03-05, P04-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P03-05, P04-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P03-02

P04-02

ID	P04-02
Title	Add attributable workspace revision and projection contracts
Repository	app
Work package	WP04 / Node 0 workspace and paid-case separation
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P04-01, P03-05
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-01, P03-05; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P04-02

P05-01

ID	P05-01
Title	Replace Pathfinder routing with deterministic entry decisions
Repository	app
Work package	WP05 / Node 0 Free, Pathfinder, and resume
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P03-01, P04-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P04-01; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P05-01

P06-01

ID	P06-01
Title	Add Node 0 Complete entitlement and versioned feature gates
Repository	app
Work package	WP06 / Node 0 Complete and GC soft onboarding
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P04-01, P07-01, P08-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-01, P07-01, P08-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P06-01

P07-02

ID	P07-02
Title	Implement commitment and enrollment-pending lifecycle
Repository	app
Work package	WP07 / Commercial configuration, enrollment, and promotion
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P07-01, P08-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P07-01, P08-01; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P07-02

P03-04

ID	P03-04
Title	Replace shared signup admission with claim-aware Cognito admission
Repository	infra
Work package	WP03 / Authentication, invitations, and resume
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-AUTH-02: Alan - later identity/admission mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P03-02, P03-03
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P03-03; technical decisions/controls D-AUTH-02; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	D-AUTH-02
Project item	P03-04

P04-03

ID	P04-03
Title	Adapt legacy Node 0 case-shaped data without semantic promotion
Repository	app
Work package	WP04 / Node 0 workspace and paid-case separation
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P04-02
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P04-03

P10-01

ID	P10-01
Title	Version the scoped intelligence context envelope
Repository	intelligence
Work package	WP10 / Intelligence, memory, proposals, and Ask safety
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P01-01, P04-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01, P04-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	D-AI-01
Project item	P10-01

P05-02

ID	P05-02
Title	Rebuild Node 0 Free candidate and outreach domain
Repository	app
Work package	WP05 / Node 0 Free, Pathfinder, and resume
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P04-02, P05-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P05-01; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P05-02

P06-02

ID	P06-02
Title	Add GC soft onboarding and attributable confirmation
Repository	app
Work package	WP06 / Node 0 Complete and GC soft onboarding
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P03-02, P04-02, P06-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P04-02, P06-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P06-02

P08-02

ID	P08-02
Title	Add processor-neutral payment command and webhook inbox
Repository	app
Work package	WP08 / Payments, invoices, credits, and refunds
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
Dependencies	P08-01, P07-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-01, P07-02; technical decisions/controls D-PAY-01; Gate A0 active and verified; Alan review
Blocking decision	D-PAY-01
Project item	P08-02

P10-02

ID	P10-02
Title	Enforce private/shared memory and evidence-bound proposal approval
Repository	app
Work package	WP10 / Intelligence, memory, proposals, and Ask safety
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P04-02, P10-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P10-01; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	D-AI-01
Project item	P10-02

P10-03

ID	P10-03
Title	Harden Ask and Learn grounding and degraded-mode behavior
Repository	intelligence
Work package	WP10 / Intelligence, memory, proposals, and Ask safety
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
Dependencies	P10-01, P11-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-01, P11-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	D-AI-01
Project item	P10-03

P05-03

ID	P05-03
Title	Ship resumable Node 0 Free participant surface
Repository	app
Work package	WP05 / Node 0 Free, Pathfinder, and resume
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P03-01, P05-02, P11-03
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P05-02, P11-03; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P05-03

P07-03

ID	P07-03
Title	Emit processor-neutral workspace promotion authorization
Repository	app
Work package	WP07 / Commercial configuration, enrollment, and promotion
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P04-02, P07-02, P08-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P07-02, P08-02; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P07-03

P06-03

ID	P06-03
Title	Add readiness report and role-scoped progress surfaces
Repository	app
Work package	WP06 / Node 0 Complete and GC soft onboarding
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P06-02, P10-02, P11-04
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P06-02, P10-02, P11-04; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P06-03

P10-04

ID	P10-04
Title	Add intelligence safety evaluations and cross-service observability
Repository	intelligence
Work package	WP10 / Intelligence, memory, proposals, and Ask safety
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P10-02, P10-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-02, P10-03; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P10-04

P13-03

ID	P13-03
Title	Automate synthetic A1 core participant journey verification
Repository	app
Work package	WP13 / Deployment, cloud gates, and PHI readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P03-02, P04-03, P05-03, P11-03, P13-02
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P04-03, P05-03, P11-03, P13-02; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P13-03

P07-04

ID	P07-04
Title	Promote workspace exactly once from authorization event
Repository	app
Work package	WP07 / Commercial configuration, enrollment, and promotion
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P03-03, P04-02, P07-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-03, P04-02, P07-03; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P07-04

P09-04

ID	P09-04
Title	Ship phase-1 case progress and metadata-only request/review
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P03-03, P03-05, P07-04, P09-01, P11-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-03, P03-05, P07-04, P09-01, P11-03; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P09-04

P08-03

ID	P08-03
Title	Implement milestone invoicing, credits, and deterministic refunds
Repository	app
Work package	WP08 / Payments, invoices, credits, and refunds
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
Dependencies	P08-02, P07-04, P09-04
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-02, P07-04, P09-04; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P08-03

P09-06

ID	P09-06
Title	Integrate content-bearing document versions into participant surfaces
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Dependencies	P09-03, P09-04
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-03, P09-04; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P09-06

P12-01

ID	P12-01
Title	Unify Ask and Learn shell with explicit context scopes
Repository	app
Work package	WP12 / Ask/Learn shell and mobile cutover
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Alan
Final approver	Luke
Dependencies	P10-03, P11-04, P09-04
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-03, P11-04, P09-04; technical decisions/controls D-AI-01; Gate A0 active and verified; Alan review
Blocking decision	D-AI-01
Project item	P12-01

P08-04

ID	P08-04
Title	Add payment reconciliation and safe service-pause operations
Repository	app
Work package	WP08 / Payments, invoices, credits, and refunds
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
Dependencies	P08-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-03; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P08-04

P09-05

ID	P09-05
Title	Implement consent-scoped provider handoff snapshots and operations
Repository	app
Work package	WP09 / Documents, active case, and provider handoffs
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-PROV-01: Alan - later provider-packet mechanism; privacy/provider validation; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Dependencies	P04-02, P09-06, P10-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P09-06, P10-02; technical decisions/controls D-PROV-01; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	D-PROV-01
Project item	P09-05

P12-02

ID	P12-02
Title	Complete mobile participant interaction and accessibility cutover
Repository	app
Work package	WP12 / Ask/Learn shell and mobile cutover
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P05-03, P06-03, P09-04, P09-06, P12-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P05-03, P06-03, P09-04, P09-06, P12-01; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P12-02

P13-05

ID	P13-05
Title	Enforce case RLS and a non-owner PostgreSQL runtime
Repository	app
Work package	WP13 / Deployment, cloud gates, and PHI readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Dependencies	P02-03, P03-05, P04-03, P08-04, P09-05, P10-02
Gate	B1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-05, P04-03, P08-04, P09-05, P10-02; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P13-05

P13-04

ID	P13-04
Title	Automate synthetic A2 commercial and document journey verification
Repository	app
Work package	WP13 / Deployment, cloud gates, and PHI readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Dependencies	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P13-04

P13-06

ID	P13-06
Title	Harden B1/B2 backups, audit, alarms, and cohort controls
Repository	infra
Work package	WP13 / Deployment, cloud gates, and PHI readiness
WP DRI	Alan
Primary implementer	Alan
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval; D-PHI-02: Alan - evidence index; qualified external reviewers - attestations
Reviewer	Luke
Final approver	Luke
Dependencies	P09-02, P13-02, P13-05
Gate	B1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-02, P13-02, P13-05; Gate A0 active and verified; Luke review/final founder approval
Blocking decision	None
Project item	P13-06

P12-03

ID	P12-03
Title	Retire legacy participant routes behind verified redirects
Repository	app
Work package	WP12 / Ask/Learn shell and mobile cutover
WP DRI	Luke
Primary implementer	Luke
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Dependencies	P12-02, P13-04
Gate	B1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P12-02, P13-04; Gate A0 active and verified; Alan review
Blocking decision	None
Project item	P12-03

P13-07

ID	P13-07
Title	Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks
Repository	.github
Work package	WP13 / Deployment, cloud gates, and PHI readiness
WP DRI	Alan
Primary implementer	Joint
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval; D-PHI-02: Alan - evidence index; qualified external reviewers - attestations
Reviewer	Luke + External/TBD
Final approver	Luke
Dependencies	P13-04, P13-05, P13-06
Gate	B1
Local preparation	Evidence-structure preparation only; technical and external evidence remains incomplete
Merge readiness	Blocked pending PR dependencies P13-04, P13-05, P13-06; Gate A0 active and verified; Luke review/final founder approval; qualified external attestations and B1 evidence
Blocking decision	None
Project item	P13-07

Topological execution order:

Order	ID	Title	Repository	Dependencies	Gate
1	P00-01	Approve programme baseline and extraction ledger	.github	P00-01	Pre-A0
2	P01-01	Establish current-test and PostgreSQL CI evidence	app	P00-01	A0
3	P11-01	Publish validated Learn artifact and provenance contract	learn	P00-01	A2
4	P11-03	Release participant design-system accessibility and interaction contract	design-system	P00-01	A1
5	P02-01	Decide migration architecture and compatibility contract	app	P01-01	A0
6	P03-01	Harden auth return-to and participant resume	app	P01-01	A1

Order	ID	Title	Repository	Dependencies	Gate
7	P13-01	Expose known-revision health, readiness, and dependency diagnostics	app	P01-01	A0
8	P11-02	Serve approved Learn catalog versions from intelligence	intelligence	P11-01	A2
9	P02-02	Introduce versioned migration runner and readiness checks	app	P02-01	A0
10	P09-01	Define document lifecycle, authorization, and retention contract	app	P00-01, P02-01	A2
11	P13-02	Harden guarded deployment, rollback, and evidence capture	infra	P01-01, P13-01	A0
12	P11-04	Adopt approved Learn and source-card recipes in app	app	P11-02, P11-03	A2
13	P02-03	Baseline canonical schema and prove expand-contract repair	app	P02-02	A1
14	P02-04	Move intelligence persistence to a revisioned migration namespace	intelligence	P02-01, P02-02	A1
15	P09-02	Provision private encrypted document and quarantine storage	infra	P09-01, P13-02	A2
16	P03-05	Implement canonical identity, participation, access, and authorization contracts	app	P02-03, P03-01	A1
17	P07-01	Add versioned commercial configuration and terms acceptance	app	P02-03	A2
18	P09-03	Implement resilient document upload, scan, and download	app	P09-02, P02-03	A2
19	P03-03	Bind paid-case invitations to identity and consent	app	P03-01, P03-05	A2
20	P04-01	Separate Node 0 workspace storage from paid cases	app	P02-03, P03-05	A1
21	P08-01	Add append-only commercial and money ledgers	app	P02-03, P07-01	A2
22	P03-02	Add low-risk workspace claim lifecycle	app	P03-01, P03-05, P04-01	A1
23	P04-02	Add attributable workspace revision and projection contracts	app	P04-01, P03-05	A1
24	P05-01	Replace Pathfinder routing with deterministic entry decisions	app	P03-01, P04-01	A1
25	P06-01	Add Node 0 Complete entitlement and versioned feature gates	app	P04-01, P07-01, P08-01	A2
26	P07-02	Implement commitment and enrollment-pending lifecycle	app	P07-01, P08-01	A2
27	P03-04	Replace shared signup admission with claim-aware Cognito admission	infra	P03-02, P03-03	A1
28	P04-03	Adapt legacy Node 0 case-shaped data without semantic promotion	app	P04-02	A1
29	P10-01	Version the scoped intelligence context envelope	intelligence	P01-01, P04-02	A2
30	P05-02	Rebuild Node 0 Free candidate and outreach domain	app	P04-02, P05-01	A1

Order	ID	Title	Repository	Dependencies	Gate
31	P06-02	Add GC soft onboarding and attributable confirmation	app	P03-02, P04-02, P06-01	A2
32	P08-02	Add processor-neutral payment command and webhook inbox	app	P08-01, P07-02	A2
33	P10-02	Enforce private/shared memory and evidence-bound proposal approval	app	P04-02, P10-01	A2
34	P10-03	Harden Ask and Learn grounding and degraded-mode behavior	intelligence	P10-01, P11-02	A2
35	P05-03	Ship resumable Node 0 Free participant surface	app	P03-01, P05-02, P11-03	A1
36	P07-03	Emit processor-neutral workspace promotion authorization	app	P04-02, P07-02, P08-02	A2
37	P06-03	Add readiness report and role-scoped progress surfaces	app	P06-02, P10-02, P11-04	A2
38	P10-04	Add intelligence safety evaluations and cross-service observability	intelligence	P10-02, P10-03	A2
39	P13-03	Automate synthetic A1 core participant journey verification	app	P03-02, P04-03, P05-03, P11-03, P13-02	A1
40	P07-04	Promote workspace exactly once from authorization event	app	P03-03, P04-02, P07-03	A2
41	P09-04	Ship phase-1 case progress and metadata-only request/review	app	P03-03, P03-05, P07-04, P09-01, P11-03	A2
42	P08-03	Implement milestone invoicing, credits, and deterministic refunds	app	P08-02, P07-04, P09-04	A2
43	P09-06	Integrate content-bearing document versions into participant surfaces	app	P09-03, P09-04	A2
44	P12-01	Unify Ask and Learn shell with explicit context scopes	app	P10-03, P11-04, P09-04	A2
45	P08-04	Add payment reconciliation and safe service-pause operations	app	P08-03	A2
46	P09-05	Implement consent-scoped provider handoff snapshots and operations	app	P04-02, P09-06, P10-02	A2
47	P12-02	Complete mobile participant interaction and accessibility cutover	app	P05-03, P06-03, P09-04, P09-06, P12-01	A2
48	P13-05	Enforce case RLS and a non-owner PostgreSQL runtime	app	P02-03, P03-05, P04-03, P08-04, P09-05, P10-02	B1
49	P13-04	Automate synthetic A2 commercial and document journey verification	app	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03	A2
50	P13-06	Harden B1/B2 backups, audit, alarms, and cohort controls	infra	P09-02, P13-02, P13-05	B1
51	P12-03	Retire legacy participant routes behind verified redirects	app	P12-02, P13-04	B1
52	P13-07	Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks	.github	P13-04, P13-05, P13-06	B1

Parallel lanes and critical path

Critical path:

P00-01 → P01-01 → P02-01 → P02-02 → P02-03 → P03-05 → P04-01 → P04-02 → P07-01 → P08-01 → P07-02 → P08-02 → P07-03 → P03-03 → P07-04 → P09-01 → P09-04 → P09-02 → P09-03 → P09-06 → P09-05 → P13-04 → P13-05 → P13-06 → P13-07

Required dependency joins:

- P03-02 and P04-01 wait for P03-05 identity/participation/access; P07-03 additionally waits for P04-02; P07-04 additionally waits for P03-03.
- P09-04 metadata/readiness is independent of content storage; P09-06 joins P09-03 content to P09-04 surfaces; P09-05 then builds provider snapshots from actual clean versions.
- P13-04 waits for the complete synthetic A2 surface; P13-05/06/07 then supply RLS, operational controls and human evidence required before B1, while B2 remains a later PHI approval.

First three implementation PRs:

1. P01-01 — Establish current-test and PostgreSQL CI evidence
2. P02-01 — Decide migration architecture and compatibility contract
3. P13-01 — Expose known-revision health, readiness, and dependency diagnostics

Lane	Lead	Items	Coordination
Foundation and live recovery	Alan	P01-01, P02-01, P02-02, P02-03, P02-04, P13-01, P13-02	Luke approves ADR and A0; no replacement participant surface required.
Completed source reconciliation	Luke	CHECK-ALAN-122-124, CLEAN-01	The review checkpoint is complete; CLEAN-01 remains future closure evidence and authorizes no cleanup.
Identity and workspace	Alan	P03-01, P03-02, P03-03, P03-04, P03-05, P04-01, P04-02, P04-03	P03-05 encodes the reconciled access contract before workspace/case schemas consume it; Luke reviews privacy and role semantics.
Participant Free	Luke	P05-01, P05-02, P05-03	Waits for WP03/WP04 and the required WP11 contract; Alan reviews technical safety and failure behavior.
Complete and GC onboarding	Alan	P06-01, P06-02, P06-03	Depends on workspace/provenance and commercial primitives; Luke reviews product and privacy boundaries.
Commercial and payment	Luke	P07-01, P07-02, P07-03, P07-04, P08-01, P08-02, P08-03, P08-04	Luke implements settled founder policy; Alan reviews and retains named technical-mechanism decisions, including D-PAY-01.
Documents and provider exchange	Alan	P09-01, P09-02, P09-03, P09-04, P09-05, P09-06	Metadata readiness can progress without bytes; content integration waits for the secure pipeline; provider handoff waits for clean content versions.
Intelligence	Alan	P10-01, P10-02, P10-03, P10-04	Alan implements the scoped service/app contracts after D-AI-01; Luke reviews truth and privacy boundaries.
Learn, participant DS and cutover	Luke	P11-01, P11-02, P11-03, P11-04, P12-01, P12-02, P12-03	P11-01/P11-03 allow isolated uncommitted preparation; app adoption and cutover wait for their named upstream dependencies.
Formal gates	Alan	P13-03, P13-04, P13-05, P13-06, P13-07	Alan is DRI for technical gate evidence; P13-07 is jointly implemented with Luke and qualified external reviewers.

Early sequencing:

1. P00-01
2. P11-01 (isolated local preparation only) + P11-03 (isolated local preparation only)
3. P01-01 + P02-01 + P13-01
4. P02-02 + P13-02

5 P02-03 + P02-04 + P03-01

6 P03-05 + P04-01

State transition contract

Common envelope:

```
event_id
entity_type
entity_id
event_type
from_state
to_state
expected_revision
result_revision
actor_type
actor_subject_or_service
authority_scope
evidence_refs[]
idempotency_key
correlation_id
causation_id
policy_version
schema_version
commercial_config_version (nullable)
reason_code
occurred_at
recorded_at
safe_metadata
```

- Reusing an idempotency key with the same canonical payload returns the original result; a different payload returns 409 idempotency_conflict.
- A wrong from_state or expected revision returns 409 stale_transition; no generic setter silently coerces state.
- The state write, append-only audit event and outbox entry commit in one transaction.
- Retry resumes from committed event/projection; recovery uses an explicit compensating or forward-repair event.
- Audit safe_metadata excludes raw PHI, PII beyond essential identifiers, invitation tokens, processor payloads and document contents.

Participant entry and routing (`entry_session`)

Owner: natalyx-app. States: `unstarted`, `education`, `identity_required`, `authenticated`, `routed`, `handed_off`, `expired`.

unstarted

From	unstarted
To	education
Actor	visitor
Evidence	GET/explicit entry choice
Idempotency key	entry:{session}:start
Invalid transitions	No participant/case write
Audit event	entry_started
Retry	Return current session
Recovery	Create fresh anonymous session

education

From	education
To	identity_required
Actor	visitor/policy
Evidence	route requires saved workspace
Idempotency key	entry:{session}:identity
Invalid transitions	Cannot grant workspace/case
Audit event	identity_required
Retry	Preserve safe return intent
Recovery	Restart at public choice

identity_required

From	identity_required
To	authenticated
Actor	Cognito/auth provider
Evidence	verified subject/session
Idempotency key	auth:{subject}:{nonce}
Invalid transitions	Invalid/expired token
Audit event	identity_bound
Retry	Replay returns authenticated session
Recovery	Rechallenge; no partial membership

education|authenticated

From	education|authenticated
To	routed
Actor	participant + deterministic policy
Evidence	validated route answers + policy version
Idempotency key	route:{session}:{answer_hash}
Invalid transitions	Unknown/unsafe route goes review/education
Audit event	entry_routed
Retry	Same input returns same decision
Recovery	Allow corrected answers/new decision revision

routed

From	routed
To	handed_off
Actor	participant
Evidence	accepted destination and authorized target
Idempotency key	handoff:{session}:{route_version}
Invalid transitions	Target not authorized/available
Audit event	entry_handed_off
Retry	Resume same safe destination
Recovery	Return to route result with explanation

any nonterminal

From	any nonterminal
To	expired
Actor	system
Evidence	TTL elapsed/revoked session
Idempotency key	entry:{session}:expire
Invalid transitions	No transition from handed_off needed
Audit event	entry_expired
Retry	Idempotent
Recovery	New session; no case/access change

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Entitlement (`workspace_entitlement`)

Owner: natalyx-app commercial domain. States: `free`, `complete_pending`, `complete_active`, `suspended`, `revoked`.

none

From	none
To	free
Actor	system
Evidence	active Node 0 workspace
Idempotency key	entitlement:{workspace}:free
Invalid transitions	No workspace
Audit event	entitlement_granted
Retry	Return current
Recovery	Repair projection from event

free

From	free
To	complete_pending
Actor	IP/commercial service
Evidence	accepted Complete offer/config
Idempotency key	entitlement:{workspace}:{commitment}
Invalid transitions	Unknown price/config/terms
Audit event	complete_requested
Retry	Return pending
Recovery	Cancel/expire commitment

complete_pending

From	complete_pending
To	complete_active
Actor	payment domain
Evidence	settled Complete ledger entry
Idempotency key	entitlement:{payment}:activate
Invalid transitions	Amount/currency/config mismatch
Audit event	complete_activated
Retry	Replay no-op
Recovery	Reconcile and retry from inbox

complete_active

From	complete_active
To	suspended
Actor	authorized commercial policy
Evidence	refund/reversal or approved pause reason
Idempotency key	entitlement:{id}:suspend:{event}
Invalid transitions	Cannot remove free/safety capability
Audit event	entitlement_suspended
Retry	No-op if same event
Recovery	Reinstate or revoke through new event

complete_active|suspended

From	complete_active|suspended
To	revoked
Actor	authorized operator/policy
Evidence	fraud/refund/terms revocation evidence
Idempotency key	entitlement:{id}:revoke:{event}
Invalid transitions	No silent case cancellation
Audit event	entitlement_revoked
Retry	Idempotent
Recovery	Manual reviewed regrant creates new entitlement event

Commercial commitment (`journey_commitment`)

Owner: natalyx-app commercial domain. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

none

From	none
To	draft
Actor	IP
Evidence	active workspace + displayed config/terms
Idempotency key	commitment:{workspace}:{config}
Invalid transitions	No active config/workspace
Audit event	commitment_drafted
Retry	Return existing draft
Recovery	Create new draft after expiry/version change

draft

From	draft
To	accepted
Actor	IP
Evidence	terms/privacy consent + payment path
Idempotency key	commitment:{id}:accept:{terms_hash}
Invalid transitions	Stale/missing acceptance
Audit event	commitment_accepted
Retry	Return accepted
Recovery	New acceptance revision if version changes

accepted

From	accepted
To	funding_pending
Actor	commercial service
Evidence	enrollment invoice/payment intent created
Idempotency key	commitment:{id}:funding
Invalid transitions	No invoice or invalid amount
Audit event	funding_requested
Retry	Resume same invoice
Recovery	Cancel/reissue under audited adjustment

funding_pending

From	funding_pending
To	funded
Actor	payment domain
Evidence	net settled enrollment funding meets versioned requirement
Idempotency key	enrollment-funding:{id}:v1
Invalid transitions	Redirect/pending/failed/reversed evidence
Audit event	commercial.enrollment_funding_confirmed.v1
Retry	Exactly-once event/outbox replay
Recovery	Reconcile inbox/ledger; no browser repair

draft|accepted|funding_pending

From	draft|accepted|funding_pending
To	cancelled
Actor	IP/authorized operator
Evidence	explicit cancellation and refund policy
Idempotency key	commitment:{id}:cancel:{request}
Invalid transitions	Already funded/promoted uses case cancellation rail
Audit event	commitment_cancelled
Retry	Idempotent
Recovery	New commitment required

draft|accepted

From	draft|accepted
To	expired
Actor	system
Evidence	offer/terms TTL or config deactivation
Idempotency key	commitment:{id}:expire
Invalid transitions	No expiry after funding
Audit event	commitment_expired
Retry	Idempotent
Recovery	Draft new version

Payment (`payment_obligation + append-only entries`)

Owner: natalyx-app payment domain. States: `created`, `processing`, `settled`, `failed`, `partially_refunded`, `refunded`, `reversed`.

none

From	none
To	created
Actor	billing policy
Evidence	valid invoice/obligation
Idempotency key	payment:{invoice}:create
Invalid transitions	Duplicate or invalid invoice
Audit event	payment_created
Retry	Return current intent
Recovery	Void/reissue through adjustment

created|failed

From	created|failed
To	processing
Actor	processor adapter
Evidence	signed provider acknowledgement
Idempotency key	processor:{provider_event}
Invalid transitions	Amount/currency/customer mismatch
Audit event	payment_processing
Retry	Inbox replay no-op
Recovery	Retry same intent or audited replacement

processing|created

From	processing|created
To	settled
Actor	payment domain
Evidence	verified settlement event
Idempotency key	processor:{provider_event}
Invalid transitions	Browser redirect or unverified event
Audit event	payment_settled
Retry	Exactly-once append
Recovery	Reconcile provider and append repair event

created|processing

From	created|processing
To	failed
Actor	payment domain
Evidence	verified failure/timeout policy
Idempotency key	processor:{provider_event}
Invalid transitions	Cannot overwrite settled
Audit event	payment_failed
Retry	Replay no-op
Recovery	Retry/new attempt linked to obligation

settled|partially_refunded

From	settled|partially_refunded
To	partially_refunded|refunded
Actor	authorized refund service
Evidence	approved refund calculation + processor confirmation
Idempotency key	refund:{refund_request}
Invalid transitions	Exceeds settled less prior refunds
Audit event	refund_settled
Retry	Replay no-op
Recovery	Reconcile/adjust; never mutate original

settled

From	settled
To	reversed
Actor	payment domain
Evidence	verified chargeback/reversal
Idempotency key	processor:{provider_event}
Invalid transitions	Unsupported/manual assertion
Audit event	payment_reversed
Retry	Replay no-op
Recovery	Open discrepancy and apply policy; case is not deleted

Paid case lifecycle (`case`)

Owner: natalyx-app. States: `enrollment_pending`, `active`, `paused`, `closed`, `cancelled`.

none

From	none
To	enrollment_pending
Actor	promotion service
Evidence	byos.workspace_promotion_authorized.v1 + projection manifest
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization/duplicate workspace
Audit event	paid_case_created
Retry	Return same case
Recovery	Forward-repair same promotion

enrollment_pending

From	enrollment_pending
To	active
Actor	authorized coordinator/domain service
Evidence	approved onboarding/compatibility/intake/education evidence
Idempotency key	case:{id}:activate:{evidence_hash}
Invalid transitions	Missing/revoked evidence or GC consent
Audit event	case_activated
Retry	No duplicate activation/invoice
Recovery	Return pending with unmet checklist

active

From	active
To	paused
Actor	authorized commercial/operations policy
Evidence	audited non-critical pause reason
Idempotency key	case:{id}:pause:{event}
Invalid transitions	Safety-critical access cannot be paused
Audit event	case_paused
Retry	Idempotent
Recovery	Resume after evidence; preserve entitled docs

paused

From	paused
To	active
Actor	authorized operator/domain service
Evidence	pause reason resolved
Idempotency key	case:{id}:resume:{event}
Invalid transitions	Unresolved safety/authorization block
Audit event	case_resumed
Retry	Idempotent
Recovery	Escalate manual review

active|paused

From	active|paused
To	closed
Actor	authorized coordinator
Evidence	birth/close or completed termination handoff evidence
Idempotency key	case:{id}:close:{evidence_hash}
Invalid transitions	Open critical obligations
Audit event	case_closed
Retry	No-op if same evidence
Recovery	Reopen only by exceptional audited event

enrollment_pending|active|paused

From	enrollment_pending|active|paused
To	cancelled
Actor	authorized operator under policy
Evidence	withdrawal/non-clearance/cancellation + financial disposition
Idempotency key	case:{id}:cancel:{request}
Invalid transitions	Cannot erase records/refund ledger
Audit event	case_cancelled
Retry	Idempotent
Recovery	Replacement is a new case/workspace policy event

Low-risk workspace identity claim (`workspace_claim`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `consented`, `claimed`, `mismatch_pending`, `revoked`, `expired`.

none

From	none
To	issued
Actor	workspace IP
Evidence	selected candidate + allowed claim scope
Idempotency key	claim:{workspace}:{candidate}:{generation}
Invalid transitions	Existing live claim superseded
Audit event	workspace_claim_issued
Retry	Return current token only at issuance
Recovery	Issue new generation; old revoked

issued

From	issued
To	pending_identity
Actor	link holder
Evidence	valid token landing
Idempotency key	claim:{id}:land
Invalid transitions	Expired/revoked token
Audit event	claim_opened
Retry	Safe repeat
Recovery	Authenticate or exit; no data grant

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email and binding policy
Idempotency key	claim:{id}:bind:{subject}
Invalid transitions	Already claimed/different subject
Audit event	claim_identity_bound|claim_mismatch
Retry	CAS no-op or mismatch state
Recovery	Issuer review/new claim

identity_bound

From	identity_bound
To	consented
Actor	GC claimant
Evidence	workspace consent version
Idempotency key	claim:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	claim_consented
Retry	Idempotent
Recovery	Display current consent

consented

From	consented
To	claimed
Actor	claim service
Evidence	authorized low-risk workspace membership transaction
Idempotency key	claim:{id}:complete
Invalid transitions	Case membership requested or conflicting claimant
Audit event	workspace_claimed
Retry	Return same membership
Recovery	Transaction rollback/forward repair

issued|pending_identity|identity_bound|consented|mismatch_pending

From	issued|pending_identity|identity_bound|consented|mismatch_pending
To	revoked|expired
Actor	issuer/system
Evidence	revocation or TTL
Idempotency key	claim:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	claim_revoked|claim_expired
Retry	Idempotent
Recovery	Fresh claim only

Paid-case membership invitation (`case_invitation`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

none

From	none
To	issued
Actor	authorized IP case member
Evidence	enrollment_pending/active case + recipient binding
Idempotency key	case-invite:{case}:{role}:{generation}
Invalid transitions	Self/member/active-GC/unauthorized
Audit event	case_invite_issued
Retry	Return current; supersede prior live GC invite
Recovery	Issue corrected invite

issued

From	issued
To	pending_identity
Actor	recipient
Evidence	valid addressed token landing
Idempotency key	case-invite:{id}:land
Invalid transitions	Expired/revoked
Audit event	case_invite_opened
Retry	Safe repeat
Recovery	Authenticate with return_to

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email
Idempotency key	case-invite:{id}:bind:{subject}
Invalid transitions	Terminal/duplicate/different subject
Audit event	case_invite_bound|mismatch
Retry	CAS protects terminal
Recovery	Issuer confirmation or corrected invite

mismatch_pending

From	mismatch_pending
To	issuer_confirmed
Actor	authorized issuer
Evidence	explicit reviewed mismatch confirmation
Idempotency key	case-invite:{id}:confirm:{issuer}
Invalid transitions	Issuer unauthorized/recipient changed
Audit event	case_invite_mismatch_confirmed
Retry	CAS no-op
Recovery	Revoke and reissue

identity_bound|issuer_confirmed

From	identity_bound|issuer_confirmed
To	consented
Actor	invitee
Evidence	paid-case role/privacy consent version
Idempotency key	case-invite:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	case_invite_consented
Retry	Idempotent
Recovery	Present current consent

consented

From	consented
To	redeemed
Actor	invitation service
Evidence	single transaction creates person/party/membership
Idempotency key	case-invite:{id}:redeem
Invalid transitions	Existing member/active GC/expired/revoked
Audit event	case_invite_redeemed
Retry	Return same membership
Recovery	Rollback entire transaction; forward repair from invite

issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented

From	issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented
To	declined|revoked|expired|superseded
Actor	invitee/issuer/system
Evidence	explicit decline/revoke/TTL/new generation
Idempotency key	case-invite:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	case_invite_terminal
Retry	Idempotent CAS
Recovery	Fresh invitation only

Document lifecycle (`document + document_version + upload`)

Owner: natalyx-app; storage in natalyx-infra. States: `upload_requested`, `uploading`, `quarantined`, `scanning`, `available`, `review`, `approved`, `replaced`, `deleted`, `failed`.

none

From	none
To	upload_requested
Actor	authorized case member
Evidence	document purpose/type/size intent
Idempotency key	upload:{case}:{client_request}
Invalid transitions	Unauthorized/type/size/retention block
Audit event	upload_requested
Retry	Return same intent
Recovery	Expire/orphan intent

upload_requested

From	upload_requested
To	uploading
Actor	client/storage service
Evidence	authorized upload starts
Idempotency key	upload:{id}:start
Invalid transitions	Expired intent/wrong key
Audit event	upload_started
Retry	Multipart resume
Recovery	Abort and mark failed/orphan cleanup

uploading

From	uploading
To	quarantined
Actor	finalize service
Evidence	object exists + size/checksum/content sniff
Idempotency key	upload:{id}:finalize:{checksum}
Invalid transitions	Mismatch/missing object
Audit event	upload_quarantined
Retry	Repeat verify
Recovery	Delete bad object; request new upload

quarantined

From	quarantined
To	scanning
Actor	scan worker
Evidence	durable queued task
Idempotency key	scan:{version}:{engine_version}
Invalid transitions	Already available/deleted
Audit event	scan_started
Retry	Leased retry
Recovery	Lease expiry requeues after ECS replacement

scanning

From	scanning
To	available|failed
Actor	scan service
Evidence	signed clean/malicious/error result
Idempotency key	scan:{version}:{result_id}
Invalid transitions	Unknown result/engine
Audit event	scan_completed|scan_failed
Retry	Inbox replay no-op
Recovery	Remain quarantined on timeout/error; security review

available

From	available
To	review|approved
Actor	authorized participant/reviewer
Evidence	purpose-specific review/evidence
Idempotency key	document:{version}:review:{event}
Invalid transitions	Wrong role/stale version
Audit event	document_reviewed
Retry	Idempotent
Recovery	New review revision

available|review|approved

From	available|review|approved
To	replaced
Actor	authorized owner
Evidence	new clean version linked
Idempotency key	document:{id}:replace:{new_version}
Invalid transitions	New version not available
Audit event	document_replaced
Retry	Return current version
Recovery	Keep prior per retention; roll current pointer forward

available|review|approved|replaced|failed

From	available|review|approved|replaced|failed
To	deleted
Actor	authorized policy job/operator
Evidence	retention/delete request and no hold
Idempotency key	document:{version}:delete:{request}
Invalid transitions	Legal hold/provider obligation
Audit event	document_deleted
Retry	Tombstone idempotent
Recovery	Object-delete retry/orphan alarm; metadata/audit retained

Programme gates

Gate A0 — Infrastructure recovery and known-revision health

Field	Contract
Prerequisite PRs/gates	P01-01, P02-02, P13-01, P13-02
Prerequisite decisions	D-MIG-01, D-CI-01, D-OPS-01
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ GitHub Team is active for NatalyxPlatform; unittest and PostgreSQL checks are mandatory, routine bypass is restricted, and negative-bypass evidence is accepted.
- □ Alan's migration ADR records the supported low-maintenance framework, PostgreSQL authority, staged compatibility and the tested recovery path; P02-02/P02-03 evidence passes.
- □ CloudFormation/ECS/ALB state is captured from refreshed dev credentials and is stable.

- □ The running app and intelligence image SHAs/task definitions are known and map to reviewed commits.
- □ Liveness and readiness are both healthy; PostgreSQL, migrations and required private intelligence dependencies pass their bounded checks.
- □ A synthetic unauthenticated health request and authenticated minimal route succeed without PII/PHI.
- □ The fail-closed recovery runbook is exercised synthetically; the prior compatible image/schema pair is identified and the 30-minute planned-beta recovery target is evidenced.

Required evidence:

- GitHub Team/ruleset readback and negative-bypass result
- Migration ADR and consequential-migration recovery evidence
- AWS account/region and stack/service snapshots
- ALB status and target health
- image SHA/task definition
- readiness JSON/log correlation
- CI/PostgreSQL artifacts
- fail-closed synthetic recovery exercise

Failure and rollback response. Do not proceed to participant validation. Freeze live updates, retain evidence, restore the known compatible image/schema or forward-repair under the approved runbook, and keep the Project gate blocked.

Gate A1 — Synthetic core participant journey

Field	Contract
Prerequisite PRs/gates	P02-04, P03-01, P03-02, P03-04, P03-05, P04-01, P04-02, P04-03, P05-01, P05-02, P05-03, P11-03, P13-03
Prerequisite decisions	D-AUTH-02, D-DATA-01, D-DATA-02
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ A new synthetic IP signs up, is deterministically routed, enters Node 0 Free and no paid case is created.
- □ Candidate/outreach data remains IP-attributed and provisional; no score or answer becomes GC/provider truth.
- □ A synthetic GC begins a low-risk claim, binds identity/consent and sees only the allowed workspace projection.
- □ Cross-role negative tests prove class-2 IP assertions and class-4 GC-private answers do not leak.
- □ The admission proof is short-lived, claim/purpose/email-bound and cannot grant workspace or case membership; replay, mismatch, expiry and revocation fail closed.
- □ Both roles log out/in/resume and complete the mobile journey at tested widths.
- □ Known revision, audit events, correlation IDs and sanitized evidence are attached to the gate item.

Required evidence:

- A1 synthetic run artifact
- separate IP/GC subject IDs (synthetic only)
- authorization matrix results
- mobile screenshots/accessibility output
- audit event trace
- database invariant queries

Failure and rollback response. Stop before Complete, commercial or paid-case validation. Revert/disable the failing participant feature flag or deploy the known A0 revision; preserve synthetic data only for defect diagnosis, then purge under the runbook.

Gate A2 — Synthetic commercial, promotion, billing and document journey

Field	Contract
Prerequisite PRs/gates	P03-03, P06-01, P06-02, P06-03, P07-01, P07-02, P07-03, P07-04, P08-01, P08-02, P08-03, P08-04, P09-01, P09-02, P09-03, P09-04, P09-05, P09-06, P10-01, P10-02, P10-03, P10-04, P11-01, P11-02, P11-04, P12-01, P12-02, P13-04
Prerequisite decisions	D-COM-01, D-COM-02, D-COM-03, D-COM-04, D-PAY-01, D-COM-05, D-COM-06, D-COM-07, D-COM-08, D-DOC-01, D-DOC-02, D-PROV-01, D-AI-01
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ Synthetic Complete entitlement does not itself create a case and records its commercial configuration version/credit.
- □ Synthetic configuration uses the founder-approved $495 Complete and $8,900 v1 journey schedules, including credits, refunds, loss policy and humane grace behavior; no real charge or live terms acceptance occurs.
- □ Only byos.workspace_promotion_authorized.v1 authorizes exactly one enrollment_pending case; callbacks/redirects/replays create none/duplicates.
- □ Recipient-bound paid-case invitation and consent grant correct membership; activation waits for approved evidence.
- □ Enrollment/activation/milestone/credit/refund ledgers reconcile, including duplicate and out-of-order provider events.
- □ Synthetic files follow upload/quarantine/scan/available/download/replacement/revocation/recovery and never become public.
- □ Provider packet contains only purpose-allowed consented sources; intelligence scopes and proposals do not upgrade truth.
- □ Financing is absent; scanning and AI stay within their $50/month and $100/month beta ceilings with alerts and fail-safe fallback.

Required evidence:

- A2 synthetic run artifact
- processor simulator event ledger
- promotion idempotency query
- invoice/refund explanations
- S3/KMS/IAM/quarantine assertions
- provider snapshot checksum
- Ask/proposal safety evals

Failure and rollback response. Disable commercial/document/provider features, retain free/safety content, prevent live money or real documents, reconcile synthetic ledger inconsistencies, and restore the A1-known revision or forward-repair as approved.

Gate B1 — Real-user and PII beta readiness

Field	Contract
Prerequisite PRs/gates	A2, P12-03, P13-05, P13-06, P13-07
Prerequisite decisions	D-PHI-01, D-PHI-02
Responsible owner	Luke
Verifier	External/TBD

Acceptance matrix:

- □ A written B1 data-classification boundary limits the invitation-only US cohort to at most five IP/GC pairs and permits only the approved nonclinical PII fields.
- □ Medical, reproductive-health, psychological, insurance-health and clinical-clearance data, real document bytes and real provider packets are expressly prohibited; AI receives only de-identified nonclinical context.
- □ The exact release uses a NOBYPASSRLS non-owner runtime, FORCE RLS on every in-scope table, verified transaction-local context and passing pool-reset/cross-org/cross-resource negative tests.

- □ Named beta users, support/on-call, access review, incident response, backup/restore, retention/deletion and complaint routes are ready.
- □ Alan's technical evidence is current; external privacy/legal advice is recorded; there is no unresolved critical finding and any high exception is written, time-limited and externally reviewed.
- □ A2 evidence is current for the exact candidate revision and rollback/feature-disable controls are operational.
- □ Consent/privacy/terms copy is approved and data-subject correction/deletion workflows are testable.

Required evidence:

- Signed B1 checklist and five-pair cohort roster
- data inventory/classification and prohibited-data tests
- runtime/migrator grants and RLS policy matrix
- cross-org/pool-reset results
- de-identification evidence
- access review
- incident tabletop
- backup/restore result
- vendor register
- external privacy/legal advice
- current A2 bundle no older than 60 days

Failure and rollback response. No real-user invitation. Continue synthetic-only, close control gaps, invalidate stale evidence and re-run the exact affected reviews/tests before a new go/no-go.

Gate B2 — PHI readiness and formal go/no-go

Field	Contract
Prerequisite PRs/gates	B1
Prerequisite decisions	D-PHI-01, D-PHI-02
Responsible owner	Luke
Verifier	External security/privacy/compliance reviewers

Acceptance matrix:

- □ Formal PHI data flows, risk assessment, policies, BAAs/vendor controls, least privilege, audit/log retention, encryption/key management, incident/breach response and backup/restore are approved.
- □ Application, infrastructure, document, provider and intelligence threat models have current evidence and no unresolved critical/high risk without accepted exception.
- □ Operational staff training, access provisioning/deprovisioning, monitoring, support and emergency stop are complete.
- □ The exact release passed A2 and B1 revalidation; a signed human go/no-go records scope, date, evidence versions and approvers.
- □ Qualified external security, privacy/compliance and legal/vendor reviewers are named; release evidence is no older than 30 days and underlying assessments are no older than 12 months.
- □ No critical exception is open; any high exception has joint Luke and relevant external-reviewer approval.
- □ The gate states explicitly that testing/deployment alone did not establish HIPAA/PHI readiness.
- □ PHI-capable upload, provider and intelligence paths remain disabled until this signed approval; B1 controls are necessary but not sufficient.

Required evidence:

- Signed go/no-go
- risk/security/privacy assessments
- BAA/vendor evidence

- access/log/incident/backup controls
- penetration/security testing
- training/operations evidence
- current technical gate bundles

Failure and rollback response. PHI remains prohibited. Keep B1 PII scope only if still independently valid; disable any PHI-capable upload/context path, remediate, and repeat formal review.

GitHub Project operating model

Project: MVP Product Flow Reshape.

Pre-change inspection

Direct GraphQL inspection found Project 3 open with only GitHub's default Title, Assignees, Status, Labels, Linked PRs, Milestone, Repository, Reviewers, Parent issue, Sub-issues progress, Created, Updated and Closed fields, and zero items. The programme sync therefore does not duplicate prior programme items.

Operating model

Convention	Rule
Project	MVP Product Flow Reshape
Number	3
Status	Benched and closed after plan reconciliation; do not use for execution until the amended Drive package is jointly accepted and the Project is separately reactivated/resynced.
Hierarchy	Draft items cannot be true GitHub sub-issues. Work-package parents are identifiable draft epics; child PR/decision/gate items link by Work package and Dependencies fields and exact IDs. Convert only when real implementation begins.
Readiness	Local preparation status is distinct from merge readiness. A PR may permit isolated uncommitted preparation while remaining blocked from commit, PR creation or merge.
Assignment	PR ownership inherits the proposed work-package split pending Alan's review; named technical-decision ownership does not transfer implementation ownership across work packages.
Reviewer Field	GitHub reserves the literal custom-field name Reviewer, so Programme reviewer holds the requested planning reviewer role without duplicating the built-in Reviewers metadata.
Source Of Truth	programme_data.py is the package generator source and Drive is the canonical review surface. The benched Project becomes an operational control plane only after a separately authorized sync; catalogue changes must not be presented as live Project state.

Fields

Field	Type	Options / use
Item type	single	Work package, Proposed PR, Existing PR, Existing issue, Decision, Review checkpoint, Programme gate, Cleanup task
Work package	single	Programme, WP00, WP01, WP02, WP03, WP04, WP05, WP06, WP07, WP08, WP09, WP10, WP11, WP12, WP13
Target repository	single	.github, natalyx-app, natalyx-infra, natalyx-intelligence, natalyx-learn-resources, natalyx-design-system, Multiple / external
Proposed PR ID	text	Free text
DRI	single	Luke, Alan, Joint, External / TBD, N/A
Implementer	single	Luke, Alan, Joint, External / TBD, N/A
Programme reviewer	single	Luke, Alan, Joint, External / TBD, N/A
Final approver	single	Luke, Alan, Joint, External / TBD, N/A
Dependencies	text	Free text
Programme gate	single	Pre-A0, A0, A1, A2, B1, B2, Post-B2, N/A
Size	single	Small, Medium, Large, Exceptional, N/A
Blocking decision	text	Free text
Source / salvage PR	text	Free text
Plan document path	text	Free text
Readiness	single	Needs plan approval, Awaiting reconciliation, Blocked by decision, Dependencies unmet, Ready to implement, In implementation, Verification, Complete, Retained for salvage
Disposition	single	Programme baseline, Proposed, Retain pending review, Direct reuse, Selective port, Rewrite, Supersede after extraction, Defer, Gate criterion, Decision needed, Checkpoint, Cleanup only

GitHub's existing Status field is reused; no duplicate status field is created. Custom fields are the minimum set needed to hold the fields explicitly required by the programme brief for draft items as well as real PRs/issues.

Item catalogue

Item type	Expected count	Naming convention
Work packages	14	[WPxx] title
Proposed PRs	52	[Pxx-yy] title
Existing PRs	9	Actual linked PR
Existing issues	2	Actual linked issue
Decisions	24	[DECISION D-*] title
Review checkpoints	1	[CHECKPOINT] title
Programme gates	5	[GATE Ax] title
Cleanup tasks	3	[CLEANUP C-*] title

Desired catalogue and last live snapshot

Check	Recorded result
Catalogue Generated At	2026-07-12 19:30 America/New_York
Live Snapshot At	2026-07-10 21:59 America/New_York
Project Id	PVT_kwDOEAWZDM4BYbai
Items	110
Draft Items	99
Linked Items	11
Live Items	106
Live Draft Items	95
Live Linked Items	11
Total Fields	29
Custom Fields	16
Statuses	Desired catalogue: checkpoint and three resolved decisions Done; implementation PRs Todo. Last live snapshot: all 106 items Todo.
Required Values	Desired fields complete on 110 items; Proposed PR ID complete on all 52 proposed PR items
Type Counts	14 WP; 52 proposed PR; 9 existing PR; 2 existing issue; 24 decision; 1 checkpoint; 5 gate; 3 cleanup
Result	Local desired catalogue validated; remote Project intentionally not resynced because it is benched and the task authorizes only canonical planning sources/generated artifacts.

The desired 110-item catalogue is validated locally. The last read-back live snapshot remains 106 items and all Todo; this reconciliation did not apply remote Project changes. All implementation items remain planning-only and no proposed PR is marked `Ready to implement`.

Dependency and hierarchy representation

Draft items cannot be true GitHub sub-issues. Work-package parents are identifiable draft epics; child PR/decision/gate items link by Work package and Dependencies fields and exact IDs. Convert only when real implementation begins.

Each proposed PR carries a Work package value and exact Dependencies text. Work-package draft bodies list child PR IDs, decisions, gates and plan path. Decision/gate/checkpoint bodies list what they block. This provides deterministic hierarchy without creating empty issues/PRs.

Status, readiness and disposition

Dimension	Meaning
Status	Todo / In Progress / Done: execution state only.
Local preparation	Whether isolated, uncommitted, synthetic-only work may occur before joint acceptance.
Merge readiness	Whether dependencies, decisions, reciprocal review, A0 and applicable external controls permit merge.
Disposition	How existing/proposed/decision/gate/cleanup material should be treated.
Ownership	Work-package DRI and primary implementer, separate technical-decision owner, reciprocal reviewer and founder approver; all remain proposed pending Alan's review.

Project sync and maintenance

Dry-run/read current state:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/sync_proje
ct.py
```

Apply idempotent updates only after separate reactivation/authorization (the script blocks apply while benched):

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/sync_proje
ct.py --apply
```

After any canonical change, regenerate Markdown, validate and re-render the PDF. Validate the desired catalogue locally; remote sync is a separate authorized action after the benched Project is reactivated.

Final Project validation checklist

- ☐ One identifiable item for every WP00-WP13.
- ☐ One proposed-PR draft item for all 52 PR IDs.
- ☐ Actual linked PRs #112/#117/#119/#121/#122/#123/#124 and DS #7/#8.
- ☐ Actual linked Learn #3 and Intelligence #14 dependency issues.
- ☐ Alan reconciliation checkpoint present and complete; normal dependencies/decisions remain.
- ☐ Every blocking decision and A0/A1/A2/B1/B2 gate present.
- ☐ Ownership, technical-decision responsibility, dependencies, local preparation, merge readiness, plan path, source/salvage and disposition populated.
- ☐ No duplicate titles/IDs and Project/document parity validator passes.

Ownership, delegation and review

Work package	DRI	Primary implementer	Reciprocal reviewer	Founder approver	Split
WP00	Luke	Luke (proposed)	Alan	Luke	Ready for Alan review and joint acceptance; no new implementation is authorized by this review package.
WP01	Alan	Alan (proposed)	Luke	Luke	Technical evidence foundation.
WP02	Alan	Alan (proposed)	Luke	Luke	Alan chooses the migration framework/mechanism; Luke approves compatibility and risk.
WP03	Alan	Alan (proposed)	Luke	Luke	Alan implements identity/admission controls; Luke reviews founder access constraints.
WP04	Alan	Alan (proposed)	Luke	Luke	Alan implements the domain/provenance contracts; Luke reviews product/privacy boundaries.
WP05	Luke	Luke (proposed)	Alan	Luke	Luke owns Free implementation; Alan reviews technical architecture, security and failure behavior.
WP06	Alan	Alan (proposed)	Luke	Luke	Alan implements Complete/GC onboarding; Luke reviews cross-role projection and product boundaries.
WP07	Luke	Luke (proposed)	Alan	Luke	Luke implements settled commercial semantics; Alan reviews technical mechanisms and idempotency.
WP08	Luke	Luke (proposed)	Alan	Luke	Luke implements payment-domain contracts after D-PAY-01; external review remains mandatory before real money.
WP09	Alan	Alan (proposed)	Luke	Luke	Alan implements document/provider controls; Luke reviews retention, purpose and PHI boundaries.
WP10	Alan	Alan (proposed)	Luke	Luke	Alan implements intelligence contracts after D-AI-01; Luke reviews truth and privacy policy.
WP11	Luke	Luke (proposed)	Alan	Luke	Luke implements the separate Learn, intelligence-catalog, participant-DS and app-adoption PRs; Alan reviews technical contracts.
WP12	Luke	Luke (proposed)	Alan	Luke	Luke implements participant shell/cutover; Alan reviews integration, authorization and failure behavior.
WP13	Alan	Alan (proposed)	Luke	Luke	Alan owns technical gate evidence; Luke accepts gates and external reviewers retain their required attestations.

Checkpoint	When	Owner	Evidence
Architecture review	Before any package PR plan is marked ready	Luke	Approved entities/contracts/decision items
Isolated local preparation	Only where the PR record expressly permits it	Named proposed implementer	Separate worktree; uncommitted; synthetic-only; protected work untouched
PR-plan approval	Before commit or PR creation	Named proposed implementer + reciprocal reviewer	Scope, non-goals, dependencies, tests, rollout/repair

Checkpoint	When	Owner	Evidence
Alan PR reconciliation	Completed 2026-07-12 before planning amendments	Luke; Alan supplied the completed technical review	Frozen matrix and CHECK-ALAN-122-124 complete
Pre-merge review	Every PR	Named reviewer/approver	CI, objective acceptance, risk review
Cloud verification	Gate-related PRs	Alan	Known revision and redacted live/synthetic artifacts
Gate acceptance	A0/A1/A2/B1/B2	Luke	Gate matrix; external sign-off where required

Delegation rule: PRs inherit the proposed work-package DRI and primary implementer pending Alan's review. Alan reviews Luke-owned work; Luke reviews Alan-owned work and retains founder approval where required. Technical-decision ownership and external attestations remain separate from implementation ownership. P13-07 is the sole Joint-implementation exception.

Existing-PR extraction and Alan checkpoint

> Read-only reconciliation record. No PR, branch or worktree may be closed, retargeted, rebased, cleaned or deleted from this ledger alone.

Inspection snapshot: 2026-07-12 19:30 America/New_York.

Preservation and extraction sequence

1. Preserve dirty worktrees and identify active branches/PRs.
2. Reinspect latest heads.
3. Complete this extraction ledger.
4. Reconcile Alan's findings.
5. Approve replacement architecture/PR boundaries.
6. Create/ready Project draft items.
7. Open replacement drafts only when implementation starts.
8. Extract/port/rewrite value.
9. Close superseded PRs only after accounting.
10. Delete branches/worktrees only after closure and team confirmation.

Summary

PR	Latest head	Useful capability	Destination	Status	Disposition
app#112	9605d88e31bd7ea5ccb0ddc524c5f2412efb242e	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P00-01, P09-04, P11-03, P11-04, P12-03	Catalogued; content remains to be reconciled into replacement PR acceptance tests.	Selective port, then supersede after extraction
app#117	5ac09f0cdd9299667fd70e845e606b1892f98ece	Deterministic candidate questionnaire, scoring/status calculation, conversation-guide resource, routes and extensive Node 0 tests.	P04-01, P04-02, P05-02, P05-03	Unextracted; algorithm/test review required after Alan checkpoint and Luke field/score decision.	Selective port and rewrite
app#119	a8cd2f428535807749 0f02089ac8f9e6f1e8bed3	Concise product-leverage framing and boundaries for the MVP flow.	P00-01, P04-01, P05-01, P06-01, P10-01, P12-03	Catalogued; reconcile terminology into approved programme.	Retain as source or merge separately after plan decision
app#121	529d377e23376268b44c95ba9974aa0cd133275f	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P05-03, P11-03, P12-02	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.	Selective UI/test port, then supersede
app#122	e85382150853cf4774948c55b91e330e7e69617e	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P01-01, P02-03, P03-01, P03-03, P03-05, P04-02, P09-04, P10-02, P11-03, P12-02	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.	Selective-port behavior and tests; supersede as programme architecture
app#123	34078f810b634aa6ddce5954763c03cb46b574d8	GC onboarding form/copy, hosted auth return, invite handoff and role-specific surface work.	P03-01, P03-02, P06-02, P06-03	Unextracted; likely partially copied into #124 but not proven complete.	Selective UX/test port and rewrite
app#124	84a5a847fea71213b6257549e5a620f654836cb1	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P03-02, P03-03, P04-01, P04-02, P04-03, P05-02, P05-03, P06-02, P06-03, P09-04, P10-02, P11-03, P11-04, P12-02	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence
ds#7	cedfcc489b1f60842aed3a9701348c85964cfd96	Documents an escaping/attrs hardening follow-up relevant to participant recipe security.	P11-03	Direct-reuse candidate; not yet merged.	Retain for direct review/merge or fold into P11-03
ds#8	4656426e493e243785f8490589bf44fc0e05b8d6	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.	P11-03, P12-02	Direct-reuse candidate; local dirty asset delta separately protected.	Prefer direct review/merge if still correct

Detailed PR records

app#112 — BYOS journey surface migration: surface map + workplan (spec handoff)

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/112
Latest inspected head	9605d88e31bd7ea5ccb0ddc524c5f2412efb242e
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.
Reusable commits/files	docs/plans/byos-journey-surface-map.md; docs/plans/byos-journey-surface-workplan.md; single documentation commit 9605d88.
Tests worth preserving	Docs-only; preserve its stated acceptance/test map, not CI results as product evidence.
Obsolete/unsafe assumptions	Predates the explicit free-workspace/paid-case/commercial/provenance split and cannot be the implementation sequence by itself.
Replacement destination	P00-01, P09-04, P11-03, P11-04, P12-03
Extraction status	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Selective port, then supersede after extraction
Exact close condition	Every unique architecture/product decision is either adopted in the programme or rejected with rationale; Alan confirms no needed review input remains; replacement plan is approved.
Exact branch/worktree delete condition	PR is closed, commit is reachable from retained history or archived reference, no attached dirty worktree/unique commit remains, and Luke/Alan explicitly confirm cleanup.

app#117 — Node 0 beta: Outreach Readiness Quiz and deterministic Conversation Guide

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/117
Latest inspected head	5ac09f0cdd9299667fd70e845e606b1892f98ece
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	Deterministic candidate questionnaire, scoring/status calculation, conversation-guide resource, routes and extensive Node 0 tests.
Reusable commits/files	app/node0.py; app/node0_conversation_guide.py; resource JSON; focused tests; only the non-ranking/safety-safe algorithms that pass D-DATA-03.
Tests worth preserving	tests/python/test_node0.py contains useful deterministic and authorization cases.
Obsolete/unsafe assumptions	Large mixed UI/style/schema PR; case-shaped persistence, 0-100 ranking and unverified third-party answers conflict with workspace/provenance contract.
Replacement destination	P04-01, P04-02, P05-02, P05-03
Extraction status	Unextracted; algorithm/test review required after Alan checkpoint and Luke field/score decision.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Selective port and rewrite
Exact close condition	Approved questionnaire/guide behavior and every valuable test are mapped to P05-02/P05-03 or deliberately rejected; replacement PRs contain provenance and Free-tier contract.
Exact branch/worktree delete condition	Close conditions met, stacked PRs #121/#123 are reconciled, branch has no unique retained worktree content, and team confirms deletion.

app#119 — Docs: align MVP app flows to process leverage points

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/119
Latest inspected head	a8cd2f4285358077490f02089ac8f9e6f1e8bed3
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	Concise product-leverage framing and boundaries for the MVP flow.
Reusable commits/files	docs/plans/mvp-process-leverage-alignment.md and its two documentation commits.
Tests worth preserving	Docs-only; no executable behavior.
Obsolete/unsafe assumptions	Does not define implementation state machines, funding, provenance or gate evidence; title similarity must not make it the programme authority.
Replacement destination	P00-01, P04-01, P05-01, P06-01, P10-01, P12-03
Extraction status	Catalogued; reconcile terminology into approved programme.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Retain as source or merge separately after plan decision
Exact close condition	Luke decides whether the source document remains useful independently; if superseded, every unique principle is cited/adopted or rejected in the approved programme.
Exact branch/worktree delete condition	Only after close/merge, source commit remains reachable, no worktree depends on branch and team confirms.

app#121 — Node 0 beta: primary participant surface

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/121
Latest inspected head	529d377e23376268b44c95ba9974aa0cd133275f
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.
Reusable commits/files	app/node0_surface.py; node0_pdf.py; route/template/asset/test slices after comparison with #124.
Tests worth preserving	Node 0 surface/rendering tests worth porting to P05-03.
Obsolete/unsafe assumptions	Stacked on #117 rather than main, inherits case-shaped candidate model, and includes broad Wave 0/DS content already landed through #115; only tip 529d377 is genuinely unique. It predates the authorship/workspace split.
Replacement destination	P05-03, P11-03, P12-02
Extraction status	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Selective UI/test port, then supersede
Exact close condition	All unique surface/PDF/test value is proven present in an approved replacement or explicitly rejected; stacked dependency is no longer needed.
Exact branch/worktree delete condition	After #117/#123/#124 reconciliation, PR closure, no dirty worktree/unique commits, and team confirmation.

app#122 — Wave 1 participant case shell surfaces

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/122
Latest inspected head	e85382150853cf4774948c55b91e330e7e69617e
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.
Reusable commits/files	Invitation behavior/tests; case_journey/case_home presenter and paired-lens/mobile tests; .github/workflows/python-tests.yml after CI hardening; DS escaping contract where still applicable.
Tests worth preserving	Reported 499-test suite; green unittest/build/contract checks observed at e853821. Preserve concurrency, logged-out invite, paired IP/GC, Learn timeout and render tests.
Obsolete/unsafe assumptions	No Node 0 workspace target, payment/promotion or separate claim model; facts/proposals lack strong authorship/approval authority; generic invitation status setter remains; case lifecycle is not the whole programme state model.
Replacement destination	P01-01, P02-03, P03-01, P03-03, P03-05, P04-02, P09-04, P10-02, P11-03, P12-02
Extraction status	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
Outstanding reviewer input	Alan's 2026-07-10 review-round comment and the completed 2026-07-12 reconciliation are captured. No formal GitHub approval is inferred.
Recommended disposition	Selective-port behavior and tests; supersede as programme architecture
Exact close condition	Every retained behavior/test is reachable in approved replacements or deliberately rejected; no team member elects a separately narrowed #122 merge.
Exact branch/worktree delete condition	Only after closure, retained commits/tests are reachable or archived, no worktree depends on the branch, and Alan/Luke explicitly confirm cleanup.

app#123 — Node 0 beta: GC onboarding and hosted handoff

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/123
Latest inspected head	34078f810b634aa6ddce5954763c03cb46b574d8
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	GC onboarding form/copy, hosted auth return, invite handoff and role-specific surface work.
Reusable commits/files	Onboarding templates/routes/tests after field-level privacy and claim separation review.
Tests worth preserving	Preserve hosted-return, invalid/decline and GC lens interaction cases after comparison with #124/dirty onboarding branch.
Obsolete/unsafe assumptions	Stacked on #121, predates PR #122 hardening, and conflates soft onboarding with case-shaped data/membership; GC answers do not revise IP assertions. It also mixes participant work with a large internal/operator redesign and vendored internal DS, which must stay out of this participant programme.
Replacement destination	P03-01, P03-02, P06-02, P06-03
Extraction status	Unextracted; likely partially copied into #124 but not proven complete.
Outstanding reviewer input	No GitHub review recorded.
Recommended disposition	Selective UX/test port and rewrite
Exact close condition	Hosted return and onboarding value is mapped to claim-aware, attributable GC replacement PRs and no unique behavior remains.
Exact branch/worktree delete condition	After stacked PR/worktree reconciliation, closure, retained commits/tests are reachable, and team confirms.

app#124 — Consolidate BYOS Node 0 handoff

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-app/pull/124
Latest inspected head	84a5a847fea71213b6257549e5a620f654836cb1
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.
Reusable commits/files	Deterministic guide/resource/tests, selected Node 0/GC/case presentation, source-aware Learn dependency notes, subject to file-level extraction.
Tests worth preserving	test_node0.py and test_case_home.py contain broad behavior/render coverage; green build/contract-smoke observed, but no main Python/PostgreSQL proof.
Obsolete/unsafe assumptions	Node 0 Free creates a synthetic canonical case; unaddressed link can lead to case membership; IP third-party assertions drive score/invite; notes become shared memory; no commercial event, payment ledger or promotion. It contains older #122 equivalents but not the later hardening commits through e853821; its checks omit the unittest workflow.
Replacement destination	P03-02, P03-03, P04-01, P04-02, P04-03, P05-02, P05-03, P06-02, P06-03, P09-04, P10-02, P11-03, P11-04, P12-02
Extraction status	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
Outstanding reviewer input	Alan's complete consolidated comment 4948149089 is independently reconciled. Its extraction-only recommendation is accepted; several salvage semantics are modified by the workspace/access/document contracts.
Recommended disposition	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence
Exact close condition	Every retained behavior/test is reachable in approved replacements or explicitly rejected and Luke/Alan confirm supersession.
Exact branch/worktree delete condition	Only after closure, replacement reachability/extraction verification, no protected worktree depends on the branch and explicit team cleanup confirmation.

ds#7 — Document Jinja attrs hardening follow-up

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-design-system/pull/7
Latest inspected head	cedfcc489b1f60842aed3a9701348c85964cfd96
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	Documents an escaping/attrs hardening follow-up relevant to participant recipe security.
Reusable commits/files	Documentation commit/contract as direct source for P11-03.
Tests worth preserving	Preserve attrs/escaping contract tests in the eventual DS release.
Obsolete/unsafe assumptions	Draft guidance alone does not prove consumer safety and should not block cloud recovery for aesthetic reasons.
Replacement destination	P11-03
Extraction status	Direct-reuse candidate; not yet merged.
Outstanding reviewer input	No recorded review in this inspection.
Recommended disposition	Retain for direct review/merge or fold into P11-03
Exact close condition	Merged as-is after review or content/tests are present in an approved DS replacement.
Exact branch/worktree delete condition	After merge/closure, no unique dirty asset changes on the branch and team confirmation.

ds#8 — Add consumer visual parity guidance

Field	Finding
URL	https://github.com/NatalyxPlatform/natalyx-design-system/pull/8
Latest inspected head	4656426e493e243785f8490589bf44fc0e05b8d6
Inspection time	2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York)
Useful capability	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.
Reusable commits/files	Guidance and validation expectations for P11-03/P12-02.
Tests worth preserving	Use concrete responsive/a11y/security requirements; do not turn every token mismatch into a blocker.
Obsolete/unsafe assumptions	Local worktree contains extra dirty brand assets not represented by remote PR; do not overwrite or assume included.
Replacement destination	P11-03, P12-02
Extraction status	Direct-reuse candidate; local dirty asset delta separately protected.
Outstanding reviewer input	Open, not draft; no recorded formal review in this inspection.
Recommended disposition	Prefer direct review/merge if still correct
Exact close condition	Merged or equivalent guidance/tests are released and app pins that release.
Exact branch/worktree delete condition	After closure/merge and explicit confirmation that local dirty assets are committed elsewhere or intentionally discarded.

Latest PR #122 reinspection

Latest inspected head: `e85382150853cf4774948c55b91e330e7e69617e` at 2026-07-12T23:30:00Z (2026-07-12 19:30 America/New_York). The earlier reviewed workflow run/comment baseline was `8df63dd`; Alan's current head is `e853821`.

Fixed or materially improved since the earlier snapshot:

- Logged-out invite delivery/resume and validated internal return_to across login/signup/confirm/resend.
- Paired IP/GC lens semantics and render-time action labels.
- CAS redemption/expiry/mismatch transitions, self-invite refusal and one-live-GC constraints.
- 14-day expiry ceiling, portable integrity handling and duplicate-live-invite forward migration.
- Blocking Python CI workflow, fail-closed Learn approval, bounded Learn timeout, SQL-limited timeline and safer cookie/dead-button behavior.

Remaining conflicts/gaps:

- No distinct Node 0 workspace or workspace claim target; it cannot be stacked unchanged with #124.
- No commercial commitment/funding/promotion or separate entitlement/payment/document lifecycle.
- Generic fact/proposal/invitation rails still need explicit author/authority/evidence transition commands.
- The case projection is useful case-stage presentation, not a universal programme state machine.

Decomposition impact: retain #122's named invitation, CI and case-surface behavior/tests, but do not collapse WP03/WP04/WP07/WP09. The reconciliation checkpoint is complete; normal dependencies and unresolved decisions now govern readiness.

Preserved Wave 2 branch

Field	Finding
Branch / exact head	feat/byos-journey-wave-2 / 5b63e3a7c3643efcd5f3e74f422d095423b55f59
Corrected #122 base	e85382150853cf4774948c55b91e330e7e69617e
Commit ancestry	3a549baf — phase-1 cluster pages and initial metadata-only request flow; e40eebaa — transactional submission, private rows and canonical case anchoring; 5b63e3a7 — role/owner checks, atomic review audit, fail-closed dispatch and N+1 removal
Useful	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.
Unsafe / superseded	Runtime ensure_column/index DDL; identity/participation/access conflation; caller-supplied role/lens authority; unguarded/global loaders; no RLS or staff capability; metadata readiness represented as an approvable empty document; restart purges case-anchored work items; stale work item can approve a resubmitted row; paused/done actions are not capability-gated.
Independent reproductions	Second init_db changed a canonical case-anchored reviewer item from present to absent because participant_id='' matched the orphan cleanup.; After v1 was resolved and v2 resubmitted on the same row, the stale v1 work item approved v2 because review CAS bound only document_id/state, not an immutable submission revision.
Validation	Controlled local run: 535 passed, 9 skipped in 12.471s; focused Wave 2 cluster suite 36/36. No branch check runs, PostgreSQL, browser/accessibility or visual evidence was present.
Destinations	P01-01, P02-03, P03-01, P03-03, P03-05, P09-01, P09-04, P09-06, P11-03, P11-04, P12-02, P13-05
Disposition	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.

Alan reconciliation checkpoint

Field	Contract
Name	Alan review reconciliation: PRs #122/#124 and preserved Wave 2
Status	Complete — independently reconciled 2026-07-12
Blocks	
May amend	WP01, WP03, WP04, WP05, WP06, WP09, WP10, WP11, WP12, WP13
Avoid wasted review	Do not ask Alan to polish code already known to be architecturally obsolete. Complete the safety/defect review at the contract/test level, identify reusable commits/tests, and stop once findings are captured. Obsolete code can be superseded while its findings become replacement acceptance tests.

Finding recording:

- Record each finding with PR/head SHA, file/line or diff anchor, severity, reproducible behavior, architecture/product implication, proposed acceptance test and destination PR ID.
- Mark whether the finding concerns reusable behavior, obsolete architecture, security/privacy, accessibility, migration, concurrency, test coverage or product semantics.
- Update the extraction ledger and canonical programme data together; regenerate all plans/manifest. The benched Project catalogue may be dry-run validated but is not remotely mutated by this reconciliation.

Resolution rule:

- Evidence beats plan text: a verified defect or missing contract amends the package.
- Alan proposes technical alternatives and records a concise dissent where needed; Luke resolves product, lifecycle, privacy, commercial and cross-repository architecture decisions.
- Security/privacy/payment/PHI disagreements that exceed internal authority go to the named external reviewer; implementation remains blocked.

Completion evidence:

- Alan's #122 changes, completed #124 findings and preserved Wave 2 branch are fully captured.

- Every finding has a disposition and destination or explicit no-action rationale.
- Luke and Alan accept the workspace/case, invitation, authorship and PR-boundary contract, or an explicit unresolved decision remains blocked.
- The checkpoint is complete; each implementation item remains independently blocked by its unresolved decisions, dependencies and gate evidence.

Frozen independent recommendation matrix

ID	Recommendation	Disposition	Rationale	Affected WPs / PRs / gates	Decision
REC-01	Treat PR #124 as WP00 extraction material, not a merge/conflict-resolution unit.	Accept	No independently mergeable unit survives the stale base and domain blockers. Preserve concepts/tests through clean replacement work.	WP00; P03-02/03; P04-01/02/03; P05-02/03; P06-02/03; P09-04; P10-02; P11-03/04; P12-02	No — disposition resolved; closure/deletion still requires later team confirmation.
REC-02	Use corrected PR #122 as the invitation/case-shell baseline.	Accept with modification	#122 is authoritative for named tested behavior and reusable tests, not for final entities, runtime DDL, generic fact/proposal authority or the whole journey model.	WP01; WP02; WP03; WP09; WP10; WP11; WP12	No.
REC-03	Node 0 must delegate invitations to shared authority and make issuance/redemption transactionally complete.	Accept with modification	One invitation authority serves separate resource targets and lifecycles; optional profile completion is resumable rather than forced into membership redemption.	WP03 P03-02/03; WP06 P06-02; A1/A2	No; D-AUTH-02 remains for Cognito admission mechanics.
REC-04	Resolve canonical identity, participation, access and authorization before schema encoding.	Accept with modification	Identity binds principal to person; participation records domain relationship; grants carry resource capabilities; authorization uses verified context, ownership and purpose. Lens is presentation only.	P03-02/03/05; P04-01/02; P09-04; P13-05; A1/B1	Resolved as D-AUTH-03.
REC-05	Restrict IP-entered pre-join GC information and remove unverified suitability ranking/gating.	Accept with modification	Allow minimum-necessary relationship/outreach/logistics context only. Health, mental health, substances, citizenship, race, pregnancy, insurance and eligibility normally wait for authenticated GC entry; no provisional value drives critical decisions.	WP04; WP05 P05-02; WP06; WP10; A1/A2	Resolved as D-DATA-03; exact cross-role fields remain D-DATA-01.
REC-06	Correct shared-memory private notes and migrate already-written rows.	Accept with modification	Run a metadata-only inventory first; record zero-row evidence if none. If rows exist, deny shared/Ask reads immediately and migrate narrowly without broadening access.	WP04 P04-02; WP10 P10-02; A1/A2	No; actual affected-row count is execution evidence, not an architecture choice.
REC-07	Preserve Wave 2 work from 5b63e3a rather than reimplementing useful behavior.	Accept with modification	Treat 5b63e3a as the strongest cumulative evidence source, not an implementation baseline. Reuse tests and semantics only after the named contract changes.	WP00; WP03; WP09; WP11; WP12; WP13; A2/B1	No.
REC-08	Separate metadata-only document requests from the full content pipeline.	Accept with modification	A request/readiness marker is not a document. Content state and review begin only after a version exists, and partial merges fail closed.	WP09 P09-01/03/04/05/06; P12-02; A2	D-DOC-01/02 and D-PHI-01 remain open for content.
REC-09	Move foundational RLS and non-owner runtime before real-user PII.	Accept	App-only checks are inadequate for real-user PII given current broad readers. No alternative isolation model has equivalent evidence at lower programme cost.	WP13 P13-05/06/07; B1/B2	D-PHI-01/02 remain for approved scope/controls/approvers, not for whether RLS precedes B1.

ID	Recommendation	Disposition	Rationale	Affected WPs / PRs / gates	Decision
REC-10	Make #122 Python CI required, stable and resistant to bypass.	Defer	Accept the workflow recommendation now, but enforcement choice/evidence is unavailable. Until D-CI-01 resolves, do not call it bypass-resistant or open real-user gates.	WP01 P01-01; A0 and every later app PR	Yes — D-CI-01, owner Luke; options and required readback/negative test are recorded.
REC-11	Add an explicit Node 0 IP-owner guard and correct non-IP behavior.	Accept with modification	Use one subject-derived capability policy across commands and pages; return explicit deny/role destination without creating a case.	WP03 P03-05; WP05 P05-03; A1	No.
REC-12	Bound and type Node 0 inputs and malformed request behavior.	Accept	Use strict enums/max lengths plus body/collection limits and controlled 4xx errors; do not silently truncate.	WP05; WP06; WP09; WP12	No; exact numerical limits are PR acceptance details and D-DOC-02 for files.
REC-13	Move candidate and invitation invariants from app checks into PostgreSQL migrations.	Accept	Business invariants that can be represented in PostgreSQL must remain true independently of application races.	WP02; WP03; WP04; WP05; A1	No.
REC-14	Replace broad public/cloud prefixes with explicit exposure.	Accept	Explicit route identity is the simplest fail-closed control for the current app.	WP03; WP05; WP06; WP09; WP13; A1/A2	No.
REC-15	Project expiry on GET and add token/PDF cache, referrer and indexing protections.	Accept	Read projection and mutation must agree, and sensitive responses must be non-cacheable with restrictive referrer/noindex policy.	WP03; WP05	No.
REC-16	Bound candidate/invitation and memory queries and remove N+1 behavior.	Accept	Filter/limit at the authoritative storage boundary rather than after loading broad rows.	WP05; WP09; WP10; WP12	No.
REC-17	Add candidate retention, archive, correction and deletion behavior.	Accept	Implement the lifecycle in the candidate domain; keep the exact duration as the named bounded privacy decision.	WP04; WP05 P05-02; A1	Yes — D-DATA-02 supplies duration and exception policy before live use.
REC-18	Move #124's large global CSS addition to route-scoped Node 0 styling.	Accept with modification	Port only necessary styles through route scope and released DS contracts; block on interaction/accessibility/security, not cosmetic literal purity.	WP05; WP11; WP12	No.
REC-19	Retain unaddressed claim semantics and one-time share-link rotation.	Accept with modification	Retain token mechanics and low-risk semantics only for a pending free-workspace claim; never layer bearer access onto a paid case.	WP03 P03-02/03; WP06	Resolved by D-AUTH-01.
REC-20	Standardize organization handling on NATALYX_ORG_ID.	Reject	Configuration unification is convenient evidence, not the final authorization model. Organization must be verified from the authenticated/granted resource context.	WP03 P03-05; WP13 P13-05	No.
REC-21	Land the GC at shared /case/home and remove a duplicate GC Node 0 home.	Accept with modification	Keep one shared participant shell pattern, but route by actual resource: free-workspace GC projection before promotion, case home only after paid-case access exists.	WP03; WP06; WP09; WP12	No.

ID	Recommendation	Disposition	Rationale	Affected WPs / PRs / gates	Decision
REC-22	Keep return_to, token-scoped landing, duplicate-home cleanup, symbol cleanup and refreshed DS assets from #124.	Accept with modification	Adopt the behavior and cleanup, but keep redirect validation, resource-specific landing and canonical DS release ownership.	WP03; WP05; WP11; WP12	No.

Assessment detail

REC-01 — Accept

Assessment field	Recorded conclusion
Recommendation	Treat PR #124 as WP00 extraction material, not a merge/conflict-resolution unit.
Supporting evidence	Alan comment 4948149089; #124@84a5a84; merge-tree against #122@e853821 produced about 93 conflict regions across 11 files.
Prior plan treatment	Already warned against merge but left final disposition pending.
Assumptions	Corrected #122 and replacement PRs remain available as source history.
Security, privacy and operational impact	Avoids stale invitation/security behavior and case-shaped Node 0 reaching users; preserves useful tests/copy/assets.
Dependency and migration impact	WP00 ledger; P03/P04/P05/P06/P09/P10/P11/P12 destinations; no branch mutation.
Alternatives considered	Merge whole branch; cherry-pick mixed commits; file-level extraction.
Rationale	No independently mergeable unit survives the stale base and domain blockers. Preserve concepts/tests through clean replacement work.
Affected WPs / PRs / gates	WP00; P03-02/03; P04-01/02/03; P05-02/03; P06-02/03; P09-04; P10-02; P11-03/04; P12-02
Explicit decision required	No — disposition resolved; closure/deletion still requires later team confirmation.

REC-02 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Use corrected PR #122 as the invitation/case-shell baseline.
Supporting evidence	e853821; 499-test report and green workflow; verified return_to, TTL ceiling, CAS, self/existing-member refusal, one-live-GC invariant, transactional join, paired lenses and fail-closed Learn.
Prior plan treatment	Selective salvage, pending reconciliation.
Assumptions	Behavioral tests can be ported without retaining startup DDL or the case-only domain model.
Security, privacy and operational impact	Retains proven concurrency/auth safety while preventing weak participation/provenance/schema semantics becoming programme authority.
Dependency and migration impact	P01-01, P03-01/03/05, P09-04, P11-03/04, P12-02; re-express schema through WP02.
Alternatives considered	Treat all #122 code/schema as canonical; discard it; behavior/test baseline only.
Rationale	#122 is authoritative for named tested behavior and reusable tests, not for final entities, runtime DDL, generic fact/proposal authority or the whole journey model.
Affected WPs / PRs / gates	WP01; WP02; WP03; WP09; WP10; WP11; WP12
Explicit decision required	No.

REC-03 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Node 0 must delegate invitations to shared authority and make issuance/redemption transactionally complete.
Supporting evidence	#124 splits supersede/create/link/timeline and redemption/profile facts across commits; corrected #122 supplies CAS/TTL guards.
Prior plan treatment	Separate workspace claim and paid-case invitation but transaction boundary was underspecified.
Assumptions	Identity-provider interaction may require an idempotent completion workflow around a database transaction/outbox.
Security, privacy and operational impact	Prevents valid orphan tokens, partial membership/onboarding and possession-only case access.
Dependency and migration impact	P03-02 atomically binds workspace claim/identity/consent; P03-03 atomically redeems invite/participation/access/consent/audit; P06-02 resumes optional onboarding idempotently.
Alternatives considered	One huge cross-system transaction; explicit idempotent saga/outbox; current sequential commits.
Rationale	One invitation authority serves separate resource targets and lifecycles; optional profile completion is resumable rather than forced into membership redemption.
Affected WPs / PRs / gates	WP03 P03-02/03; WP06 P06-02; A1/A2
Explicit decision required	No; D-AUTH-02 remains for Cognito admission mechanics.

REC-04 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Resolve canonical identity, participation, access and authorization before schema encoding.
Supporting evidence	#122/Wave 2 account_memberships roles double as party role, lens and authorization; first-active membership selection; no party/person alignment; callers pass viewer role/lens; RLS absent.
Prior plan treatment	Roles, membership and projections were present but their ownership was not explicit.
Assumptions	A principal can be staff globally without participant-data access and a person can participate without an active grant.
Security, privacy and operational impact	Eliminates confused role authority and gives application/RLS one auditable policy input.
Dependency and migration impact	P03-05 adds identity binding, participation and access-grant constraints; later domain migrations and P13-05 consume the contract.
Alternatives considered	Single membership table/role enum; participation-only authorization; separate grants/capabilities.
Rationale	Identity binds principal to person; participation records domain relationship; grants carry resource capabilities; authorization uses verified context, ownership and purpose. Lens is presentation only.
Affected WPs / PRs / gates	P03-02/03/05; P04-01/02; P09-04; P13-05; A1/B1
Explicit decision required	Resolved as D-AUTH-03.

REC-05 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Restrict IP-entered pre-join GC information and remove unverified suitability ranking/gating.
Supporting evidence	#124 scores mental health, substance, citizenship, pregnancy and support claims into 0-100/high_viability/critical_flag and sorts/gates guidance.
Prior plan treatment	Provisional attribution existed but still allowed a preliminary hard-criteria check.
Assumptions	Before GC joins, the legitimate purpose is low-pressure outreach preparation, not eligibility or professional assessment.
Security, privacy and operational impact	Reduces sensitive third-party processing, discrimination and false clinical/eligibility authority.
Dependency and migration impact	P04-02 provenance; P05-02 strict allowlist; P06-02 GC-authored data; P10 scopes; legacy scores retained only as source-labelled evidence or excluded.
Alternatives considered	Keep ranking with disclaimer; professional approval later; neutral questions only.
Rationale	Allow minimum-necessary relationship/outreach/logistics context only. Health, mental health, substances, citizenship, race, pregnancy, insurance and eligibility normally wait for authenticated GC entry; no provisional value drives critical decisions.
Affected WPs / PRs / gates	WP04; WP05 P05-02; WP06; WP10; A1/A2
Explicit decision required	Resolved as D-DATA-03; exact cross-role fields remain D-DATA-01.

REC-06 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Correct shared-memory private notes and migrate already-written rows.
Supporting evidence	#124 saves Conversation Guide notes with scope=shared and Ask loads shared memory; local inspected SQLite had zero affected rows; #124 was not deployed; remote database state was not verified.
Prior plan treatment	New private/shared ledger planned; explicit legacy remediation was missing.
Assumptions	Source kind identifies guide notes and owner can normally be recovered from subject/person provenance.
Security, privacy and operational impact	Prevents IP reflection notes reaching GC/provider/intelligence while preserving audit and avoiding content in logs.
Dependency and migration impact	P04-02 person-owned contract; P10-02 inventory, immediate exclusion, idempotent private/quarantine migration; ambiguous owner quarantines.
Alternatives considered	Behavior-only fix; delete rows; owner-safe migration/quarantine.
Rationale	Run a metadata-only inventory first; record zero-row evidence if none. If rows exist, deny shared/Ask reads immediately and migrate narrowly without broadening access.
Affected WPs / PRs / gates	WP04 P04-02; WP10 P10-02; A1/A2
Explicit decision required	No; actual affected-row count is execution evidence, not an architecture choice.

REC-07 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Preserve Wave 2 work from 5b63e3a rather than reimplementing useful behavior.
Supporting evidence	Exact three-commit stack on e853821; 535 passed/9 skipped and 36 focused tests; two independent defects reproduced; runtime DDL, caller roles, broad loaders and no RLS remain.
Prior plan treatment	Wave 2 was previously missing and blocked final mapping.
Assumptions	The branch remains paused and available for later extraction.
Security, privacy and operational impact	Saves tested presentation/transaction semantics without importing restart loss, stale approval, metadata/content conflation or weak access.
Dependency and migration impact	P03-05, P09-01/04/06, P11-03/04, P12-02, P13-05; schema re-expressed as migrations.
Alternatives considered	Direct merge/cherry-pick; discard; selective adaptation/test port.
Rationale	Treat 5b63e3a as the strongest cumulative evidence source, not an implementation baseline. Reuse tests and semantics only after the named contract changes.
Affected WPs / PRs / gates	WP00; WP03; WP09; WP11; WP12; WP13; A2/B1
Explicit decision required	No.

REC-08 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Separate metadata-only document requests from the full content pipeline.
Supporting evidence	Wave 2 creates an approvable documents row with storage_uri empty; request labels may contain sensitive data; P09 baseline coupled active-case UI to full upload pipeline.
Prior plan treatment	P09-04 depended on P09-03 and mixed progress/documents; no independent requirement/work-item model.
Assumptions	Phase-1 readiness coordination can operate synthetically without receiving file bytes.
Security, privacy and operational impact	Avoids false artifact approval and permits safe partial delivery while content remains quarantined/disabled.
Dependency and migration impact	P09-01 defines both machines; P09-04 owns metadata requirements/work items and phase-1 surfaces without P09-03; P09-06 integrates content versions after P09-03; P09-05 depends P09-06.
Alternatives considered	Single documents table; metadata-only workflow first; full pipeline first.
Rationale	A request/readiness marker is not a document. Content state and review begin only after a version exists, and partial merges fail closed.
Affected WPs / PRs / gates	WP09 P09-01/03/04/05/06; P12-02; A2
Explicit decision required	D-DOC-01/02 and D-PHI-01 remain open for content.

REC-09 — Accept

Assessment field	Recorded conclusion
Recommendation	Move foundational RLS and non-owner runtime before real-user PII.
Supporting evidence	Runtime uses owner-level startup DDL; app/Wave 2 storage has broad/global loaders; no DB policy enforcement or pool-context proof.
Prior plan treatment	P13-05 was a B2 prerequisite only.
Assumptions	Synthetic A1/A2 may validate features before B1, but no real subject data is admitted.
Security, privacy and operational impact	Database denies cross-org/resource/owner access if an app guard fails; reduces B1 breach blast radius.
Dependency and migration impact	P13-05/06/07 move to B1; FORCE RLS, NOBYPASSRLS runtime, transaction-local verified context and pool reset tests; B2 remains separate evidence approval.
Alternatives considered	App-only isolation; separate databases per cohort; RLS at B2.
Rationale	App-only checks are inadequate for real-user PII given current broad readers. No alternative isolation model has equivalent evidence at lower programme cost.
Affected WPs / PRs / gates	WP13 P13-05/06/07; B1/B2
Explicit decision required	D-PHI-01/02 remain for approved scope/controls/approvers, not for whether RLS precedes B1.

REC-10 — Defer

Assessment field	Recorded conclusion
Recommendation	Make #122 Python CI required, stable and resistant to bypass.
Supporting evidence	#122 workflow passed; #124 ran only build/contract checks; main lacks Python/PostgreSQL CI; private-repo protection/ruleset API returned plan-level 403.
Prior plan treatment	P01-01 added tests but did not define stable contexts, workflow hardening or enforcement limitation.
Assumptions	Repository billing/visibility may change independently of code.
Security, privacy and operational impact	Prevents test-path bypass and false governance claims; adds PostgreSQL evidence.
Dependency and migration impact	P01-01: no path filters, stable names, pinned action SHAs, supported runtime, constraints, timeouts, PostgreSQL job; D-CI-01 governs repository enforcement/readback.
Alternatives considered	Procedural review only; GitHub plan/ruleset upgrade; externally enforced merge queue.
Rationale	Accept the workflow recommendation now, but enforcement choice/evidence is unavailable. Until D-CI-01 resolves, do not call it bypass-resistant or open real-user gates.
Affected WPs / PRs / gates	WP01 P01-01; A0 and every later app PR
Explicit decision required	Yes — D-CI-01, owner Luke; options and required readback/negative test are recorded.

REC-11 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Add an explicit Node 0 IP-owner guard and correct non-IP behavior.
Supporting evidence	#124 ensure endpoint lets any authenticated subject mint a primary-IP case; viewer/provider/staff paths can fall into storage errors/500.
Prior plan treatment	General workspace authorization existed without a shared participant/API guard acceptance matrix.
Assumptions	Staff tools are separate from participant Node 0 routes.
Security, privacy and operational impact	Prevents privilege creation, role confusion and avoidable errors.
Dependency and migration impact	P03-05 subject-derived capabilities; P05-03 shared IP-owner guard and negative role tests.
Alternatives considered	Per-route checks; shared router dependency/service policy; redirect all roles.
Rationale	Use one subject-derived capability policy across commands and pages; return explicit deny/role destination without creating a case.
Affected WPs / PRs / gates	WP03 P03-05; WP05 P05-03; A1
Explicit decision required	No.

REC-12 — Accept

Assessment field	Recorded conclusion
Recommendation	Bound and type Node 0 inputs and malformed request behavior.
Supporting evidence	#124 hand-parses unbounded names/notes/questions/JSON/forms/collections/PDF content; Wave 2 silently truncates labels/notes.
Prior plan treatment	Validation was implicit and spread across feature PRs.
Assumptions	Limits can be product-specific but must reject rather than silently alter user data.
Security, privacy and operational impact	Controls denial-of-service, log/storage amplification and ambiguous stored content.
Dependency and migration impact	P05-02 typed candidate commands/collections; P05-03 form/PDF/body limits; P06-02 onboarding; P09-04 structured metadata/reviewer note; P12-01 Ask.
Alternatives considered	Global proxy limit only; silent slicing; layered strict validation.
Rationale	Use strict enums/max lengths plus body/collection limits and controlled 4xx errors; do not silently truncate.
Affected WPs / PRs / gates	WP05; WP06; WP09; WP12
Explicit decision required	No; exact numerical limits are PR acceptance details and D-DOC-02 for files.

REC-13 — Accept

Assessment field	Recorded conclusion
Recommendation	Move candidate and invitation invariants from app checks into PostgreSQL migrations.
Supporting evidence	#124 lacks candidate-case FK, invitation-link uniqueness, one-active-outreach constraint and checks for enums/scores; concurrent writes can violate product rules.
Prior plan treatment	WP04/WP05 proposed new schema but did not enumerate all reviewed invariants.
Assumptions	Versioned migrations and expand/contract compatibility precede domain tables.
Security, privacy and operational impact	Prevents orphan/duplicate/contradictory state under concurrency.
Dependency and migration impact	P04-01 FKs/uniqueness; P05-02 outreach constraints and no score bounds because ranking is removed; P03-02/03 claim/invite uniqueness; P02-03 migrations.
Alternatives considered	App locks only; advisory locks; database constraints plus commands.
Rationale	Business invariants that can be represented in PostgreSQL must remain true independently of application races.
Affected WPs / PRs / gates	WP02; WP03; WP04; WP05; A1
Explicit decision required	No.

REC-14 — Accept

Assessment field	Recorded conclusion
Recommendation	Replace broad public/cloud prefixes with explicit exposure.
Supporting evidence	#124 exposes /node0/onboarding/ publicly and /api/node0/ by prefix; Wave 2 demonstrates exact phase-1 allowlisting.
Prior plan treatment	Fail-closed route policy existed but did not assign this finding.
Assumptions	New routes require explicit security review before cloud/public reachability.
Security, privacy and operational impact	Prevents future routes inheriting token/public or cloud exposure accidentally.
Dependency and migration impact	P03-02, P05-03, P06-02 exact registries/dependencies; P09-04 exact cluster paths; P13-03 route-negative inventory.
Alternatives considered	Prefix allowlist; per-route registry; separate public app.
Rationale	Explicit route identity is the simplest fail-closed control for the current app.
Affected WPs / PRs / gates	WP03; WP05; WP06; WP09; WP13; A1/A2
Explicit decision required	No.

REC-15 — Accept

Assessment field	Recorded conclusion
Recommendation	Project expiry on GET and add token/PDF cache, referrer and indexing protections.
Supporting evidence	#124 expired links look active until POST; token-bearing pages/PDFs lack no-store and restrictive referrer policy.
Prior plan treatment	Token transitions were covered, response exposure was not.
Assumptions	Tokens remain URL-borne only where unavoidable and never appear in telemetry.
Security, privacy and operational impact	Reduces stale-action confusion and token/sensitive-output leakage via browser cache, referrer or indexing.
Dependency and migration impact	P03-02/03 GET+command expiry and response headers; P05-03 outreach PDF headers; security tests.
Alternatives considered	POST-only check; session exchange; header hardening plus one-time token.
Rationale	Read projection and mutation must agree, and sensitive responses must be non-cacheable with restrictive referrer/noindex policy.
Affected WPs / PRs / gates	WP03; WP05
Explicit decision required	No.

REC-16 — Accept

Assessment field	Recorded conclusion
Recommendation	Bound candidate/invitation and memory queries and remove N+1 behavior.
Supporting evidence	#124 looks up invitations per candidate and loads all shared memory before Python filtering; Wave 2 removes one reviewer-inbox N+1 but retains broad loaders.
Prior plan treatment	Performance was a general risk rather than explicit acceptance criteria.
Assumptions	Pagination and purpose-scoped limits are compatible with Node 0 and Ask UX.
Security, privacy and operational impact	Bounds latency/memory, narrows accidental data access and improves tenant isolation.
Dependency and migration impact	P05-02 batched query-shaped hydration/indexes; P10-02 scope/subject limits at persistence; P12-01 bounded context; P09-04 owner-scoped query shapes.
Alternatives considered	Cache current readers; batch only; purpose-scoped SQL.
Rationale	Filter/limit at the authoritative storage boundary rather than after loading broad rows.
Affected WPs / PRs / gates	WP05; WP09; WP10; WP12
Explicit decision required	No.

REC-17 — Accept

Assessment field	Recorded conclusion
Recommendation	Add candidate retention, archive, correction and deletion behavior.
Supporting evidence	#124 stores sensitive third-party assessments without a lifecycle; current plan has D-DATA-02 but implementation mapping was weak.
Prior plan treatment	Retention decision and legacy backfill existed.
Assumptions	Legal/privacy may set the exact period; source/audit may require tombstones rather than raw-content retention.
Security, privacy and operational impact	Reduces unnecessary third-party data retention while preserving correction/audit needs.
Dependency and migration impact	D-DATA-02; P04-02/03 provenance/tombstones; P05-02 archive/anonymize/delete command/job and evidence.
Alternatives considered	Indefinite retention; immediate deletion; shortest justified period with tombstone/audit.
Rationale	Implement the lifecycle in the candidate domain; keep the exact duration as the named bounded privacy decision.
Affected WPs / PRs / gates	WP04; WP05 P05-02; A1
Explicit decision required	Yes — D-DATA-02 supplies duration and exception policy before live use.

REC-18 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Move #124's large global CSS addition to route-scoped Node 0 styling.
Supporting evidence	About 3,030 lines were added to global style.css; participant DS and route-specific recipes exist.
Prior plan treatment	P05-03 and P11-03 mentioned DS adoption but not the global-style finding.
Assumptions	Participant and internal/operator design systems remain separate.
Security, privacy and operational impact	Limits cross-surface regressions and clarifies style ownership without making aesthetic purity a release gate.
Dependency and migration impact	P05-03 node0.css; P11-03 released tokens/recipes; P12-02 visual/a11y verification.
Alternatives considered	Keep global block; CSS modules/build; route stylesheet plus shared DS.
Rationale	Port only necessary styles through route scope and released DS contracts; block on interaction/accessibility/security, not cosmetic literal purity.
Affected WPs / PRs / gates	WP05; WP11; WP12
Explicit decision required	No.

REC-19 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Retain unaddressed claim semantics and one-time share-link rotation.
Supporting evidence	#124 gates allow_unaddressed to GC intent, stores hash only, rotates/supersedes, and displays plaintext once; corrected #122 supplies case-invite guards.
Prior plan treatment	Workspace claim separated conceptually but shared engine/resource-target distinction was incomplete.
Assumptions	Bearer possession cannot reveal sensitive data or create paid-case membership.
Security, privacy and operational impact	Keeps low-friction outreach while bounding replay, forwarding and privilege escalation.
Dependency and migration impact	P03-02 workspace claim target; P03-03 recipient-bound case target; shared token/CAS primitives, distinct state machines and consent.
Alternatives considered	Addressed links only; unaddressed case invite; unaddressed low-risk workspace claim.
Rationale	Retain token mechanics and low-risk semantics only for a pending free-workspace claim; never layer bearer access onto a paid case.
Affected WPs / PRs / gates	WP03 P03-02/03; WP06
Explicit decision required	Resolved by D-AUTH-01.

REC-20 — Reject

Assessment field	Recorded conclusion
Recommendation	Standardize organization handling on NATALYX_ORG_ID.
Supporting evidence	#124 restores a configured organization constant, but the intended product is multi-resource/multi-tenant and Wave 2 reviewers/readers are insufficiently org-scoped.
Prior plan treatment	Organization scope is part of case/workspace contracts and RLS.
Assumptions	A fixed organization may remain a synthetic/dev bootstrap default only.
Security, privacy and operational impact	Prevents a deployment constant becoming tenant/access authority.
Dependency and migration impact	P03-05 and P13-05 derive organization from verified identity/grant/resource context; config default permitted only in synthetic bootstrap.
Alternatives considered	Global configured org; identity claim; resource/grant-derived org context.
Rationale	Configuration unification is convenient evidence, not the final authorization model. Organization must be verified from the authenticated/granted resource context.
Affected WPs / PRs / gates	WP03 P03-05; WP13 P13-05
Explicit decision required	No.

REC-21 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Land the GC at shared /case/home and remove a duplicate GC Node 0 home.
Supporting evidence	#122 has paired case lenses; #124 routes GC to case home after bearer redemption even though Node 0 should not yet create a case.
Prior plan treatment	Free workspace and paid case are separate but landing wording was not explicit.
Assumptions	The GC may join a free workspace before any promotion and later receive a paid-case invitation.
Security, privacy and operational impact	Avoids duplicate participant shells and prevents pre-promotion users from being shown a canonical-case fiction.
Dependency and migration impact	P03-02/P06 workspace lens before promotion; P03-03/P09-04 shared case home only after canonical paid-case grant; P12 cutover removes duplicates.
Alternatives considered	Always case home; duplicate GC Node 0 home; shared shell with resource-specific projection.
Rationale	Keep one shared participant shell pattern, but route by actual resource: free-workspace GC projection before promotion, case home only after paid-case access exists.
Affected WPs / PRs / gates	WP03; WP06; WP09; WP12
Explicit decision required	No.

REC-22 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Keep return_to, token-scoped landing, duplicate-home cleanup, symbol cleanup and refreshed DS assets from #124.
Supporting evidence	Corrected #122 validates internal return_to; #124 provides one-time token landing and presentation cleanup; canonical participant DS owns released assets/tokens.
Prior plan treatment	Most items were salvage candidates without precise ownership.
Assumptions	DS assets are ported from an approved release, not vendored ad hoc.
Security, privacy and operational impact	Preserves UX value and resume behavior without weakening redirect, token or repository boundaries.
Dependency and migration impact	P03-01 return_to; P03-02 token landing; P11-03 DS release; P05-03/P12-02 selective UI cleanup.
Alternatives considered	Copy branch files; discard polish; selective port from approved contracts.
Rationale	Adopt the behavior and cleanup, but keep redirect validation, resource-specific landing and canonical DS release ownership.
Affected WPs / PRs / gates	WP03; WP05; WP11; WP12
Explicit decision required	No.

Protected local worktrees

Repository	Path	Branch/head	Protected content	Destination	Safety rule
app	`/Users/lukerhodes/Natalyx/natalyx-app`	feature/provider-exchange-layer-v0 / 84a5a84	substantial uncommitted provider handoff/ops/document work	P09-01/P09-03/P09-05 after provenance/security review	Do not stage, reset, stash, switch, commit, clean or overwrite.
app	`/Users/lukerhodes/.codex/worktrees/1b3f/natalyx-app`	feature/node0-surrogate-onboarding-link / 529d377	substantial uncommitted onboarding/invitation work	P03-02/P03-03/P06-02	Preserve until file/test extraction is signed off.
app	`/Users/lukerhodes/.codex/worktrees/39b2/natalyx-app`	detached candidate rewrite / 9605d88	substantial uncommitted Node 0 quiz rewrite	P04-02/P05-02	Preserve; compare with predecessor and #117/#124 before any discard.
app	`/Users/lukerhodes/.codex/worktrees/6233/natalyx-app`	detached predecessor candidate model / dc6a2f0	substantial uncommitted predecessor model	P04-02/P05-02 if unique	Preserve until rewrite diff/extraction ledger is complete.
learn	`/Users/lukerhodes/Natalyx/natalyx-learn-resources`	feature/learn-resource-rendering-contract / 9c9a9c2	substantial uncommitted builder/schema/templates/resource work	P11-01	Do not update/switch/clean; create a future isolated extraction worktree.
learn	`/Users/lukerhodes/.codex/worktrees/9dfc/natalyx-learn-resources`	feature/node0-beta-learn-resources / e1fb2d6	uncommitted Node 0 Learn implementation	P11-01 after provenance/no-fake-content review	Preserve until compared with rendering-contract branch.
intelligence	`/Users/lukerhodes/Natalyx/natalyx-intelligence`	main (behind origin/main) / 9d824e8	dirty Learn registry/corpus plus uv.lock	P11-02/P10-03 as appropriate	Do not pull/reset/stash; extract in isolated worktree later.
design-system	`/Users/lukerhodes/Natalyx/Natalyx Design System`	feature/ds-consumer-parity-guidance / 4656426	dirty brand assets beyond PR #8	P11-03 after asset review	Remote PR and local asset delta are separate evidence; do not overwrite.
internal-design-system	`/Users/lukerhodes/Natalyx/Natalyx Internal View Design System`	main / current local	substantial uncommitted operator shell/design-system work	Not part of participant DS; review only if P09-05 operator UI needs approved internal recipes	Preserve repo boundary and dirty worktree.

Cleanup tasks

ID	Task	Dependencies	Owner
CLEAN-01	Verify extraction completeness before any PR closure	CHECK-ALAN-122-124	Joint
CLEAN-02	Close superseded PRs only after replacement evidence is reachable	CLEAN-01	Luke
CLEAN-03	Delete remote branches/worktrees only after closure and team confirmation	CLEAN-02	Alan

Data authorship and provenance

Class definitions:

Class	Meaning
1	IP-owned observation or relationship context
2	Provisional IP-supplied candidate information
3	GC-authored or GC-confirmed information
4	GC-only private information
5	Provider- or reviewer-confirmed information

Pathfinder known-carrier/route answers

Element	Pathfinder known-carrier/route answers
Class	1
Subject	IP journey
Current author	IP
Storage	participants.intake_data and pathfinder_context
Provenance	participant settings; inconsistent per-field source
Visibility	IP/app routing; legacy reviewer surfaces
Verification	Not applicable as GC truth
Correction/replacement	IP appends/amends route-answer revision
GC sees it?	No by default
IP sees GC answer?	N/A
Downstream influence	Entry routing only
Promotion behavior	Route/source reference only
Recommended contract	Never create a GC fact, entitlement, membership or case.

IP preferences about a GC/journey (location, communication, involvement, reduction/twins, relationship)

Element	IP preferences about a GC/journey (location, communication, involvement, reduction/twins, relationship)
Class	1
Subject	IP preferences
Current author	IP
Storage	legacy participant intake_data
Provenance	participant-supplied, no field history
Visibility	IP; some selected axes in frozen matching UI
Verification	IP preference, not eligibility
Correction/replacement	IP revision
GC sees it?	Only an approved purpose-limited projection
IP sees GC answer?	Only separate GC preference projection
Downstream influence	Routing/compatibility discussion
Promotion behavior	As IP preference with author
Recommended contract	Race-based preference is prohibited; sensitive/clinical preferences do not enter pre-join candidate ranking or gates.

Candidate nickname/name and pronouns

Element	Candidate nickname/name and pronouns
Class	2
Subject	Prospective GC
Current author	IP/co-IP
Storage	node0_candidates columns
Provenance	row creator only; no field history
Visibility	IP/co-IP only
Verification	Unverified
Correction/replacement	IP revision; GC identity later supersedes reference
GC sees it?	No current path
IP sees GC answer?	GC-authored identity after explicit share
Downstream influence	Candidate list/guide
Promotion behavior	Link selected candidate reference only
Recommended contract	Paid identity comes from authenticated GC/person record.

Relationship to prospective GC

Element	Relationship to prospective GC
Class	1
Subject	IP-GC relationship
Current author	IP
Storage	node0_candidates.responses_json q1_2
Provenance	Implicit IP
Visibility	IP private
Verification	Observation
Correction/replacement	IP revision; GC may provide separate relationship answer
GC sees it?	No by default
IP sees GC answer?	Approved relationship projection
Downstream influence	Outreach route
Promotion behavior	Optional IP-attributed context
Recommended contract	Conflicting answers coexist; neither silently overwrites.

Pregnancy history, family-complete, citizenship and age

Element	Pregnancy history, family-complete, citizenship and age
Class	2 prohibited pre-join
Subject	Prospective GC
Current author	IP
Storage	legacy responses_json q2_1-q2_4 only
Provenance	Implicit IP; copied to guide packet
Visibility	Quarantined legacy evidence
Verification	Unverified third-party assertion
Correction/replacement	GC class-3/4 revision; reviewer class-5 later
GC sees it?	No pre-join prompt derived from IP assertion
IP sees GC answer?	Only purpose-limited reviewed status
Downstream influence	None in replacement flow
Promotion behavior	Exclude raw values
Recommended contract	Do not collect from the IP in the replacement pre-join flow; never score, rank, gate or project.

Gestational-health or systemic-health assertions

Element	Gestational-health or systemic-health assertions
Class	2 prohibited pre-join
Subject	Prospective GC health
Current author	IP
Storage	legacy responses_json q3_1-q3_2 and guide packet only
Provenance	Implicit IP
Visibility	Quarantined legacy evidence
Verification	Unverified/sensitive
Correction/replacement	GC private self-report; provider evidence class 5
GC sees it?	No pre-join prompt derived from IP assertion
IP sees GC answer?	Only reviewed coordination outcome
Downstream influence	None in replacement flow
Promotion behavior	Exclude
Recommended contract	Do not collect from the IP; never use for guide, provider material, scoring, ranking or critical decisions.

Substance, schedule, support and stability assertions

Element	Substance, schedule, support and stability assertions
Class	1 logistics / 2 prohibited sensitive
Subject	Prospective GC circumstances
Current author	IP
Storage	legacy responses_json q4_1/q4_3/q4_4/q5_5
Provenance	Implicit IP
Visibility	IP private; sensitive legacy values quarantined
Verification	Unverified
Correction/replacement	GC self-report revision
GC sees it?	No sensitive pre-join prompt
IP sees GC answer?	Only explicit progress/status projection
Downstream influence	Neutral outreach/logistics questions only
Promotion behavior	Exclude raw
Recommended contract	Schedule/outreach logistics may produce neutral questions; substance, mental-health, household and eligibility assertions are not collected from the IP.

Candidate state and derived proximity

Element	Candidate state and derived proximity
Class	2
Subject	Prospective GC location
Current author	IP/system derivative
Storage	responses_json q4_2 plus IP profile state
Provenance	IP and derived-from missing
Visibility	IP private
Verification	Unverified
Correction/replacement	GC class-3 state revision
GC sees it?	Optional labelled confirmation
IP sees GC answer?	GC-shared state as separate revision
Downstream influence	Score/route
Promotion behavior	GC-authored state only when permitted
Recommended contract	Derived value inherits weakest source status.

Worldview, communication, power, tested relationship and privacy observations

Element	Worldview, communication, power, tested relationship and privacy observations
Class	1
Subject	Relationship
Current author	IP
Storage	responses_json q5_1-q5_6
Provenance	Implicit IP
Visibility	IP private
Verification	Personal observation
Correction/replacement	IP revision; GC separate answer
GC sees it?	No automatic visibility
IP sees GC answer?	Only GC-approved projection
Downstream influence	Private outreach guide
Promotion behavior	Exclude unless expressly shared
Recommended contract	Suitable for private reflection, not eligibility.

GC spouse/household assertion

Element	GC spouse/household assertion
Class	2
Subject	GC/third party
Current author	IP
Storage	responses_json q5_7
Provenance	Implicit IP
Visibility	IP private
Verification	Unverified
Correction/replacement	GC private revision
GC sees it?	No default
IP sees GC answer?	Purpose-limited status only
Downstream influence	Conversation prompt
Promotion behavior	Exclude
Recommended contract	No provider/case truth.

Observed openness/signals

Element	Observed openness/signals
Class	1
Subject	IP-GC interaction
Current author	IP
Storage	responses_json q5_8
Provenance	Implicit IP
Visibility	IP private
Verification	Observation
Correction/replacement	IP revision
GC sees it?	No default
IP sees GC answer?	N/A
Downstream influence	Low-pressure outreach route
Promotion behavior	Exclude or retain IP-private
Recommended contract	Never imply GC consent/willingness.

Legacy archetype, score, status, score suppression, things-to-uncover and guide route

Element	Legacy archetype, score, status, score suppression, things-to-uncover and guide route
Class	2 derived legacy
Subject	Prospective GC
Current author	System from IP input
Storage	legacy node0_candidates derived columns/JSON
Provenance	No derived_from/source version
Visibility	Excluded from replacement projections
Verification	Unverified derivative
Correction/replacement	Do not recompute in replacement flow
GC sees it?	No
IP sees GC answer?	N/A
Downstream influence	None in replacement flow
Promotion behavior	Exclude
Recommended contract	Remove ranking, viability/status flags and eligibility/guide gating; only allow neutral questions from approved non-sensitive inputs.

Guide evidence/scaffold/snapshot/PDF

Element	Guide evidence/scaffold/snapshot/PDF
Class	2 derived
Subject	Outreach about GC
Current author	System for IP
Storage	candidate guide JSON/PDF
Provenance	Raw IP responses copied
Visibility	IP private
Verification	Unverified
Correction/replacement	New immutable version on source change
GC sees it?	No automatic visibility
IP sees GC answer?	N/A
Downstream influence	Conversation preparation/Ask
Promotion behavior	Exclude by default
Recommended contract	Workspace-private; never provider/canonical truth.

Candidate ordering/outreach state/decline

Element	Candidate ordering/outreach state/decline
Class	1/2
Subject	IP workflow and GC response
Current author	IP, GC or system
Storage	candidate row plus timeline
Provenance	Single enum obscures actor/reason
Visibility	IP; GC only own flow
Verification	Workflow evidence
Correction/replacement	Append actor/reason transition
GC sees it?	Own invitation status
IP sees GC answer?	Purpose-limited response status
Downstream influence	Outreach workflow
Promotion behavior	Selected relationship history only
Recommended contract	Decline is not eligibility; retain append-only actor.

Invited email and IP message

Element	Invited email and IP message
Class	2/1
Subject	Prospective GC contact / IP message
Current author	IP
Storage	invitations
Provenance	Created-by subject
Visibility	Issuer/internal; token landing
Verification	Email unverified until identity bind
Correction/replacement	Revoke/reissue; mismatch confirmation
GC sees it?	Own invitation/message
IP sees GC answer?	Claim status, not private answers
Downstream influence	Identity claim
Promotion behavior	Membership audit reference only
Recommended contract	Original recipient assertion remains in audit; do not silently rewrite.

Redeemed subject/email and GC decline

Element	Redeemed subject/email and GC decline
Class	3
Subject	GC/account
Current author	Authenticated GC
Storage	invitation claim/redemption
Provenance	Verified account event
Visibility	Issuer sees bounded status; operator audit
Verification	Identity-bound, not professional truth
Correction/replacement	New claim/identity-review event
GC sees it?	Own status
IP sees GC answer?	Bounded status/email per policy
Downstream influence	Workspace/case membership
Promotion behavior	Purpose/membership event only
Recommended contract	No clinical/eligibility meaning.

GC name, preferred name and state

Element	GC name, preferred name and state
Class	3
Subject	GC
Current author	Authenticated GC
Storage	people plus gc_onboarding:first_run_profile
Provenance	Broad user-supplied
Visibility	Currently shared fact
Verification	Self-attested
Correction/replacement	GC revision/history UI
GC sees it?	Yes
IP sees GC answer?	Approved identity/state projection
Downstream influence	Profile/routing/onboarding
Promotion behavior	Preserve GC author and self-attested status
Recommended contract	Revises, never erases, IP assertion.

GC relationship/familiarity/readiness

Element	GC relationship/familiarity/readiness
Class	3
Subject	GC/relationship
Current author	GC
Storage	shared first_run_profile
Provenance	User-supplied
Visibility	Shared latent; no clear current UI
Verification	Self-attested familiarity, not eligibility
Correction/replacement	GC revision
GC sees it?	Yes
IP sees GC answer?	Only approved progress/relationship projection
Downstream influence	Onboarding/readiness presentation
Promotion behavior	Preserve source/meaning
Recommended contract	Do not label as medical/overall readiness.

GC email, phone, best time and contact preference

Element	GC email, phone, best time and contact preference
Class	4
Subject	GC
Current author	GC/account
Storage	private contact_preferences; preference duplicated shared
Provenance	User/account supplied
Visibility	GC owner/internal purpose
Verification	Self-attested; email account-bound
Correction/replacement	GC revision
GC sees it?	Yes
IP sees GC answer?	Minimal operational projection only
Downstream influence	Contact coordination
Promotion behavior	Private; project only explicit contact channel
Recommended contract	Remove shared duplication and audit every projection.

GC case-profile consent

Element	GC case-profile consent
Class	3
Subject	GC
Current author	GC
Storage	case_profile_consent/shared fact in PR #124
Provenance	User-supplied
Visibility	Status may be shared; evidence private/audit
Verification	Authenticated acceptance
Correction/replacement	Revoke/new consent version
GC sees it?	Yes
IP sees GC answer?	Purpose/status only
Downstream influence	Projection/membership authorization
Promotion behavior	Carry consent ID/version/status, not form content
Recommended contract	Consent is purpose/version specific and revocable.

GC direct clinical/social screening answers

Element	GC direct clinical/social screening answers
Class	4 then 5
Subject	GC
Current author	GC; later reviewer/provider
Storage	legacy GC intake_data
Provenance	No per-field provenance/history
Visibility	GC-private/reviewer; some legacy summaries
Verification	Self-report until evidence review
Correction/replacement	GC revision; dedicated class-5 evidence revision
GC sees it?	Yes
IP sees GC answer?	Reviewed coordination status only
Downstream influence	Eligibility/readiness/provider review
Promotion behavior	Private self-report with separate verified revisions
Recommended contract	Never collected/answered authoritatively by IP or upgraded by generic proposal.

GC preferences and boundaries

Element	GC preferences and boundaries
Class	3 or 4
Subject	GC
Current author	GC
Storage	legacy intake_data / future authored assertions
Provenance	User-supplied
Visibility	Per field/purpose
Verification	Self-attested
Correction/replacement	GC revision
GC sees it?	Yes
IP sees GC answer?	Only explicit compatibility/coordination projection
Downstream influence	Relationship discussion/routing
Promotion behavior	Preserve field-level visibility
Recommended contract	No blanket sharing.

GC insurance identifiers and answers

Element	GC insurance identifiers and answers
Class	4 then 5
Subject	GC
Current author	GC; reviewer/provider confirms
Storage	legacy intake_data / future document-fact rail
Provenance	Currently broad
Visibility	GC/private reviewer
Verification	Self-report then reviewed
Correction/replacement	GC/provider evidence revision
GC sees it?	Yes
IP sees GC answer?	Coverage/review status only
Downstream influence	Insurance coordination/provider packet
Promotion behavior	Raw identifiers stay private
Recommended contract	Never promise coverage; class 5 evidence linked.

IP Conversation Guide notes

Element	IP Conversation Guide notes
Class	1
Subject	IP relationship intent
Current author	IP
Storage	currently memory_records.scope=shared
Provenance	IP but scope unsafe
Visibility	Future all-case-member risk
Verification	Private note
Correction/replacement	IP revision/delete per retention
GC sees it?	No by default
IP sees GC answer?	N/A
Downstream influence	Ask/guide
Promotion behavior	Off by default
Recommended contract	Change default to IP-private; explicit share creates a new projection.

Ask context and AI proposals about GC

Element	Ask context and AI proposals about GC
Class	2 until 3/5 evidence
Subject	GC/case
Current author	AI from mixed sources
Storage	known_intake_state, proposals, memory/facts
Provenance	Insufficiently typed
Visibility	Depends caller/payload
Verification	Proposal only
Correction/replacement	Reject or evidence-linked revision
GC sees it?	Only scope-authorized output
IP sees GC answer?	Only scope-authorized output
Downstream influence	Guidance, case facts, provider prep
Promotion behavior	Never upgrades class
Recommended contract	Every datum carries subject/author/class/source/allowed use; generic approval cannot create class 5.

GC self-authored case facts

Element	GC self-authored case facts
Class	3/4
Subject	GC
Current author	GC
Storage	facts with user-supplied provenance
Provenance	Owner/created_by but no author role/supersedes
Visibility	Private/shared broad
Verification	Self-attested
Correction/replacement	Append superseding revision
GC sees it?	Yes
IP sees GC answer?	Only shared purpose projection
Downstream influence	Case/Ask/readiness
Promotion behavior	Preserve visibility/verification
Recommended contract	Add field-level author, revision and verification.

Reviewer/provider conclusions

Element	Reviewer/provider conclusions
Class	5
Subject	GC/case
Current author	Authorized reviewer/provider
Storage	future evidence rail; current broad fact provenance allows it
Provenance	Must include evidence/source/authority
Visibility	Purpose/role scoped
Verification	Confirmed
Correction/replacement	New evidence-linked revision
GC sees it?	Per policy
IP sees GC answer?	Approved coordination status
Downstream influence	Eligibility, routing, provider and milestone evidence
Promotion behavior	Retain class/evidence
Recommended contract	Only dedicated role-gated evidence rail can create class 5.

Provider packet facts/documents

Element	Provider packet facts/documents
Class	3/5 plus consent
Subject	GC/provider purpose
Current author	GC/operator/provider sources
Storage	dirty handoff packet/snapshot work
Provenance	Current selector too broad
Visibility	Named recipient/purpose only
Verification	Per included item
Correction/replacement	New snapshot/revision/revocation
GC sees it?	Preview/consent
IP sees GC answer?	No raw packet unless policy permits
Downstream influence	Provider handoff
Promotion behavior	Immutable purpose-bound snapshot
Recommended contract	Class 1/2 excluded; class 4 explicit GC selection; exact source/evidence required.

Global contract:

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

Concise answer for Alan

The intended parent answers today because Node 0 was built as a private pre-contact reflection tool: relationship context and observed signals can tailor a low-pressure conversation guide before the prospective GC has an account. That limited use is reasonable. What is not reasonable is treating those answers as if the GC made or confirmed them, especially health, age, citizenship, family status, insurance, consent, preferences or eligibility. Those must be GC-authored, with reviewer/provider confirmation where required. When the GC joins, her answer appends an attributable revision; the earlier IP statement remains labelled as IP-reported/unconfirmed, GC-private data stays private, and only an explicit purpose-limited projection reaches the IP. The current score/invite gating and missing reconciliation should be replaced.

Migration architecture

Options:

1. Alembic with reviewed operations/raw SQL
2. Custom ordered Python migrations with a demonstrated reason not to use an established framework
3. A standalone established runner such as Flyway/dbmate if its runtime/deployment burden fits

Provisional recommendation: Founder constraints are final: Alan should prefer Alembic or another well-supported low-maintenance framework, while retaining authority to choose a different established runner or justify a custom runner in the ADR.

- Zero intended data loss is the governing reliability objective.
- PostgreSQL is production authority; SQLite may remain a convenient synthetic local-development projection without full production parity.
- Fingerprint existing databases before baseline stamping; never blindly stamp unknown schema.

- App and intelligence use separate revision namespaces and PostgreSQL advisory-lock keys.
- Migration identities hold DDL; app/intelligence runtime identities do not after adoption.
- Run migrations in a deploy-controlled one-off task before app rollout, not opportunistically from web startup.
- Stage consequential changes and keep them backward-compatible wherever reasonably possible; use expand in release N, bounded/backpressure-aware backfill, dual-compatible N+1, then contract only when N cannot run.
- CI covers fresh, current, one-revision-behind, N-1 app on expanded schema, concurrent runner, checksum drift and interrupted backfill.
- Every consequential migration has tested recovery or rollback evidence before use.
- Scheduled and communicated beta downtime may be up to 30 minutes when a tested recovery plan exists.
- Irreversible changes that could affect real-user data require explicit evidence and Luke approval plus a forward-repair plan. Schema rollback is not automatic.

Promotion, commercial and payment contract

Sole authorization: `byos.workspace_promotion_authorized.v1`. Producer: natalyx-app commitment/commercial domain after deterministic validation. Consumer: natalyx-app promotion service.

Payload:

```
event_id
idempotency_key
workspace_id
commitment_id
owner_person_id
owner_subject
commercial_config_version
terms_acceptance_ids[]
payment_plan_id
funding_evidence.kind
funding_evidence.commercial_event_id
funding_evidence.settled_amount_minor
funding_evidence.currency
credit_allocation_ids[]
initial_case_state=enrollment_pending
occurred_at
correlation_id
causation_id
schema_version
```

Conditions:

- Workspace is active/match-ready, unarchived and not already promoted.
- Commitment references the same workspace, has current terms/privacy acceptances and is in funding_pending/funded per the approved state contract.
- Payment path and activated commercial configuration are pinned.
- Required enrollment funding has a verified settled ledger entry or separately approved financing/manual-funding evidence; a processor webhook is first normalized and cannot itself authorize promotion.
- Credit allocation is non-negative, reproducible and cannot exceed the applicable obligation.
- No cancellation, compliance hold, reversal that invalidates funding, or existing promotion is present.

Constraints:

- one promotion per workspace
- one case per promotion event
- one result per idempotency key
- one primary owner membership per created case

No browser success/return route and no raw processor callback can emit or substitute for this event.

Append-only ledgers:

- `commercial_events`

- `processor_events`
- `invoice_events`
- `payment_ledger_entries`
- `credit_ledger_entries`
- `refund_ledger_entries`
- `adjustment_ledger_entries`
- `performance_obligation_events`
- `promotion_events`

Rules:

- Every record carries currency in integer minor units, commercial configuration version, source/idempotency/correlation identifiers and immutable timestamps.
- Mutable balances/statuses are projections only; reversals, chargebacks, credits, refunds and adjustments append entries.
- Processor events are signature-verified, stored once and normalized before domain use.
- Node 0 Complete credit cannot create a negative JMF obligation and stays distinct from cash/refund entries.
- Pass-through clinic/legal/psych/background/escrow/insurance/GC funds never enter Natalyx revenue ledgers.
- Clinical-linked invoices require durable evidence plus dignity-of-timing delay and can never be the first clinical notification.
- Refundable cash = max(0, settled JMF cash - earned JMF - prior cash refunds - chargebacks/reversals), with policy hooks for unresolved cases.

Invitation safety

Retain from #122:

- hashed tokens
- purpose-specific TTL with a 14-day ceiling
- CAS terminal transitions
- atomic membership creation
- one-live-GC constraint
- self/existing-member refusal
- safe return_to
- portable integrity/concurrency tests

Required changes:

- Exactly one target: workspace_id or case_id; include purpose, role, access_class, recipient_binding_mode and policy version.
- Email-less possession may only open pending low-risk workspace claim; it cannot reveal sensitive data or grant paid-case membership.
- Paid-case invitation is addressed, authenticated, consented and atomically redeems invitation plus participation, access grant and audit; mismatch requires explicit policy/issuer handling and preferably a fresh bound token.
- Workspace-claim issuance commits supersession, hashed claim, candidate/workspace linkage and required audit atomically; external identity steps use an explicit idempotent completion state.
- Cognito admission and app membership are separate decisions; account creation never equals workspace/case access.
- Verified identity comes from auth claims, not form-entered email; duplicate claims return the original result.
- Never put raw token/email/PHI in general participant timeline payloads; revoke/expire/supersede fail closed.
- Remove generic unconditional status mutation; expose authorized transition commands only.

Documents and S3

Metadata

- Requirement/readiness submissions and reviewer work items are separate entities from document versions and objects.
- A metadata-only marker cannot carry file bytes/storage_uri, enter document-approved state or claim that a clinic received/reviewed content.
- Work items bind organization, case, requirement, owner scope and immutable submission revision; decision plus audit is atomic and completion is durable/idempotent.
- Free text is minimum-necessary and strictly bounded; sensitive categories remain synthetic through A2 and content/PHI paths remain disabled until their gates.

Storage

- Private versioned quarantine bucket and clean vault with block-public-access, SSL-only and KMS encryption.
- App can create short-lived unique upload targets in quarantine but cannot read unscanned content; scanner moves/copies only clean versions to vault.
- Object key: env/{org_id}/{case_id}/{document_id}/{version_id}/{generated_object_id}; no names, emails, filenames or clinical labels.
- Authorization is always a PostgreSQL access-grant/ownership/purpose/consent decision independent of the key; participation or presentation lens alone is insufficient.
- CloudTrail S3 data events and app document_access_event records provide purpose-aware access evidence.

Upload

- Configurable maximum size; provisional 25 MiB awaits D-DOC-02.
- Initial proposed allowlist PDF/JPEG/PNG; DOCX only after scanner/archive review; no macro-enabled Office formats.
- Unicode-normalize/sanitize display filename; byte-sniff MIME, expected length and required SHA-256 checksum.
- Never reuse a key. Finalize verifies object version, HeadObject size/checksum/type and upload-session ownership.
- S3 event -> durable queue -> scanner -> signed clean/rejected/error result; unavailable/timed-out scan remains quarantined.

Download

- App authenticates and authorizes exact version/purpose, records access event, then returns a very short-lived clean-vault GET with safe content disposition.
- App proxy is reserved for justified transformation/audit cases, not the default durable path.

Recovery

- Upload sessions/tasks live in Postgres/queue so ECS replacement does not lose them; leases expire and retry idempotently.
- Replacement creates a new immutable version and supersedes the prior current pointer.
- Deletion is an audited tombstone followed by retention/legal-hold controlled purge; object-delete failures alert and retry.
- Orphan reconciliation handles expired intents and objects without document versions; lifecycle aborts incomplete multipart uploads.
- Provider share uses a purpose-bound snapshot/checksum/expiry, not direct object-key authority.

Design-system release rubric

Class	Items
Release-blocking	XSS/escaping or unsafe raw slots; keyboard/focus/screen-reader failures; unusable responsive layout; misleading or nonfunctional interaction; motion without reduced-motion safety; contrast or error-state defects that block task completion
Desirable	token/recipe consistency; visual parity; removal of duplicate spacing/color literals; component naming/adoption
Deferred	zero raw hex values; zero inline geometry styles; full legacy strangler cleanup; broad participant redesign; internal/operator DS convergence

Style purity blocks a release only when tied to a concrete security, accessibility, interaction or material maintainability risk.

Decision register

The register preserves all 24 IDs while separating founder outcome, technical control state and external activation evidence.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-MIG-01	Founder constraints final; Alan immediate; operationally pending	Luke sets constraints; Alan decides immediately	Alan - immediate	P02-01, P02-02, P02-03; Before P02-02	P02-01 decision work may proceed; P02-02, P02-03 and Gate A0 remain blocked until Alan's mechanism and evidence are accepted.
D-AUTH-01	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P03-02, P03-03; Resolved 2026-07-12	No longer blocks P03-02 or P03-03.
D-AUTH-02	Founder constraints final; Alan later	Alan decides later within founder constraints	Alan - later at P03-04	P03-04; Before P03-04	P03-04 and Gate A1 remain blocked until the technical mechanism and evidence exist.
D-AUTH-03	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12	No longer blocks the reconciled access/workspace PR sequence.
D-CI-01	Founder final; operationally pending	Luke decides commercial outcome; Alan configures immediately	Alan - immediate	P01-01; Founder decision resolved 2026-07-13; control required before P01-01 merge or any later app implementation PR	P01-01 merge, Gate A0 and dependent implementation PRs remain blocked until the live control is active and verified.
D-DATA-01	Founder decision final; implementation evidence pending	Luke decides product/privacy outcome	Alan - implementation	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-DATA-02	Founder default final; external validation before B1	Luke decides default; external specialist validates	Alan - implementation; privacy/legal - validation	P04-02, P04-03, P06-02; Before P04-02	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.
D-DATA-03	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P05-02, P06-03; Resolved 2026-07-12	No longer blocks Node 0 implementation planning.
D-COM-01	Founder decision final for beta; tax externally dependent	Luke decides commercial outcome	Alan - implementation	P06-01, P07-01, P08-01; Before activation of a Complete configuration	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
D-COM-02	Founder decision final for v1; live activation gated	Luke decides commercial outcome	Alan - implementation	P07-01, P08-01, P08-03; Before P07-01 activation data	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
D-COM-03	Externally dependent; founder policy inputs settled	External specialist / later evidence required	External legal/privacy/finance; Luke final business approval	P07-01, P07-02, P08-03; Before P07-02 user acceptance	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
D-COM-04	Founder decision final; implementation evidence pending	Luke decides commercial /product outcome	Alan - implementation	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	P07-02 through P09-04 may proceed under the settled semantics.
D-PAY-01	Founder constraints final; Alan later	Luke sets constraints; Alan decides later	Alan - later before P08-02	P08-02, P08-03, P08-04; Before P08-02 adapter	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
D-COM-05	Founder decision final for beta	Luke decides beta scope	None during beta	P07-02, P08-02; Before financed option is exposed	Financing no longer blocks P08-02; the feature is deliberately out of scope.
D-COM-06	Founder policy final; legal/accounting externally dependent	Luke decides policy; specialists translate	Alan - implementation; legal/accounting - wording	P08-03; Before automated pregnancy-stage refund logic	Synthetic P08-03 implementation may proceed; live policy activation remains gated.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-COM-07	Founder policy final; legal/accounting externally dependent	Luke decides policy; specialists translate	Alan - implementation; legal/accounting - wording	P08-03; Before real billing	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
D-COM-08	Founder decision final; implementation evidence pending	Luke decides customer treatment	Alan - implementation	P08-04; Before P08-04	P08-04 may proceed under the settled policy.
D-DOC-01	Founder default final; Alan later; external validation before B1/B2	Luke sets policy; Alan decides implementation later	Alan - later before P09-03; privacy/legal - validation	P09-01, P09-03, P09-06; Before P09-03	P09-01 ADR work may proceed; P09-03/P09-06 and real-content activation wait for Alan's delivery choice and evidence.
D-DOC-02	Founder constraints final; Alan later	Luke sets constraints; Alan decides later	Alan - later before P09-02	P09-01, P09-02, P09-03, P09-06; Before P09-02	P09-01 contract work may proceed; P09-02/P09-03/P09-06 wait for Alan's technical selection and evidence.
D-PROV-01	Founder decision final; Alan later	Luke decides product/privacy outcome; Alan implements later	Alan - later before P09-05	P09-05; Before P09-05	Synthetic provider-snapshot design may proceed; P09-05 completion and real handoff require Alan's evidence and later external review.
D-AI-01	Founder scope/cost final; Alan later; PHI externally gated	Luke sets scope and cost; Alan decides implementation later	Alan - later before P10-01	P10-01, P10-02, P10-03, P12-01; Before P10-01	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.
D-OPS-01	Founder constraints final; Alan immediate; operationally pending	Luke sets constraints; Alan decides immediately	Alan - immediate	P13-02; Before Gate A0 recovery action	P13-02 and Gate A0 remain blocked until the runbook and synthetic exercise pass; A1 must exercise the control.
D-PHI-01	Founder beta scope final; externally gated	Luke decides beta scope; external evidence required	Alan - B1/B2 evidence; external reviewers - approval	P09-01, P09-02, P09-05, P09-06, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.
D-PHI-02	Founder governance final; external reviewers pending	Luke decides governance; external evidence required	Alan - evidence; external reviewers - attestations; Luke - go/no-go	P13-06, P13-07; Before B1	B1 and B2 remain blocked until the corresponding evidence is current and the required humans sign off.

D-MIG-01 - Migration framework and SQLite support

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Reliability-first migrations with zero intended data loss; PostgreSQL is authoritative for production and SQLite may remain a convenient synthetic local-development projection without full production parity.
Founder constraints	Stage consequential changes and keep them backward-compatible wherever reasonably possible; Test recovery or rollback evidence before every consequential migration is used; Prefer a well-supported low-maintenance approach over a clever or bespoke runner; Planned beta downtime may be up to 30 minutes when scheduled, communicated and backed by a tested recovery plan; Destructive or irreversible real-user-data changes require explicit evidence and Luke approval
What remains for Alan	Choose the migration framework; Choose sequencing/locking and expand-contract mechanics; Choose the detailed rollback, restore or forward-repair mechanism
Evidence required	Migration ADR; Fresh/current/upgrade compatibility matrix; Tested recovery or rollback evidence; Destructive-change approval path
Approval rule	Luke approves only destructive or irreversible actions that could affect real-user data; ordinary technical choices stay with Alan.
External dependencies	None
Affected PRs / latest point	P02-01, P02-02, P02-03; Before P02-02
Programme effect	P02-01 decision work may proceed; P02-02, P02-03 and Gate A0 remain blocked until Alan's mechanism and evidence are accepted.
Independent recommendation	Prefer an established framework unless the ADR proves current SQL compatibility makes it unsafe or disproportionate.

D-AUTH-01 - Workspace claim and paid-case identity policy

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	A bearer link creates only a pending low-risk workspace claim; paid-case access requires authenticated binding, explicit consent and recipient/issuer confirmation.
Founder constraints	Minimum-necessary relationship, outreach and logistics context only; No bearer-link case membership
What remains for Alan	Implement the settled claim and paid-case invitation contracts
Evidence required	Forwarded-link, expiry, mismatch, consent and membership negative tests
Approval rule	No further founder decision unless access is broadened.
External dependencies	None
Affected PRs / latest point	P03-02, P03-03; Resolved 2026-07-12
Programme effect	No longer blocks P03-02 or P03-03.
Independent recommendation	A bearer link creates only a pending free-workspace claim with minimum-necessary relationship/outreach/logistics context; paid-case access always requires authenticated binding, explicit consent and recipient/issuer confirmation.

D-AUTH-02 - Claim-aware Cognito admission mechanism

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Participant admission uses short-lived signed or server-verified proof bound to email, purpose, claim and expiry; it admits account creation only and never grants workspace or case membership. Internal admission remains a separate allowlist path.
Founder constraints	Short lived; Replay, mismatch, expiry and revocation fail closed; No membership or resource authorization from admission alone
What remains for Alan	Choose the proof format, signer/verifier, key lifecycle and claim lifecycle
Evidence required	Expiry, replay, wrong-email, wrong-purpose, mismatch and revocation tests; Separate internal allowlist evidence
Approval rule	Luke approval is not required unless Alan proposes broader access or materially weaker identity assurance.
External dependencies	None
Affected PRs / latest point	P03-04; Before P03-04
Programme effect	P03-04 and Gate A1 remain blocked until the technical mechanism and evidence exist.
Independent recommendation	Bind proof to email, purpose, claim ID and expiry; it admits account creation only and never grants workspace/case membership.

D-AUTH-03 - Identity, participation and access authorization contract

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Identity, participation, access grants, ownership and operation authorization remain separate; participation or lens alone never authorizes.
Founder constraints	Subject-derived authorization context; Application and RLS policy parity; Default-deny cross-resource access
What remains for Alan	Encode the settled contract in application services and RLS
Evidence required	Participation-without-grant and grant-without-capability denials; Application/RLS parity tests
Approval rule	No further founder decision unless the boundary changes.
External dependencies	None
Affected PRs / latest point	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12
Programme effect	No longer blocks the reconciled access/workspace PR sequence.
Independent recommendation	Identity binds principal to person; participation records domain relationship; grants carry resource-scoped capabilities; application and RLS consume a subject-derived context plus ownership and purpose. Participation/lens alone never authorizes.

D-CI-01 - Required-check enforcement for the private app repository

Field	Recorded decision or requirement
Decision state	Final founder decision; control pending
Founder outcome	Option A — GitHub Team approved as the private-repository enforcement control.
Founder constraints	Approximately $12 USD/month before tax for three billable users; Commercial decision is final; Expenditure is authorized but not yet incurred; No claim of bypass-resistant enforcement before live configuration and readback
What remains for Alan	Activate GitHub Team when enforcement is required; Configure required unittest and PostgreSQL checks; Restrict routine bypass; Supply ruleset readback and negative-bypass evidence
Evidence required	GitHub Team activation/readback; Mandatory unittest and PostgreSQL checks; Restricted routine bypass; Negative-bypass evidence
Approval rule	Luke accepts the enforcement evidence; the approved purchase itself needs no further founder decision.
External dependencies	GitHub Team activation; expenditure authorized but not yet incurred
Affected PRs / latest point	P01-01; Founder decision resolved 2026-07-13; control required before P01-01 merge or any later app implementation PR
Programme effect	P01-01 merge, Gate A0 and dependent implementation PRs remain blocked until the live control is active and verified.
Independent recommendation	Approved: GitHub Team for three billable users at approximately $12 USD/month before tax. Expenditure is authorized but not yet incurred. Alan owns activation-time ruleset design and evidence; Gate A0 and dependent implementation PRs remain blocked until Team is active, both checks are mandatory, routine bypass is restricted, and negative-bypass evidence is accepted.

D-DATA-01 - IP-GC field projection allowlist

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.
Founder constraints	Default-deny field-and-purpose allowlists; Source-labelled corrections; Named-purpose provider projection only
What remains for Alan	Define explicit field/purpose allowlists in code and projection contracts; Implement cross-role and provider negative tests
Evidence required	Cross-role denial matrix; Provider projection allowlist tests; Correction/source-label evidence
Approval rule	Luke approval is required for a broader sharing policy, not for compliant implementation detail.
External dependencies	None
Affected PRs / latest point	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02
Programme effect	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
Independent recommendation	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.

D-DATA-02 - Correction, supersession and unselected-candidate retention

Field	Recorded decision or requirement
Decision state	Final founder default; externally dependent activation
Founder outcome	Retain candidate/workspace material for a 90-day minimum after explicit rejection, withdrawal or last legitimate outreach, honor earlier valid deletion, retain only non-identifying funnel metrics and a minimal audit tombstone, and preserve legal holds. Corrections append source-labelled revisions and private IP notes remain author-only.
Founder constraints	No silent overwrite; Deletion and anonymization are auditable; Legal holds override ordinary deletion
What remains for Alan	Implement lifecycle, deletion, anonymization, correction and legal-hold behavior
Evidence required	Retention/deletion tests; Anonymization and tombstone evidence; Privacy/legal approval before B1
Approval rule	Luke accepts the live policy after privacy/legal review.
External dependencies	Privacy/legal validation before B1
Affected PRs / latest point	P04-02, P04-03, P06-02; Before P04-02
Programme effect	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.
Independent recommendation	Append revisions with supersedes/derived-from; never silently erase source; shortest justified retention for unselected candidates.

D-DATA-03 - Candidate scoring and sensitive preference policy

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	No candidate ranking, viability/eligibility flag or race preference survives; provisional pre-join input can generate neutral source-labelled questions only.
Founder constraints	No sensitive third-party scoring or gating; No provisional assertion becomes GC/provider truth
What remains for Alan	Remove legacy ranking/gating and preserve neutral-question behavior
Evidence required	Prohibited-field and no-ranking tests
Approval rule	No further founder decision unless ranking or eligibility behavior is proposed.
External dependencies	None
Affected PRs / latest point	P05-02, P06-03; Resolved 2026-07-12
Programme effect	No longer blocks Node 0 implementation planning.
Independent recommendation	No ranking, viability/eligibility flag or race preference survives. Pre-join IP input is limited to minimum-necessary relationship/outreach/logistics context and may produce neutral source-labelled questions only; it never gates claim, onboarding, provider use or critical decisions.

D-COM-01 - Node 0 Complete price

Field	Recorded decision or requirement
Decision state	Final beta policy; external tax wording pending
Founder outcome	Node 0 Complete is a voluntary $495 USD one-time upgrade paid by the IP or authorized IP account holder, never the GC, collected before Complete features and fully credited against the later Phase 1/JMF without gating matching or safety support.
Founder constraints	Transparent one-time feature/credit wording; Versioned prospective changes only; No forced payment to identify, approach or reach agreement with a GC
What remains for Alan	Implement configurable price, payer ownership, durable payment/credit ledger and idempotent flows
Evidence required	Synthetic success, failure, retry, duplicate and credit tests; Tax wording before live activation
Approval rule	Luke approves future commercial-policy changes; tax treatment requires specialist review.
External dependencies	Tax treatment and customer-facing wording before live activation
Affected PRs / latest point	P06-01, P07-01, P08-01; Before activation of a Complete configuration
Programme effect	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
Independent recommendation	Keep no active production value until explicitly approved; test with synthetic configuration.

D-COM-02 - Ratify JMF price book and effective date

Field	Recorded decision or requirement
Decision state	Final v1 policy; external activation dependencies
Founder outcome	Configuration v1 is $8,900 USD: a $750 deposit followed by $1,630, $2,445, $1,630, $1,630 and $815 milestones. Pay-in-full is $7,830 total, leaving $7,080 after the deposit; the separate $495 Complete credit must not be double-counted.
Founder constraints	Versioned configuration; Prospective changes only; First live beta requires A2 plus approved terms and tax treatment
What remains for Alan	Implement versioned configuration, milestone and reconciliation behavior
Evidence required	Synthetic schedule, pay-in-full and no-double-credit reconciliation tests
Approval rule	Luke approves future price-book versions and first live activation.
External dependencies	Approved terms and tax treatment before live activation
Affected PRs / latest point	P07-01, P08-01, P08-03; Before P07-01 activation data
Programme effect	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
Independent recommendation	Use the document anchors but require explicit version activation rather than code literals.

D-COM-03 - Terms, privacy and refund language versions

Field	Recorded decision or requirement
Decision state	Externally dependent
Founder outcome	No live acceptance or charge may rely on the contradictory current terms draft. D-COM-06 and D-COM-07 supply the approved business policy for a new versioned terms package.
Founder constraints	Resolve refundable-until-onboarding versus non-refundable contradiction; No retrospective policy changes; Versioned terms/privacy/refund acceptance
What remains for Alan	Implement versioned acceptance and keep real charging disabled until an approved version is active
Evidence required	Approved paid-service terms; Approved privacy/consent disclosure; Approved tax/refund/accounting wording
Approval rule	Luke approves the final business policy and customer-facing version after specialist review.
External dependencies	Legal, privacy, tax and finance review before real-user acceptance or charging
Affected PRs / latest point	P07-01, P07-02, P08-03; Before P07-02 user acceptance
Programme effect	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
Independent recommendation	External legal/privacy/finance review before any real acceptance.

D-COM-04 - Enrollment funding and activation evidence

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use two-step commitment: settled $750 plus current accepted terms proves commercial enrollment, but payment alone never promotes. Exactly one enrollment-pending case follows explicit IP promotion and authenticated GC consent. The $1,630 activation tranche is earned only after defined IP/GC onboarding and screening coordination has begun; it never implies medical clearance.
Founder constraints	Redirects/webhooks never promote; Typed promotion event only; Source-labelled facts and explicit evidence checklist
What remains for Alan	Define typed events, activation checklist, idempotency and audit mechanics
Evidence required	Duplicate, replay, partial-failure, promotion and activation evidence tests
Approval rule	Luke approves any change to commercial enrollment or activation semantics.
External dependencies	None
Affected PRs / latest point	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02
Programme effect	P07-02 through P09-04 may proceed under the settled semantics.
Independent recommendation	Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.

D-PAY-01 - Payment processor and card/ACH scope

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Offer card and US ACH, encourage but do not require ACH above $1,000, absorb ordinary beta fees without surcharging, and keep processor metadata free of GC, journey, clinical or document detail.
Founder constraints	Card cost ceiling near 3% plus $0.30; ACH cost ceiling near 1% capped at $10; Processor-neutral domain and ledger contract
What remains for Alan	Choose the processor/vendor, tokenization, webhook, settlement, return/refund and reconciliation mechanisms
Evidence required	Provider security and cost comparison; Synthetic settlement, return, refund, duplicate and reconciliation tests
Approval rule	Luke approval is required only if the proposed processor exceeds the cost/privacy constraints or changes the payment experience.
External dependencies	Payment/security/finance review before real money
Affected PRs / latest point	P08-02, P08-03, P08-04; Before P08-02 adapter
Programme effect	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
Independent recommendation	Choose only after domain/ledger contract; retain processor-neutral ports.

D-COM-05 - Financing path

Field	Recorded decision or requirement
Decision state	Final beta scope
Founder outcome	Financing is excluded and hidden from beta; Natalyx does not advertise, recommend or integrate a lender or treat lender status as funding evidence.
Founder constraints	Card and ACH remain unaffected; Reconsider only after demand evidence plus legal/privacy review
What remains for Alan	Keep financing paths disabled and absent from beta UI/configuration
Evidence required	Negative configuration/UI tests proving financing is unavailable
Approval rule	A future financing proposal requires a new founder decision.
External dependencies	None
Affected PRs / latest point	P07-02, P08-02; Before financed option is exposed
Programme effect	Financing no longer blocks P08-02; the feature is deliberately out of scope.
Independent recommendation	Do not block card/ACH MVP; keep optional and disabled.

D-COM-06 - Pregnancy-loss credit or retained-revenue policy

Field	Recorded decision or requirement
Decision state	Final founder policy; external wording pending
Founder outcome	After post-heartbeat loss, never charge the $815 birth tranche; retain the delivered $1,630 pregnancy-support tranche and grant an equal $1,630 future credit to the same IP household for 36 months, with compassionate manual extension and defined offboarding support.
Founder constraints	No automatic birth charge; Auditable credit expiry/extension; Compassionate support workflow
What remains for Alan	Implement credit ledger, expiry, extension and support workflow
Evidence required	Loss, credit, expiry, extension and duplicate tests; Approved legal/accounting treatment before live use
Approval rule	Luke approves final customer-facing policy after specialist review.
External dependencies	Legal and accounting review before live use
Affected PRs / latest point	P08-03; Before automated pregnancy-stage refund logic
Programme effect	Synthetic P08-03 implementation may proceed; live policy activation remains gated.
Independent recommendation	Manual/counsel-reviewed hook until policy is approved; never hard-code.

D-COM-07 - Cancellation and non-clearance policy sign-off

Field	Recorded decision or requirement
Decision state	Final founder policy; external wording pending
Founder outcome	Protect delivered work: the $750 is refundable until onboarding is delivered and then earned; activation follows D-COM-04. Non-clearance or pre-clearance GC withdrawal stops later charges. Unearned prepayment is refunded within ten business days, with refund or rollover choice and one replacement within 12 months without a second $750 deposit.
Founder constraints	A new activation charge may apply for genuinely new onboarding; No double charge or silent retained unearned value; Refund and rollover are auditable
What remains for Alan	Implement reconciliation, refund, rollover and replacement behavior
Evidence required	Withdrawal, non-clearance, refund, rollover, replacement and duplicate tests; Approved legal/accounting treatment before live use
Approval rule	Luke approves final customer-facing policy after specialist review.
External dependencies	Legal and accounting review before live use
Affected PRs / latest point	P08-03; Before real billing
Programme effect	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
Independent recommendation	Use document schedule as test anchor; require legal/finance approval for live policy.

D-COM-08 - Grace period and non-critical service pause

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use a humane 30-day grace period with reminders seven days before due, on the due date, and at 7, 14 and 30 days overdue; no beta late fee and human outreach before pause. Only future non-critical paid work may pause after 30 days.
Founder constraints	Never block safety content, existing documents/data, correction/export/deletion, consent/revocation, urgent support or already-required provider coordination
What remains for Alan	Implement reminder, escalation, grace and capability-specific pause states
Evidence required	Reminder cadence and protected-access tests
Approval rule	Luke approves changes to grace or protected access.
External dependencies	None
Affected PRs / latest point	P08-04; Before P08-04
Programme effect	P08-04 may proceed under the settled policy.
Independent recommendation	Default to continued safety and entitled-document access; pause only clearly non-critical future services.

D-DOC-01 - Document retention, deletion, replacement and download mode

Field	Recorded decision or requirement
Decision state	Final founder default; externally dependent activation
Founder outcome	Retain clean active-case documents while active plus two years after close; signed consents, payment/access audits and deletion tombstones for seven years; failed/quarantine orphans seven days; superseded content 90 days unless required; honor valid deletion within 30 days subject to legal hold. Downloads require fresh authorization and links expire within five minutes.
Founder constraints	Never public; Never withheld for non-payment; Legal hold and deletion remain auditable
What remains for Alan	Choose proxy versus presigned delivery by class; Implement hold, deletion, tombstone, retry and authorization behavior
Evidence required	Fresh-authorization and five-minute-link tests; Retention/hold/deletion exercise; Privacy/legal approval before real content
Approval rule	Luke accepts live policy after privacy/legal review.
External dependencies	Privacy/legal validation before B1/B2 real-content use
Affected PRs / latest point	P09-01, P09-03, P09-06; Before P09-03
Programme effect	P09-01 ADR work may proceed; P09-03/P09-06 and real-content activation wait for Alan's delivery choice and evidence.
Independent recommendation	Purpose/role authorization independent of key; short-lived downloads; deletion tombstone/audit; legal/privacy review.

D-DOC-02 - Document limits and malware/quarantine provider

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Allow PDF, JPEG, PNG and non-macro DOCX up to 50 MiB. Reject spoofed, macro-enabled and archive-like input; quarantine until clean; treat password-protected, skipped, error and timeout results as unavailable.
Founder constraints	Normal scan target no more than two minutes; Transient retries no more than 15 minutes; $50/month beta scanning ceiling with alerts; Synthetic only at A2 and no PHI before B2
What remains for Alan	Choose the AWS-native scanner, queue, retry, timeout, recovery and cost-control mechanisms
Evidence required	Type/sniffing/spoof tests; Quarantine, error, timeout and retry tests; Cost estimate and alerts
Approval rule	Luke approval is required if the solution exceeds the cost or safety constraints.
External dependencies	None
Affected PRs / latest point	P09-01, P09-02, P09-03, P09-06; Before P09-02
Programme effect	P09-01 contract work may proceed; P09-02/P09-03/P09-06 wait for Alan's technical selection and evidence.
Independent recommendation	Narrow allowlist, checksum and content sniffing; unavailable scanner remains quarantined.

D-PROV-01 - Provider handoff purposes and field allowlists

Field	Recorded decision or requirement
Decision state	Final founder policy
Founder outcome	Provider handoff is a purpose-bound immutable snapshot for one verified recipient and named purpose, with participant preview/consent, clean documents and appropriately verified source-labelled facts. Unopened access expires after 14 days and can be revoked before access; downloaded copies cannot be recalled.
Founder constraints	No ambient provider case access in MVP; Audit every issue/open/download/revoke/expire event; Reissue after expiry
What remains for Alan	Choose recipient verification, snapshot checksum, expiry, revocation and audit mechanisms
Evidence required	Wrong-recipient, expiry, revoke-before-access, reissue and checksum tests
Approval rule	Luke approves broader recipient or ambient-access scope.
External dependencies	Privacy/provider review before real handoff
Affected PRs / latest point	P09-05; Before P09-05
Programme effect	Synthetic provider-snapshot design may proceed; P09-05 completion and real handoff require Alan's evidence and later external review.
Independent recommendation	Class 1/2 excluded; class 4 only if GC explicitly selects for recipient/purpose; class 5 evidence linked.

D-AI-01 - Intelligence context and approval authority matrix

Field	Recorded decision or requirement
Decision state	Final founder policy; PHI externally dependent
Founder outcome	AI is bounded and advisory: role-scoped education, retrieval, summaries, question drafts, next steps and provider preparation using approved Learn and authorized source-labelled context. It never diagnoses, ranks, approves, clears or writes durable truth; authorized human approval precedes memory, case or provider material.
Founder constraints	Deterministic or curated fallback; $10 per active case/month and $100 total beta/month ceilings with alerts and graceful fallback; No PHI before B2
What remains for Alan	Choose Bedrock model, prompts, retrieval, schemas, fallback, human-review rails and cost controls
Evidence required	Scope/authority negative tests; Fallback and malformed-response tests; Safety and cost evidence
Approval rule	Luke approves broader AI authority, higher cost or any PHI use; B2 external approval is mandatory for PHI.
External dependencies	B2 external security/privacy/compliance approval before PHI
Affected PRs / latest point	P10-01, P10-02, P10-03, P12-01; Before P10-01
Programme effect	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.
Independent recommendation	Generic case members cannot create reviewed/provider truth; intelligence remains advisory and scope explicit.

D-OPS-01 - Known-good recovery revision and rollback sequence

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Use fail-closed recovery: protect data integrity over uptime, restore only a reviewed proven-compatible application-image/schema pair, use safe read-only or unavailability when uncertain, and forward-repair when rollback is unsafe. Target 30-minute beta recovery; allow up to two hours emergency downtime and notify Luke when more than one hour is expected.
Founder constraints	Zero intended loss or corruption; No expensive 24/7 coverage for synthetic/early controlled beta without evidence; Real-user, destructive or uncertain-loss recovery requires explicit human approval
What remains for Alan	Choose exact compatibility, rollback/forward-repair and automation mechanisms; Name the evidence owner and write the runbook
Evidence required	Documented recovery procedure; Successful synthetic recovery exercise; Known-compatible image/schema evidence
Approval rule	Luke accepts A0 evidence and explicitly approves risky real-user/destructive recovery.
External dependencies	None
Affected PRs / latest point	P13-02; Before Gate A0 recovery action
Programme effect	P13-02 and Gate A0 remain blocked until the runbook and synthetic exercise pass; A1 must exercise the control.
Independent recommendation	Use a tested immutable image and compatible schema; do not deploy from dirty or draft branches.

D-PHI-01 - PII/PHI boundary, vendors and control set

Field	Recorded decision or requirement
Decision state	Final founder scope; externally dependent activation
Founder outcome	B1 is an invitation-only US beta of at most five IP/GC pairs using names, verified account/contact data, relationship/outreach/logistics, consent versions, nonclinical progress, feedback and processor references only. Medical, reproductive-health, psychological, insurance-health and clinical-clearance data are prohibited; no real document bytes or provider packets; AI receives de-identified nonclinical context only.
Founder constraints	RLS/non-owner runtime and cross-tenant negative evidence; Access review plus incident and restore exercises; No PHI before B2
What remains for Alan	Implement inventory, cohort controls, RLS, cross-tenant denials, de-identification and prohibited-data tests
Evidence required	Five-pair cohort configuration; RLS/cross-tenant/de-identification/prohibited-data tests; External privacy/legal approval
Approval rule	Luke accepts B1 only after required evidence and external privacy/legal advice; B2 requires qualified external approval.
External dependencies	External privacy/legal approval before B1; Qualified external security/privacy/compliance/legal approval before B2
Affected PRs / latest point	P09-01, P09-02, P09-05, P09-06, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI
Programme effect	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.
Independent recommendation	B1 remains PII-only and content controls stay synthetic; B2 requires formal external security/privacy/compliance evidence before any PHI content path is enabled.

D-PHI-02 - B1/B2 approvers and evidence expiry

Field	Recorded decision or requirement
Decision state	Final founder governance; externally dependent activation
Founder outcome	Luke accepts B1 from Alan's technical evidence plus external privacy/legal advice, without separate independent technical sign-off. B1 evidence is at most 60 days old and reruns after material change or 180 days. B2 requires qualified external security, privacy/compliance and legal/vendor review with release evidence at most 30 days old and assessments at most 12 months old.
Founder constraints	No critical exceptions; High B1 exceptions are written, time-limited and externally reviewed; High B2 exceptions require joint Luke and relevant external approval
What remains for Alan	Maintain evidence recency, change-triggered reruns and exception records
Evidence required	B1/B2 evidence index with dates; Named reviewers; Exception register; Signed go/no-go
Approval rule	Luke is final product go/no-go within the specified external-review model.
External dependencies	Named privacy/legal advisers for B1; Named qualified security/privacy/compliance/legal reviewers for B2
Affected PRs / latest point	P13-06, P13-07; Before B1
Programme effect	B1 and B2 remain blocked until the corresponding evidence is current and the required humans sign off.
Independent recommendation	Luke is product go/no-go; independent qualified reviewers own their control attestations.

Alan decision timing and external dependencies

Immediate Alan decisions

These block current foundation work or Gate A0; founder constraints are final but the controls are not active until the evidence passes.

ID	Alan owns	Affected / latest	Evidence	Founder approval rule
D-CI-01	Activate GitHub Team when enforcement is required; Configure required unittest and PostgreSQL checks; Restrict routine bypass; Supply ruleset readback and negative-bypass evidence	P01-01; Founder decision resolved 2026-07-13; control required before P01-01 merge or any later app implementation PR	GitHub Team activation/readback; Mandatory unittest and PostgreSQL checks; Restricted routine bypass; Negative-bypass evidence	Luke accepts the enforcement evidence; the approved purchase itself needs no further founder decision.

ID	Alan owns	Affected / latest	Evidence	Founder approval rule
D-MIG-01	Choose the migration framework; Choose sequencing/locking and expand-contract mechanics; Choose the detailed rollback, restore or forward-repair mechanism	P02-01, P02-02, P02-03; Before P02-02	Migration ADR; Fresh/current/upgrade compatibility matrix; Tested recovery or rollback evidence; Destructive-change approval path	Luke approves only destructive or irreversible actions that could affect real-user data; ordinary technical choices stay with Alan.
D-OPS-01	Choose exact compatibility, rollback/forward-repair and automation mechanisms; Name the evidence owner and write the runbook	P13-02; Before Gate A0 recovery action	Documented recovery procedure; Successful synthetic recovery exercise; Known-compatible image/schema evidence	Luke accepts A0 evidence and explicitly approves risky real-user/destructive recovery.

Later Alan decisions

These are resolved only at the named WP/checkpoint; Alan should not be forced to choose their mechanisms prematurely.

ID	Alan owns	Affected / latest	Evidence	Founder approval rule
D-AUTH-02	Choose the proof format, signer/verifier, key lifecycle and claim lifecycle	P03-04; Before P03-04	Expiry, replay, wrong-email, wrong-purpose, mismatch and revocation tests; Separate internal allowlist evidence	Luke approval is not required unless Alan proposes broader access or materially weaker identity assurance.
D-PAY-01	Choose the processor/vendor, tokenization, webhook, settlement, return/refund and reconciliation mechanisms	P08-02, P08-03, P08-04; Before P08-02 adapter	Provider security and cost comparison; Synthetic settlement, return, refund, duplicate and reconciliation tests	Luke approval is required only if the proposed processor exceeds the cost/privacy constraints or changes the payment experience.
D-DOC-01	Choose proxy versus presigned delivery by class; Implement hold, deletion, tombstone, retry and authorization behavior	P09-01, P09-03, P09-06; Before P09-03	Fresh-authorization and five-minute-link tests; Retention/hold/deletion exercise; Privacy/legal approval before real content	Luke accepts live policy after privacy/legal review.
D-DOC-02	Choose the AWS-native scanner, queue, retry, timeout, recovery and cost-control mechanisms	P09-01, P09-02, P09-03, P09-06; Before P09-02	Type/sniffing/spoof tests; Quarantine, error, timeout and retry tests; Cost estimate and alerts	Luke approval is required if the solution exceeds the cost or safety constraints.
D-PROV-01	Choose recipient verification, snapshot checksum, expiry, revocation and audit mechanisms	P09-05; Before P09-05	Wrong-recipient, expiry, revoke-before-access, reissue and checksum tests	Luke approves broader recipient or ambient-access scope.
D-AI-01	Choose Bedrock model, prompts, retrieval, schemas, fallback, human-review rails and cost controls	P10-01, P10-02, P10-03, P12-01; Before P10-01	Scope/authority negative tests; Fallback and malformed-response tests; Safety and cost evidence	Luke approves broader AI authority, higher cost or any PHI use; B2 external approval is mandatory for PHI.
D-PHI-01	Implement inventory, cohort controls, RLS, cross-tenant denials, de-identification and prohibited-data tests	P09-01, P09-02, P09-05, P09-06, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI	Five-pair cohort configuration; RLS/cross-tenant/de-identification /prohibited-data tests; External privacy/legal approval	Luke accepts B1 only after required evidence and external privacy/legal advice; B2 requires qualified external approval.
D-PHI-02	Maintain evidence recency, change-triggered reruns and exception records	P13-06, P13-07; Before B1	B1/B2 evidence index with dates; Named reviewers; Exception register; Signed go/no-go	Luke is final product go/no-go within the specified external-review model.

External dependencies

ID	Dependency	Activation boundary
D-DATA-02	Privacy/legal validation before B1	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.
D-COM-01	Tax treatment and customer-facing wording before live activation	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
D-COM-02	Approved terms and tax treatment before live activation	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
D-COM-03	Legal, privacy, tax and finance review before real-user acceptance or charging	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
D-PAY-01	Payment/security/finance review before real money	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
D-COM-06	Legal and accounting review before live use	Synthetic P08-03 implementation may proceed; live policy activation remains gated.
D-COM-07	Legal and accounting review before live use	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
D-DOC-01	Privacy/legal validation before B1/B2 real-content use	P09-01 ADR work may proceed; P09-03/P09-06 and real-content activation wait for Alan's delivery choice and evidence.
D-PROV-01	Privacy/provider review before real handoff	Synthetic provider-snapshot design may proceed; P09-05 completion and real handoff require Alan's evidence and later external review.
D-AI-01	B2 external security/privacy/compliance approval before PHI	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.
D-PHI-01	External privacy/legal approval before B1; Qualified external security/privacy/compliance/legal approval before B2	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.
D-PHI-02	Named privacy/legal advisers for B1; Named qualified security/privacy/compliance/legal reviewers for B2	B1 and B2 remain blocked until the corresponding evidence is current and the required humans sign off.

Active decision blockers

Only the following decision IDs may still appear in PR blocking fields: D-AI-01, D-AUTH-02, D-CI-01, D-DOC-01, D-DOC-02, D-MIG-01, D-OPS-01, D-PAY-01, D-PROV-01. Founder-final policy rows without a remaining technical mechanism are not PR blockers.

Risk register

ID	Class	Risk	Owner	Mitigation	Gate
R-01	Implementation risk	Runtime DDL or incompatible migration leaves dev unavailable.	Alan	WP01/WP02, advisory lock, one-off migrator, expand/contract, forward repair.	A0
R-02	Product/privacy risk	IP assertions are presented as GC truth or leak across roles.	Luke	Authorship matrix, D-DATA decisions, WP04/WP06 negative tests.	A1/A2
R-03	Security risk	Forwarded invitation or shared signup code grants sensitive access.	Alan	Separate claim/invite machines, claim-aware Cognito admission, consent/membership transaction.	A1/A2
R-04	Financial risk	Duplicate charge/case/invoice/refund or unreproducible balance.	Luke	Append-only ledgers, processor inbox, typed promotion event, idempotency/concurrency tests.	A2
R-05	Security/compliance risk	Metadata is mistaken for a document or unscanned/public/over-shared content/provider packets expose sensitive data.	Luke	Separate metadata/content state machines, quarantine/vault, KMS/IAM, purpose consent, access audit, RLS before B1 and formal B2.	A2/B1/B2
R-06	AI safety risk	Private memory or provisional/AI fact becomes shared/provider truth.	Luke	Scoped envelopes, evidence rail, proposal authority, safety evals.	A2
R-07	Delivery risk	Useful #122/#124/Wave 2/worktree evidence is deleted or obsolete architecture is continued.	Joint	Completed matrix/extraction ledger, destination tests and no cleanup before replacement reachability/team confirmation.	Pre-A0
R-08	Operational risk	Green CI is mistaken for live or PHI readiness.	Luke	Known-revision cloud gates and formal B1/B2 evidence/approvers.	A0-B2
R-09	UX/accessibility risk	Mobile or assistive users cannot complete the participant journey.	Alan	Release-blocking DS criteria and end-to-end mobile/keyboard tests.	A1/A2
R-10	Repository-boundary risk	Participant/internal DS, Learn/intelligence/app truth or app/infra ownership is blurred.	Luke	Pinned contracts, separate PRs/repositories, architecture-after-merge statements.	All

Release strategy and real-user/PHI boundaries

- The corrected work-package split is proposed pending Alan's review. Only PRs expressly marked for isolated local preparation may be worked on before joint acceptance, and that work remains uncommitted, synthetic-only and outside protected worktrees; commit, PR creation and merge wait for reciprocal review and the named controls.
- Use additive schemas, feature flags and dual-compatible reads/writes; contract only after telemetry/evidence proves the prior revision is unavailable.
- Recover and pass A0 first, then validate A1 core participant behavior before adding commercial/document complexity, then A2 before any real user.
- Keep all development synthetic through A2. B1 requires forced RLS/non-owner runtime plus the explicitly approved PII beta scope. PHI-capable paths remain disabled until B2's formal human go/no-go.
- Every gate release identifies exact app/intelligence/infra revisions, migration heads, artifact pins and rollback/forward-repair response.
- Existing PRs/branches are closed or deleted only after extraction, replacement reachability, review reconciliation and explicit team confirmation.

A0/A1/A2 remain synthetic. B1 is a named, approved PII-only beta boundary. B2 is a formal PHI go/no-go that requires organizational, contractual, operational and technical evidence; successful tests or deployment alone are insufficient.

Document index and maintenance

Document	Purpose
README.md	Programme entrypoint, index, operating/maintenance/render instructions
existing-pr-extraction-ledger.md	PR/worktree preservation, extraction and closure/delete conditions
project-structure.md	GitHub Project fields, items and conventions
pr-manifest.md	Canonical proposed PR inventory and execution order
consolidated-plan.md	Full programme source for the PDF
consolidated-plan.pdf	Rendered review copy generated only from consolidated-plan.md
00-preservation-cloud-diagnosis-and-pr-reconciliation.md	Standalone execution plan for WP00 — Preservation, cloud diagnosis, and PR reconciliation
01-ci-and-postgres-evidence-gate.md	Standalone execution plan for WP01 — CI and PostgreSQL evidence gate
02-versioned-migrations-and-readiness.md	Standalone execution plan for WP02 — Versioned migrations and readiness
03-auth-invitations-and-resume.md	Standalone execution plan for WP03 — Authentication, invitations, and resume
04-node0-workspace-case-separation.md	Standalone execution plan for WP04 — Node 0 workspace and paid-case separation
05-node0-free-pathfinder-and-resume.md	Standalone execution plan for WP05 — Node 0 Free, Pathfinder, and resume
06-node0-complete-and-soft-onboarding.md	Standalone execution plan for WP06 — Node 0 Complete and GC soft onboarding
07-commercial-configuration-enrollment-and-promotion.md	Standalone execution plan for WP07 — Commercial configuration, enrollment, and promotion
08-payments-invoices-credits-and-refunds.md	Standalone execution plan for WP08 — Payments, invoices, credits, and refunds
09-documents-active-case-and-provider-handoffs.md	Standalone execution plan for WP09 — Documents, active case, and provider handoffs
10-intelligence-memory-proposals-and-ask-safety.md	Standalone execution plan for WP10 — Intelligence, memory, proposals, and Ask safety
11-learn-release-and-design-system-contract.md	Standalone execution plan for WP11 — Learn release and design-system contract
12-ask-learn-shell-and-mobile-cutover.md	Standalone execution plan for WP12 — Ask/Learn shell and mobile cutover
13-deployment-cloud-gates-and-phi-readiness.md	Standalone execution plan for WP13 — Deployment, cloud gates, and PHI readiness
programme_data.py	Canonical structured source used to keep IDs/contracts consistent
scripts/render_programme.py	Mechanical Markdown renderer
scripts/render_pdf.py	Markdown-to-PDF renderer
scripts/validate_programme.py	Cross-file/dependency/gate validator
scripts/sync_project.py	Idempotent GitHub Project inspector/synchronizer

Regenerate all Markdown:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/render_pro
gramme.py
```

Validate parity/dependencies:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/validate_p
rogramme.py
```

Render this exact Markdown to PDF:

```
/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/render_pdf
.py consolidated-plan.md consolidated-plan.pdf
```

Detailed work packages

Work package 00 — Preservation, cloud diagnosis, and PR reconciliation

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP00

Plan	Value
Plan file	`00-preservation-cloud-diagnosis-and-pr-reconciliation.md`
Proposed PRs	P00-01
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. The live app is returning ALB 503 responses while overlapping draft PRs and protected dirty worktrees contain useful but contradictory implementations.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A source-of-truth baseline with Alan's completed review independently dispositioned, existing work mapped, and explicit no-delete/no-close conditions before replacement work begins.

Explicit non-goals.

- No code porting
- No PR closure or retargeting
- No branch/worktree cleanup
- No deployment or AWS mutation

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-001	verified	live dev / 2026-07-12 19:30 America/New_York	https://dev.app.natalyx.health/ and /health	Both requests returned HTTP 503 from awselb/2.0; the deployed participant app is not presently usable.
E-002	verified limitation	AWS dev / 2026-07-12 19:30 America/New_York	aws sts get-caller-identity --profile natalyx-dev	Local AWS SSO was expired, so current ECS desired/running counts and task revision were not independently verified. Gate A0 must refresh access and capture them.
E-003	verified	GitHub / 2026-07-12 19:30 America/New_York	NatalyxPlatform open PR query	App PRs #112, #117, #119, #121, #122, #123 and #124 and design-system PRs #7/#8 are open; all app PRs are drafts.
E-004	verified	.github / main@6072ce9	GitHub Project 3 GraphQL inspection	MVP Product Flow Reshape existed with only default fields and zero items before this planning update.
E-028	verified review	app / PR #124 comment 4948149089	Consolidated review — findings for the WP00 extraction ledger	Alan's completed review classifies #124 as high-value extraction material rather than a merge unit and identifies transaction, privacy, scoring, access, validation, invariant, exposure, retention, performance, CSS and CI findings.
E-029	verified preserved branch	app / feat/byos-journey-wave-2 @5b63e3a7	exact three-commit stack on e853821	Wave 2 contains phase-1 cluster pages, metadata-only request/review, case anchoring and role/owner fixes. Controlled local evidence passed 535 tests with 9 skipped and 36/36 focused tests, but no branch checks, PostgreSQL, browser/accessibility or visual run exists.

3. Architecture and contracts

- Treat every existing branch and worktree as read-only evidence until its valuable files, tests, and reviewer findings have a named destination.
- Use Drive as the canonical review surface; the locally validated Project catalogue remains benched until separately reactivated and synced.
- Do not infer live health from GitHub build success; Gate A0 requires a known revision and live evidence.

Contract area	Package rule
Ownership boundary	`.github`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	No broad public contract; use the focused commands/events in each PR.
Storage	No new storage; preserve current state read-only for this package.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Frozen independent recommendation matrix

This matrix was completed from read-only repository/Drive evidence before any canonical planning source was edited.

ID	Recommendation	Disposition	Rationale	Affected WPs / PRs / gates	Decision
REC-01	Treat PR #124 as WP00 extraction material, not a merge/conflict-resolution unit.	Accept	No independently mergeable unit survives the stale base and domain blockers. Preserve concepts/tests through clean replacement work.	WP00; P03-02/03; P04-01/02/03; P05-02/03; P06-02/03; P09-04; P10-02; P11-03/04; P12-02	No — disposition resolved; closure/deletion still requires later team confirmation.
REC-02	Use corrected PR #122 as the invitation/case-shell baseline.	Accept with modification	#122 is authoritative for named tested behavior and reusable tests, not for final entities, runtime DDL, generic fact/proposal authority or the whole journey model.	WP01; WP02; WP03; WP09; WP10; WP11; WP12	No.
REC-03	Node 0 must delegate invitations to shared authority and make issuance/redemption transactionally complete.	Accept with modification	One invitation authority serves separate resource targets and lifecycles; optional profile completion is resumable rather than forced into membership redemption.	WP03 P03-02/03; WP06 P06-02; A1/A2	No; D-AUTH-02 remains for Cognito admission mechanics.
REC-04	Resolve canonical identity, participation, access and authorization before schema encoding.	Accept with modification	Identity binds principal to person; participation records domain relationship; grants carry resource capabilities; authorization uses verified context, ownership and purpose. Lens is presentation only.	P03-02/03/05; P04-01/02; P09-04; P13-05; A1/B1	Resolved as D-AUTH-03.
REC-05	Restrict IP-entered pre-join GC information and remove unverified suitability ranking/gating.	Accept with modification	Allow minimum-necessary relationship/outreach/logistics context only. Health, mental health, substances, citizenship, race, pregnancy, insurance and eligibility normally wait for authenticated GC entry; no provisional value drives critical decisions.	WP04; WP05 P05-02; WP06; WP10; A1/A2	Resolved as D-DATA-03; exact cross-role fields remain D-DATA-01.

ID	Recommendation	Disposition	Rationale	Affected WPs / PRs / gates	Decision
REC-06	Correct shared-memory private notes and migrate already-written rows.	Accept with modification	Run a metadata-only inventory first; record zero-row evidence if none. If rows exist, deny shared/Ask reads immediately and migrate narrowly without broadening access.	WP04 P04-02; WP10 P10-02; A1/A2	No; actual affected-row count is execution evidence, not an architecture choice.
REC-07	Preserve Wave 2 work from 5b63e3a rather than reimplementing useful behavior.	Accept with modification	Treat 5b63e3a as the strongest cumulative evidence source, not an implementation baseline. Reuse tests and semantics only after the named contract changes.	WP00; WP03; WP09; WP11; WP12; WP13; A2/B1	No.
REC-08	Separate metadata-only document requests from the full content pipeline.	Accept with modification	A request/readiness marker is not a document. Content state and review begin only after a version exists, and partial merges fail closed.	WP09 P09-01/03/04/05/06; P12-02; A2	D-DOC-01/02 and D-PHI-01 remain open for content.
REC-09	Move foundational RLS and non-owner runtime before real-user PII.	Accept	App-only checks are inadequate for real-user PII given current broad readers. No alternative isolation model has equivalent evidence at lower programme cost.	WP13 P13-05/06/07; B1/B2	D-PHI-01/02 remain for approved scope/controls/approvers, not for whether RLS precedes B1.
REC-10	Make #122 Python CI required, stable and resistant to bypass.	Defer	Accept the workflow recommendation now, but enforcement choice/evidence is unavailable. Until D-CI-01 resolves, do not call it bypass-resistant or open real-user gates.	WP01 P01-01; A0 and every later app PR	Yes — D-CI-01, owner Luke; options and required readback/negative test are recorded.
REC-11	Add an explicit Node 0 IP-owner guard and correct non-IP behavior.	Accept with modification	Use one subject-derived capability policy across commands and pages; return explicit deny/role destination without creating a case.	WP03 P03-05; WP05 P05-03; A1	No.
REC-12	Bound and type Node 0 inputs and malformed request behavior.	Accept	Use strict enums/max lengths plus body/collection limits and controlled 4xx errors; do not silently truncate.	WP05; WP06; WP09; WP12	No; exact numerical limits are PR acceptance details and D-DOC-02 for files.
REC-13	Move candidate and invitation invariants from app checks into PostgreSQL migrations.	Accept	Business invariants that can be represented in PostgreSQL must remain true independently of application races.	WP02; WP03; WP04; WP05; A1	No.
REC-14	Replace broad public/cloud prefixes with explicit exposure.	Accept	Explicit route identity is the simplest fail-closed control for the current app.	WP03; WP05; WP06; WP09; WP13; A1/A2	No.
REC-15	Project expiry on GET and add token/PDF cache, referrer and indexing protections.	Accept	Read projection and mutation must agree, and sensitive responses must be non-cacheable with restrictive referrer/noindex policy.	WP03; WP05	No.
REC-16	Bound candidate/invitation and memory queries and remove N+1 behavior.	Accept	Filter/limit at the authoritative storage boundary rather than after loading broad rows.	WP05; WP09; WP10; WP12	No.
REC-17	Add candidate retention, archive, correction and deletion behavior.	Accept	Implement the lifecycle in the candidate domain; keep the exact duration as the named bounded privacy decision.	WP04; WP05 P05-02; A1	Yes — D-DATA-02 supplies duration and exception policy before live use.

ID	Recommendation	Disposition	Rationale	Affected WPs / PRs / gates	Decision
REC-18	Move #124's large global CSS addition to route-scoped Node 0 styling.	Accept with modification	Port only necessary styles through route scope and released DS contracts; block on interaction/accessibility/security, not cosmetic literal purity.	WP05; WP11; WP12	No.
REC-19	Retain unaddressed claim semantics and one-time share-link rotation.	Accept with modification	Retain token mechanics and low-risk semantics only for a pending free-workspace claim; never layer bearer access onto a paid case.	WP03 P03-02/03; WP06	Resolved by D-AUTH-01.
REC-20	Standardize organization handling on NATALYX_ORG_ID.	Reject	Configuration unification is convenient evidence, not the final authorization model. Organization must be verified from the authenticated/granted resource context.	WP03 P03-05; WP13 P13-05	No.
REC-21	Land the GC at shared /case/home and remove a duplicate GC Node 0 home.	Accept with modification	Keep one shared participant shell pattern, but route by actual resource: free-workspace GC projection before promotion, case home only after paid-case access exists.	WP03; WP06; WP09; WP12	No.
REC-22	Keep return_to, token-scoped landing, duplicate-home cleanup, symbol cleanup and refreshed DS assets from #124.	Accept with modification	Adopt the behavior and cleanup, but keep redirect validation, resource-specific landing and canonical DS release ownership.	WP03; WP05; WP11; WP12	No.

Assessment detail

REC-01 — Accept

Assessment field	Recorded conclusion
Recommendation	Treat PR #124 as WP00 extraction material, not a merge/conflict-resolution unit.
Supporting evidence	Alan comment 4948149089; #124@84a5a84; merge-tree against #122@e853821 produced about 93 conflict regions across 11 files.
Prior plan treatment	Already warned against merge but left final disposition pending.
Assumptions	Corrected #122 and replacement PRs remain available as source history.
Security, privacy and operational impact	Avoids stale invitation/security behavior and case-shaped Node 0 reaching users; preserves useful tests/copy/assets.
Dependency and migration impact	WP00 ledger; P03/P04/P05/P06/P09/P10/P11/P12 destinations; no branch mutation.
Alternatives considered	Merge whole branch; cherry-pick mixed commits; file-level extraction.
Rationale	No independently mergeable unit survives the stale base and domain blockers. Preserve concepts/tests through clean replacement work.
Affected WPs / PRs / gates	WP00; P03-02/03; P04-01/02/03; P05-02/03; P06-02/03; P09-04; P10-02; P11-03/04; P12-02
Explicit decision required	No — disposition resolved; closure/deletion still requires later team confirmation.

REC-02 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Use corrected PR #122 as the invitation/case-shell baseline.
Supporting evidence	e853821; 499-test report and green workflow; verified return_to, TTL ceiling, CAS, self/existing-member refusal, one-live-GC invariant, transactional join, paired lenses and fail-closed Learn.
Prior plan treatment	Selective salvage, pending reconciliation.
Assumptions	Behavioral tests can be ported without retaining startup DDL or the case-only domain model.
Security, privacy and operational impact	Retains proven concurrency/auth safety while preventing weak participation/provenance/schema semantics becoming programme authority.
Dependency and migration impact	P01-01, P03-01/03/05, P09-04, P11-03/04, P12-02; re-express schema through WP02.
Alternatives considered	Treat all #122 code/schema as canonical; discard it; behavior/test baseline only.
Rationale	#122 is authoritative for named tested behavior and reusable tests, not for final entities, runtime DDL, generic fact/proposal authority or the whole journey model.
Affected WPs / PRs / gates	WP01; WP02; WP03; WP09; WP10; WP11; WP12
Explicit decision required	No.

REC-03 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Node 0 must delegate invitations to shared authority and make issuance/redemption transactionally complete.
Supporting evidence	#124 splits supersede/create/link/timeline and redemption/profile facts across commits; corrected #122 supplies CAS/TTL guards.
Prior plan treatment	Separate workspace claim and paid-case invitation but transaction boundary was underspecified.
Assumptions	Identity-provider interaction may require an idempotent completion workflow around a database transaction/outbox.
Security, privacy and operational impact	Prevents valid orphan tokens, partial membership/onboarding and possession-only case access.
Dependency and migration impact	P03-02 atomically binds workspace claim/identity/consent; P03-03 atomically redeems invite/participation/access/consent/audit; P06-02 resumes optional onboarding idempotently.
Alternatives considered	One huge cross-system transaction; explicit idempotent saga/outbox; current sequential commits.
Rationale	One invitation authority serves separate resource targets and lifecycles; optional profile completion is resumable rather than forced into membership redemption.
Affected WPs / PRs / gates	WP03 P03-02/03; WP06 P06-02; A1/A2
Explicit decision required	No; D-AUTH-02 remains for Cognito admission mechanics.

REC-04 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Resolve canonical identity, participation, access and authorization before schema encoding.
Supporting evidence	#122/Wave 2 account_memberships roles double as party role, lens and authorization; first-active membership selection; no party/person alignment; callers pass viewer role/lens; RLS absent.
Prior plan treatment	Roles, membership and projections were present but their ownership was not explicit.
Assumptions	A principal can be staff globally without participant-data access and a person can participate without an active grant.
Security, privacy and operational impact	Eliminates confused role authority and gives application/RLS one auditable policy input.
Dependency and migration impact	P03-05 adds identity binding, participation and access-grant constraints; later domain migrations and P13-05 consume the contract.
Alternatives considered	Single membership table/role enum; participation-only authorization; separate grants/capabilities.
Rationale	Identity binds principal to person; participation records domain relationship; grants carry resource capabilities; authorization uses verified context, ownership and purpose. Lens is presentation only.
Affected WPs / PRs / gates	P03-02/03/05; P04-01/02; P09-04; P13-05; A1/B1
Explicit decision required	Resolved as D-AUTH-03.

REC-05 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Restrict IP-entered pre-join GC information and remove unverified suitability ranking/gating.
Supporting evidence	#124 scores mental health, substance, citizenship, pregnancy and support claims into 0-100/high_viability/critical_flag and sorts/gates guidance.
Prior plan treatment	Provisional attribution existed but still allowed a preliminary hard-criteria check.
Assumptions	Before GC joins, the legitimate purpose is low-pressure outreach preparation, not eligibility or professional assessment.
Security, privacy and operational impact	Reduces sensitive third-party processing, discrimination and false clinical/eligibility authority.
Dependency and migration impact	P04-02 provenance; P05-02 strict allowlist; P06-02 GC-authored data; P10 scopes; legacy scores retained only as source-labelled evidence or excluded.
Alternatives considered	Keep ranking with disclaimer; professional approval later; neutral questions only.
Rationale	Allow minimum-necessary relationship/outreach/logistics context only. Health, mental health, substances, citizenship, race, pregnancy, insurance and eligibility normally wait for authenticated GC entry; no provisional value drives critical decisions.
Affected WPs / PRs / gates	WP04; WP05 P05-02; WP06; WP10; A1/A2
Explicit decision required	Resolved as D-DATA-03; exact cross-role fields remain D-DATA-01.

REC-06 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Correct shared-memory private notes and migrate already-written rows.
Supporting evidence	#124 saves Conversation Guide notes with scope=shared and Ask loads shared memory; local inspected SQLite had zero affected rows; #124 was not deployed; remote database state was not verified.
Prior plan treatment	New private/shared ledger planned; explicit legacy remediation was missing.
Assumptions	Source kind identifies guide notes and owner can normally be recovered from subject/person provenance.
Security, privacy and operational impact	Prevents IP reflection notes reaching GC/provider/intelligence while preserving audit and avoiding content in logs.
Dependency and migration impact	P04-02 person-owned contract; P10-02 inventory, immediate exclusion, idempotent private/quarantine migration; ambiguous owner quarantines.
Alternatives considered	Behavior-only fix; delete rows; owner-safe migration/quarantine.
Rationale	Run a metadata-only inventory first; record zero-row evidence if none. If rows exist, deny shared/Ask reads immediately and migrate narrowly without broadening access.
Affected WPs / PRs / gates	WP04 P04-02; WP10 P10-02; A1/A2
Explicit decision required	No; actual affected-row count is execution evidence, not an architecture choice.

REC-07 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Preserve Wave 2 work from 5b63e3a rather than reimplementing useful behavior.
Supporting evidence	Exact three-commit stack on e853821; 535 passed/9 skipped and 36 focused tests; two independent defects reproduced; runtime DDL, caller roles, broad loaders and no RLS remain.
Prior plan treatment	Wave 2 was previously missing and blocked final mapping.
Assumptions	The branch remains paused and available for later extraction.
Security, privacy and operational impact	Saves tested presentation/transaction semantics without importing restart loss, stale approval, metadata/content conflation or weak access.
Dependency and migration impact	P03-05, P09-01/04/06, P11-03/04, P12-02, P13-05; schema re-expressed as migrations.
Alternatives considered	Direct merge/cherry-pick; discard; selective adaptation/test port.
Rationale	Treat 5b63e3a as the strongest cumulative evidence source, not an implementation baseline. Reuse tests and semantics only after the named contract changes.
Affected WPs / PRs / gates	WP00; WP03; WP09; WP11; WP12; WP13; A2/B1
Explicit decision required	No.

REC-08 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Separate metadata-only document requests from the full content pipeline.
Supporting evidence	Wave 2 creates an approvable documents row with storage_uri empty; request labels may contain sensitive data; P09 baseline coupled active-case UI to full upload pipeline.
Prior plan treatment	P09-04 depended on P09-03 and mixed progress/documents; no independent requirement/work-item model.
Assumptions	Phase-1 readiness coordination can operate synthetically without receiving file bytes.
Security, privacy and operational impact	Avoids false artifact approval and permits safe partial delivery while content remains quarantined/disabled.
Dependency and migration impact	P09-01 defines both machines; P09-04 owns metadata requirements/work items and phase-1 surfaces without P09-03; P09-06 integrates content versions after P09-03; P09-05 depends P09-06.
Alternatives considered	Single documents table; metadata-only workflow first; full pipeline first.
Rationale	A request/readiness marker is not a document. Content state and review begin only after a version exists, and partial merges fail closed.
Affected WPs / PRs / gates	WP09 P09-01/03/04/05/06; P12-02; A2
Explicit decision required	D-DOC-01/02 and D-PHI-01 remain open for content.

REC-09 — Accept

Assessment field	Recorded conclusion
Recommendation	Move foundational RLS and non-owner runtime before real-user PII.
Supporting evidence	Runtime uses owner-level startup DDL; app/Wave 2 storage has broad/global loaders; no DB policy enforcement or pool-context proof.
Prior plan treatment	P13-05 was a B2 prerequisite only.
Assumptions	Synthetic A1/A2 may validate features before B1, but no real subject data is admitted.
Security, privacy and operational impact	Database denies cross-org/resource/owner access if an app guard fails; reduces B1 breach blast radius.
Dependency and migration impact	P13-05/06/07 move to B1; FORCE RLS, NOBYPASSRLS runtime, transaction-local verified context and pool reset tests; B2 remains separate evidence approval.
Alternatives considered	App-only isolation; separate databases per cohort; RLS at B2.
Rationale	App-only checks are inadequate for real-user PII given current broad readers. No alternative isolation model has equivalent evidence at lower programme cost.
Affected WPs / PRs / gates	WP13 P13-05/06/07; B1/B2
Explicit decision required	D-PHI-01/02 remain for approved scope/controls/approvers, not for whether RLS precedes B1.

REC-10 — Defer

Assessment field	Recorded conclusion
Recommendation	Make #122 Python CI required, stable and resistant to bypass.
Supporting evidence	#122 workflow passed; #124 ran only build/contract checks; main lacks Python/PostgreSQL CI; private-repo protection/ruleset API returned plan-level 403.
Prior plan treatment	P01-01 added tests but did not define stable contexts, workflow hardening or enforcement limitation.
Assumptions	Repository billing/visibility may change independently of code.
Security, privacy and operational impact	Prevents test-path bypass and false governance claims; adds PostgreSQL evidence.
Dependency and migration impact	P01-01: no path filters, stable names, pinned action SHAs, supported runtime, constraints, timeouts, PostgreSQL job; D-CI-01 governs repository enforcement/readback.
Alternatives considered	Procedural review only; GitHub plan/ruleset upgrade; externally enforced merge queue.
Rationale	Accept the workflow recommendation now, but enforcement choice/evidence is unavailable. Until D-CI-01 resolves, do not call it bypass-resistant or open real-user gates.
Affected WPs / PRs / gates	WP01 P01-01; A0 and every later app PR
Explicit decision required	Yes — D-CI-01, owner Luke; options and required readback/negative test are recorded.

REC-11 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Add an explicit Node 0 IP-owner guard and correct non-IP behavior.
Supporting evidence	#124 ensure endpoint lets any authenticated subject mint a primary-IP case; viewer/provider/staff paths can fall into storage errors/500.
Prior plan treatment	General workspace authorization existed without a shared participant/API guard acceptance matrix.
Assumptions	Staff tools are separate from participant Node 0 routes.
Security, privacy and operational impact	Prevents privilege creation, role confusion and avoidable errors.
Dependency and migration impact	P03-05 subject-derived capabilities; P05-03 shared IP-owner guard and negative role tests.
Alternatives considered	Per-route checks; shared router dependency/service policy; redirect all roles.
Rationale	Use one subject-derived capability policy across commands and pages; return explicit deny/role destination without creating a case.
Affected WPs / PRs / gates	WP03 P03-05; WP05 P05-03; A1
Explicit decision required	No.

REC-12 — Accept

Assessment field	Recorded conclusion
Recommendation	Bound and type Node 0 inputs and malformed request behavior.
Supporting evidence	#124 hand-parses unbounded names/notes/questions/JSON/forms/collections/PDF content; Wave 2 silently truncates labels/notes.
Prior plan treatment	Validation was implicit and spread across feature PRs.
Assumptions	Limits can be product-specific but must reject rather than silently alter user data.
Security, privacy and operational impact	Controls denial-of-service, log/storage amplification and ambiguous stored content.
Dependency and migration impact	P05-02 typed candidate commands/collections; P05-03 form/PDF/body limits; P06-02 onboarding; P09-04 structured metadata/reviewer note; P12-01 Ask.
Alternatives considered	Global proxy limit only; silent slicing; layered strict validation.
Rationale	Use strict enums/max lengths plus body/collection limits and controlled 4xx errors; do not silently truncate.
Affected WPs / PRs / gates	WP05; WP06; WP09; WP12
Explicit decision required	No; exact numerical limits are PR acceptance details and D-DOC-02 for files.

REC-13 — Accept

Assessment field	Recorded conclusion
Recommendation	Move candidate and invitation invariants from app checks into PostgreSQL migrations.
Supporting evidence	#124 lacks candidate-case FK, invitation-link uniqueness, one-active-outreach constraint and checks for enums/scores; concurrent writes can violate product rules.
Prior plan treatment	WP04/WP05 proposed new schema but did not enumerate all reviewed invariants.
Assumptions	Versioned migrations and expand/contract compatibility precede domain tables.
Security, privacy and operational impact	Prevents orphan/duplicate/contradictory state under concurrency.
Dependency and migration impact	P04-01 FKs/uniqueness; P05-02 outreach constraints and no score bounds because ranking is removed; P03-02/03 claim/invite uniqueness; P02-03 migrations.
Alternatives considered	App locks only; advisory locks; database constraints plus commands.
Rationale	Business invariants that can be represented in PostgreSQL must remain true independently of application races.
Affected WPs / PRs / gates	WP02; WP03; WP04; WP05; A1
Explicit decision required	No.

REC-14 — Accept

Assessment field	Recorded conclusion
Recommendation	Replace broad public/cloud prefixes with explicit exposure.
Supporting evidence	#124 exposes /node0/onboarding/ publicly and /api/node0/ by prefix; Wave 2 demonstrates exact phase-1 allowlisting.
Prior plan treatment	Fail-closed route policy existed but did not assign this finding.
Assumptions	New routes require explicit security review before cloud/public reachability.
Security, privacy and operational impact	Prevents future routes inheriting token/public or cloud exposure accidentally.
Dependency and migration impact	P03-02, P05-03, P06-02 exact registries/dependencies; P09-04 exact cluster paths; P13-03 route-negative inventory.
Alternatives considered	Prefix allowlist; per-route registry; separate public app.
Rationale	Explicit route identity is the simplest fail-closed control for the current app.
Affected WPs / PRs / gates	WP03; WP05; WP06; WP09; WP13; A1/A2
Explicit decision required	No.

REC-15 — Accept

Assessment field	Recorded conclusion
Recommendation	Project expiry on GET and add token/PDF cache, referrer and indexing protections.
Supporting evidence	#124 expired links look active until POST; token-bearing pages/PDFs lack no-store and restrictive referrer policy.
Prior plan treatment	Token transitions were covered, response exposure was not.
Assumptions	Tokens remain URL-borne only where unavoidable and never appear in telemetry.
Security, privacy and operational impact	Reduces stale-action confusion and token/sensitive-output leakage via browser cache, referrer or indexing.
Dependency and migration impact	P03-02/03 GET+command expiry and response headers; P05-03 outreach PDF headers; security tests.
Alternatives considered	POST-only check; session exchange; header hardening plus one-time token.
Rationale	Read projection and mutation must agree, and sensitive responses must be non-cacheable with restrictive referrer/noindex policy.
Affected WPs / PRs / gates	WP03; WP05
Explicit decision required	No.

REC-16 — Accept

Assessment field	Recorded conclusion
Recommendation	Bound candidate/invitation and memory queries and remove N+1 behavior.
Supporting evidence	#124 looks up invitations per candidate and loads all shared memory before Python filtering; Wave 2 removes one reviewer-inbox N+1 but retains broad loaders.
Prior plan treatment	Performance was a general risk rather than explicit acceptance criteria.
Assumptions	Pagination and purpose-scoped limits are compatible with Node 0 and Ask UX.
Security, privacy and operational impact	Bounds latency/memory, narrows accidental data access and improves tenant isolation.
Dependency and migration impact	P05-02 batched query-shaped hydration/indexes; P10-02 scope/subject limits at persistence; P12-01 bounded context; P09-04 owner-scoped query shapes.
Alternatives considered	Cache current readers; batch only; purpose-scoped SQL.
Rationale	Filter/limit at the authoritative storage boundary rather than after loading broad rows.
Affected WPs / PRs / gates	WP05; WP09; WP10; WP12
Explicit decision required	No.

REC-17 — Accept

Assessment field	Recorded conclusion
Recommendation	Add candidate retention, archive, correction and deletion behavior.
Supporting evidence	#124 stores sensitive third-party assessments without a lifecycle; current plan has D-DATA-02 but implementation mapping was weak.
Prior plan treatment	Retention decision and legacy backfill existed.
Assumptions	Legal/privacy may set the exact period; source/audit may require tombstones rather than raw-content retention.
Security, privacy and operational impact	Reduces unnecessary third-party data retention while preserving correction/audit needs.
Dependency and migration impact	D-DATA-02; P04-02/03 provenance/tombstones; P05-02 archive/anonymize/delete command/job and evidence.
Alternatives considered	Indefinite retention; immediate deletion; shortest justified period with tombstone/audit.
Rationale	Implement the lifecycle in the candidate domain; keep the exact duration as the named bounded privacy decision.
Affected WPs / PRs / gates	WP04; WP05 P05-02; A1
Explicit decision required	Yes — D-DATA-02 supplies duration and exception policy before live use.

REC-18 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Move #124's large global CSS addition to route-scoped Node 0 styling.
Supporting evidence	About 3,030 lines were added to global style.css; participant DS and route-specific recipes exist.
Prior plan treatment	P05-03 and P11-03 mentioned DS adoption but not the global-style finding.
Assumptions	Participant and internal/operator design systems remain separate.
Security, privacy and operational impact	Limits cross-surface regressions and clarifies style ownership without making aesthetic purity a release gate.
Dependency and migration impact	P05-03 node0.css; P11-03 released tokens/recipes; P12-02 visual/a11y verification.
Alternatives considered	Keep global block; CSS modules/build; route stylesheet plus shared DS.
Rationale	Port only necessary styles through route scope and released DS contracts; block on interaction/accessibility/security, not cosmetic literal purity.
Affected WPs / PRs / gates	WP05; WP11; WP12
Explicit decision required	No.

REC-19 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Retain unaddressed claim semantics and one-time share-link rotation.
Supporting evidence	#124 gates allow_unaddressed to GC intent, stores hash only, rotates/supersedes, and displays plaintext once; corrected #122 supplies case-invite guards.
Prior plan treatment	Workspace claim separated conceptually but shared engine/resource-target distinction was incomplete.
Assumptions	Bearer possession cannot reveal sensitive data or create paid-case membership.
Security, privacy and operational impact	Keeps low-friction outreach while bounding replay, forwarding and privilege escalation.
Dependency and migration impact	P03-02 workspace claim target; P03-03 recipient-bound case target; shared token/CAS primitives, distinct state machines and consent.
Alternatives considered	Addressed links only; unaddressed case invite; unaddressed low-risk workspace claim.
Rationale	Retain token mechanics and low-risk semantics only for a pending free-workspace claim; never layer bearer access onto a paid case.
Affected WPs / PRs / gates	WP03 P03-02/03; WP06
Explicit decision required	Resolved by D-AUTH-01.

REC-20 — Reject

Assessment field	Recorded conclusion
Recommendation	Standardize organization handling on NATALYX_ORG_ID.
Supporting evidence	#124 restores a configured organization constant, but the intended product is multi-resource/multi-tenant and Wave 2 reviewers/readers are insufficiently org-scoped.
Prior plan treatment	Organization scope is part of case/workspace contracts and RLS.
Assumptions	A fixed organization may remain a synthetic/dev bootstrap default only.
Security, privacy and operational impact	Prevents a deployment constant becoming tenant/access authority.
Dependency and migration impact	P03-05 and P13-05 derive organization from verified identity/grant/resource context; config default permitted only in synthetic bootstrap.
Alternatives considered	Global configured org; identity claim; resource/grant-derived org context.
Rationale	Configuration unification is convenient evidence, not the final authorization model. Organization must be verified from the authenticated/granted resource context.
Affected WPs / PRs / gates	WP03 P03-05; WP13 P13-05
Explicit decision required	No.

REC-21 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Land the GC at shared /case/home and remove a duplicate GC Node 0 home.
Supporting evidence	#122 has paired case lenses; #124 routes GC to case home after bearer redemption even though Node 0 should not yet create a case.
Prior plan treatment	Free workspace and paid case are separate but landing wording was not explicit.
Assumptions	The GC may join a free workspace before any promotion and later receive a paid-case invitation.
Security, privacy and operational impact	Avoids duplicate participant shells and prevents pre-promotion users from being shown a canonical-case fiction.
Dependency and migration impact	P03-02/P06 workspace lens before promotion; P03-03/P09-04 shared case home only after canonical paid-case grant; P12 cutover removes duplicates.
Alternatives considered	Always case home; duplicate GC Node 0 home; shared shell with resource-specific projection.
Rationale	Keep one shared participant shell pattern, but route by actual resource: free-workspace GC projection before promotion, case home only after paid-case access exists.
Affected WPs / PRs / gates	WP03; WP06; WP09; WP12
Explicit decision required	No.

REC-22 — Accept with modification

Assessment field	Recorded conclusion
Recommendation	Keep return_to, token-scoped landing, duplicate-home cleanup, symbol cleanup and refreshed DS assets from #124.
Supporting evidence	Corrected #122 validates internal return_to; #124 provides one-time token landing and presentation cleanup; canonical participant DS owns released assets/tokens.
Prior plan treatment	Most items were salvage candidates without precise ownership.
Assumptions	DS assets are ported from an approved release, not vendored ad hoc.
Security, privacy and operational impact	Preserves UX value and resume behavior without weakening redirect, token or repository boundaries.
Dependency and migration impact	P03-01 return_to; P03-02 token landing; P11-03 DS release; P05-03/P12-02 selective UI cleanup.
Alternatives considered	Copy branch files; discard polish; selective port from approved contracts.
Rationale	Adopt the behavior and cleanup, but keep redirect validation, resource-specific landing and canonical DS release ownership.
Affected WPs / PRs / gates	WP03; WP05; WP11; WP12
Explicit decision required	No.

Preserved Wave 2 branch

Field	Finding
Branch / head	feat/byos-journey-wave-2 / 5b63e3a7c3643efcd5f3e74f422d095423b55f59
Corrected #122 base	e85382150853cf4774948c55b91e330e7e69617e
Commit chain	3a549baf — phase-1 cluster pages and initial metadata-only request flow; e40eebaa — transactional submission, private rows and canonical case anchoring; 5b63e3a7 — role/owner checks, atomic review audit, fail-closed dispatch and N+1 removal
Useful	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.
Unsafe	Runtime ensure_column/index DDL; identity/participation/access conflation; caller-supplied role/lens authority; unguarded/global loaders; no RLS or staff capability; metadata readiness represented as an approvable empty document; restart purges case-anchored work items; stale work item can approve a resubmitted row; paused/done actions are not capability-gated.
Defect reproductions	Second init_db changed a canonical case-anchored reviewer item from present to absent because participant_id='' matched the orphan cleanup.; After v1 was resolved and v2 resubmitted on the same row, the stale v1 work item approved v2 because review CAS bound only document_id/state, not an immutable submission revision.
Validation	Controlled local run: 535 passed, 9 skipped in 12.471s; focused Wave 2 cluster suite 36/36. No branch check runs, PostgreSQL, browser/accessibility or visual evidence was present.
Destinations	P01-01, P02-03, P03-01, P03-03, P03-05, P09-01, P09-04, P09-06, P11-03, P11-04, P12-02, P13-05
Disposition	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.

4. Explicit PR sequence

1

Order	1
ID	P00-01
Title	Approve programme baseline and extraction ledger
Repository	.github
Dependencies	None
Gate	Pre-A0
Local preparation	Ready for Alan review and joint acceptance; no new implementation
Merge readiness	Blocked pending Alan review and joint programme acceptance

P00-01 — Approve programme baseline and extraction ledger

Field	Plan
Repository / base	`.github` (`main`)
Project item	P00-01
Gate / size	Pre-A0 / Medium
Local preparation status	Ready for Alan review and joint acceptance; no new implementation
Merge readiness	Blocked pending Alan review and joint programme acceptance
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P00-01 plan sign-off only
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	All listed draft PRs and protected worktrees

Objective

Land the reviewed planning package and extraction baseline only after Alan has reviewed the consolidated document and Luke has accepted the programme contract.

Scope

- Record sign-off, unresolved decisions, source SHAs, and reconciliation procedure
- Freeze programme IDs used by the Project

Explicit non-goals

- No implementation code
- No assignment becomes mandatory before Alan sign-off

Expected files or modules

- `natalyx-repair-program/*.md`
- `natalyx-repair-program/consolidated-plan.pdf`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Repository-neutral planning and programme-control documents only.

Tests and validation

Objective tests:

- git diff --check
- document/project parity validator
- PDF visual QA

Local validation:

- `git diff --check`
- `/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/validate_programme.py`

Cloud validation: run only through Gate Pre-A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Merge only after document/sign-off checks; it changes no runtime. Re-sync the Project from the same canonical data.

Rollback or forward repair

Correct the canonical data and regenerate; do not rewrite implementation history.

Dependencies and parallel work

Hard dependencies: approved programme baseline. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Land the reviewed planning package and extraction baseline only after Alan has reviewed the consolidated document and Luke has accepted the programme contract.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.

- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	programme architecture, technical programme management
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`.github`	programme/governance evidence and Project control plane

Merge order inside this package: `P00-01`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	SHA and timestamp reproducibility; Project/document identifier parity; No dirty-worktree mutation; HTTP and workflow evidence captured separately	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect

Implementation risk	Owner	Treatment
Review findings arrive after decomposition	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
A protected worktree contains the only copy of useful code	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
A green PR is mistaken for live readiness	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P00-01	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P00-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
feat/byos-journey-wave-2	5b63e3a7c364	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.	P01-01, P02-03, P03-01, P03-03, P03-05, P09-01, P09-04, P09-06, P11-03, P11-04, P12-02, P13-05	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.	Paused exact-head evidence; no direct merge

Package source inventory:

- App PRs #112, #117, #119, #121, #122, #123, #124
- Design-system PRs #7 and #8
- Provider-exchange, Node 0 onboarding/quiz, Learn-rendering, and intelligence dirty worktrees

9. Ownership recommendation

Luke owns product/architecture acceptance and the closure decision. Alan's completed technical recommendations are recorded with independent dispositions; implementation ownership remains proposed until the amended package is jointly accepted.

P00-01

PR	P00-01
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 01 — CI and PostgreSQL evidence gate

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP01
Plan file	`01-ci-and-postgres-evidence-gate.md`
Proposed PRs	P01-01
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Main builds containers and runs a deploy-contract smoke, but does not run the Python suite or prove current schema behavior against PostgreSQL.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A first, deliberately CI-only PR that runs the current tests, exercises the existing schema on PostgreSQL, and publishes diagnosable evidence under stable required-check names without introducing the future migration framework.

Explicit non-goals.

- No migration runner
- No product behavior changes
- No schema redesign
- No deployment

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-005	verified	app / origin/main@fdf2bb4	.github/workflows/build.yml:1-42; aws-app-deploy.yml:1-38	Main builds images and runs a local deploy-contract smoke but has no Python test workflow or PostgreSQL evidence job.
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.

ID	Class	Repository / ref	File, construct or location	Finding
E-009	verified	app / origin/main@fdf2bb4	app/storage/connection.py:14-57,93-123	The app supports pooled PostgreSQL and SQLite with a SQL compatibility adapter; PostgreSQL is the cloud backend but the current suite is predominantly local.
E-032	verified limitation	GitHub / 2026-07-12	private-repository branch-protection/ruleset API	The API returned a plan-level 403 requiring GitHub Pro or public visibility, so a stable workflow can be planned but bypass-resistant required-check enforcement cannot be claimed until D-CI-01 is resolved and read back.

3. Architecture and contracts

- PostgreSQL is production authority; SQLite remains a local compatibility target, not production evidence.
- Separate current-schema evidence from tests that require the future migration mechanism.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	No broad public contract; use the focused commands/events in each PR.
Storage	None; exercise the schema that main owns today.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

4. Explicit PR sequence

1

Order	1
ID	P01-01
Title	Establish current-test and PostgreSQL CI evidence
Repository	app
Dependencies	P00-01
Gate	A0
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P00-01; technical decisions/controls D-CI-01; Luke review/final founder approval

P01-01 — Establish current-test and PostgreSQL CI evidence

Field	Plan
Repository / base	`app` (`main`)
Project item	P01-01
Gate / size	A0 / Medium
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P00-01; technical decisions/controls D-CI-01; Luke review/final founder approval
Technical-decision owner	D-CI-01: Alan - immediate control configuration/evidence
Dependencies	P00-01
Blocking decisions/checkpoints	D-CI-01
Source / salvage	PR #122 unittest workflow and backend-neutral fixes after reconciliation

Objective

Run the current suite on pull requests/main and prove the existing schema boots and behaves on an ephemeral PostgreSQL service before any migration redesign.

Scope

- Stable, unfiltered unittest and PostgreSQL check names
- Pinned action SHAs and supported Python runtime
- Locked/constraint-backed dependencies, timeouts and cancellation
- Ephemeral PostgreSQL current-schema evidence
- Sanitized failure artifacts

Explicit non-goals

- No migration-version table
- No new-schema tests
- No application behavior

Expected files or modules

- `.github/workflows/python-tests.yml`
- `tests/python/test_postgres_current_schema.py`
- `scripts/ci_postgres_evidence.py`

Change map

Area	Planned change
Schema/model	None; exercise the schema that main owns today.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Add stable unittest and PostgreSQL evidence jobs; repository-side required-check enforcement remains D-CI-01 until independently read back.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Current Python suite
- workflow trigger/path-bypass contract
- init_db twice on PostgreSQL

- representative create/read/authorization path
- intentional failure log-redaction check

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P00-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Run the current suite on pull requests/main and prove the existing schema boots and behaves on an ephemeral PostgreSQL service before any migration redesign.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-CI-01: Alan - immediate control configuration/evidence
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P01-01`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Current unit suite; Fresh PostgreSQL database; Current init_db idempotency; Failure artifact and log clarity	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-CI-01	Founder final; operationally pending	Luke decides commercial outcome; Alan configures immediately	Alan - immediate	P01-01; Founder decision resolved 2026-07-13; control required before P01-01 merge or any later app implementation PR	P01-01 merge, Gate A0 and dependent implementation PRs remain blocked until the live control is active and verified.

D-CI-01 - Required-check enforcement for the private app repository

Field	Recorded decision or requirement
Decision state	Final founder decision; control pending
Founder outcome	Option A — GitHub Team approved as the private-repository enforcement control.
Founder constraints	Approximately $12 USD/month before tax for three billable users; Commercial decision is final; Expenditure is authorized but not yet incurred; No claim of bypass-resistant enforcement before live configuration and readback
What remains for Alan	Activate GitHub Team when enforcement is required; Configure required unittest and PostgreSQL checks; Restrict routine bypass; Supply ruleset readback and negative-bypass evidence
Evidence required	GitHub Team activation/readback; Mandatory unittest and PostgreSQL checks; Restricted routine bypass; Negative-bypass evidence
Approval rule	Luke accepts the enforcement evidence; the approved purchase itself needs no further founder decision.
External dependencies	GitHub Team activation; expenditure authorized but not yet incurred
Affected PRs / latest point	P01-01; Founder decision resolved 2026-07-13; control required before P01-01 merge or any later app implementation PR
Programme effect	P01-01 merge, Gate A0 and dependent implementation PRs remain blocked until the live control is active and verified.
Independent recommendation	Approved: GitHub Team for three billable users at approximately $12 USD/month before tax. Expenditure is authorized but not yet incurred. Alan owns activation-time ruleset design and evidence; Gate A0 and dependent implementation PRs remain blocked until Team is active, both checks are mandatory, routine bypass is restricted, and negative-bypass evidence is accepted.

Implementation risk	Owner	Treatment
CI exposes existing backend drift	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Tests accidentally depend on protected branch code	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Secrets or DB credentials leak in logs	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P01-01	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
feat/byos-journey-wave-2	5b63e3a7c364	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.	P01-01	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.	Paused exact-head evidence; no direct merge

Package source inventory:

- PR #122 Python-test workflow and PostgreSQL-compatible test fixes only after reconciliation

9. Ownership recommendation

Alan proposes the workflow and test matrix; Luke approves the evidence threshold. Repository-enforced required-check governance remains bounded by D-CI-01, but the workflow itself can land after programme acceptance.

P01-01

PR	P01-01
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-CI-01: Alan - immediate control configuration/evidence
Reviewer	Luke
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 02 — Versioned migrations and readiness

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP02
Plan file	`02-versioned-migrations-and-readiness.md`
Proposed PRs	P02-01, P02-02, P02-03, P02-04
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Startup calls a monolithic CREATE/ALTER routine without an ordered migration ledger, advisory lock, expand/contract contract, or reliable partial-failure recovery.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. An approved migration ADR, versioned runner, production-safe locking/readiness behavior, baseline migration, and forward-repair procedure proven on PostgreSQL and compatible with supported SQLite development.

Explicit non-goals.

- No commercial schema yet
- No Node 0 semantic migration before WP04
- No destructive rollback promise
- No application feature work

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-007	verified	app / feature/byos-node0-consolidation@84a5a84	app/storage/schema.py:1-33,353-880	Schema ownership is a monolithic init_db/ensure_column routine with inline CREATE TABLE/ALTER TABLE and no ordered migration/version/checksum ledger.
E-008	verified	app / origin/main@fdf2bb4	app/main.py:295-313	Cloud startup validates auth/storage then calls storage.init_db in-process; readiness is not separated from migration execution.
E-009	verified	app / origin/main@fdf2bb4	app/storage/connection.py:14-57,93-123	The app supports pooled PostgreSQL and SQLite with a SQL compatibility adapter; PostgreSQL is the cloud backend but the current suite is predominantly local.
E-030	independently reproduced defect	app / feat/byos-journey-wave-2@5b63e3a7	app/storage/schema.py orphan cleanup; document review CAS/work-item payload	A second init_db deletes canonical case-anchored reviewer work because participant_id is empty. Separately, a stale v1 work item can approve v2 after same-row resubmission because it is not bound to an immutable submission revision.

3. Architecture and contracts

- Alan evaluates Alembic versus a custom ordered runner; a custom runner is not the default without ADR evidence.
- Migrations are forward-only by default, transaction-scoped where PostgreSQL permits, advisory-locked, checksum-recorded, and expand/contract compatible with the previous app revision.

Contract area	Package rule
Ownership boundary	`app`, `intelligence`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Add internal readiness detail behind safe health surface.
Storage	Add migration metadata only. Existing canonical schema becomes a checksummed baseline. Intelligence-owned tables enter a checksummed revision chain.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Migration ADR contract

Options Alan must assess:

1. Alembic with reviewed operations/raw SQL
2. Custom ordered Python migrations with a demonstrated reason not to use an established framework
3. A standalone established runner such as Flyway/dbmate if its runtime/deployment burden fits

Founder constraints and independent recommendation. Founder constraints are final: Alan should prefer Alembic or another well-supported low-maintenance framework, while retaining authority to choose a different established runner or justify a custom runner in the ADR.

- Zero intended data loss is the governing reliability objective.
- PostgreSQL is production authority; SQLite may remain a convenient synthetic local-development projection without full production parity.
- Fingerprint existing databases before baseline stamping; never blindly stamp unknown schema.
- App and intelligence use separate revision namespaces and PostgreSQL advisory-lock keys.
- Migration identities hold DDL; app/intelligence runtime identities do not after adoption.
- Run migrations in a deploy-controlled one-off task before app rollout, not opportunistically from web startup.
- Stage consequential changes and keep them backward-compatible wherever reasonably possible; use expand in release N, bounded/backpressure-aware backfill, dual-compatible N+1, then contract only when N cannot run.
- CI covers fresh, current, one-revision-behind, N-1 app on expanded schema, concurrent runner, checksum drift and interrupted backfill.
- Every consequential migration has tested recovery or rollback evidence before use.
- Scheduled and communicated beta downtime may be up to 30 minutes when a tested recovery plan exists.
- Irreversible changes that could affect real-user data require explicit evidence and Luke approval plus a forward-repair plan. Schema rollback is not automatic.

4. Explicit PR sequence

1

Order	1
ID	P02-01
Title	Decide migration architecture and compatibility contract
Repository	app
Dependencies	P01-01
Gate	A0
Local preparation	Technical-decision preparation only; implementation waits on named dependencies
Merge readiness	Blocked pending PR dependencies P01-01; Luke review/final founder approval

2

Order	2
ID	P02-02
Title	Introduce versioned migration runner and readiness checks
Repository	app
Dependencies	P02-01
Gate	A0
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-01; technical decisions/controls D-MIG-01; Luke review/final founder approval

3

Order	3
ID	P02-03
Title	Baseline canonical schema and prove expand-contract repair
Repository	app
Dependencies	P02-02
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-02; technical decisions/controls D-MIG-01; Gate A0 active and verified; Luke review/final founder approval

4

Order	4
ID	P02-04
Title	Move intelligence persistence to a revisioned migration namespace
Repository	intelligence
Dependencies	P02-01, P02-02
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-01, P02-02; Gate A0 active and verified; Luke review/final founder approval

P02-01 — Decide migration architecture and compatibility contract

Field	Plan
Repository / base	`app` (`main`)
Project item	P02-01
Gate / size	A0 / Small
Local preparation status	Technical-decision preparation only; implementation waits on named dependencies
Merge readiness	Blocked pending PR dependencies P01-01; Luke review/final founder approval
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Dependencies	P01-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	app/storage/schema.py, app/storage/connection.py

Objective

Record an ADR choosing an established migration framework or justified ordered runner, including PostgreSQL authority, SQLite support, locks, expand/contract, backfills, rollback, and forward repair.

Scope

- Compare Alembic and custom runner against current SQL compatibility layer
- Specify old/new revision compatibility and irreversible-change policy

Explicit non-goals

- No migration code
- No schema change

Expected files or modules

- `docs/adr/ADR-00x-versioned-migrations.md`
- `docs/agents/storage-data-layer.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Decision record only; select framework before P02-02.

Tests and validation

Objective tests:

- ADR decision checklist
- example expand/contract and interrupted-backfill walkthrough

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P01-01. Parallel opportunities: P02-03

Objective acceptance criteria

- □ Record an ADR choosing an established migration framework or justified ordered runner, including PostgreSQL authority, SQLite support, locks, expand/contract, backfills, rollback, and forward repair.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	PostgreSQL migrations, zero-downtime deployment
Higher-capability review	Mandatory before merge

P02-02 — Introduce versioned migration runner and readiness checks

Field	Plan
Repository / base	`app` (`main`)
Project item	P02-02
Gate / size	A0 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-01; technical decisions/controls D-MIG-01; Luke review/final founder approval
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Dependencies	P02-01
Blocking decisions/checkpoints	D-MIG-01
Source / salvage	Current init_db startup path

Objective

Implement the chosen ordered migration mechanism, checksum ledger, advisory lock, startup mode, and readiness signal without moving product tables yet.

Scope

- Version/checksum ledger and PostgreSQL advisory locking
- Separate migrate job/startup fail-closed policy
- Readiness reports pending/failed migration without sensitive detail

Explicit non-goals

- No Node 0/commercial tables
- No destructive down migration

Expected files or modules

- `migrations/`
- `app/storage/migrations.py`
- `app/readiness.py`
- `app/main.py`
- `tests/python/test_migrations.py`

Change map

Area	Planned change
Schema/model	Add migration metadata only.
API/routes	Add internal readiness detail behind safe health surface.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Bootstrap current database at a verified baseline.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Two concurrent migrators
- Checksum mismatch
- Partial failure and forward repair
- old app during expand window

- SQLite mode per ADR

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Implement the chosen ordered migration mechanism, checksum ledger, advisory lock, startup mode, and readiness signal without moving product tables yet.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	PostgreSQL, migration framework, runtime operations
Higher-capability review	Mandatory before merge

P02-03 — Baseline canonical schema and prove expand-contract repair

Field	Plan
Repository / base	`app` (`main`)
Project item	P02-03
Gate / size	A1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-02; technical decisions/controls D-MIG-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Dependencies	P02-02
Blocking decisions/checkpoints	D-MIG-01
Source / salvage	app/storage/schema.py current definitions, PR #122 forward-migration tests after reconciliation

Objective

Move the existing canonical schema under version control, preserve compatible reads/writes, and supply rehearsal/backfill/repair tooling for future packages.

Scope

- Baseline existing tables and indexes
- Remove startup DDL only after parity is proved
- Document restore/forward-repair procedure

Explicit non-goals

- No semantic Node 0 split
- No commercial schema

Expected files or modules

- `migrations/versions/*_canonical_baseline.py`
- `app/storage/schema.py`
- `scripts/rehearse_migrations.py`
- `tests/migrations/`

Change map

Area	Planned change
Schema/model	Existing canonical schema becomes a checksummed baseline.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Expand/contract rehearsal from previous main and representative legacy fixtures.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Fresh install
- Upgrade from captured pre-baseline schema
- previous app compatibility
- large-table backfill rehearsal
- failure injection

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-02. Parallel opportunities: P02-01, P02-04

Objective acceptance criteria

- □ Move the existing canonical schema under version control, preserve compatible reads/writes, and supply rehearsal/backfill/repair tooling for future packages.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P02-04 — Move intelligence persistence to a revisioned migration namespace

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P02-04
Gate / size	A1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-01, P02-02; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P02-01, P02-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	src/natalyx_intelligence/shared/persistence.py runtime _ensure_schema

Objective

Replace intelligence runtime _ensure_schema DDL with the chosen framework, a separate revision namespace/advisory lock, deploy-time migration entrypoint, and cross-service compatibility evidence.

Scope

- Fingerprint and baseline intelligence-owned tables
- Separate revision/lock key from app
- Remove runtime DDL authority after adoption
- Shared-Postgres compatibility matrix

Explicit non-goals

- No context/product feature
- No app-owned table mutation

Expected files or modules

- `migrations/`
- `src/natalyx_intelligence/shared/persistence.py`
- `scripts/migrate.py`
- `tests/test_migrations.py`

Change map

Area	Planned change
Schema/model	Intelligence-owned tables enter a checksummed revision chain.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Deploy-controlled one-off task; runtime verifies head only.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Fresh/legacy PostgreSQL
- concurrent app/intelligence migrators
- old/new service compatibility
- interrupted migration
- runtime non-owner

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-01, P02-02. Parallel opportunities: P02-03

Objective acceptance criteria

- □ Replace intelligence runtime _ensure_schema DDL with the chosen framework, a separate revision namespace/advisory lock, deploy-time migration entrypoint, and cross-service compatibility evidence.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	PostgreSQL migrations, FastAPI service operations
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`intelligence`	advisory inference, scoped response/citation contract; no case truth

Merge order inside this package: `P02-01` → `P02-02` → `P02-03` → `P02-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Fresh and upgraded PostgreSQL; Concurrent migrators; Old/new app compatibility; Interrupted migration and forward repair; SQLite support decision	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-MIG-01	Founder constraints final; Alan immediate; operationally pending	Luke sets constraints; Alan decides immediately	Alan - immediate	P02-01, P02-02, P02-03; Before P02-02	P02-01 decision work may proceed; P02-02, P02-03 and Gate A0 remain blocked until Alan's mechanism and evidence are accepted.

D-MIG-01 - Migration framework and SQLite support

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Reliability-first migrations with zero intended data loss; PostgreSQL is authoritative for production and SQLite may remain a convenient synthetic local-development projection without full production parity.
Founder constraints	Stage consequential changes and keep them backward-compatible wherever reasonably possible; Test recovery or rollback evidence before every consequential migration is used; Prefer a well-supported low-maintenance approach over a clever or bespoke runner; Planned beta downtime may be up to 30 minutes when scheduled, communicated and backed by a tested recovery plan; Destructive or irreversible real-user-data changes require explicit evidence and Luke approval
What remains for Alan	Choose the migration framework; Choose sequencing/locking and expand-contract mechanics; Choose the detailed rollback, restore or forward-repair mechanism
Evidence required	Migration ADR; Fresh/current/upgrade compatibility matrix; Tested recovery or rollback evidence; Destructive-change approval path
Approval rule	Luke approves only destructive or irreversible actions that could affect real-user data; ordinary technical choices stay with Alan.
External dependencies	None
Affected PRs / latest point	P02-01, P02-02, P02-03; Before P02-02
Programme effect	P02-01 decision work may proceed; P02-02, P02-03 and Gate A0 remain blocked until Alan's mechanism and evidence are accepted.
Independent recommendation	Prefer an established framework unless the ADR proves current SQL compatibility makes it unsafe or disproportionate.

Implementation risk	Owner	Treatment
Irreversible DDL	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Concurrent deploy race	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Long backfill lock	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
SQLite semantics hide PostgreSQL failure	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P02-03	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
feat/byos-journey-wave-2	5b63e3a7c364	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.	P02-03	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.	Paused exact-head evidence; no direct merge

Package source inventory:

- Current app/storage/schema.py definitions as baseline evidence; no direct copy of ad-hoc ensure_column sequencing

9. Ownership recommendation

Alan owns the technical proposal and implementation. Luke approves the architecture/compatibility contract. A higher-capability review is mandatory before merge.

P02-01

PR	P02-01
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P02-02

PR	P02-02
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P02-03

PR	P02-03
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-MIG-01: Alan - immediate framework, sequencing and recovery mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P02-04

PR	P02-04
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.

- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 03 — Authentication, invitations, and resume

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP03
Plan file	`03-auth-invitations-and-resume.md`
Proposed PRs	P03-01, P03-02, P03-03, P03-04, P03-05
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Current invitation code mixes low-risk onboarding links with case membership, and existing flows need explicit recipient binding, return-to safety, consent, expiry, revocation, duplicate-claim, and mismatch semantics.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Separate workspace-claim and paid-case invitation lifecycles with safe authentication/resume behavior and no link-possession path to sensitive case access.

Explicit non-goals.

- No automatic paid-case membership
- No email deliverability product
- No commercial promotion
- No PHI enablement

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.
E-011	verified	app / PR #124@84a5a84	app/storage/node0.py:400-462; app/routes/node0.py:605-652	Node 0 can issue an unaddressed GC invitation and redemption ultimately creates case membership, conflating low-risk onboarding possession with case access.
E-012	verified	app / PR #122@e853821	app/storage/invitations.py and PR review-round comment	The updated case invitation work adds 14-day ceiling, CAS transitions, mismatch confirmation, self-invite refusal, return_to preservation and paired lens tests; it has no Node 0 workspace target.
E-028	verified review	app / PR #124 comment 4948149089	Consolidated review — findings for the WP00 extraction ledger	Alan's completed review classifies #124 as high-value extraction material rather than a merge unit and identifies transaction, privacy, scoring, access, validation, invariant, exposure, retention, performance, CSS and CI findings.

ID	Class	Repository / ref	File, construct or location	Finding
E-029	verified preserved branch	app / feat/byos-journey-wave-2 @5b63e3a7	exact three-commit stack on e853821	Wave 2 contains phase-1 cluster pages, metadata-only request/review, case anchoring and role/owner fixes. Controlled local evidence passed 535 tests with 9 skipped and 36/36 focused tests, but no branch checks, PostgreSQL, browser/accessibility or visual run exists.
E-031	verified	app / feat/byos-journey-wave-2 @5b63e3a7	case_members.py, case_documents.py, storage/documents.py, routes/agency.py	Membership roles still conflate participation, lens and access; document/reviewer loaders are broad or global; participant role/lens can be caller-supplied; operator actions lack a canonical organization-scoped staff capability; no RLS/non-owner runtime enforces denial.

3. Architecture and contracts

- An email-less link may create a pending low-risk workspace claim only.
- Identity binds an authenticated principal to a person; participation records the person's domain relationship; access grants carry resource-scoped capabilities; authorization consumes all three plus ownership and purpose.
- Paid-case membership requires authenticated identity binding, recipient or issuer confirmation as appropriate, explicit consent, and auditable redemption.
- Return targets are internal allowlisted paths and survive login/signup/confirmation without becoming redirect authority.

Contract area	Package rule
Ownership boundary	`app`, `infra`; repository-specific PRs remain separate.
Entities/state	`case_invitation`, `entry_session`, `workspace_claim`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Auth routes accept only allowlisted relative return targets. Issue, inspect, bind, consent, claim, revoke endpoints. Separate paid-case redeem/confirm flow. Client metadata carries opaque admission proof only.
Storage	Workspace claims and audit/idempotency fields. Case-invitation identity, consent, expiry, revocation, idempotency fields. Unique identity binding, participation and access-grant tables with lifecycle and scope constraints.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Participant entry and routing (`entry_session`)

Owner: natalyx-app. States: `unstarted`, `education`, `identity_required`, `authenticated`, `routed`, `handed_off`, `expired`.

unstarted

From	unstarted
To	education
Actor	visitor
Evidence	GET/explicit entry choice
Idempotency key	entry:{session}:start
Invalid transitions	No participant/case write
Audit event	entry_started
Retry	Return current session
Recovery	Create fresh anonymous session

education

From	education
To	identity_required
Actor	visitor/policy
Evidence	route requires saved workspace
Idempotency key	entry:{session}:identity
Invalid transitions	Cannot grant workspace/case
Audit event	identity_required
Retry	Preserve safe return intent
Recovery	Restart at public choice

identity_required

From	identity_required
To	authenticated
Actor	Cognito/auth provider
Evidence	verified subject/session
Idempotency key	auth:{subject}:{nonce}
Invalid transitions	Invalid/expired token
Audit event	identity_bound
Retry	Replay returns authenticated session
Recovery	Rechallenge; no partial membership

education|authenticated

From	education|authenticated
To	routed
Actor	participant + deterministic policy
Evidence	validated route answers + policy version
Idempotency key	route:{session}:{answer_hash}
Invalid transitions	Unknown/unsafe route goes review/education
Audit event	entry_routed
Retry	Same input returns same decision
Recovery	Allow corrected answers/new decision revision

routed

From	routed
To	handed_off
Actor	participant
Evidence	accepted destination and authorized target
Idempotency key	handoff:{session}:{route_version}
Invalid transitions	Target not authorized/available
Audit event	entry_handed_off
Retry	Resume same safe destination
Recovery	Return to route result with explanation

any nonterminal

From	any nonterminal
To	expired
Actor	system
Evidence	TTL elapsed/revoked session
Idempotency key	entry:{session}:expire
Invalid transitions	No transition from handed_off needed
Audit event	entry_expired
Retry	Idempotent
Recovery	New session; no case/access change

Low-risk workspace identity claim (`workspace_claim`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `consented`, `claimed`, `mismatch_pending`, `revoked`, `expired`.

none

From	none
To	issued
Actor	workspace IP
Evidence	selected candidate + allowed claim scope
Idempotency key	claim:{workspace}:{candidate}:{generation}
Invalid transitions	Existing live claim superseded
Audit event	workspace_claim_issued
Retry	Return current token only at issuance
Recovery	Issue new generation; old revoked

issued

From	issued
To	pending_identity
Actor	link holder
Evidence	valid token landing
Idempotency key	claim:{id}:land
Invalid transitions	Expired/revoked token
Audit event	claim_opened
Retry	Safe repeat
Recovery	Authenticate or exit; no data grant

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email and binding policy
Idempotency key	claim:{id}:bind:{subject}
Invalid transitions	Already claimed/different subject
Audit event	claim_identity_bound|claim_mismatch
Retry	CAS no-op or mismatch state
Recovery	Issuer review/new claim

identity_bound

From	identity_bound
To	consented
Actor	GC claimant
Evidence	workspace consent version
Idempotency key	claim:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	claim_consented
Retry	Idempotent
Recovery	Display current consent

consented

From	consented
To	claimed
Actor	claim service
Evidence	authorized low-risk workspace membership transaction
Idempotency key	claim:{id}:complete
Invalid transitions	Case membership requested or conflicting claimant
Audit event	workspace_claimed
Retry	Return same membership
Recovery	Transaction rollback/forward repair

issued|pending_identity|identity_bound|consented|mismatch_pending

From	issued|pending_identity|identity_bound|consented|mismatch_pending
To	revoked|expired
Actor	issuer/system
Evidence	revocation or TTL
Idempotency key	claim:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	claim_revoked|claim_expired
Retry	Idempotent
Recovery	Fresh claim only

Paid-case membership invitation (`case_invitation`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

none

From	none
To	issued
Actor	authorized IP case member
Evidence	enrollment_pending/active case + recipient binding
Idempotency key	case-invite:{case}:{role}:{generation}
Invalid transitions	Self/member/active-GC/unauthorized
Audit event	case_invite_issued
Retry	Return current; supersede prior live GC invite
Recovery	Issue corrected invite

issued

From	issued
To	pending_identity
Actor	recipient
Evidence	valid addressed token landing
Idempotency key	case-invite:{id}:land
Invalid transitions	Expired/revoked
Audit event	case_invite_opened
Retry	Safe repeat
Recovery	Authenticate with return_to

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email
Idempotency key	case-invite:{id}:bind:{subject}
Invalid transitions	Terminal/duplicate/different subject
Audit event	case_invite_bound|mismatch
Retry	CAS protects terminal
Recovery	Issuer confirmation or corrected invite

mismatch_pending

From	mismatch_pending
To	issuer_confirmed
Actor	authorized issuer
Evidence	explicit reviewed mismatch confirmation
Idempotency key	case-invite:{id}:confirm:{issuer}
Invalid transitions	Issuer unauthorized/recipient changed
Audit event	case_invite_mismatch_confirmed
Retry	CAS no-op
Recovery	Revoke and reissue

identity_bound|issuer_confirmed

From	identity_bound|issuer_confirmed
To	consented
Actor	invitee
Evidence	paid-case role/privacy consent version
Idempotency key	case-invite:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	case_invite_consented
Retry	Idempotent
Recovery	Present current consent

consented

From	consented
To	redeemed
Actor	invitation service
Evidence	single transaction creates person/party/membership
Idempotency key	case-invite:{id}:redeem
Invalid transitions	Existing member/active GC/expired/revoked
Audit event	case_invite_redeemed
Retry	Return same membership
Recovery	Rollback entire transaction; forward repair from invite

issued\|pending_identity\|identity_bound\|mismatch_pending\|issuer_confirmed\|consented

From	issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented
To	declined|revoked|expired|superseded
Actor	invitee/issuer/system
Evidence	explicit decline/revoke/TTL/new generation
Idempotency key	case-invite:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	case_invite_terminal
Retry	Idempotent CAS
Recovery	Fresh invitation only

Invitation and identity safety

Retain from updated PR #122:

- hashed tokens
- purpose-specific TTL with a 14-day ceiling
- CAS terminal transitions
- atomic membership creation
- one-live-GC constraint
- self/existing-member refusal
- safe return_to
- portable integrity/concurrency tests

Required replacement contract:

- Exactly one target: workspace_id or case_id; include purpose, role, access_class, recipient_binding_mode and policy version.
- Email-less possession may only open pending low-risk workspace claim; it cannot reveal sensitive data or grant paid-case membership.

- Paid-case invitation is addressed, authenticated, consented and atomically redeems invitation plus participation, access grant and audit; mismatch requires explicit policy/issuer handling and preferably a fresh bound token.
- Workspace-claim issuance commits supersession, hashed claim, candidate/workspace linkage and required audit atomically; external identity steps use an explicit idempotent completion state.
- Cognito admission and app membership are separate decisions; account creation never equals workspace/case access.
- Verified identity comes from auth claims, not form-entered email; duplicate claims return the original result.
- Never put raw token/email/PHI in general participant timeline payloads; revoke/expire/supersede fail closed.
- Remove generic unconditional status mutation; expose authorized transition commands only.

4. Explicit PR sequence

1

Order	1
ID	P03-01
Title	Harden auth return-to and participant resume
Repository	app
Dependencies	P01-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01; Gate A0 active and verified; Luke review/final founder approval

2

Order	2
ID	P03-02
Title	Add low-risk workspace claim lifecycle
Repository	app
Dependencies	P03-01, P03-05, P04-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P03-05, P04-01; Gate A0 active and verified; Luke review/final founder approval

3

Order	3
ID	P03-03
Title	Bind paid-case invitations to identity and consent
Repository	app
Dependencies	P03-01, P03-05
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P03-05; Gate A0 active and verified; Luke review/final founder approval

4

Order	4
ID	P03-04
Title	Replace shared signup admission with claim-aware Cognito admission
Repository	infra
Dependencies	P03-02, P03-03
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P03-03; technical decisions/controls D-AUTH-02; Gate A0 active and verified; Luke review/final founder approval

5

Order	5
ID	P03-05
Title	Implement canonical identity, participation, access, and authorization contracts
Repository	app
Dependencies	P02-03, P03-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-01; Gate A0 active and verified; Luke review/final founder approval

P03-01 — Harden auth return-to and participant resume

Field	Plan
Repository / base	`app` (`main`)
Project item	P03-01
Gate / size	A1 / Medium
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P01-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 return_to fixes

Objective

Make signup/login/confirmation/logout preserve a validated internal destination and restore the correct role-scoped workspace without query parameters becoming authority.

Scope

- Central safe_return_to parser
- Resume intent across login/signup/confirm/resend
- Session/cookie security and error states

Explicit non-goals

- No invitation issuance
- No case access grant

Expected files or modules

- `app/auth/routes.py`
- `app/auth/cloud.py`
- `app/auth/cognito.py`

- `tests/python/test_auth_routes.py`
- `tests/python/test_auth_cloud_integration.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Auth routes accept only allowlisted relative return targets.
UI/product surface	Clear expired/failed session recovery.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Open redirect corpus
- logged-out deep link
- signup confirmation
- logout/login resume
- secure/samesite cookie matrix

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P01-01. Parallel opportunities: P03-04

Objective acceptance criteria

- □ Make signup/login/confirmation/logout preserve a validated internal destination and restore the correct role-scoped workspace without query parameters becoming authority.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Cognito, web session security, FastAPI
Higher-capability review	Risk-proportionate; not mandatory by programme default

P03-02 — Add low-risk workspace claim lifecycle

Field	Plan
Repository / base	`app` (`main`)
Project item	P03-02
Gate / size	A1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P03-05, P04-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-01, P03-05, P04-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PRs #123/#124, dirty onboarding-link worktree

Objective

Create a pending Node 0 workspace claim that can start from an unaddressed link but requires identity binding, consent, expiry, revocation, and audit before low-risk access.

Scope

- Hashed one-time token and claim state machine
- Identity bind/consent/revoke/duplicate/mismatch behavior
- No sensitive case scope

Explicit non-goals

- No paid-case membership
- No possession-only private access

Expected files or modules

- `app/storage/workspace_claims.py`
- `app/routes/node0_claims.py`
- `app/templates/node0/claim*.html`
- `tests/python/test_workspace_claims.py`

Change map

Area	Planned change
Schema/model	Workspace claims and audit/idempotency fields.
API/routes	Issue, inspect, bind, consent, claim, revoke endpoints.
UI/product surface	Low-risk claim and mismatch screens.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Forwarded link
- duplicate/concurrent claim
- expiry/revocation
- identity mismatch
- issuer visibility

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-01, P03-05, P04-01. Parallel opportunities: P03-03

Objective acceptance criteria

- □ Create a pending Node 0 workspace claim that can start from an unaddressed link but requires identity binding, consent, expiry, revocation, and audit before low-risk access.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P03-03 — Bind paid-case invitations to identity and consent

Field	Plan
Repository / base	`app` (`main`)
Project item	P03-03
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P03-05; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-01, P03-05
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 CAS/TTL/self-invite fixes, current invitations.py

Objective

Implement a separate recipient-bound paid-case invitation that grants membership only after authenticated binding, required confirmation, consent, and case/role checks.

Scope

- Case invite CAS transitions
- Recipient/issuer mismatch confirmation
- Consent-version capture and membership transaction

Explicit non-goals

- No workspace token reuse
- No automatic GC migration from soft onboarding

Expected files or modules

- `app/storage/case_invitations.py`
- `app/case_members.py`
- `app/routes/case.py`
- `tests/python/test_case_invitations.py`

Change map

Area	Planned change
Schema/model	Case-invitation identity, consent, expiry, revocation, idempotency fields.
API/routes	Separate paid-case redeem/confirm flow.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Membership grant is a single authorized transaction.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Replay
- self-invite
- active-member duplicate
- mismatch
- expired/revoked
- concurrent redemption rollback

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-01, P03-05. Parallel opportunities: P03-02

Objective acceptance criteria

- □ Implement a separate recipient-bound paid-case invitation that grants membership only after authenticated binding, required confirmation, consent, and case/role checks.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P03-04 — Replace shared signup admission with claim-aware Cognito admission

Field	Plan
Repository / base	`infra` (`main`)
Project item	P03-04
Gate / size	A1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P03-03; technical decisions/controls D-AUTH-02; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-AUTH-02: Alan - later identity/admission mechanism
Dependencies	P03-02, P03-03
Blocking decisions/checkpoints	D-AUTH-02
Source / salvage	infra platform-stack.ts current allowlist + shared invite code

Objective

Keep internal/dev allowlisting but add a short-lived, recipient-bound admission proof for approved workspace/case claimants so a shared invite code is never the participant trust boundary.

Scope

- One-time signed or server-verified admission proof
- Email/purpose/claim/expiry binding
- Replay/revocation/audit
- No membership grant in Cognito

Explicit non-goals

- No open self-signup
- No Cognito group as case authority
- No shared user-entered master code

Expected files or modules

- `lib/stacks/platform-stack.ts`
- `lib/constructs/auth-admission.ts`
- `test/auth-admission.test.ts`
- `docs/agents/platform-stack.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Client metadata carries opaque admission proof only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Harden pre-signup Lambda and key/claim verification contract; keep missing config fail closed.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Wrong email/purpose
- expired/replayed proof
- revoked claim
- internal allowlist path
- missing key/config
- membership still denied before app redemption

Local validation:

- `npm test`
- `npm run build`
- `Synthesize the affected CDK stack with billable/live changes disabled.`

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Synthesize and review first; deploy only through the guarded dev workflow with billable/desired-count defaults unchanged, exact revision recorded, and explicit confirmation outside this planning task.

Rollback or forward repair

Return to the prior reviewed stack/task definition when compatible; otherwise keep resources fail-closed and forward-repair under the runbook. Never delete data-bearing resources as an automatic rollback.

Dependencies and parallel work

Hard dependencies: P03-02, P03-03. Parallel opportunities: P03-01, P03-05

Objective acceptance criteria

- □ Keep internal/dev allowlisting but add a short-lived, recipient-bound admission proof for approved workspace/case claimants so a shared invite code is never the participant trust boundary.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-AUTH-02: Alan - later identity/admission mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Cognito, AWS Lambda/KMS, identity security
Higher-capability review	Mandatory before merge

P03-05 — Implement canonical identity, participation, access, and authorization contracts

Field	Plan
Repository / base	`app` (`main`)
Project item	P03-05
Gate / size	A1 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P02-03, P03-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Alan consolidated #124 review, corrected #122 role/lens behavior, Wave 2 person/case-side ownership tests

Objective

Encode the reconciled separation between authenticated identity, domain participation, resource access grants, operation policy and ownership before workspace/case schemas consume role semantics.

Scope

- Subject-to-person identity binding
- Separate workspace/case participation lifecycle
- Explicit resource/capability access grants
- Operation policy consumed by application services and RLS
- Person/case-side/case ownership independent of visibility

Explicit non-goals

- No global role grants participant data access
- No lens or participation label as standalone authorization
- No Cognito group as case authority

Expected files or modules

- `docs/adr/ADR-00x-participation-access.md`
- `migrations/versions/*_participation_access.py`
- `app/authz/context.py`
- `app/storage/access_grants.py`
- `tests/security/test_participation_access.py`

Change map

Area	Planned change
Schema/model	Unique identity binding, participation and access-grant tables with lifecycle and scope constraints.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Every protected command derives context from the authenticated principal rather than caller-supplied role/lens.
Architecture/docs/config	Identity, participation, access, authorization and ownership become separate contracts.

Tests and validation

Objective tests:

- Duplicate identity/participation is unrepresentable
- Participation without grant denies
- Grant without operation capability denies
- ownership and cross-org negative matrix
- application/RLS policy parity

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03, P03-01. Parallel opportunities: P03-04

Objective acceptance criteria

- □ Encode the reconciled separation between authenticated identity, domain participation, resource access grants, operation policy and ownership before workspace/case schemas consume role semantics.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	domain modelling, authorization, PostgreSQL RLS, identity security
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`infra`	AWS resources, IAM/KMS/network/deployment/backup/audit contracts

Merge order inside this package: `P03-01` → `P03-05` → `P03-03` → `P03-02` → `P03-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Logged-out resume; Token replay/expiry/revocation; Duplicate and concurrent claims; Identity mismatch; Workspace-to-case access escalation	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-AUTH-01	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P03-02, P03-03; Resolved 2026-07-12	No longer blocks P03-02 or P03-03.
D-AUTH-02	Founder constraints final; Alan later	Alan decides later within founder constraints	Alan - later at P03-04	P03-04; Before P03-04	P03-04 and Gate A1 remain blocked until the technical mechanism and evidence exist.
D-AUTH-03	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12	No longer blocks the reconciled access/workspace PR sequence.

D-AUTH-01 - Workspace claim and paid-case identity policy

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	A bearer link creates only a pending low-risk workspace claim; paid-case access requires authenticated binding, explicit consent and recipient/issuer confirmation.
Founder constraints	Minimum-necessary relationship, outreach and logistics context only; No bearer-link case membership
What remains for Alan	Implement the settled claim and paid-case invitation contracts
Evidence required	Forwarded-link, expiry, mismatch, consent and membership negative tests
Approval rule	No further founder decision unless access is broadened.
External dependencies	None
Affected PRs / latest point	P03-02, P03-03; Resolved 2026-07-12
Programme effect	No longer blocks P03-02 or P03-03.
Independent recommendation	A bearer link creates only a pending free-workspace claim with minimum-necessary relationship/outreach/logistics context; paid-case access always requires authenticated binding, explicit consent and recipient/issuer confirmation.

D-AUTH-02 - Claim-aware Cognito admission mechanism

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Participant admission uses short-lived signed or server-verified proof bound to email, purpose, claim and expiry; it admits account creation only and never grants workspace or case membership. Internal admission remains a separate allowlist path.
Founder constraints	Short lived; Replay, mismatch, expiry and revocation fail closed; No membership or resource authorization from admission alone
What remains for Alan	Choose the proof format, signer/verifier, key lifecycle and claim lifecycle
Evidence required	Expiry, replay, wrong-email, wrong-purpose, mismatch and revocation tests; Separate internal allowlist evidence
Approval rule	Luke approval is not required unless Alan proposes broader access or materially weaker identity assurance.
External dependencies	None
Affected PRs / latest point	P03-04; Before P03-04
Programme effect	P03-04 and Gate A1 remain blocked until the technical mechanism and evidence exist.
Independent recommendation	Bind proof to email, purpose, claim ID and expiry; it admits account creation only and never grants workspace/case membership.

D-AUTH-03 - Identity, participation and access authorization contract

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Identity, participation, access grants, ownership and operation authorization remain separate; participation or lens alone never authorizes.
Founder constraints	Subject-derived authorization context; Application and RLS policy parity; Default-deny cross-resource access
What remains for Alan	Encode the settled contract in application services and RLS
Evidence required	Participation-without-grant and grant-without-capability denials; Application/RLS parity tests
Approval rule	No further founder decision unless the boundary changes.
External dependencies	None
Affected PRs / latest point	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12
Programme effect	No longer blocks the reconciled access/workspace PR sequence.
Independent recommendation	Identity binds principal to person; participation records domain relationship; grants carry resource-scoped capabilities; application and RLS consume a subject-derived context plus ownership and purpose. Participation/lens alone never authorizes.

Implementation risk	Owner	Treatment
Shared links forwarded	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Account mismatch	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Confused-deputy redirects	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Workspace data exposed during upgrade	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P03-01, P03-03, P03-05	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
app#123	34078f810b63	GC onboarding form/copy, hosted auth return, invite handoff and role-specific surface work.	P03-01, P03-02	Selective UX/test port and rewrite	Unextracted; likely partially copied into #124 but not proven complete.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P03-02, P03-03	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
feat/byos-journey-wave-2	5b63e3a7c364	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.	P03-01, P03-03, P03-05	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.	Paused exact-head evidence; no direct merge
feature/node0-surrogate-onboarding-link	529d377	substantial uncommitted onboarding/invitation work	P03-02/P03-03/P06-02	Selective port from protected worktree	Preserve until file/test extraction is signed off.

Package source inventory:

- PR #122 return_to, CAS, TTL, self-invite and mismatch fixes
- PRs #123/#124 hosted GC onboarding UX
- Dirty onboarding-link worktree

9. Ownership recommendation

Alan proposes implementation and threat-model details; Luke approves role/privacy policy. The reconciliation checkpoint is complete; replacement work still waits for package acceptance and its own dependencies/decisions.

P03-01

PR	P03-01
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P03-02

PR	P03-02
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P03-03

PR	P03-03
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P03-04

PR	P03-04
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-AUTH-02: Alan - later identity/admission mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P03-05

PR	P03-05
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 04 — Node 0 workspace and paid-case separation

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP04
Plan file	`04-node0-workspace-case-separation.md`
Proposed PRs	P04-01, P04-02, P04-03
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. PR #124 creates a synthetic canonical case as soon as Node 0 is opened, collapsing free exploration, workspace membership, commercial commitment, and paid-case truth.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A distinct Node 0 workspace aggregate with attributable revisions and an idempotent, purpose-limited projection into one paid case only after the authorized commercial event.

Explicit non-goals.

- No payment processor
- No full Node 0 UI
- No clinical or legal truth promotion
- No deletion of legacy rows during compatibility window

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-010	verified	app / PR #124@84a5a84	app/routes/node0.py:63-90	Opening Node 0 calls create_byos_case_for_ip with is_synthetic=True; there is no separate free workspace or authorized promotion.
E-011	verified	app / PR #124@84a5a84	app/storage/node0.py:400-462; app/routes/node0.py:605-652	Node 0 can issue an unaddressed GC invitation and redemption ultimately creates case membership, conflating low-risk onboarding possession with case access.
E-013	verified	app / PR #124@84a5a84	app/node0.py:252-482,622-878,975-1001	Nineteen IP-entered candidate answers about relationship and third-party circumstances/health derive a 0-100 score, readiness/status, ordering, guide route and evidence packet without per-field provenance or verification.
E-014	verified	app / PR #124@84a5a84	app/storage/node0.py:42-200,509-600; tests/python/test_node0.py:444-490	IP/co-IP own candidate reads; GC onboarding writes separate shared/private facts and does not revise the IP assertion. The GC cannot see or correct the IP candidate record.
E-015	verified	app / PR #122@e853821	app/storage/facts.py:19-69,107-127	Facts have broad provenance and visibility values but no subject-versus-author role, verification state, derived-from, or supersedes link.
E-028	verified review	app / PR #124 comment 4948149089	Consolidated review — findings for the WP00 extraction ledger	Alan's completed review classifies #124 as high-value extraction material rather than a merge unit and identifies transaction, privacy, scoring, access, validation, invariant, exposure, retention, performance, CSS and CI findings.

3. Architecture and contracts

- Workspace, entitlement, commercial, payment, and case lifecycles are separate entities and state machines.
- Promotion copies only allowlisted projections with provenance, visibility, and verification intact.
- Compatibility adapters read old case-shaped Node 0 records while new writes target the workspace model.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`node0_workspace`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	No broad public contract; use the focused commands/events in each PR.
Storage	Add node0_workspaces, workspace_memberships, candidate assertion linkage and constraints. Append-only assertion/revision and projection-audit records. Idempotent backfill with source identifiers and exception report.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.

Contract area	Package rule
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

4. Explicit PR sequence

1

Order	1
ID	P04-01
Title	Separate Node 0 workspace storage from paid cases
Repository	app
Dependencies	P02-03, P03-05
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-05; Gate A0 active and verified; Luke review/final founder approval

2

Order	2
ID	P04-02
Title	Add attributable workspace revision and projection contracts
Repository	app
Dependencies	P04-01, P03-05
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-01, P03-05; Gate A0 active and verified; Luke review/final founder approval

3

Order	3
ID	P04-03
Title	Adapt legacy Node 0 case-shaped data without semantic promotion
Repository	app
Dependencies	P04-02
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02; Gate A0 active and verified; Luke review/final founder approval

P04-01 — Separate Node 0 workspace storage from paid cases

Field	Plan
Repository / base	`app` (`main`)
Project item	P04-01
Gate / size	A1 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-05; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P02-03, P03-05
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #124 _ensure_node0_case as behavior to replace, candidate worktrees

Objective

Introduce workspace, workspace membership, candidate, and selection entities so opening Node 0 never creates a canonical paid case.

Scope

- Workspace aggregate and memberships

- Candidate rows linked to workspace
- Case tables remain paid/manageable aggregate

Explicit non-goals

- No UI rewrite
- No promotion
- No legacy deletion

Expected files or modules

- `migrations/versions/*_node0_workspaces.py`
- `app/storage/workspaces.py`
- `app/storage/workspace_memberships.py`
- `app/storage/node0.py`

Change map

Area	Planned change
Schema/model	Add node0_workspaces, workspace_memberships, candidate assertion linkage and constraints.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Expand only; old records remain readable.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Opening Node 0 creates no case
- workspace authorization
- candidate isolation
- concurrent workspace creation
- old app compatibility

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03, P03-05. Parallel opportunities: P04-03

Objective acceptance criteria

- □ Introduce workspace, workspace membership, candidate, and selection entities so opening Node 0 never creates a canonical paid case.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	domain modelling, PostgreSQL, authorization
Higher-capability review	Mandatory before merge

P04-02 — Add attributable workspace revision and projection contracts

Field	Plan
Repository / base	`app` (`main`)
Project item	P04-02
Gate / size	A1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-01, P03-05; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P04-01, P03-05
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current facts/memory models, GC authorship audit

Objective

Represent IP assertions, GC confirmations, private notes, and reviewer/provider confirmations as attributable append-only revisions with purpose-limited projections.

Scope

- Authorship classes and supersedes links
- Verification/visibility/purpose metadata
- Projection allowlists for IP, GC, case, intelligence, provider
- Person-owned private-note contract and explicit-share projection

Explicit non-goals

- No provider transmission
- No silent in-place overwrite

Expected files or modules

- `migrations/versions/*_authored_assertions.py`

- `app/storage/assertions.py`
- `app/domain/projections.py`
- `tests/python/test_authorship_projections.py`

Change map

Area	Planned change
Schema/model	Append-only assertion/revision and projection-audit records.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Default deny between roles; explicit purpose projection.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- IP provisional remains attributed
- GC revision does not erase source
- private cross-role denial
- AI cannot upgrade verification
- provider projection allowlist

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-01, P03-05. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Represent IP assertions, GC confirmations, private notes, and reviewer/provider confirmations as attributable append-only revisions with purpose-limited projections.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	data provenance, privacy, authorization
Higher-capability review	Mandatory before merge

P04-03 — Adapt legacy Node 0 case-shaped data without semantic promotion

Field	Plan
Repository / base	`app` (`main`)
Project item	P04-03
Gate / size	A1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P04-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PRs #117/#121/#123/#124 and dirty candidate worktrees

Objective

Provide a compatibility reader and controlled backfill that maps existing Node 0 rows to workspaces while preserving provenance and never treating legacy scores as verified truth.

Scope

- Dry-run audit and deterministic mapping
- Dual-read compatibility window
- Forward repair and per-row exception ledger

Explicit non-goals

- No destructive legacy cleanup
- No case creation

Expected files or modules

- `migrations/versions/*_node0_legacy_backfill.py`
- `app/storage/node0_compat.py`
- `scripts/audit_node0_backfill.py`
- `tests/fixtures/node0_legacy/`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Idempotent backfill with source identifiers and exception report.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Legacy fixtures
- retry after interruption
- duplicate prevention
- mixed old/new writes
- privacy/provenance preservation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Counts/checksums and unmapped-row alerts.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-02. Parallel opportunities: P04-01

Objective acceptance criteria

- □ Provide a compatibility reader and controlled backfill that maps existing Node 0 rows to workspaces while preserving provenance and never treating legacy scores as verified truth.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P04-01` → `P04-02` → `P04-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	One workspace/one promotion; Retry/concurrency; No implicit fact upgrade; Private candidate isolation; Old/new revision coexistence	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-AUTH-03	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12	No longer blocks the reconciled access/workspace PR sequence.
D-DATA-01	Founder decision final; implementation evidence pending	Luke decides product/privacy outcome	Alan - implementation	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-DATA-02	Founder default final; external validation before B1	Luke decides default; external specialist validates	Alan - implementation; privacy/legal - validation	P04-02, P04-03, P06-02; Before P04-02	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.

D-AUTH-03 - Identity, participation and access authorization contract

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Identity, participation, access grants, ownership and operation authorization remain separate; participation or lens alone never authorizes.
Founder constraints	Subject-derived authorization context; Application and RLS policy parity; Default-deny cross-resource access
What remains for Alan	Encode the settled contract in application services and RLS
Evidence required	Participation-without-grant and grant-without-capability denials; Application/RLS parity tests
Approval rule	No further founder decision unless the boundary changes.
External dependencies	None
Affected PRs / latest point	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12
Programme effect	No longer blocks the reconciled access/workspace PR sequence.
Independent recommendation	Identity binds principal to person; participation records domain relationship; grants carry resource-scoped capabilities; application and RLS consume a subject-derived context plus ownership and purpose. Participation/lens alone never authorizes.

D-DATA-01 - IP-GC field projection allowlist

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.
Founder constraints	Default-deny field-and-purpose allowlists; Source-labelled corrections; Named-purpose provider projection only
What remains for Alan	Define explicit field/purpose allowlists in code and projection contracts; Implement cross-role and provider negative tests
Evidence required	Cross-role denial matrix; Provider projection allowlist tests; Correction/source-label evidence
Approval rule	Luke approval is required for a broader sharing policy, not for compliant implementation detail.
External dependencies	None
Affected PRs / latest point	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02
Programme effect	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
Independent recommendation	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.

D-DATA-02 - Correction, supersession and unselected-candidate retention

Field	Recorded decision or requirement
Decision state	Final founder default; externally dependent activation
Founder outcome	Retain candidate/workspace material for a 90-day minimum after explicit rejection, withdrawal or last legitimate outreach, honor earlier valid deletion, retain only non-identifying funnel metrics and a minimal audit tombstone, and preserve legal holds. Corrections append source-labelled revisions and private IP notes remain author-only.
Founder constraints	No silent overwrite; Deletion and anonymization are auditable; Legal holds override ordinary deletion
What remains for Alan	Implement lifecycle, deletion, anonymization, correction and legal-hold behavior
Evidence required	Retention/deletion tests; Anonymization and tombstone evidence; Privacy/legal approval before B1
Approval rule	Luke accepts the live policy after privacy/legal review.
External dependencies	Privacy/legal validation before B1
Affected PRs / latest point	P04-02, P04-03, P06-02; Before P04-02
Programme effect	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.
Independent recommendation	Append revisions with supersedes/derived-from; never silently erase source; shortest justified retention for unselected candidates.

Implementation risk	Owner	Treatment
Semantic data corruption during backfill	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Duplicate cases	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Private candidate leakage	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Older app revision writes legacy shape	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#117	5ac09f0cdd92	Deterministic candidate questionnaire, scoring/status calculation, conversation-guide resource, routes and extensive Node 0 tests.	P04-01, P04-02	Selective port and rewrite	Unextracted; algorithm/test review required after Alan checkpoint and Luke field/score decision.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P04-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P04-02	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P04-01, P04-02, P04-03	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
detached candidate rewrite	9605d88	substantial uncommitted Node 0 quiz rewrite	P04-02/P05-02	Selective port from protected worktree	Preserve; compare with predecessor and #117/#124 before any discard.
detached predecessor candidate model	dc6a2f0	substantial uncommitted predecessor model	P04-02/P05-02 if unique	Selective port from protected worktree	Preserve until rewrite diff/extraction ledger is complete.

Package source inventory:

- PR #124 candidate/outreach behavior and tests by selective port
- PR #122 case projection primitives after reconciliation
- Older candidate-model and rewrite worktrees

9. Ownership recommendation

Alan owns domain/schema implementation; Luke approves entity boundaries and promotion contract. Joint high-capability review is mandatory.

P04-01

PR	P04-01
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P04-02

PR	P04-02
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P04-03

PR	P04-03
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 05 — Node 0 Free, Pathfinder, and resume

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP05
Plan file	`05-node0-free-pathfinder-and-resume.md`
Proposed PRs	P05-01, P05-02, P05-03
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Pathfinder contains service/LLM-era routing complexity and Node 0 Free must satisfy the complete-the-match boundary without payment while preserving explicit IP authorship.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A deterministic entry router and resumable Node 0 Free flow for education, candidate tracking, preliminary safety checks, outreach guidance, and a willing candidate without a paid case.

Explicit non-goals.

- No weighted ranking
- No GC authenticated soft onboarding
- No paid case
- No final Learn content

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-010	verified	app / PR #124@84a5a84	app/routes/node0.py:63-90	Opening Node 0 calls create_byos_case_for_ip with is_synthetic=True; there is no separate free workspace or authorized promotion.
E-013	verified	app / PR #124@84a5a84	app/node0.py:252-482,622-878, 975-1001	Nineteen IP-entered candidate answers about relationship and third-party circumstances/health derive a 0-100 score, readiness/status, ordering, guide route and evidence packet without per-field provenance or verification.
E-028	verified review	app / PR #124 comment 4948149089	Consolidated review — findings for the WP00 extraction ledger	Alan's completed review classifies #124 as high-value extraction material rather than a merge unit and identifies transaction, privacy, scoring, access, validation, invariant, exposure, retention, performance, CSS and CI findings.

3. Architecture and contracts

- Routing is deterministic policy, not LLM authority.
- Candidate responses are provisional IP assertions, never GC-authored or medically verified facts.
- Safety-critical information and complete-the-match capability remain free.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.

Contract area	Package rule
Entities/state	`entry_session`, `node0_workspace`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Deterministic route decision response with policy version. Typed workspace-scoped commands reject malformed/oversize input and return neutral questions to clarify, never eligibility flags. UI calls workspace-scoped commands only.
Storage	Use workspace candidates/assertions from WP04 with FKs, uniqueness and one-active-outreach invariants.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Participant entry and routing (`entry_session`)

Owner: natalyx-app. States: `unstarted`, `education`, `identity_required`, `authenticated`, `routed`, `handed_off`, `expired`.

unstarted

From	unstarted
To	education
Actor	visitor
Evidence	GET/explicit entry choice
Idempotency key	entry:{session}:start
Invalid transitions	No participant/case write
Audit event	entry_started
Retry	Return current session
Recovery	Create fresh anonymous session

education

From	education
To	identity_required
Actor	visitor/policy
Evidence	route requires saved workspace
Idempotency key	entry:{session}:identity
Invalid transitions	Cannot grant workspace/case
Audit event	identity_required
Retry	Preserve safe return intent
Recovery	Restart at public choice

identity_required

From	identity_required
To	authenticated
Actor	Cognito/auth provider
Evidence	verified subject/session
Idempotency key	auth:{subject}:{nonce}
Invalid transitions	Invalid/expired token
Audit event	identity_bound
Retry	Replay returns authenticated session
Recovery	Rechallenge; no partial membership

education|authenticated

From	education|authenticated
To	routed
Actor	participant + deterministic policy
Evidence	validated route answers + policy version
Idempotency key	route:{session}:{answer_hash}
Invalid transitions	Unknown/unsafe route goes review/education
Audit event	entry_routed
Retry	Same input returns same decision
Recovery	Allow corrected answers/new decision revision

routed

From	routed
To	handed_off
Actor	participant
Evidence	accepted destination and authorized target
Idempotency key	handoff:{session}:{route_version}
Invalid transitions	Target not authorized/available
Audit event	entry_handed_off
Retry	Resume same safe destination
Recovery	Return to route result with explanation

any nonterminal

From	any nonterminal
To	expired
Actor	system
Evidence	TTL elapsed/revoked session
Idempotency key	entry:{session}:expire
Invalid transitions	No transition from handed_off needed
Audit event	entry_expired
Retry	Idempotent
Recovery	New session; no case/access change

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

4. Explicit PR sequence

1

Order	1
ID	P05-01
Title	Replace Pathfinder routing with deterministic entry decisions
Repository	app
Dependencies	P03-01, P04-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P04-01; Gate A0 active and verified; Alan review

2

Order	2
ID	P05-02
Title	Rebuild Node 0 Free candidate and outreach domain
Repository	app
Dependencies	P04-02, P05-01
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P05-01; Gate A0 active and verified; Alan review

3

Order	3
ID	P05-03
Title	Ship resumable Node 0 Free participant surface
Repository	app
Dependencies	P03-01, P05-02, P11-03
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P05-02, P11-03; Gate A0 active and verified; Alan review

P05-01 — Replace Pathfinder routing with deterministic entry decisions

Field	Plan
Repository / base	`app` (`main`)
Project item	P05-01
Gate / size	A1 / Medium
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P04-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-01, P04-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current pathfinder_destinations.py, PR #119 product-alignment doc

Objective

Route users predictably to Node 0, full IP setup, GC claim, Learn/Ask education, or provider/partner paths using explicit inputs and policy versioning.

Scope

- Small versioned decision table
- Clear reason/next-step output
- No LLM dependency for route selection

Explicit non-goals

- No final educational content
- No eligibility clearance

Expected files or modules

- `app/pathfinder_destinations.py`
- `app/pathfinder_policy.py`
- `app/routes/participant.py`

- `tests/python/test_pathfinder_policy.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Deterministic route decision response with policy version.
UI/product surface	Clear entry choices and safe review states.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Decision table coverage
- ambiguous/malformed inputs
- GC invite deep link
- service unavailable
- mobile entry

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P03-01, P04-01. Parallel opportunities: P05-03

Objective acceptance criteria

- □ Route users predictably to Node 0, full IP setup, GC claim, Learn/Ask education, or provider/partner paths using explicit inputs and policy versioning.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

P05-02 — Rebuild Node 0 Free candidate and outreach domain

Field	Plan
Repository / base	`app` (`main`)
Project item	P05-02
Gate / size	A1 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P05-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P04-02, P05-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #117 direct algorithm/test salvage, PR #124, quiz rewrite worktree

Objective

Port the useful deterministic candidate, preliminary safety, and outreach behavior onto workspace/provenance contracts while preserving the complete-the-match free boundary.

Scope

- Unlimited candidates under bounded command/collection sizes
- Minimum-necessary relationship, outreach and logistics questions only
- Basic outreach sequencing and shareable intro packet
- Migration-backed case/workspace/candidate/outreach constraints
- Batched hydration plus archive/anonymize/delete behavior under D-DATA-02

Explicit non-goals

- No weighted ranking or suitability score
- No IP-entered health, substance, mental-health, citizenship, race or eligibility data
- No medical clearance
- No paid feature

Expected files or modules

- `app/domain/node0_candidates.py`
- `app/storage/node0.py`
- `app/node0_conversation_guide.py`
- `app/resources/node0_conversation_guide_v1.json`
- `tests/python/test_node0_domain.py`

Change map

Area	Planned change
Schema/model	Use workspace candidates/assertions from WP04 with FKs, uniqueness and one-active-outreach invariants.
API/routes	Typed workspace-scoped commands reject malformed/oversize input and return neutral questions to clarify, never eligibility flags.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Allowed/blocked field matrix
- no ranking/gating
- strict enum/body/collection limits
- author attribution
- candidate isolation
- concurrent outreach invariant
- retention/deletion
- query-count bound
- free feature contract

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-02, P05-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Port the useful deterministic candidate, preliminary safety, and outreach behavior onto workspace/provenance contracts while preserving the complete-the-match free boundary.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.

- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P05-03 — Ship resumable Node 0 Free participant surface

Field	Plan
Repository / base	`app` (`main`)
Project item	P05-03
Gate / size	A1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-01, P05-02, P11-03; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-01, P05-02, P11-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PRs #121/#124 selective UI port, DS PR #8

Objective

Provide a coherent mobile-capable Node 0 Free workspace shell for candidate tracking, outreach, guide use, and resume without paid-case language.

Scope

- Workspace home and candidate flow
- Shared IP-owner guard across participant and API routes
- Exact public/cloud exposure registry
- Expiry-aware token views and no-store/referrer/noindex response policy
- Route-scoped Node 0 stylesheet
- Honest preliminary labels/empty/error states
- Logout/login resume and mobile navigation

Explicit non-goals

- No Complete-only features
- No paid case/journey hero

Expected files or modules

- `app/routes/node0.py`
- `app/node0_surface.py`
- `app/templates/node0/*.html`
- `app/static/node0.*`
- `tests/python/test_node0_ui.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	UI calls workspace-scoped commands only.
UI/product surface	Participant DS recipes with responsive interaction and no fake content.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Clickable Free path
- GC/viewer/staff/nonowner denial without 500
- exact route exposure negatives
- expired-token GET
- cache/referrer headers
- malformed/oversize form/PDF input
- resume
- no-JS/keyboard
- 320/375/768px
- dependency failure

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P03-01, P05-02, P11-03. Parallel opportunities: P05-01

Objective acceptance criteria

- □ Provide a coherent mobile-capable Node 0 Free workspace shell for candidate tracking, outreach, guide use, and resume without paid-case language.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.

- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P05-01` → `P05-02` → `P05-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	All deterministic route outcomes; Free candidate/outreach boundary; Resume across sessions; Critical-flag fail-safe; Mobile interaction	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-DATA-03	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P05-02, P06-03; Resolved 2026-07-12	No longer blocks Node 0 implementation planning.

D-DATA-03 - Candidate scoring and sensitive preference policy

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	No candidate ranking, viability/eligibility flag or race preference survives; provisional pre-join input can generate neutral source-labelled questions only.
Founder constraints	No sensitive third-party scoring or gating; No provisional assertion becomes GC/provider truth
What remains for Alan	Remove legacy ranking/gating and preserve neutral-question behavior
Evidence required	Prohibited-field and no-ranking tests
Approval rule	No further founder decision unless ranking or eligibility behavior is proposed.
External dependencies	None
Affected PRs / latest point	P05-02, P06-03; Resolved 2026-07-12
Programme effect	No longer blocks Node 0 implementation planning.
Independent recommendation	No ranking, viability/eligibility flag or race preference survives. Pre-join IP input is limited to minimum-necessary relationship/outreach/logistics context and may produce neutral source-labelled questions only; it never gates claim, onboarding, provider use or critical decisions.

Implementation risk	Owner	Treatment
Paid features leak into Free	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Provisional score is overclaimed	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Route drift	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
LLM dependency reintroduced	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#117	5ac09f0cdd92	Deterministic candidate questionnaire, scoring/status calculation, conversation-guide resource, routes and extensive Node 0 tests.	P05-02, P05-03	Selective port and rewrite	Unextracted; algorithm/test review required after Alan checkpoint and Luke field/score decision.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P05-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#121	529d377e2337	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P05-03	Selective UI/test port, then supersede	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P05-02, P05-03	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
detached candidate rewrite	9605d88	substantial uncommitted Node 0 quiz rewrite	P04-02/P05-02	Selective port from protected worktree	Preserve; compare with predecessor and #117/#124 before any discard.
detached predecessor candidate model	dc6a2f0	substantial uncommitted predecessor model	P04-02/P05-02 if unique	Selective port from protected worktree	Preserve until rewrite diff/extraction ledger is complete.

Package source inventory:

- PR #117 quiz/guide algorithms and tests
- PR #121 primary surface
- PR #124 consolidated UI
- Quiz rewrite worktree

9. Ownership recommendation

Luke is the proposed DRI and primary implementer; Alan reviews the technical architecture, security and failure behavior. Reconciliation is complete; replacement work waits for WP03/WP04 contracts, and commit/PR/merge waits for Alan's review and joint package acceptance.

P05-01

PR	P05-01
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P05-02

PR	P05-02
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P05-03

PR	P05-03
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.

- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 06 — Node 0 Complete and GC soft onboarding

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP06
Plan file	`06-node0-complete-and-soft-onboarding.md`
Proposed PRs	P06-01, P06-02, P06-03
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Optional paid depth, GC soft onboarding, and readiness reporting need clear entitlement gates without creating a managed case or treating IP assertions as GC answers.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned Node 0 Complete entitlement, authenticated low-risk GC onboarding, attributable GC confirmation/replacement, and a readiness report that remains preliminary.

Explicit non-goals.

- No managed case activation
- No provider access
- No medical/legal clearance
- No hard-coded price

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-010	verified	app / PR #124@84a5a84	app/routes/node0.py:63-90	Opening Node 0 calls create_byos_case_for_ip with is_synthetic=True; there is no separate free workspace or authorized promotion.
E-011	verified	app / PR #124@84a5a84	app/storage/node0.py:400-462; app/routes/node0.py:605-652	Node 0 can issue an unaddressed GC invitation and redemption ultimately creates case membership, conflating low-risk onboarding possession with case access.
E-013	verified	app / PR #124@84a5a84	app/node0.py:252-482,622-878,975-1001	Nineteen IP-entered candidate answers about relationship and third-party circumstances/health derive a 0-100 score, readiness/status, ordering, guide route and evidence packet without per-field provenance or verification.
E-014	verified	app / PR #124@84a5a84	app/storage/node0.py:42-200,509-600; tests/python/test_node0.py:444-490	IP/co-IP own candidate reads; GC onboarding writes separate shared/private facts and does not revise the IP assertion. The GC cannot see or correct the IP candidate record.

3. Architecture and contracts

- Node 0 Complete is a flat configurable entitlement fully creditable against the JMF.

- GC confirmation appends an attributable revision; it never silently overwrites the IP source.
- GC-private answers stay private; only explicit purpose-limited projections reach the IP.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`node0_workspace`, `workspace_claim`, `workspace_entitlement`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Read-only readiness projection with policy version.
Storage	Append-only entitlement events plus current projection. Use assertion/revision contract; onboarding progress events.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Entitlement (`workspace_entitlement`)

Owner: natalyx-app commercial domain. States: `free`, `complete_pending`, `complete_active`, `suspended`, `revoked`.

none

From	none
To	free
Actor	system
Evidence	active Node 0 workspace
Idempotency key	entitlement:{workspace}:free
Invalid transitions	No workspace
Audit event	entitlement_granted
Retry	Return current
Recovery	Repair projection from event

free

From	free
To	complete_pending
Actor	IP/commercial service
Evidence	accepted Complete offer/config
Idempotency key	entitlement:{workspace}:{commitment}
Invalid transitions	Unknown price/config/terms
Audit event	complete_requested
Retry	Return pending
Recovery	Cancel/expire commitment

complete_pending

From	complete_pending
To	complete_active
Actor	payment domain
Evidence	settled Complete ledger entry
Idempotency key	entitlement:{payment}:activate
Invalid transitions	Amount/currency/config mismatch
Audit event	complete_activated
Retry	Replay no-op
Recovery	Reconcile and retry from inbox

complete_active

From	complete_active
To	suspended
Actor	authorized commercial policy
Evidence	refund/reversal or approved pause reason
Idempotency key	entitlement:{id}:suspend:{event}
Invalid transitions	Cannot remove free/safety capability
Audit event	entitlement_suspended
Retry	No-op if same event
Recovery	Reinstate or revoke through new event

complete_active|suspended

From	complete_active|suspended
To	revoked
Actor	authorized operator/policy
Evidence	fraud/refund/terms revocation evidence
Idempotency key	entitlement:{id}:revoke:{event}
Invalid transitions	No silent case cancellation
Audit event	entitlement_revoked
Retry	Idempotent
Recovery	Manual reviewed regrant creates new entitlement event

Low-risk workspace identity claim (`workspace_claim`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `consented`, `claimed`, `mismatch_pending`, `revoked`, `expired`.

none

From	none
To	issued
Actor	workspace IP
Evidence	selected candidate + allowed claim scope
Idempotency key	claim:{workspace}:{candidate}:{generation}
Invalid transitions	Existing live claim superseded
Audit event	workspace_claim_issued
Retry	Return current token only at issuance
Recovery	Issue new generation; old revoked

issued

From	issued
To	pending_identity
Actor	link holder
Evidence	valid token landing
Idempotency key	claim:{id}:land
Invalid transitions	Expired/revoked token
Audit event	claim_opened
Retry	Safe repeat
Recovery	Authenticate or exit; no data grant

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email and binding policy
Idempotency key	claim:{id}:bind:{subject}
Invalid transitions	Already claimed/different subject
Audit event	claim_identity_bound|claim_mismatch
Retry	CAS no-op or mismatch state
Recovery	Issuer review/new claim

identity_bound

From	identity_bound
To	consented
Actor	GC claimant
Evidence	workspace consent version
Idempotency key	claim:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	claim_consented
Retry	Idempotent
Recovery	Display current consent

consented

From	consented
To	claimed
Actor	claim service
Evidence	authorized low-risk workspace membership transaction
Idempotency key	claim:{id}:complete
Invalid transitions	Case membership requested or conflicting claimant
Audit event	workspace_claimed
Retry	Return same membership
Recovery	Transaction rollback/forward repair

issued|pending_identity|identity_bound|consented|mismatch_pending

From	issued|pending_identity|identity_bound|consented|mismatch_pending
To	revoked|expired
Actor	issuer/system
Evidence	revocation or TTL
Idempotency key	claim:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	claim_revoked|claim_expired
Retry	Idempotent
Recovery	Fresh claim only

Invitation and identity safety

Retain from updated PR #122:

- hashed tokens
- purpose-specific TTL with a 14-day ceiling
- CAS terminal transitions
- atomic membership creation
- one-live-GC constraint
- self/existing-member refusal
- safe return_to
- portable integrity/concurrency tests

Required replacement contract:

- Exactly one target: workspace_id or case_id; include purpose, role, access_class, recipient_binding_mode and policy version.
- Email-less possession may only open pending low-risk workspace claim; it cannot reveal sensitive data or grant paid-case membership.

- Paid-case invitation is addressed, authenticated, consented and atomically redeems invitation plus participation, access grant and audit; mismatch requires explicit policy/issuer handling and preferably a fresh bound token.
- Workspace-claim issuance commits supersession, hashed claim, candidate/workspace linkage and required audit atomically; external identity steps use an explicit idempotent completion state.
- Cognito admission and app membership are separate decisions; account creation never equals workspace/case access.
- Verified identity comes from auth claims, not form-entered email; duplicate claims return the original result.
- Never put raw token/email/PHI in general participant timeline payloads; revoke/expire/supersede fail closed.
- Remove generic unconditional status mutation; expose authorized transition commands only.

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

4. Explicit PR sequence

1

Order	1
ID	P06-01
Title	Add Node 0 Complete entitlement and versioned feature gates
Repository	app
Dependencies	P04-01, P07-01, P08-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-01, P07-01, P08-01; Gate A0 active and verified; Luke review/final founder approval

2

Order	2
ID	P06-02
Title	Add GC soft onboarding and attributable confirmation
Repository	app
Dependencies	P03-02, P04-02, P06-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P04-02, P06-01; Gate A0 active and verified; Luke review/final founder approval

3

Order	3
ID	P06-03
Title	Add readiness report and role-scoped progress surfaces
Repository	app
Dependencies	P06-02, P10-02, P11-04
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P06-02, P10-02, P11-04; Gate A0 active and verified; Luke review/final founder approval

P06-01 — Add Node 0 Complete entitlement and versioned feature gates

Field	Plan
Repository / base	`app` (`main`)
Project item	P06-01
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-01, P07-01, P08-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P04-01, P07-01, P08-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Business-model Node 0 feature table

Objective

Model optional Complete access independently from workspace and case, pin the commercial configuration, and expose server-side gates for the approved paid-depth features.

Scope

- Grant/suspend/revoke entitlement events
- Commercial-version and payment evidence reference
- Credit accrual link

Explicit non-goals

- No hard-coded price
- No paid-case creation

Expected files or modules

- `migrations/versions/*_entitlements.py`
- `app/domain/entitlements.py`
- `app/storage/entitlements.py`
- `tests/python/test_entitlements.py`

Change map

Area	Planned change
Schema/model	Append-only entitlement events plus current projection.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Every Complete feature checks server entitlement.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Grant replay
- suspend/revoke
- config version
- free-feature unaffected
- no case created

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-01, P07-01, P08-01. Parallel opportunities: P06-03

Objective acceptance criteria

- □ Model optional Complete access independently from workspace and case, pin the commercial configuration, and expose server-side gates for the approved paid-depth features.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P06-02 — Add GC soft onboarding and attributable confirmation

Field	Plan
Repository / base	`app` (`main`)
Project item	P06-02
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P04-02, P06-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-02, P04-02, P06-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PRs #123/#124 UX only, GC authorship audit

Objective

Let an authenticated GC claim low-risk workspace access, provide GC-owned answers, confirm/replace selected provisional facts, and preserve private boundaries.

Scope

- GC-authored profile/intake revisions
- Field-by-field confirmation/replacement
- Purpose-limited IP progress projection

Explicit non-goals

- No paid-case membership
- No provider share
- No medical eligibility determination

Expected files or modules

- `app/routes/gc_workspace.py`
- `app/storage/gc_onboarding.py`
- `app/templates/node0/gc_onboarding*.html`
- `tests/python/test_gc_soft_onboarding.py`

Change map

Area	Planned change
Schema/model	Use assertion/revision contract; onboarding progress events.
API/routes	No public API/route contract change.
UI/product surface	GC consent and privacy explanations.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Paired IP/GC visibility
- private contact/health fields
- revision lineage
- claim replay/mismatch
- mobile resume

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-02, P04-02, P06-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Let an authenticated GC claim low-risk workspace access, provide GC-owned answers, confirm/replace selected provisional facts, and preserve private boundaries.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	privacy, accessible forms, authorization
Higher-capability review	Mandatory before merge

P06-03 — Add readiness report and role-scoped progress surfaces

Field	Plan
Repository / base	`app` (`main`)
Project item	P06-03
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P06-02, P10-02, P11-04; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P06-02, P10-02, P11-04
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #124 score/report concepts

Objective

Produce a preliminary, source-labelled Node 0 readiness report and separate IP/GC progress views without overclaiming clearance or exposing private answers.

Scope

- Source/verification labels
- IP and GC lenses
- Promotion-ready projection manifest

Explicit non-goals

- No provider-confirmed status
- No final case activation

Expected files or modules

- `app/domain/node0_readiness.py`
- `app/templates/node0/readiness*.html`
- `app/node0_surface.py`
- `tests/python/test_node0_readiness.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Read-only readiness projection with policy version.
UI/product surface	Role-specific report/progress cards.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Provisional labels
- critical missing data
- cross-role redaction
- stale revision
- print/mobile

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P06-02, P10-02, P11-04. Parallel opportunities: P06-01

Objective acceptance criteria

- □ Produce a preliminary, source-labelled Node 0 readiness report and separate IP/GC progress views without overclaiming clearance or exposing private answers.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P06-01` → `P06-02` → `P06-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Entitlement versioning; GC-authored revisions; IP/GC visibility pairs; Revoked/expired claims; No paid-case creation	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-DATA-01	Founder decision final; implementation evidence pending	Luke decides product/privacy outcome	Alan - implementation	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-DATA-02	Founder default final; external validation before B1	Luke decides default; external specialist validates	Alan - implementation; privacy/legal - validation	P04-02, P04-03, P06-02; Before P04-02	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.
D-DATA-03	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P05-02, P06-03; Resolved 2026-07-12	No longer blocks Node 0 implementation planning.
D-COM-01	Founder decision final for beta; tax externally dependent	Luke decides commercial outcome	Alan - implementation	P06-01, P07-01, P08-01; Before activation of a Complete configuration	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.

D-DATA-01 - IP-GC field projection allowlist

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.
Founder constraints	Default-deny field-and-purpose allowlists; Source-labelled corrections; Named-purpose provider projection only
What remains for Alan	Define explicit field/purpose allowlists in code and projection contracts; Implement cross-role and provider negative tests
Evidence required	Cross-role denial matrix; Provider projection allowlist tests; Correction/source-label evidence
Approval rule	Luke approval is required for a broader sharing policy, not for compliant implementation detail.
External dependencies	None
Affected PRs / latest point	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02
Programme effect	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
Independent recommendation	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.

D-DATA-02 - Correction, supersession and unselected-candidate retention

Field	Recorded decision or requirement
Decision state	Final founder default; externally dependent activation
Founder outcome	Retain candidate/workspace material for a 90-day minimum after explicit rejection, withdrawal or last legitimate outreach, honor earlier valid deletion, retain only non-identifying funnel metrics and a minimal audit tombstone, and preserve legal holds. Corrections append source-labelled revisions and private IP notes remain author-only.
Founder constraints	No silent overwrite; Deletion and anonymization are auditable; Legal holds override ordinary deletion
What remains for Alan	Implement lifecycle, deletion, anonymization, correction and legal-hold behavior
Evidence required	Retention/deletion tests; Anonymization and tombstone evidence; Privacy/legal approval before B1
Approval rule	Luke accepts the live policy after privacy/legal review.
External dependencies	Privacy/legal validation before B1
Affected PRs / latest point	P04-02, P04-03, P06-02; Before P04-02
Programme effect	Synthetic implementation may proceed; real-user B1 activation remains externally dependent.
Independent recommendation	Append revisions with supersedes/derived-from; never silently erase source; shortest justified retention for unselected candidates.

D-DATA-03 - Candidate scoring and sensitive preference policy

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	No candidate ranking, viability/eligibility flag or race preference survives; provisional pre-join input can generate neutral source-labelled questions only.
Founder constraints	No sensitive third-party scoring or gating; No provisional assertion becomes GC/provider truth
What remains for Alan	Remove legacy ranking/gating and preserve neutral-question behavior
Evidence required	Prohibited-field and no-ranking tests
Approval rule	No further founder decision unless ranking or eligibility behavior is proposed.
External dependencies	None
Affected PRs / latest point	P05-02, P06-03; Resolved 2026-07-12
Programme effect	No longer blocks Node 0 implementation planning.
Independent recommendation	No ranking, viability/eligibility flag or race preference survives. Pre-join IP input is limited to minimum-necessary relationship/outreach/logistics context and may produce neutral source-labelled questions only; it never gates claim, onboarding, provider use or critical decisions.

D-COM-01 - Node 0 Complete price

Field	Recorded decision or requirement
Decision state	Final beta policy; external tax wording pending
Founder outcome	Node 0 Complete is a voluntary $495 USD one-time upgrade paid by the IP or authorized IP account holder, never the GC, collected before Complete features and fully credited against the later Phase 1/JMF without gating matching or safety support.
Founder constraints	Transparent one-time feature/credit wording; Versioned prospective changes only; No forced payment to identify, approach or reach agreement with a GC
What remains for Alan	Implement configurable price, payer ownership, durable payment/credit ledger and idempotent flows
Evidence required	Synthetic success, failure, retry, duplicate and credit tests; Tax wording before live activation
Approval rule	Luke approves future commercial-policy changes; tax treatment requires specialist review.
External dependencies	Tax treatment and customer-facing wording before live activation
Affected PRs / latest point	P06-01, P07-01, P08-01; Before activation of a Complete configuration
Programme effect	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
Independent recommendation	Keep no active production value until explicitly approved; test with synthetic configuration.

Implementation risk	Owner	Treatment
Entitlement mistaken for case access	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
GC answers exposed to IP	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
IP assertions silently converted	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Price hard-coded	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P06-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#123	34078f810b63	GC onboarding form/copy, hosted auth return, invite handoff and role-specific surface work.	P06-02, P06-03	Selective UX/test port and rewrite	Unextracted; likely partially copied into #124 but not proven complete.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P06-02, P06-03	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
feature/node0-surrogate-onboarding-link	529d377	substantial uncommitted onboarding/invitation work	P03-02/P03-03/P06-02	Selective port from protected worktree	Preserve until file/test extraction is signed off.

Package source inventory:

- PRs #123/#124 onboarding screens
- Dirty onboarding-link branch
- PR #122 paired-lens tests

9. Ownership recommendation

Luke decides product visibility and entitlement boundaries; Alan implements. External privacy review is recommended for field-level projections.

P06-01

PR	P06-01
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P06-02

PR	P06-02
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P06-03

PR	P06-03
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- ☐ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- ☐ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- ☐ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- ☐ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- ☐ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- ☐ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- ☐ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 07 — Commercial configuration, enrollment, and promotion

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP07
Plan file	`07-commercial-configuration-enrollment-and-promotion.md`
Proposed PRs	P07-01, P07-02, P07-03, P07-04
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. The business model requires versioned price/terms, explicit commitment, and one precise processor-neutral event before workspace-to-case promotion; none is currently authoritative.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned commercial configuration and acceptances, a separate commitment/enrollment lifecycle, one processor-neutral byos.workspace_promotion_authorized.v1 event, and exactly-once paid-case creation by its consumer.

Explicit non-goals.

- No processor-specific webhook
- No automated unresolved pricing policy
- No activation before onboarding evidence
- No browser-redirect authority

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-025	design authority	business model / Natalyx_Payment_and_Business_Model_BYOS_MVP.docx v1.0	Sections 3-12 and implementation tables	Node 0 Free must complete the match; Complete is optional/creditable; $8,900 JMF and $750 enrollment anchor the configurable model; promotion follows explicit commitment and funding, not willingness or redirect.

3. Architecture and contracts

- Every commitment, invoice, payment, credit, refund, and promotion records the activated commercial-configuration version.
- Only the app-owned commitment/commercial domain emits byos.workspace_promotion_authorized.v1 after durable funding/terms/credit validation; a processor callback or redirect never emits it.
- The consumer creates/fetches one enrollment_pending case and restricted IP membership; GC case access follows the paid-case invitation contract.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`case`, `case_invitation`, `journey_commitment`, `node0_workspace`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Read active config and record acceptance commands. Create, accept, cancel, inspect enrollment. Internal event producer only. Internal event handler only.
Storage	Commercial configuration and acceptance ledgers. Commitment events/current projection. Promotion ledger and source-workspace unique constraint.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Node 0 workspace (`node0_workspace`)

Owner: natalyx-app. States: `draft`, `active`, `match_ready`, `promotion_pending`, `promoted`, `archived`.

none

From	none
To	draft
Actor	authenticated IP
Evidence	first explicit Node 0 start
Idempotency key	workspace:{owner}:create
Invalid transitions	Existing active workspace returns existing
Audit event	workspace_created
Retry	Unique owner/idempotency key
Recovery	Forward repair orphan membership

draft

From	draft
To	active
Actor	IP
Evidence	minimum workspace consent/profile
Idempotency key	workspace:{id}:activate:{revision}
Invalid transitions	Missing consent/owner
Audit event	workspace_activated
Retry	Return current active revision
Recovery	Correct inputs and retry

active

From	active
To	match_ready
Actor	IP + selected GC evidence policy
Evidence	selected candidate + required provisional/GC-authored fields
Idempotency key	workspace:{id}:ready:{revision}
Invalid transitions	Critical unknown/decline/revoked claim
Audit event	workspace_match_ready
Retry	No duplicate event
Recovery	Reopen active with reason revision

match_ready

From	match_ready
To	promotion_pending
Actor	promotion event consumer
Evidence	valid byos.workspace_promotion_authorized.v1
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization; already promoted; stale config
Audit event	workspace_promotion_requested
Retry	Replay resumes/finds same promotion
Recovery	Forward-repair incomplete transaction

promotion_pending

From	promotion_pending
To	promoted
Actor	promotion service
Evidence	paid case + manifest committed
Idempotency key	promotion:{commitment}:v1
Invalid transitions	Projection invariant/privacy failure
Audit event	workspace_promoted
Retry	Return same case ID
Recovery	Repair from immutable manifest; never duplicate

draft|active|match_ready

From	draft|active|match_ready
To	archived
Actor	IP/operator per policy
Evidence	explicit archive + retention policy
Idempotency key	workspace:{id}:archive:{request}
Invalid transitions	Promotion pending/promoted
Audit event	workspace_archived
Retry	Idempotent
Recovery	Restore only through new audited revision if policy permits

Commercial commitment (`journey_commitment`)

Owner: natalyx-app commercial domain. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

none

From	none
To	draft
Actor	IP
Evidence	active workspace + displayed config/terms
Idempotency key	commitment:{workspace}:{config}
Invalid transitions	No active config/workspace
Audit event	commitment_drafted
Retry	Return existing draft
Recovery	Create new draft after expiry/version change

draft

From	draft
To	accepted
Actor	IP
Evidence	terms/privacy consent + payment path
Idempotency key	commitment:{id}:accept:{terms_hash}
Invalid transitions	Stale/missing acceptance
Audit event	commitment_accepted
Retry	Return accepted
Recovery	New acceptance revision if version changes

accepted

From	accepted
To	funding_pending
Actor	commercial service
Evidence	enrollment invoice/payment intent created
Idempotency key	commitment:{id}:funding
Invalid transitions	No invoice or invalid amount
Audit event	funding_requested
Retry	Resume same invoice
Recovery	Cancel/reissue under audited adjustment

funding_pending

From	funding_pending
To	funded
Actor	payment domain
Evidence	net settled enrollment funding meets versioned requirement
Idempotency key	enrollment-funding:{id}:v1
Invalid transitions	Redirect/pending/failed/reversed evidence
Audit event	commercial.enrollment_funding_confirmed.v1
Retry	Exactly-once event/outbox replay
Recovery	Reconcile inbox/ledger; no browser repair

draft|accepted|funding_pending

From	draft|accepted|funding_pending
To	cancelled
Actor	IP/authorized operator
Evidence	explicit cancellation and refund policy
Idempotency key	commitment:{id}:cancel:{request}
Invalid transitions	Already funded/promoted uses case cancellation rail
Audit event	commitment_cancelled
Retry	Idempotent
Recovery	New commitment required

draft|accepted

From	draft|accepted
To	expired
Actor	system
Evidence	offer/terms TTL or config deactivation
Idempotency key	commitment:{id}:expire
Invalid transitions	No expiry after funding
Audit event	commitment_expired
Retry	Idempotent
Recovery	Draft new version

Paid case lifecycle (`case`)

Owner: natalyx-app. States: `enrollment_pending`, `active`, `paused`, `closed`, `cancelled`.

none

From	none
To	enrollment_pending
Actor	promotion service
Evidence	byos.workspace_promotion_authorized.v1 + projection manifest
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization/duplicate workspace
Audit event	paid_case_created
Retry	Return same case
Recovery	Forward-repair same promotion

enrollment_pending

From	enrollment_pending
To	active
Actor	authorized coordinator/domain service
Evidence	approved onboarding/compatibility/intake/education evidence
Idempotency key	case:{id}:activate:{evidence_hash}
Invalid transitions	Missing/revoked evidence or GC consent
Audit event	case_activated
Retry	No duplicate activation/invoice
Recovery	Return pending with unmet checklist

active

From	active
To	paused
Actor	authorized commercial/operations policy
Evidence	audited non-critical pause reason
Idempotency key	case:{id}:pause:{event}
Invalid transitions	Safety-critical access cannot be paused
Audit event	case_paused
Retry	Idempotent
Recovery	Resume after evidence; preserve entitled docs

paused

From	paused
To	active
Actor	authorized operator/domain service
Evidence	pause reason resolved
Idempotency key	case:{id}:resume:{event}
Invalid transitions	Unresolved safety/authorization block
Audit event	case_resumed
Retry	Idempotent
Recovery	Escalate manual review

active|paused

From	active|paused
To	closed
Actor	authorized coordinator
Evidence	birth/close or completed termination handoff evidence
Idempotency key	case:{id}:close:{evidence_hash}
Invalid transitions	Open critical obligations
Audit event	case_closed
Retry	No-op if same evidence
Recovery	Reopen only by exceptional audited event

enrollment_pending|active|paused

From	enrollment_pending|active|paused
To	cancelled
Actor	authorized operator under policy
Evidence	withdrawal/non-clearance/cancellation + financial disposition
Idempotency key	case:{id}:cancel:{request}
Invalid transitions	Cannot erase records/refund ledger
Audit event	case_cancelled
Retry	Idempotent
Recovery	Replacement is a new case/workspace policy event

Paid-case membership invitation (`case_invitation`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

none

From	none
To	issued
Actor	authorized IP case member
Evidence	enrollment_pending/active case + recipient binding
Idempotency key	case-invite:{case}:{role}:{generation}
Invalid transitions	Self/member/active-GC/unauthorized
Audit event	case_invite_issued
Retry	Return current; supersede prior live GC invite
Recovery	Issue corrected invite

issued

From	issued
To	pending_identity
Actor	recipient
Evidence	valid addressed token landing
Idempotency key	case-invite:{id}:land
Invalid transitions	Expired/revoked
Audit event	case_invite_opened
Retry	Safe repeat
Recovery	Authenticate with return_to

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email
Idempotency key	case-invite:{id}:bind:{subject}
Invalid transitions	Terminal/duplicate/different subject
Audit event	case_invite_bound|mismatch
Retry	CAS protects terminal
Recovery	Issuer confirmation or corrected invite

mismatch_pending

From	mismatch_pending
To	issuer_confirmed
Actor	authorized issuer
Evidence	explicit reviewed mismatch confirmation
Idempotency key	case-invite:{id}:confirm:{issuer}
Invalid transitions	Issuer unauthorized/recipient changed
Audit event	case_invite_mismatch_confirmed
Retry	CAS no-op
Recovery	Revoke and reissue

identity_bound|issuer_confirmed

From	identity_bound|issuer_confirmed
To	consented
Actor	invitee
Evidence	paid-case role/privacy consent version
Idempotency key	case-invite:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	case_invite_consented
Retry	Idempotent
Recovery	Present current consent

consented

From	consented
To	redeemed
Actor	invitation service
Evidence	single transaction creates person/party/membership
Idempotency key	case-invite:{id}:redeem
Invalid transitions	Existing member/active GC/expired/revoked
Audit event	case_invite_redeemed
Retry	Return same membership
Recovery	Rollback entire transaction; forward repair from invite

issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented

From	issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented
To	declined|revoked|expired|superseded
Actor	invitee/issuer/system
Evidence	explicit decline/revoke/TTL/new generation
Idempotency key	case-invite:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	case_invite_terminal
Retry	Idempotent CAS
Recovery	Fresh invitation only

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

Sole promotion-authorizing event

```
event_type: byos.workspace_promotion_authorized.v1
```

Producer: natalyx-app commitment/commercial domain after deterministic validation. Consumer: natalyx-app promotion service.

Payload:

```
event_id
idempotency_key
workspace_id
commitment_id
owner_person_id
owner_subject
commercial_config_version
terms_acceptance_ids[]
payment_plan_id
funding_evidence.kind
funding_evidence.commercial_event_id
funding_evidence.settled_amount_minor
funding_evidence.currency
credit_allocation_ids[]
initial_case_state=enrollment_pending
occurred_at
correlation_id
causation_id
schema_version
```

Authorization conditions:

- Workspace is active/match-ready, unarchived and not already promoted.
- Commitment references the same workspace, has current terms/privacy acceptances and is in funding_pending/funded per the approved state contract.
- Payment path and activated commercial configuration are pinned.
- Required enrollment funding has a verified settled ledger entry or separately approved financing/manual-funding evidence; a processor webhook is first normalized and cannot itself authorize promotion.
- Credit allocation is non-negative, reproducible and cannot exceed the applicable obligation.
- No cancellation, compliance hold, reversal that invalidates funding, or existing promotion is present.

Unique constraints:

- one promotion per workspace
- one case per promotion event
- one result per idempotency key
- one primary owner membership per created case

No browser success/return route and no raw processor callback can emit or substitute for this event.

4. Explicit PR sequence

1

Order	1
ID	P07-01
Title	Add versioned commercial configuration and terms acceptance
Repository	app
Dependencies	P02-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03; Gate A0 active and verified; Alan review

2

Order	2
ID	P07-02
Title	Implement commitment and enrollment-pending lifecycle
Repository	app
Dependencies	P07-01, P08-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P07-01, P08-01; Gate A0 active and verified; Alan review

3

Order	3
ID	P07-03
Title	Emit processor-neutral workspace promotion authorization
Repository	app
Dependencies	P04-02, P07-02, P08-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P07-02, P08-02; Gate A0 active and verified; Alan review

4

Order	4
ID	P07-04
Title	Promote workspace exactly once from authorization event
Repository	app
Dependencies	P03-03, P04-02, P07-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-03, P04-02, P07-03; Gate A0 active and verified; Alan review

P07-01 — Add versioned commercial configuration and terms acceptance

Field	Plan
Repository / base	`app` (`main`)
Project item	P07-01
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P02-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Payment/business-model DOCX

Objective

Store activated price books, milestone schedules, Node 0 credit rules, and immutable terms/privacy acceptance versions without scattering literals.

Scope

- Immutable activated config versions
- JMF/deposit/discount/milestones/credit policy

- Terms and privacy acceptance records

Explicit non-goals

- No unresolved Node 0 price value
- No payment provider

Expected files or modules

- `migrations/versions/*_commercial_configuration.py`
- `app/domain/commercial_config.py`
- `app/storage/commercial_config.py`
- `tests/python/test_commercial_config.py`

Change map

Area	Planned change
Schema/model	Commercial configuration and acceptance ledgers.
API/routes	Read active config and record acceptance commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Activation immutability
- currency/amount validation
- acceptance version
- no retroactive mutation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03. Parallel opportunities: P07-03, P07-04

Objective acceptance criteria

- □ Store activated price books, milestone schedules, Node 0 credit rules, and immutable terms/privacy acceptance versions without scattering literals.

- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	commercial systems, PostgreSQL
Higher-capability review	Mandatory before merge

P07-02 — Implement commitment and enrollment-pending lifecycle

Field	Plan
Repository / base	`app` (`main`)
Project item	P07-02
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P07-01, P08-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P07-01, P08-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Business-model conversion trigger

Objective

Create a deliberate managed-journey commitment with accepted terms, selected payment path, pinned config, and an enrollment invoice request independent from case state.

Scope

- Draft/accepted/cancelled/expired commitment transitions
- Payment-path choice
- Enrollment invoice command

Explicit non-goals

- No browser-triggered promotion
- No activation

Expected files or modules

- `app/domain/commitments.py`
- `app/storage/commitments.py`
- `app/routes/enrollment.py`
- `tests/python/test_commitments.py`

Change map

Area	Planned change
Schema/model	Commitment events/current projection.
API/routes	Create, accept, cancel, inspect enrollment.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Acceptance prerequisites
- duplicate accept
- stale config
- cancel/expire
- resume

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P07-01, P08-01. Parallel opportunities: P07-04

Objective acceptance criteria

- □ Create a deliberate managed-journey commitment with accepted terms, selected payment path, pinned config, and an enrollment invoice request independent from case state.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Mandatory before merge

P07-03 — Emit processor-neutral workspace promotion authorization

Field	Plan
Repository / base	`app` (`main`)
Project item	P07-03
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P07-02, P08-02; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P04-02, P07-02, P08-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Business-model promotion invariants

Objective

Validate commitment, terms, commercial version, settled funding and credit evidence, then emit exactly one byos.workspace_promotion_authorized.v1 event; processor callbacks and browser redirects cannot emit it.

Scope

- Deterministic authorization predicate
- Canonical payload and idempotency key
- Transactional event/outbox
- Funding/credit/config evidence references

Explicit non-goals

- No case creation
- No redirect/webhook-as-authority
- No GC membership

Expected files or modules

- `app/domain/promotion_authorization.py`
- `app/event_handlers/commercial.py`
- `app/storage/outbox.py`
- `tests/python/test_promotion_authorization.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Internal event producer only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Only commercial-domain authority can emit the typed event.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Pending/failed/reversed funding
- stale terms/config
- duplicate/replayed inputs
- credit mismatch
- event payload/checksum

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P04-02, P07-02, P08-02. Parallel opportunities: P07-01

Objective acceptance criteria

- □ Validate commitment, terms, commercial version, settled funding and credit evidence, then emit exactly one byos.workspace_promotion_authorized.v1 event; processor callbacks and browser redirects cannot emit it.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	event-driven systems, payments, commercial policy
Higher-capability review	Mandatory before merge

P07-04 — Promote workspace exactly once from authorization event

Field	Plan
Repository / base	`app` (`main`)
Project item	P07-04
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-03, P04-02, P07-03; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-03, P04-02, P07-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	byos.workspace_promotion_authorized.v1 contract, PR #124 source linkage concepts

Objective

Consume byos.workspace_promotion_authorized.v1 to create/find one enrollment_pending paid case, link the source workspace, project allowlisted data, and grant only restricted IP membership.

Scope

- Unique promotion idempotency key
- Case/source link and config version
- Purpose-limited data manifest and audit event

Explicit non-goals

- No redirect authority
- No GC case membership transfer
- No activation

Expected files or modules

- `app/domain/promotion.py`
- `app/storage/promotions.py`
- `app/event_handlers/enrollment.py`
- `tests/python/test_workspace_promotion.py`

Change map

Area	Planned change
Schema/model	Promotion ledger and source-workspace unique constraint.
API/routes	Internal event handler only.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Projection defaults deny and preserves provenance.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Replay/concurrency
- partial transaction
- funding event validation
- private candidate exclusion
- GC invitation required

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-03, P04-02, P07-03. Parallel opportunities: P07-01, P07-02

Objective acceptance criteria

- □ Consume byos.workspace_promotion_authorized.v1 to create/find one enrollment_pending paid case, link the source workspace, project allowlisted data, and grant only restricted IP membership.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	event-driven systems, privacy, PostgreSQL
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P07-01` → `P07-02` → `P07-03` → `P07-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Terms/config version pinning; Commitment retries; Exact-once promotion; Funding reversal handling; No GC access escalation	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-DATA-01	Founder decision final; implementation evidence pending	Luke decides product/privacy outcome	Alan - implementation	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-COM-01	Founder decision final for beta; tax externally dependent	Luke decides commercial outcome	Alan - implementation	P06-01, P07-01, P08-01; Before activation of a Complete configuration	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
D-COM-02	Founder decision final for v1; live activation gated	Luke decides commercial outcome	Alan - implementation	P07-01, P08-01, P08-03; Before P07-01 activation data	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
D-COM-03	Externally dependent; founder policy inputs settled	External specialist / later evidence required	External legal/privacy/finance; Luke final business approval	P07-01, P07-02, P08-03; Before P07-02 user acceptance	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
D-COM-04	Founder decision final; implementation evidence pending	Luke decides commercial/product outcome	Alan - implementation	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	P07-02 through P09-04 may proceed under the settled semantics.
D-COM-05	Founder decision final for beta	Luke decides beta scope	None during beta	P07-02, P08-02; Before financed option is exposed	Financing no longer blocks P08-02; the feature is deliberately out of scope.

D-DATA-01 - IP-GC field projection allowlist

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.
Founder constraints	Default-deny field-and-purpose allowlists; Source-labelled corrections; Named-purpose provider projection only
What remains for Alan	Define explicit field/purpose allowlists in code and projection contracts; Implement cross-role and provider negative tests
Evidence required	Cross-role denial matrix; Provider projection allowlist tests; Correction/source-label evidence
Approval rule	Luke approval is required for a broader sharing policy, not for compliant implementation detail.
External dependencies	None
Affected PRs / latest point	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02
Programme effect	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
Independent recommendation	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.

D-COM-01 - Node 0 Complete price

Field	Recorded decision or requirement
Decision state	Final beta policy; external tax wording pending
Founder outcome	Node 0 Complete is a voluntary $495 USD one-time upgrade paid by the IP or authorized IP account holder, never the GC, collected before Complete features and fully credited against the later Phase 1/JMF without gating matching or safety support.
Founder constraints	Transparent one-time feature/credit wording; Versioned prospective changes only; No forced payment to identify, approach or reach agreement with a GC
What remains for Alan	Implement configurable price, payer ownership, durable payment/credit ledger and idempotent flows
Evidence required	Synthetic success, failure, retry, duplicate and credit tests; Tax wording before live activation
Approval rule	Luke approves future commercial-policy changes; tax treatment requires specialist review.
External dependencies	Tax treatment and customer-facing wording before live activation
Affected PRs / latest point	P06-01, P07-01, P08-01; Before activation of a Complete configuration
Programme effect	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
Independent recommendation	Keep no active production value until explicitly approved; test with synthetic configuration.

D-COM-02 - Ratify JMF price book and effective date

Field	Recorded decision or requirement
Decision state	Final v1 policy; external activation dependencies
Founder outcome	Configuration v1 is $8,900 USD: a $750 deposit followed by $1,630, $2,445, $1,630, $1,630 and $815 milestones. Pay-in-full is $7,830 total, leaving $7,080 after the deposit; the separate $495 Complete credit must not be double-counted.
Founder constraints	Versioned configuration; Prospective changes only; First live beta requires A2 plus approved terms and tax treatment
What remains for Alan	Implement versioned configuration, milestone and reconciliation behavior
Evidence required	Synthetic schedule, pay-in-full and no-double-credit reconciliation tests
Approval rule	Luke approves future price-book versions and first live activation.
External dependencies	Approved terms and tax treatment before live activation
Affected PRs / latest point	P07-01, P08-01, P08-03; Before P07-01 activation data
Programme effect	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
Independent recommendation	Use the document anchors but require explicit version activation rather than code literals.

D-COM-03 - Terms, privacy and refund language versions

Field	Recorded decision or requirement
Decision state	Externally dependent
Founder outcome	No live acceptance or charge may rely on the contradictory current terms draft. D-COM-06 and D-COM-07 supply the approved business policy for a new versioned terms package.
Founder constraints	Resolve refundable-until-onboarding versus non-refundable contradiction; No retrospective policy changes; Versioned terms/privacy/refund acceptance
What remains for Alan	Implement versioned acceptance and keep real charging disabled until an approved version is active
Evidence required	Approved paid-service terms; Approved privacy/consent disclosure; Approved tax/refund/accounting wording
Approval rule	Luke approves the final business policy and customer-facing version after specialist review.
External dependencies	Legal, privacy, tax and finance review before real-user acceptance or charging
Affected PRs / latest point	P07-01, P07-02, P08-03; Before P07-02 user acceptance
Programme effect	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
Independent recommendation	External legal/privacy/finance review before any real acceptance.

D-COM-04 - Enrollment funding and activation evidence

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use two-step commitment: settled $750 plus current accepted terms proves commercial enrollment, but payment alone never promotes. Exactly one enrollment-pending case follows explicit IP promotion and authenticated GC consent. The $1,630 activation tranche is earned only after defined IP/GC onboarding and screening coordination has begun; it never implies medical clearance.
Founder constraints	Redirects/webhooks never promote; Typed promotion event only; Source-labelled facts and explicit evidence checklist
What remains for Alan	Define typed events, activation checklist, idempotency and audit mechanics
Evidence required	Duplicate, replay, partial-failure, promotion and activation evidence tests
Approval rule	Luke approves any change to commercial enrollment or activation semantics.
External dependencies	None
Affected PRs / latest point	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02
Programme effect	P07-02 through P09-04 may proceed under the settled semantics.
Independent recommendation	Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.

D-COM-05 - Financing path

Field	Recorded decision or requirement
Decision state	Final beta scope
Founder outcome	Financing is excluded and hidden from beta; Natalyx does not advertise, recommend or integrate a lender or treat lender status as funding evidence.
Founder constraints	Card and ACH remain unaffected; Reconsider only after demand evidence plus legal/privacy review
What remains for Alan	Keep financing paths disabled and absent from beta UI/configuration
Evidence required	Negative configuration/UI tests proving financing is unavailable
Approval rule	A future financing proposal requires a new founder decision.
External dependencies	None
Affected PRs / latest point	P07-02, P08-02; Before financed option is exposed
Programme effect	Financing no longer blocks P08-02; the feature is deliberately out of scope.
Independent recommendation	Do not block card/ACH MVP; keep optional and disabled.

Implementation risk	Owner	Treatment
Configuration drift	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Duplicate promotion	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Redirect spoofing	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Unresolved policy encoded as default	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
No direct source	—	Build from approved contracts/evidence	Package PRs	New implementation	N/A

Package source inventory:

- Payment/business-model DOCX as authority
- PR #124 source workspace linkage concepts only

9. Ownership recommendation

Luke is the proposed DRI and primary implementer for the settled commercial contract. Alan reviews and retains the named technical-mechanism decisions; finance/legal review remains required where the live policy names it.

P07-01

PR	P07-01
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P07-02

PR	P07-02
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P07-03

PR	P07-03
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P07-04

PR	P07-04
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.

- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 08 — Payments, invoices, credits, and refunds

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP08
Plan file	`08-payments-invoices-credits-and-refunds.md`
Proposed PRs	P08-01, P08-02, P08-03, P08-04
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. No append-only commercial ledger or processor-neutral payment boundary currently supports the $8,900 JMF, $750 enrollment, Node 0 credit, milestones, deterministic refunds, or failure recovery.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Immutable commercial ledgers, idempotent provider inbox, milestone engine, reproducible credits/refunds, reconciliation, and safe non-critical pause behavior.

Explicit non-goals.

- No custody of pass-through funds
- No financing assumption
- No pregnancy-loss automation before policy
- No accounting-general-ledger replacement

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-025	design authority	business model / Natalyx_Payment_and_Business_Model_BYOS_MVP.docx v1.0	Sections 3-12 and implementation tables	Node 0 Free must complete the match; Complete is optional/creditable; $8,900 JMF and $750 enrollment anchor the configurable model; promotion follows explicit commitment and funding, not willingness or redirect.

3. Architecture and contracts

- Payments, credits, refunds, adjustments, invoices, and commercial events are append-only; mutable balances are derived projections.
- Processor events enter an idempotent inbox and become internal commands/events only after signature and amount/currency checks.
- Billing follows durable case evidence and never acts as first notification of clinical events.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	`journey_commitment`, `payment_obligation + append-only entries`, `workspace_entitlement`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.

Contract area	Package rule
API/service	Processor webhook and internal payment commands. Internal evidence-to-invoice commands and explainable refund quote.
Storage	Append-only financial ledgers and unique idempotency constraints. Processor-event inbox/outbox and mapping records.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Entitlement (`workspace_entitlement`)

Owner: natalyx-app commercial domain. States: `free`, `complete_pending`, `complete_active`, `suspended`, `revoked`.

none

From	none
To	free
Actor	system
Evidence	active Node 0 workspace
Idempotency key	entitlement:{workspace}:free
Invalid transitions	No workspace
Audit event	entitlement_granted
Retry	Return current
Recovery	Repair projection from event

free

From	free
To	complete_pending
Actor	IP/commercial service
Evidence	accepted Complete offer/config
Idempotency key	entitlement:{workspace}:{commitment}
Invalid transitions	Unknown price/config/terms
Audit event	complete_requested
Retry	Return pending
Recovery	Cancel/expire commitment

complete_pending

From	complete_pending
To	complete_active
Actor	payment domain
Evidence	settled Complete ledger entry
Idempotency key	entitlement:{payment}:activate
Invalid transitions	Amount/currency/config mismatch
Audit event	complete_activated
Retry	Replay no-op
Recovery	Reconcile and retry from inbox

complete_active

From	complete_active
To	suspended
Actor	authorized commercial policy
Evidence	refund/reversal or approved pause reason
Idempotency key	entitlement:{id}:suspend:{event}
Invalid transitions	Cannot remove free/safety capability
Audit event	entitlement_suspended
Retry	No-op if same event
Recovery	Reinstate or revoke through new event

complete_active|suspended

From	complete_active|suspended
To	revoked
Actor	authorized operator/policy
Evidence	fraud/refund/terms revocation evidence
Idempotency key	entitlement:{id}:revoke:{event}
Invalid transitions	No silent case cancellation
Audit event	entitlement_revoked
Retry	Idempotent
Recovery	Manual reviewed regrant creates new entitlement event

Commercial commitment (`journey_commitment`)

Owner: natalyx-app commercial domain. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

none

From	none
To	draft
Actor	IP
Evidence	active workspace + displayed config/terms
Idempotency key	commitment:{workspace}:{config}
Invalid transitions	No active config/workspace
Audit event	commitment_drafted
Retry	Return existing draft
Recovery	Create new draft after expiry/version change

draft

From	draft
To	accepted
Actor	IP
Evidence	terms/privacy consent + payment path
Idempotency key	commitment:{id}:accept:{terms_hash}
Invalid transitions	Stale/missing acceptance
Audit event	commitment_accepted
Retry	Return accepted
Recovery	New acceptance revision if version changes

accepted

From	accepted
To	funding_pending
Actor	commercial service
Evidence	enrollment invoice/payment intent created
Idempotency key	commitment:{id}:funding
Invalid transitions	No invoice or invalid amount
Audit event	funding_requested
Retry	Resume same invoice
Recovery	Cancel/reissue under audited adjustment

funding_pending

From	funding_pending
To	funded
Actor	payment domain
Evidence	net settled enrollment funding meets versioned requirement
Idempotency key	enrollment-funding:{id}:v1
Invalid transitions	Redirect/pending/failed/reversed evidence
Audit event	commercial.enrollment_funding_confirmed.v1
Retry	Exactly-once event/outbox replay
Recovery	Reconcile inbox/ledger; no browser repair

draft|accepted|funding_pending

From	draft|accepted|funding_pending
To	cancelled
Actor	IP/authorized operator
Evidence	explicit cancellation and refund policy
Idempotency key	commitment:{id}:cancel:{request}
Invalid transitions	Already funded/promoted uses case cancellation rail
Audit event	commitment_cancelled
Retry	Idempotent
Recovery	New commitment required

draft|accepted

From	draft|accepted
To	expired
Actor	system
Evidence	offer/terms TTL or config deactivation
Idempotency key	commitment:{id}:expire
Invalid transitions	No expiry after funding
Audit event	commitment_expired
Retry	Idempotent
Recovery	Draft new version

Payment (`payment_obligation + append-only entries`)

Owner: natalyx-app payment domain. States: `created`, `processing`, `settled`, `failed`, `partially_refunded`, `refunded`, `reversed`.

none

From	none
To	created
Actor	billing policy
Evidence	valid invoice/obligation
Idempotency key	payment:{invoice}:create
Invalid transitions	Duplicate or invalid invoice
Audit event	payment_created
Retry	Return current intent
Recovery	Void/reissue through adjustment

created|failed

From	created|failed
To	processing
Actor	processor adapter
Evidence	signed provider acknowledgement
Idempotency key	processor:{provider_event}
Invalid transitions	Amount/currency/customer mismatch
Audit event	payment_processing
Retry	Inbox replay no-op
Recovery	Retry same intent or audited replacement

processing|created

From	processing|created
To	settled
Actor	payment domain
Evidence	verified settlement event
Idempotency key	processor:{provider_event}
Invalid transitions	Browser redirect or unverified event
Audit event	payment_settled
Retry	Exactly-once append
Recovery	Reconcile provider and append repair event

created|processing

From	created|processing
To	failed
Actor	payment domain
Evidence	verified failure/timeout policy
Idempotency key	processor:{provider_event}
Invalid transitions	Cannot overwrite settled
Audit event	payment_failed
Retry	Replay no-op
Recovery	Retry/new attempt linked to obligation

settled|partially_refunded

From	settled|partially_refunded
To	partially_refunded|refunded
Actor	authorized refund service
Evidence	approved refund calculation + processor confirmation
Idempotency key	refund:{refund_request}
Invalid transitions	Exceeds settled less prior refunds
Audit event	refund_settled
Retry	Replay no-op
Recovery	Reconcile/adjust; never mutate original

settled

From	settled
To	reversed
Actor	payment domain
Evidence	verified chargeback/reversal
Idempotency key	processor:{provider_event}
Invalid transitions	Unsupported/manual assertion
Audit event	payment_reversed
Retry	Replay no-op
Recovery	Open discrepancy and apply policy; case is not deleted

Append-only commercial contract

Authoritative ledgers:

- `commercial_events`
- `processor_events`
- `invoice_events`
- `payment_ledger_entries`
- `credit_ledger_entries`
- `refund_ledger_entries`
- `adjustment_ledger_entries`
- `performance_obligation_events`
- `promotion_events`

Derived projections:

- payment plan state
- invoice status
- settled cash
- available credit
- recognized/deferred amount
- refundable amount
- delinquency/pause status

Rules:

- Every record carries currency in integer minor units, commercial configuration version, source/idempotency/correlation identifiers and immutable timestamps.
- Mutable balances/statuses are projections only; reversals, chargebacks, credits, refunds and adjustments append entries.

- Processor events are signature-verified, stored once and normalized before domain use.
- Node 0 Complete credit cannot create a negative JMF obligation and stays distinct from cash/refund entries.
- Pass-through clinic/legal/psych/background/escrow/insurance/GC funds never enter Natalyx revenue ledgers.
- Clinical-linked invoices require durable evidence plus dignity-of-timing delay and can never be the first clinical notification.
- Refundable cash = max(0, settled JMF cash - earned JMF - prior cash refunds - chargebacks/reversals), with policy hooks for unresolved cases.

4. Explicit PR sequence

1

Order	1
ID	P08-01
Title	Add append-only commercial and money ledgers
Repository	app
Dependencies	P02-03, P07-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P07-01; Gate A0 active and verified; Alan review

2

Order	2
ID	P08-02
Title	Add processor-neutral payment command and webhook inbox
Repository	app
Dependencies	P08-01, P07-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-01, P07-02; technical decisions/controls D-PAY-01; Gate A0 active and verified; Alan review

3

Order	3
ID	P08-03
Title	Implement milestone invoicing, credits, and deterministic refunds
Repository	app
Dependencies	P08-02, P07-04, P09-04
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-02, P07-04, P09-04; Gate A0 active and verified; Alan review

4

Order	4
ID	P08-04
Title	Add payment reconciliation and safe service-pause operations
Repository	app
Dependencies	P08-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-03; Gate A0 active and verified; Alan review

P08-01 — Add append-only commercial and money ledgers

Field	Plan
Repository / base	`app` (`main`)
Project item	P08-01
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P07-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P02-03, P07-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Payment/business-model DOCX

Objective

Create immutable commercial-event, invoice, payment, credit, refund, and adjustment entries with deterministic projections and config-version references.

Scope

- Ledger invariants and currency precision
- Derived balances/status projections
- Audit/source/idempotency references

Explicit non-goals

- No processor
- No mutable balance authority
- No pass-through custody

Expected files or modules

- `migrations/versions/*_commercial_ledgers.py`
- `app/storage/ledgers.py`
- `app/domain/money.py`
- `tests/python/test_ledgers.py`

Change map

Area	Planned change
Schema/model	Append-only financial ledgers and unique idempotency constraints.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Least-privilege write paths and redacted logs.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Double-entry-like conservation checks
- duplicate key
- credit floor
- refund totals

- concurrent append

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03, P07-01. Parallel opportunities: P08-03, P08-04

Objective acceptance criteria

- □ Create immutable commercial-event, invoice, payment, credit, refund, and adjustment entries with deterministic projections and config-version references.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	payments, financial ledgers, PostgreSQL
Higher-capability review	Mandatory before merge

P08-02 — Add processor-neutral payment command and webhook inbox

Field	Plan
Repository / base	`app` (`main`)
Project item	P08-02
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-01, P07-02; technical decisions/controls D-PAY-01; Gate A0 active and verified; Alan review
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Dependencies	P08-01, P07-02
Blocking decisions/checkpoints	D-PAY-01
Source / salvage	Business-model processor-neutral requirement

Objective

Define provider ports and a signed, idempotent webhook inbox that emits internal settlement/failure/refund events, including commercial.enrollment_funding_confirmed.v1 when domain conditions hold.

Scope

- Provider-neutral intents and event mapping
- Signature/replay/order handling
- Settlement predicate and outbox

Explicit non-goals

- No browser redirect settlement
- No financing implementation unless separately approved

Expected files or modules

- `app/payments/ports.py`
- `app/payments/webhook_inbox.py`
- `app/routes/payment_webhooks.py`
- `tests/python/test_payment_webhooks.py`

Change map

Area	Planned change
Schema/model	Processor-event inbox/outbox and mapping records.
API/routes	Processor webhook and internal payment commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Secrets/endpoints deferred to guarded environment configuration.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Signature failure
- replay/out-of-order
- amount/currency mismatch
- timeout/retry
- settlement event exactly once

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P08-01, P07-02. Parallel opportunities: P08-04

Objective acceptance criteria

- □ Define provider ports and a signed, idempotent webhook inbox that emits internal settlement/failure/refund events, including commercial.enrollment_funding_confirmed.v1 when domain conditions hold.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	payment security, webhooks, event systems
Higher-capability review	Mandatory before merge

P08-03 — Implement milestone invoicing, credits, and deterministic refunds

Field	Plan
Repository / base	`app` (`main`)
Project item	P08-03
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-02, P07-04, P09-04; Gate A0 active and verified; Alan review
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Dependencies	P08-02, P07-04, P09-04
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Payment/business-model milestone/refund tables

Objective

Turn explicit evidence-backed case transitions into one invoice each, apply Node 0 credit, and calculate reproducible earned/uncharged/refunded amounts from immutable state.

Scope

- Enrollment/activation/clearance/legal-escrow/pregnancy/birth milestones
- Credit allocation
- Non-clearance and cancellation tables
- Dignity delay for clinical-linked invoices

Explicit non-goals

- No unresolved pregnancy-loss automatic disposition
- No promise of reimbursement

Expected files or modules

- `app/domain/billing_policy.py`
- `app/domain/refunds.py`
- `app/event_handlers/case_billing.py`
- `tests/python/test_billing_policy.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Internal evidence-to-invoice commands and explainable refund quote.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Only authorized domain events trigger billing.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- All milestone/refund rows
- duplicate/stale transition
- Node 0 credit
- pay-in-full/deposit math
- clinical notification separation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P08-02, P07-04, P09-04. Parallel opportunities: P08-01

Objective acceptance criteria

- □ Turn explicit evidence-backed case transitions into one invoice each, apply Node 0 credit, and calculate reproducible earned/uncharged/refunded amounts from immutable state.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	payments, commercial policy, case state
Higher-capability review	Mandatory before merge

P08-04 — Add payment reconciliation and safe service-pause operations

Field	Plan
Repository / base	`app` (`main`)
Project item	P08-04
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P08-03; Gate A0 active and verified; Alan review
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Dependencies	P08-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Business-model failure handling

Objective

Expose operator-safe reconciliation, discrepancy, receipt, retry, and non-critical pause controls without blocking safety information or altering clinical truth.

Scope

- Ledger-vs-processor reconciliation
- Itemized receipt data
- Grace/reminder/soft-pause policy
- Audited manual adjustments

Explicit non-goals

- No organization-wide ambient access

- No critical-service lockout

Expected files or modules

- `app/payments/reconciliation.py`
- `app/routes/operator_billing.py`
- `app/templates/internal/billing/`
- `tests/python/test_reconciliation.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	Internal role-gated discrepancy and adjustment surface.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Role denial
- mismatch workflow
- manual adjustment audit
- critical content remains accessible
- provider outage

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Metrics/alerts for settlement, refund, mismatch and reserve inputs.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P08-03. Parallel opportunities: P08-01, P08-02

Objective acceptance criteria

- □ Expose operator-safe reconciliation, discrepancy, receipt, retry, and non-critical pause controls without blocking safety information or altering clinical truth.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.

- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	payment operations, authorization, observability
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P08-01` → `P08-02` → `P08-03` → `P08-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Webhook replay/order; Ledger invariants; Milestone idempotency; Refund table; Node 0 credit floor; Reconciliation mismatch	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-COM-01	Founder decision final for beta; tax externally dependent	Luke decides commercial outcome	Alan - implementation	P06-01, P07-01, P08-01; Before activation of a Complete configuration	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
D-COM-02	Founder decision final for v1; live activation gated	Luke decides commercial outcome	Alan - implementation	P07-01, P08-01, P08-03; Before P07-01 activation data	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
D-COM-03	Externally dependent; founder policy inputs settled	External specialist / later evidence required	External legal/privacy/finance; Luke final business approval	P07-01, P07-02, P08-03; Before P07-02 user acceptance	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
D-COM-04	Founder decision final; implementation evidence pending	Luke decides commercial/ product outcome	Alan - implementation	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	P07-02 through P09-04 may proceed under the settled semantics.
D-PAY-01	Founder constraints final; Alan later	Luke sets constraints; Alan decides later	Alan - later before P08-02	P08-02, P08-03, P08-04; Before P08-02 adapter	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
D-COM-05	Founder decision final for beta	Luke decides beta scope	None during beta	P07-02, P08-02; Before financed option is exposed	Financing no longer blocks P08-02; the feature is deliberately out of scope.
D-COM-06	Founder policy final; legal/accounting externally dependent	Luke decides policy; specialists translate	Alan - implementation; legal/accounting - wording	P08-03; Before automated pregnancy-stage refund logic	Synthetic P08-03 implementation may proceed; live policy activation remains gated.
D-COM-07	Founder policy final; legal/accounting externally dependent	Luke decides policy; specialists translate	Alan - implementation; legal/accounting - wording	P08-03; Before real billing	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
D-COM-08	Founder decision final; implementation evidence pending	Luke decides customer treatment	Alan - implementation	P08-04; Before P08-04	P08-04 may proceed under the settled policy.

D-COM-01 - Node 0 Complete price

Field	Recorded decision or requirement
Decision state	Final beta policy; external tax wording pending
Founder outcome	Node 0 Complete is a voluntary $495 USD one-time upgrade paid by the IP or authorized IP account holder, never the GC, collected before Complete features and fully credited against the later Phase 1/JMF without gating matching or safety support.
Founder constraints	Transparent one-time feature/credit wording; Versioned prospective changes only; No forced payment to identify, approach or reach agreement with a GC
What remains for Alan	Implement configurable price, payer ownership, durable payment/credit ledger and idempotent flows
Evidence required	Synthetic success, failure, retry, duplicate and credit tests; Tax wording before live activation
Approval rule	Luke approves future commercial-policy changes; tax treatment requires specialist review.
External dependencies	Tax treatment and customer-facing wording before live activation
Affected PRs / latest point	P06-01, P07-01, P08-01; Before activation of a Complete configuration
Programme effect	P06-01, P07-01 and P08-01 may proceed synthetically; live activation remains gated.
Independent recommendation	Keep no active production value until explicitly approved; test with synthetic configuration.

D-COM-02 - Ratify JMF price book and effective date

Field	Recorded decision or requirement
Decision state	Final v1 policy; external activation dependencies
Founder outcome	Configuration v1 is $8,900 USD: a $750 deposit followed by $1,630, $2,445, $1,630, $1,630 and $815 milestones. Pay-in-full is $7,830 total, leaving $7,080 after the deposit; the separate $495 Complete credit must not be double-counted.
Founder constraints	Versioned configuration; Prospective changes only; First live beta requires A2 plus approved terms and tax treatment
What remains for Alan	Implement versioned configuration, milestone and reconciliation behavior
Evidence required	Synthetic schedule, pay-in-full and no-double-credit reconciliation tests
Approval rule	Luke approves future price-book versions and first live activation.
External dependencies	Approved terms and tax treatment before live activation
Affected PRs / latest point	P07-01, P08-01, P08-03; Before P07-01 activation data
Programme effect	Synthetic commercial configuration and ledgers may proceed; real charging remains gated.
Independent recommendation	Use the document anchors but require explicit version activation rather than code literals.

D-COM-03 - Terms, privacy and refund language versions

Field	Recorded decision or requirement
Decision state	Externally dependent
Founder outcome	No live acceptance or charge may rely on the contradictory current terms draft. D-COM-06 and D-COM-07 supply the approved business policy for a new versioned terms package.
Founder constraints	Resolve refundable-until-onboarding versus non-refundable contradiction; No retrospective policy changes; Versioned terms/privacy/refund acceptance
What remains for Alan	Implement versioned acceptance and keep real charging disabled until an approved version is active
Evidence required	Approved paid-service terms; Approved privacy/consent disclosure; Approved tax/refund/accounting wording
Approval rule	Luke approves the final business policy and customer-facing version after specialist review.
External dependencies	Legal, privacy, tax and finance review before real-user acceptance or charging
Affected PRs / latest point	P07-01, P07-02, P08-03; Before P07-02 user acceptance
Programme effect	Synthetic A2 preparation may proceed; live P07-02/P08 charging and B1 customer acceptance remain blocked.
Independent recommendation	External legal/privacy/finance review before any real acceptance.

D-COM-04 - Enrollment funding and activation evidence

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use two-step commitment: settled $750 plus current accepted terms proves commercial enrollment, but payment alone never promotes. Exactly one enrollment-pending case follows explicit IP promotion and authenticated GC consent. The $1,630 activation tranche is earned only after defined IP/GC onboarding and screening coordination has begun; it never implies medical clearance.
Founder constraints	Redirects/webhooks never promote; Typed promotion event only; Source-labelled facts and explicit evidence checklist
What remains for Alan	Define typed events, activation checklist, idempotency and audit mechanics
Evidence required	Duplicate, replay, partial-failure, promotion and activation evidence tests
Approval rule	Luke approves any change to commercial enrollment or activation semantics.
External dependencies	None
Affected PRs / latest point	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02
Programme effect	P07-02 through P09-04 may proceed under the settled semantics.
Independent recommendation	Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.

D-PAY-01 - Payment processor and card/ACH scope

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Offer card and US ACH, encourage but do not require ACH above $1,000, absorb ordinary beta fees without surcharging, and keep processor metadata free of GC, journey, clinical or document detail.
Founder constraints	Card cost ceiling near 3% plus $0.30; ACH cost ceiling near 1% capped at $10; Processor-neutral domain and ledger contract
What remains for Alan	Choose the processor/vendor, tokenization, webhook, settlement, return/refund and reconciliation mechanisms
Evidence required	Provider security and cost comparison; Synthetic settlement, return, refund, duplicate and reconciliation tests
Approval rule	Luke approval is required only if the proposed processor exceeds the cost/privacy constraints or changes the payment experience.
External dependencies	Payment/security/finance review before real money
Affected PRs / latest point	P08-02, P08-03, P08-04; Before P08-02 adapter
Programme effect	P08-01 ledger work may proceed; P08-02 and A2 payment completion wait for Alan's processor decision and evidence.
Independent recommendation	Choose only after domain/ledger contract; retain processor-neutral ports.

D-COM-05 - Financing path

Field	Recorded decision or requirement
Decision state	Final beta scope
Founder outcome	Financing is excluded and hidden from beta; Natalyx does not advertise, recommend or integrate a lender or treat lender status as funding evidence.
Founder constraints	Card and ACH remain unaffected; Reconsider only after demand evidence plus legal/privacy review
What remains for Alan	Keep financing paths disabled and absent from beta UI/configuration
Evidence required	Negative configuration/UI tests proving financing is unavailable
Approval rule	A future financing proposal requires a new founder decision.
External dependencies	None
Affected PRs / latest point	P07-02, P08-02; Before financed option is exposed
Programme effect	Financing no longer blocks P08-02; the feature is deliberately out of scope.
Independent recommendation	Do not block card/ACH MVP; keep optional and disabled.

D-COM-06 - Pregnancy-loss credit or retained-revenue policy

Field	Recorded decision or requirement
Decision state	Final founder policy; external wording pending
Founder outcome	After post-heartbeat loss, never charge the $815 birth tranche; retain the delivered $1,630 pregnancy-support tranche and grant an equal $1,630 future credit to the same IP household for 36 months, with compassionate manual extension and defined offboarding support.
Founder constraints	No automatic birth charge; Auditable credit expiry/extension; Compassionate support workflow
What remains for Alan	Implement credit ledger, expiry, extension and support workflow
Evidence required	Loss, credit, expiry, extension and duplicate tests; Approved legal/accounting treatment before live use
Approval rule	Luke approves final customer-facing policy after specialist review.
External dependencies	Legal and accounting review before live use
Affected PRs / latest point	P08-03; Before automated pregnancy-stage refund logic
Programme effect	Synthetic P08-03 implementation may proceed; live policy activation remains gated.
Independent recommendation	Manual/counsel-reviewed hook until policy is approved; never hard-code.

D-COM-07 - Cancellation and non-clearance policy sign-off

Field	Recorded decision or requirement
Decision state	Final founder policy; external wording pending
Founder outcome	Protect delivered work: the $750 is refundable until onboarding is delivered and then earned; activation follows D-COM-04. Non-clearance or pre-clearance GC withdrawal stops later charges. Unearned prepayment is refunded within ten business days, with refund or rollover choice and one replacement within 12 months without a second $750 deposit.
Founder constraints	A new activation charge may apply for genuinely new onboarding; No double charge or silent retained unearned value; Refund and rollover are auditable
What remains for Alan	Implement reconciliation, refund, rollover and replacement behavior
Evidence required	Withdrawal, non-clearance, refund, rollover, replacement and duplicate tests; Approved legal/accounting treatment before live use
Approval rule	Luke approves final customer-facing policy after specialist review.
External dependencies	Legal and accounting review before live use
Affected PRs / latest point	P08-03; Before real billing
Programme effect	Synthetic P08-03 implementation may proceed; live refund/cancellation terms remain gated.
Independent recommendation	Use document schedule as test anchor; require legal/finance approval for live policy.

D-COM-08 - Grace period and non-critical service pause

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use a humane 30-day grace period with reminders seven days before due, on the due date, and at 7, 14 and 30 days overdue; no beta late fee and human outreach before pause. Only future non-critical paid work may pause after 30 days.
Founder constraints	Never block safety content, existing documents/data, correction/export/deletion, consent/revocation, urgent support or already-required provider coordination
What remains for Alan	Implement reminder, escalation, grace and capability-specific pause states
Evidence required	Reminder cadence and protected-access tests
Approval rule	Luke approves changes to grace or protected access.
External dependencies	None
Affected PRs / latest point	P08-04; Before P08-04
Programme effect	P08-04 may proceed under the settled policy.
Independent recommendation	Default to continued safety and entitled-document access; pause only clearly non-critical future services.

Implementation risk	Owner	Treatment
Double charge/refund	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Processor lock-in	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Clinical timing harm	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Pass-through commingling	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Policy gaps	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
No direct source	—	Build from approved contracts/evidence	Package PRs	New implementation	N/A

Package source inventory:

- Business-model schedules and smoke paths; no current code is authoritative

9. Ownership recommendation

Luke is the proposed DRI and primary implementer for payment-domain work. Alan reviews and owns D-PAY-01's later processor/security mechanism; payment, finance, security and legal review remain required before live money movement.

P08-01

PR	P08-01
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P08-02

PR	P08-02
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
Exception	None

P08-03

PR	P08-03
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
Exception	None

P08-04

PR	P08-04
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	D-PAY-01: Alan - later processor, security and settlement mechanism
Reviewer	Alan
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.

- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 09 — Documents, active case, and provider handoffs

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP09
Plan file	`09-documents-active-case-and-provider-handoffs.md`
Proposed PRs	P09-01, P09-02, P09-03, P09-04, P09-05, P09-06
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Current documents are metadata-only; the dirty provider branch is unreviewed and can over-project facts. There is no durable private S3/KMS/quarantine/recovery pipeline.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A metadata-only requirement/work-item lifecycle distinct from the secure document-content lifecycle, resilient storage/scan/download, usable active-case surfaces, and consent-scoped immutable provider snapshots with operational review.

Explicit non-goals.

- No ambient provider case access
- No clinical approval by Natalyx
- No direct public objects
- No automatic promotion of AI/IP assertions

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.
E-012	verified	app / PR #122@e853821	app/storage/invitations.py and PR review-round comment	The updated case invitation work adds 14-day ceiling, CAS transitions, mismatch confirmation, self-invite refusal, return_to preservation and paired lens tests; it has no Node 0 workspace target.
E-018	verified	app / PR #124@84a5a84	app/storage/documents.py:1-114; app/storage/schema.py:785-801	Documents are explicitly metadata-only; storage_uri/checksum fields exist but no durable upload, quarantine, scan, authorization-by-purpose, retention or recovery pipeline.

ID	Class	Repository / ref	File, construct or location	Finding
E-019	verified protected work	app / dirty feature/provider-exchange-layer-v0@84a5a84	app/provider_handoffs.py:109-133 and uncommitted handoff modules	The protected provider work is not a PR. Its broad GC-owner fact selection can include private/provisional data, so only selective port after provenance/purpose review is safe.
E-028	verified review	app / PR #124 comment 4948149089	Consolidated review — findings for the WP00 extraction ledger	Alan's completed review classifies #124 as high-value extraction material rather than a merge unit and identifies transaction, privacy, scoring, access, validation, invariant, exposure, retention, performance, CSS and CI findings.
E-029	verified preserved branch	app / feat/byos-journey-wave-2@5b63e3a7	exact three-commit stack on e853821	Wave 2 contains phase-1 cluster pages, metadata-only request/review, case anchoring and role/owner fixes. Controlled local evidence passed 535 tests with 9 skipped and 36/36 focused tests, but no branch checks, PostgreSQL, browser/accessibility or visual run exists.
E-030	independently reproduced defect	app / feat/byos-journey-wave-2@5b63e3a7	app/storage/schema.py orphan cleanup; document review CAS/work-item payload	A second init_db deletes canonical case-anchored reviewer work because participant_id is empty. Separately, a stale v1 work item can approve v2 after same-row resubmission because it is not bound to an immutable submission revision.
E-031	verified	app / feat/byos-journey-wave-2@5b63e3a7	case_members.py, case_documents.py, storage/documents.py, routes/agency.py	Membership roles still conflate participation, lens and access; document/reviewer loaders are broad or global; participant role/lens can be caller-supplied; operator actions lack a canonical organization-scoped staff capability; no RLS/non-owner runtime enforces denial.

3. Architecture and contracts

- A requirement, readiness marker or review work item is not a document and cannot be approved as if content exists.
- Private bucket, block-public-access, KMS, opaque keys, application authorization, size/type/checksum gates, quarantine/scan, audit, retention, and orphan cleanup are mandatory for content.
- Provider handoffs are purpose-bound snapshots with checksum/revision/expiry and consent; provider-visible items are allowlisted, not inferred from broad visibility.
- ECS replacement must not lose upload/task state; recovery resumes from durable metadata and object state.

Contract area	Package rule
Ownership boundary	`app`, `infra`; repository-specific PRs remain separate.
Entities/state	`case`, `case_invitation`, `document + document_version + upload`
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Intent/finalize/status/download/replace/delete commands. Participant document commands join requirement IDs to immutable content versions.
Storage	Versioned document/upload/scan/object metadata. Requirement submissions and durable typed work items use constrained case/org/owner anchors and immutable expected revisions. Handoff/packet/snapshot/transmission/evidence ledgers under migrations.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.

Contract area	Package rule
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Applicable state machines

Paid case lifecycle (`case`)

Owner: natalyx-app. States: `enrollment_pending`, `active`, `paused`, `closed`, `cancelled`.

none

From	none
To	enrollment_pending
Actor	promotion service
Evidence	byos.workspace_promotion_authorized.v1 + projection manifest
Idempotency key	promotion:{commitment}:v1
Invalid transitions	No authorization/duplicate workspace
Audit event	paid_case_created
Retry	Return same case
Recovery	Forward-repair same promotion

enrollment_pending

From	enrollment_pending
To	active
Actor	authorized coordinator/domain service
Evidence	approved onboarding/compatibility/intake/education evidence
Idempotency key	case:{id}:activate:{evidence_hash}
Invalid transitions	Missing/revoked evidence or GC consent
Audit event	case_activated
Retry	No duplicate activation/invoice
Recovery	Return pending with unmet checklist

active

From	active
To	paused
Actor	authorized commercial/operations policy
Evidence	audited non-critical pause reason
Idempotency key	case:{id}:pause:{event}
Invalid transitions	Safety-critical access cannot be paused
Audit event	case_paused
Retry	Idempotent
Recovery	Resume after evidence; preserve entitled docs

paused

From	paused
To	active
Actor	authorized operator/domain service
Evidence	pause reason resolved
Idempotency key	case:{id}:resume:{event}
Invalid transitions	Unresolved safety/authorization block
Audit event	case_resumed
Retry	Idempotent
Recovery	Escalate manual review

active|paused

From	active|paused
To	closed
Actor	authorized coordinator
Evidence	birth/close or completed termination handoff evidence
Idempotency key	case:{id}:close:{evidence_hash}
Invalid transitions	Open critical obligations
Audit event	case_closed
Retry	No-op if same evidence
Recovery	Reopen only by exceptional audited event

enrollment_pending|active|paused

From	enrollment_pending|active|paused
To	cancelled
Actor	authorized operator under policy
Evidence	withdrawal/non-clearance/cancellation + financial disposition
Idempotency key	case:{id}:cancel:{request}
Invalid transitions	Cannot erase records/refund ledger
Audit event	case_cancelled
Retry	Idempotent
Recovery	Replacement is a new case/workspace policy event

Paid-case membership invitation (`case_invitation`)

Owner: natalyx-app. States: `issued`, `pending_identity`, `identity_bound`, `mismatch_pending`, `issuer_confirmed`, `consented`, `redeemed`, `declined`, `revoked`, `expired`, `superseded`.

none

From	none
To	issued
Actor	authorized IP case member
Evidence	enrollment_pending/active case + recipient binding
Idempotency key	case-invite:{case}:{role}:{generation}
Invalid transitions	Self/member/active-GC/unauthorized
Audit event	case_invite_issued
Retry	Return current; supersede prior live GC invite
Recovery	Issue corrected invite

issued

From	issued
To	pending_identity
Actor	recipient
Evidence	valid addressed token landing
Idempotency key	case-invite:{id}:land
Invalid transitions	Expired/revoked
Audit event	case_invite_opened
Retry	Safe repeat
Recovery	Authenticate with return_to

pending_identity

From	pending_identity
To	identity_bound|mismatch_pending
Actor	auth provider
Evidence	verified subject/email
Idempotency key	case-invite:{id}:bind:{subject}
Invalid transitions	Terminal/duplicate/different subject
Audit event	case_invite_bound|mismatch
Retry	CAS protects terminal
Recovery	Issuer confirmation or corrected invite

mismatch_pending

From	mismatch_pending
To	issuer_confirmed
Actor	authorized issuer
Evidence	explicit reviewed mismatch confirmation
Idempotency key	case-invite:{id}:confirm:{issuer}
Invalid transitions	Issuer unauthorized/recipient changed
Audit event	case_invite_mismatch_confirmed
Retry	CAS no-op
Recovery	Revoke and reissue

identity_bound|issuer_confirmed

From	identity_bound|issuer_confirmed
To	consented
Actor	invitee
Evidence	paid-case role/privacy consent version
Idempotency key	case-invite:{id}:consent:{version}
Invalid transitions	Stale/missing consent
Audit event	case_invite_consented
Retry	Idempotent
Recovery	Present current consent

consented

From	consented
To	redeemed
Actor	invitation service
Evidence	single transaction creates person/party/membership
Idempotency key	case-invite:{id}:redeem
Invalid transitions	Existing member/active GC/expired/revoked
Audit event	case_invite_redeemed
Retry	Return same membership
Recovery	Rollback entire transaction; forward repair from invite

issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented

From	issued|pending_identity|identity_bound|mismatch_pending|issuer_confirmed|consented
To	declined|revoked|expired|superseded
Actor	invitee/issuer/system
Evidence	explicit decline/revoke/TTL/new generation
Idempotency key	case-invite:{id}:terminal:{event}
Invalid transitions	No terminal overwrite
Audit event	case_invite_terminal
Retry	Idempotent CAS
Recovery	Fresh invitation only

Document lifecycle (`document + document_version + upload`)

Owner: natalyx-app; storage in natalyx-infra. States: `upload_requested`, `uploading`, `quarantined`, `scanning`, `available`, `review`, `approved`, `replaced`, `deleted`, `failed`.

none

From	none
To	upload_requested
Actor	authorized case member
Evidence	document purpose/type/size intent
Idempotency key	upload:{case}:{client_request}
Invalid transitions	Unauthorized/type/size/retention block
Audit event	upload_requested
Retry	Return same intent
Recovery	Expire/orphan intent

upload_requested

From	upload_requested
To	uploading
Actor	client/storage service
Evidence	authorized upload starts
Idempotency key	upload:{id}:start
Invalid transitions	Expired intent/wrong key
Audit event	upload_started
Retry	Multipart resume
Recovery	Abort and mark failed/orphan cleanup

uploading

From	uploading
To	quarantined
Actor	finalize service
Evidence	object exists + size/checksum/content sniff
Idempotency key	upload:{id}:finalize:{checksum}
Invalid transitions	Mismatch/missing object
Audit event	upload_quarantined
Retry	Repeat verify
Recovery	Delete bad object; request new upload

quarantined

From	quarantined
To	scanning
Actor	scan worker
Evidence	durable queued task
Idempotency key	scan:{version}:{engine_version}
Invalid transitions	Already available/deleted
Audit event	scan_started
Retry	Leased retry
Recovery	Lease expiry requeues after ECS replacement

scanning

From	scanning
To	available|failed
Actor	scan service
Evidence	signed clean/malicious/error result
Idempotency key	scan:{version}:{result_id}
Invalid transitions	Unknown result/engine
Audit event	scan_completed|scan_failed
Retry	Inbox replay no-op
Recovery	Remain quarantined on timeout/error; security review

available

From	available
To	review|approved
Actor	authorized participant/reviewer
Evidence	purpose-specific review/evidence
Idempotency key	document:{version}:review:{event}
Invalid transitions	Wrong role/stale version
Audit event	document_reviewed
Retry	Idempotent
Recovery	New review revision

available|review|approved

From	available|review|approved
To	replaced
Actor	authorized owner
Evidence	new clean version linked
Idempotency key	document:{id}:replace:{new_version}
Invalid transitions	New version not available
Audit event	document_replaced
Retry	Return current version
Recovery	Keep prior per retention; roll current pointer forward

available|review|approved|replaced|failed

From	available|review|approved|replaced|failed
To	deleted
Actor	authorized policy job/operator
Evidence	retention/delete request and no hold
Idempotency key	document:{version}:delete:{request}
Invalid transitions	Legal hold/provider obligation
Audit event	document_deleted
Retry	Tombstone idempotent
Recovery	Object-delete retry/orphan alarm; metadata/audit retained

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

Durable document and S3 contract

Metadata

- Requirement/readiness submissions and reviewer work items are separate entities from document versions and objects.
- A metadata-only marker cannot carry file bytes/storage_uri, enter document-approved state or claim that a clinic received/reviewed content.
- Work items bind organization, case, requirement, owner scope and immutable submission revision; decision plus audit is atomic and completion is durable/idempotent.
- Free text is minimum-necessary and strictly bounded; sensitive categories remain synthetic through A2 and content/PHI paths remain disabled until their gates.

Storage

- Private versioned quarantine bucket and clean vault with block-public-access, SSL-only and KMS encryption.
- App can create short-lived unique upload targets in quarantine but cannot read unscanned content; scanner moves/copies only clean versions to vault.

- Object key: env/{org_id}/{case_id}/{document_id}/{version_id}/{generated_object_id}; no names, emails, filenames or clinical labels.
- Authorization is always a PostgreSQL access-grant/ownership/purpose/consent decision independent of the key; participation or presentation lens alone is insufficient.
- CloudTrail S3 data events and app document_access_event records provide purpose-aware access evidence.

Upload

- Configurable maximum size; provisional 25 MiB awaits D-DOC-02.
- Initial proposed allowlist PDF/JPEG/PNG; DOCX only after scanner/archive review; no macro-enabled Office formats.
- Unicode-normalize/sanitize display filename; byte-sniff MIME, expected length and required SHA-256 checksum.
- Never reuse a key. Finalize verifies object version, HeadObject size/checksum/type and upload-session ownership.
- S3 event -> durable queue -> scanner -> signed clean/rejected/error result; unavailable/timed-out scan remains quarantined.

Download

- App authenticates and authorizes exact version/purpose, records access event, then returns a very short-lived clean-vault GET with safe content disposition.
- App proxy is reserved for justified transformation/audit cases, not the default durable path.

Recovery

- Upload sessions/tasks live in Postgres/queue so ECS replacement does not lose them; leases expire and retry idempotently.
- Replacement creates a new immutable version and supersedes the prior current pointer.
- Deletion is an audited tombstone followed by retention/legal-hold controlled purge; object-delete failures alert and retry.
- Orphan reconciliation handles expired intents and objects without document versions; lifecycle aborts incomplete multipart uploads.
- Provider share uses a purpose-bound snapshot/checksum/expiry, not direct object-key authority.

4. Explicit PR sequence

1

Order	1
ID	P09-01
Title	Define document lifecycle, authorization, and retention contract
Repository	app
Dependencies	P00-01, P02-01
Gate	A2
Local preparation	Technical-decision preparation only; implementation waits on named dependencies
Merge readiness	Blocked pending PR dependencies P00-01, P02-01; Gate A0 active and verified; Luke review/final founder approval

2

Order	2
ID	P09-02
Title	Provision private encrypted document and quarantine storage
Repository	infra
Dependencies	P09-01, P13-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-01, P13-02; technical decisions/controls D-DOC-02; Gate A0 active and verified; Luke review/final founder approval

3

Order	3
ID	P09-03
Title	Implement resilient document upload, scan, and download
Repository	app
Dependencies	P09-02, P02-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-02, P02-03; technical decisions/controls D-DOC-01, D-DOC-02; Gate A0 active and verified; Luke review/final founder approval

4

Order	4
ID	P09-04
Title	Ship phase-1 case progress and metadata-only request/review
Repository	app
Dependencies	P03-03, P03-05, P07-04, P09-01, P11-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-03, P03-05, P07-04, P09-01, P11-03; Gate A0 active and verified; Luke review/final founder approval

5

Order	5
ID	P09-05
Title	Implement consent-scoped provider handoff snapshots and operations
Repository	app
Dependencies	P04-02, P09-06, P10-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P09-06, P10-02; technical decisions/controls D-PROV-01; Gate A0 active and verified; Luke review/final founder approval

6

Order	6
ID	P09-06
Title	Integrate content-bearing document versions into participant surfaces
Repository	app
Dependencies	P09-03, P09-04
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-03, P09-04; Gate A0 active and verified; Luke review/final founder approval

P09-01 — Define document lifecycle, authorization, and retention contract

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-01
Gate / size	A2 / Medium
Local preparation status	Technical-decision preparation only; implementation waits on named dependencies
Merge readiness	Blocked pending PR dependencies P00-01, P02-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Dependencies	P00-01, P02-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current metadata-only documents.py, Wave 2@5b63e3a, provider branch

Objective

Record separate metadata-requirement/work-item and content-bearing document/version state machines, the authorization matrix, object-key independence, retention/deletion/replacement rules, and failure recovery before implementation.

Scope

- Requirement/readiness/work-item states that never imply content approval
- Document/version/upload/scan/available/review/replace/delete states
- Owner/person/case-side/case authorization and staff capabilities
- Size/type/checksum and structured metadata policy
- Presigned versus proxied download decision
- Orphan/retention jobs

Explicit non-goals

- No bucket or application code

Expected files or modules

- `docs/adr/ADR-00x-durable-documents.md`
- `docs/agents/documents.md`
- `docs/security/document-threat-model.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Decision and threat model only.

Tests and validation

Objective tests:

- Threat-model abuse cases
- metadata-versus-content transition table review
- retention/legal-hold walkthrough

- PHI-bearing free-text negative cases

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P00-01, P02-01. Parallel opportunities: P09-03, P09-05, P09-06

Objective acceptance criteria

- □ Record separate metadata-requirement/work-item and content-bearing document/version state machines, the authorization matrix, object-key independence, retention/deletion/replacement rules, and failure recovery before implementation.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	application security, S3/KMS, privacy
Higher-capability review	Mandatory before merge

P09-02 — Provision private encrypted document and quarantine storage

Field	Plan
Repository / base	`infra` (`main`)
Project item	P09-02
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-01, P13-02; technical decisions/controls D-DOC-02; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Dependencies	P09-01, P13-02
Blocking decisions/checkpoints	D-DOC-02
Source / salvage	Existing infra platform stack patterns

Objective

Provision dev/private document buckets, KMS policy, block-public-access, lifecycle/quarantine events, least-privilege task roles, and audit logging behind PHI/billable guardrails.

Scope

- Private primary/quarantine storage
- KMS and task-role policy
- Lifecycle/orphan signals
- No public ACL/policy

Explicit non-goals

- No production/PHI activation
- No application upload

Expected files or modules

- `lib/stacks/platform-stack.ts`
- `lib/constructs/document-storage.ts`
- `test/document-storage.test.ts`
- `docs/agents/platform-stack.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	CDK resources and SSM outputs in dev, desired counts unchanged by default.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- CDK assertions
- IAM negative cases
- public-access-block
- encryption/lifecycle/event contract

Local validation:

- `npm test`
- `npm run build`
- `Synthesize the affected CDK stack with billable/live changes disabled.`

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Synthesize and review first; deploy only through the guarded dev workflow with billable/desired-count defaults unchanged, exact revision recorded, and explicit confirmation outside this planning task.

Rollback or forward repair

Return to the prior reviewed stack/task definition when compatible; otherwise keep resources fail-closed and forward-repair under the runbook. Never delete data-bearing resources as an automatic rollback.

Dependencies and parallel work

Hard dependencies: P09-01, P13-02. Parallel opportunities: P09-04, P09-05, P09-06

Objective acceptance criteria

- □ Provision dev/private document buckets, KMS policy, block-public-access, lifecycle/quarantine events, least-privilege task roles, and audit logging behind PHI/billable guardrails.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	AWS S3, KMS/IAM, CDK, security
Higher-capability review	Mandatory before merge

P09-03 — Implement resilient document upload, scan, and download

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-03
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-02, P02-03; technical decisions/controls D-DOC-01, D-DOC-02; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism
Dependencies	P09-02, P02-03
Blocking decisions/checkpoints	D-DOC-01, D-DOC-02
Source / salvage	Current metadata scaffold

Objective

Implement multipart-safe upload intent/finalize, checksum/type/size validation, quarantine/scan transitions, authorized download, replacement, deletion, and task-recovery behavior.

Scope

- Opaque object keys and sanitized display names
- Durable upload states
- Malware/quarantine integration
- Presigned/proxy path per ADR
- Audit events

Explicit non-goals

- No public object URL
- No provider share by key

Expected files or modules

- `app/documents/service.py`
- `app/documents/storage.py`
- `app/routes/documents.py`
- `app/jobs/document_recovery.py`
- `tests/python/test_documents_e2e.py`

Change map

Area	Planned change
Schema/model	Versioned document/upload/scan/object metadata.
API/routes	Intent/finalize/status/download/replace/delete commands.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Oversize/type spoof
- checksum mismatch

- scan fail/timeout
- ECS task replacement
- orphan cleanup
- authorization

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P09-02, P02-03. Parallel opportunities: P09-01, P09-04, P09-05

Objective acceptance criteria

- □ Implement multipart-safe upload intent/finalize, checksum/type/size validation, quarantine/scan transitions, authorized download, replacement, deletion, and task-recovery behavior.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	S3, secure uploads, background jobs, authorization
Higher-capability review	Mandatory before merge

P09-04 — Ship phase-1 case progress and metadata-only request/review

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-04
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-03, P03-05, P07-04, P09-01, P11-03; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-03, P03-05, P07-04, P09-01, P11-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 case projection behavior/tests, Wave 2@5b63e3a semantics/tests; implementation superseded, PR #124 presentation evidence

Objective

Expose projection-driven phase-1 cluster progress and a revision-bound metadata requirement/readiness/reviewer-work workflow without creating or approving a document object.

Scope

- Versioned phase-1 workflow/request definitions
- Projection-issued action capabilities for active/waiting/attention/paused/done states
- Person versus case-side ownership
- Atomic submission plus durable case work item
- Revision-bound/idempotent staff decision plus audit
- Exact phase-1 route exposure

Explicit non-goals

- No file bytes, storage URI, upload, scan, download or artifact approval
- No caller-supplied role/lens authority
- No client-owned state transition
- No operator/provider ambient access

Expected files or modules

- `migrations/versions/*_case_requirements.py`
- `app/case_home.py`
- `app/case_journey.py`
- `app/case_requirements.py`
- `app/routes/case.py`
- `app/routes/operator_case_work.py`
- `app/templates/case/`
- `tests/python/test_case_requirements.py`

Change map

Area	Planned change
Schema/model	Requirement submissions and durable typed work items use constrained case/org/owner anchors and immutable expected revisions.
API/routes	No public API/route contract change.
UI/product surface	Participant progress and readiness surfaces remain projection-driven.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Subject-derived participant capabilities and explicit organization-scoped staff capability gate every read/command.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- IP/GC paired lenses
- viewer/nonmember/spoofed-role denial
- restart durability
- stale work item cannot approve resubmission
- atomic submission/task and review/audit rollback
- wrong case/org and unsupported type retained
- paused/done action suppression
- later-phase 404
- mobile/no-JS/reduced-motion

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P03-03, P03-05, P07-04, P09-01, P11-03. Parallel opportunities: P09-02, P09-03, P09-05

Objective acceptance criteria

- □ Expose projection-driven phase-1 cluster progress and a revision-bound metadata requirement/readiness/reviewer-work workflow without creating or approving a document object.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.

- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	FastAPI/Jinja, accessibility, authorization, PostgreSQL transactions, state projection
Higher-capability review	Mandatory before merge

P09-05 — Implement consent-scoped provider handoff snapshots and operations

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-05
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P09-06, P10-02; technical decisions/controls D-PROV-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-PROV-01: Alan - later provider-packet mechanism; privacy/provider validation; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Dependencies	P04-02, P09-06, P10-02
Blocking decisions/checkpoints	D-PROV-01
Source / salvage	Protected provider-exchange worktree: selective port only after provenance fix

Objective

Create purpose-bound, reviewed, checksummed provider packets from explicitly allowed facts/documents and operate send/confirm/revoke/reroute without ambient provider access.

Scope

- Packet revisions/snapshots/checksums
- Consent, recipient, purpose, expiry
- Operator route review and transmission outcomes
- Inbound evidence becomes proposal

Explicit non-goals

- No raw private/provisional bulk projection
- No provider truth without review

Expected files or modules

- `app/provider_handoffs.py`
- `app/storage/handoffs.py`
- `app/routes/operator_handoffs.py`
- `app/templates/internal/handoffs/`
- `tests/python/test_provider_handoffs.py`

Change map

Area	Planned change
Schema/model	Handoff/packet/snapshot/transmission/evidence ledgers under migrations.
API/routes	No public API/route contract change.
UI/product surface	Case-scoped operator queue and GC review.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Purpose allowlist
- revoked consent
- stale route
- snapshot immutability
- delivery retry
- private/provisional exclusion

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-02, P09-06, P10-02. Parallel opportunities: P09-01, P09-02, P09-03, P09-04

Objective acceptance criteria

- □ Create purpose-bound, reviewed, checksummed provider packets from explicitly allowed facts/documents and operate send/confirm/revoke/reroute without ambient provider access.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-PROV-01: Alan - later provider-packet mechanism; privacy/provider validation; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	provider workflows, privacy, document security, operations
Higher-capability review	Mandatory before merge

P09-06 — Integrate content-bearing document versions into participant surfaces

Field	Plan
Repository / base	`app` (`main`)
Project item	P09-06
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-03, P09-04; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Dependencies	P09-03, P09-04
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	P09-03 durable pipeline, Wave 2 metadata UI as contrast evidence, PR #122 participant case surfaces

Objective

Connect metadata requirements to upload intents, quarantined/scanned content versions, replacement and authorized download without collapsing requirement state into object state.

Scope

- Requirement-to-upload intent linkage
- Quarantine/scan/available and replacement projections
- Owner/role-scoped upload and download controls
- Content-version audit and restart recovery
- Fail-closed partial merge when content pipeline is unavailable

Explicit non-goals

- No metadata endpoint accepts bytes
- No readiness marker becomes document approval
- No provider share by object key
- No PHI before B2

Expected files or modules

- `app/documents/participant_service.py`
- `app/routes/case_documents.py`
- `app/templates/case/documents/`
- `tests/python/test_case_document_integration.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Participant document commands join requirement IDs to immutable content versions.
UI/product surface	Content controls appear only when the content capability and approved gate state are present.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Authorization is re-evaluated at intent, finalize, status and download.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Metadata endpoint rejects bytes
- owner/role/viewer denial
- quarantine/scan/replacement
- restart recovery
- multi-IP isolation
- feature-disabled/partial-merge failure
- mobile/no-JS

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P09-03, P09-04. Parallel opportunities: P09-01, P09-02

Objective acceptance criteria

- □ Connect metadata requirements to upload intents, quarantined/scanned content versions, replacement and authorized download without collapsing requirement state into object state.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	secure documents, FastAPI/Jinja, authorization, resilient workflows
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`infra`	AWS resources, IAM/KMS/network/deployment/backup/audit contracts

Merge order inside this package: `P09-01` → `P09-02` → `P09-03` → `P09-04` → `P09-06` → `P09-05`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Unauthorized object access; Oversize/type/spoof; Scan failure/quarantine; Partial upload/task loss; Snapshot immutability/revocation; Mobile documents	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-AUTH-03	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12	No longer blocks the reconciled access/workspace PR sequence.
D-DATA-01	Founder decision final; implementation evidence pending	Luke decides product/privacy outcome	Alan - implementation	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-COM-04	Founder decision final; implementation evidence pending	Luke decides commercial/product outcome	Alan - implementation	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02	P07-02 through P09-04 may proceed under the settled semantics.
D-DOC-01	Founder default final; Alan later; external validation before B1/B2	Luke sets policy; Alan decides implementation later	Alan - later before P09-03; privacy/legal - validation	P09-01, P09-03, P09-06; Before P09-03	P09-01 ADR work may proceed; P09-03/P09-06 and real-content activation wait for Alan's delivery choice and evidence.
D-DOC-02	Founder constraints final; Alan later	Luke sets constraints; Alan decides later	Alan - later before P09-02	P09-01, P09-02, P09-03, P09-06; Before P09-02	P09-01 contract work may proceed; P09-02/P09-03/P09-06 wait for Alan's technical selection and evidence.
D-PROV-01	Founder decision final; Alan later	Luke decides product/privacy outcome; Alan implements later	Alan - later before P09-05	P09-05; Before P09-05	Synthetic provider-snapshot design may proceed; P09-05 completion and real handoff require Alan's evidence and later external review.
D-PHI-01	Founder beta scope final; externally gated	Luke decides beta scope; external evidence required	Alan - B1/B2 evidence; external reviewers - approval	P09-01, P09-02, P09-05, P09-06, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.

D-AUTH-03 - Identity, participation and access authorization contract

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Identity, participation, access grants, ownership and operation authorization remain separate; participation or lens alone never authorizes.
Founder constraints	Subject-derived authorization context; Application and RLS policy parity; Default-deny cross-resource access
What remains for Alan	Encode the settled contract in application services and RLS
Evidence required	Participation-without-grant and grant-without-capability denials; Application/RLS parity tests
Approval rule	No further founder decision unless the boundary changes.
External dependencies	None
Affected PRs / latest point	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12
Programme effect	No longer blocks the reconciled access/workspace PR sequence.
Independent recommendation	Identity binds principal to person; participation records domain relationship; grants carry resource-scoped capabilities; application and RLS consume a subject-derived context plus ownership and purpose. Participation/lens alone never authorizes.

D-DATA-01 - IP-GC field projection allowlist

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.
Founder constraints	Default-deny field-and-purpose allowlists; Source-labelled corrections; Named-purpose provider projection only
What remains for Alan	Define explicit field/purpose allowlists in code and projection contracts; Implement cross-role and provider negative tests
Evidence required	Cross-role denial matrix; Provider projection allowlist tests; Correction/source-label evidence
Approval rule	Luke approval is required for a broader sharing policy, not for compliant implementation detail.
External dependencies	None
Affected PRs / latest point	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02
Programme effect	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
Independent recommendation	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.

D-COM-04 - Enrollment funding and activation evidence

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use two-step commitment: settled $750 plus current accepted terms proves commercial enrollment, but payment alone never promotes. Exactly one enrollment-pending case follows explicit IP promotion and authenticated GC consent. The $1,630 activation tranche is earned only after defined IP/GC onboarding and screening coordination has begun; it never implies medical clearance.
Founder constraints	Redirects/webhooks never promote; Typed promotion event only; Source-labelled facts and explicit evidence checklist
What remains for Alan	Define typed events, activation checklist, idempotency and audit mechanics
Evidence required	Duplicate, replay, partial-failure, promotion and activation evidence tests
Approval rule	Luke approves any change to commercial enrollment or activation semantics.
External dependencies	None
Affected PRs / latest point	P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02
Programme effect	P07-02 through P09-04 may proceed under the settled semantics.
Independent recommendation	Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event.

D-DOC-01 - Document retention, deletion, replacement and download mode

Field	Recorded decision or requirement
Decision state	Final founder default; externally dependent activation
Founder outcome	Retain clean active-case documents while active plus two years after close; signed consents, payment/access audits and deletion tombstones for seven years; failed/quarantine orphans seven days; superseded content 90 days unless required; honor valid deletion within 30 days subject to legal hold. Downloads require fresh authorization and links expire within five minutes.
Founder constraints	Never public; Never withheld for non-payment; Legal hold and deletion remain auditable
What remains for Alan	Choose proxy versus presigned delivery by class; Implement hold, deletion, tombstone, retry and authorization behavior
Evidence required	Fresh-authorization and five-minute-link tests; Retention/hold/deletion exercise; Privacy/legal approval before real content
Approval rule	Luke accepts live policy after privacy/legal review.
External dependencies	Privacy/legal validation before B1/B2 real-content use
Affected PRs / latest point	P09-01, P09-03, P09-06; Before P09-03
Programme effect	P09-01 ADR work may proceed; P09-03/P09-06 and real-content activation wait for Alan's delivery choice and evidence.
Independent recommendation	Purpose/role authorization independent of key; short-lived downloads; deletion tombstone/audit; legal/privacy review.

D-DOC-02 - Document limits and malware/quarantine provider

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Allow PDF, JPEG, PNG and non-macro DOCX up to 50 MiB. Reject spoofed, macro-enabled and archive-like input; quarantine until clean; treat password-protected, skipped, error and timeout results as unavailable.
Founder constraints	Normal scan target no more than two minutes; Transient retries no more than 15 minutes; $50/month beta scanning ceiling with alerts; Synthetic only at A2 and no PHI before B2
What remains for Alan	Choose the AWS-native scanner, queue, retry, timeout, recovery and cost-control mechanisms
Evidence required	Type/sniffing/spoof tests; Quarantine, error, timeout and retry tests; Cost estimate and alerts
Approval rule	Luke approval is required if the solution exceeds the cost or safety constraints.
External dependencies	None
Affected PRs / latest point	P09-01, P09-02, P09-03, P09-06; Before P09-02
Programme effect	P09-01 contract work may proceed; P09-02/P09-03/P09-06 wait for Alan's technical selection and evidence.
Independent recommendation	Narrow allowlist, checksum and content sniffing; unavailable scanner remains quarantined.

D-PROV-01 - Provider handoff purposes and field allowlists

Field	Recorded decision or requirement
Decision state	Final founder policy
Founder outcome	Provider handoff is a purpose-bound immutable snapshot for one verified recipient and named purpose, with participant preview/consent, clean documents and appropriately verified source-labelled facts. Unopened access expires after 14 days and can be revoked before access; downloaded copies cannot be recalled.
Founder constraints	No ambient provider case access in MVP; Audit every issue/open/download/revoke/expire event; Reissue after expiry
What remains for Alan	Choose recipient verification, snapshot checksum, expiry, revocation and audit mechanisms
Evidence required	Wrong-recipient, expiry, revoke-before-access, reissue and checksum tests
Approval rule	Luke approves broader recipient or ambient-access scope.
External dependencies	Privacy/provider review before real handoff
Affected PRs / latest point	P09-05; Before P09-05
Programme effect	Synthetic provider-snapshot design may proceed; P09-05 completion and real handoff require Alan's evidence and later external review.
Independent recommendation	Class 1/2 excluded; class 4 only if GC explicitly selects for recipient/purpose; class 5 evidence linked.

D-PHI-01 - PII/PHI boundary, vendors and control set

Field	Recorded decision or requirement
Decision state	Final founder scope; externally dependent activation
Founder outcome	B1 is an invitation-only US beta of at most five IP/GC pairs using names, verified account/contact data, relationship/outreach/logistics, consent versions, nonclinical progress, feedback and processor references only. Medical, reproductive-health, psychological, insurance-health and clinical-clearance data are prohibited; no real document bytes or provider packets; AI receives de-identified nonclinical context only.
Founder constraints	RLS/non-owner runtime and cross-tenant negative evidence; Access review plus incident and restore exercises; No PHI before B2
What remains for Alan	Implement inventory, cohort controls, RLS, cross-tenant denials, de-identification and prohibited-data tests
Evidence required	Five-pair cohort configuration; RLS/cross-tenant/de-identification/prohibited-data tests; External privacy/legal approval
Approval rule	Luke accepts B1 only after required evidence and external privacy/legal advice; B2 requires qualified external approval.
External dependencies	External privacy/legal approval before B1; Qualified external security/privacy/compliance/legal approval before B2
Affected PRs / latest point	P09-01, P09-02, P09-05, P09-06, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI
Programme effect	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.
Independent recommendation	B1 remains PII-only and content controls stay synthetic; B2 requires formal external security/privacy/compliance evidence before any PHI content path is enabled.

Implementation risk	Owner	Treatment
PHI exposure	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Malware	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Orphaned objects	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Stale provider packet	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Dirty branch leakage	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P09-04	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P09-04	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P09-04	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
feat/byos-journey-wave-2	5b63e3a7c364	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.	P09-01, P09-04, P09-06	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.	Paused exact-head evidence; no direct merge
feature/provider-exchange-layer-v0	84a5a84	substantial uncommitted provider handoff/ops/document work	P09-01/P09-03/P09-05 after provenance/security review	Selective port from protected worktree	Do not stage, reset, stash, switch, commit, clean or overwrite.
main	current local	substantial uncommitted operator shell/design-system work	Not part of participant DS; review only if P09-05 operator UI needs approved internal recipes	Selective port from protected worktree	Preserve repo boundary and dirty worktree.

Package source inventory:

- Protected provider-exchange branch by selective port after review
- Current document metadata scaffold
- PR #122 case shell UX

9. Ownership recommendation

Alan owns implementation; Luke approves retention, provider purpose, and PHI boundary. External security/privacy and provider workflow review required.

P09-01

PR	P09-01
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Exception	None

P09-02

PR	P09-02
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Exception	None

P09-03

PR	P09-03
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P09-04

PR	P09-04
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P09-05

PR	P09-05
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-PROV-01: Alan - later provider-packet mechanism; privacy/provider validation; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Exception	None

P09-06

PR	P09-06
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-DOC-01: Alan - later download/recovery mechanism; privacy/legal validation; D-DOC-02: Alan - later scanner/quarantine mechanism; D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 10 — Intelligence, memory, proposals, and Ask safety

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP10
Plan file	`10-intelligence-memory-proposals-and-ask-safety.md`
Proposed PRs	P10-01, P10-02, P10-03, P10-04
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Private memory, shared case truth, Learn context, and AI proposals have partial contracts but current app code can place IP guide notes into shared memory and provider projections lack provenance enforcement.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Versioned scoped context envelopes, separate private/shared memory revisions, evidence-bound proposal approval, grounded Ask behavior, and cross-service safety evaluation/observability.

Explicit non-goals.

- No LLM writes to case truth
- No medical/legal advice
- No silent fallback success
- No final Learn corpus authoring

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-014	verified	app / PR #124@84a5a84	app/storage/node0.py:42-200,509-600; tests/python/test_node0.py:444-490	IP/co-IP own candidate reads; GC onboarding writes separate shared/private facts and does not revise the IP assertion. The GC cannot see or correct the IP candidate record.
E-015	verified	app / PR #122@e853821	app/storage/facts.py:19-69,107-127	Facts have broad provenance and visibility values but no subject-versus-author role, verification state, derived-from, or supersedes link.

ID	Class	Repository / ref	File, construct or location	Finding
E-016	verified	app / PR #122@e853821	app/storage/proposals.py:117-142	Generic proposal approval permits any case member and can pass caller-selected provenance to create_fact; a dedicated evidence/role rail is required before reviewed/provider-confirmed truth.
E-017	verified	app / PR #124@84a5a84	app/routes/node0.py:686-719	IP Conversation Guide notes are saved as shared memory and sent to intelligence, creating a future cross-role leakage risk.
E-019	verified protected work	app / dirty feature/provider-exchange-layer-v0@84a5a84	app/provider_handoffs.py:109-133 and uncommitted handoff modules	The protected provider work is not a PR. Its broad GC-owner fact selection can include private/provisional data, so only selective port after provenance/purpose review is safe.
E-020	verified	intelligence / origin/main@56f6c4b	src/natalyx_intelligence/case_context.py:8-147; docs/agents/case-context.md	The service returns advisory-only proposals and says durable writes are app-owned; this is a useful boundary but the surrounding context provenance remains incomplete.
E-021	verified	intelligence / origin/main@56f6c4b	src/natalyx_intelligence/ask_natalyx.py:34-42,365-378	Ask accepts known_intake_state without an authorship-class contract and serializes it into model context.
E-028	verified review	app / PR #124 comment 4948149089	Consolidated review — findings for the WP00 extraction ledger	Alan's completed review classifies #124 as high-value extraction material rather than a merge unit and identifies transaction, privacy, scoring, access, validation, invariant, exposure, retention, performance, CSS and CI findings.

3. Architecture and contracts

- The app owns case truth, authorization, PHI policy, and durable approval; intelligence returns advisory proposals only.
- Every request declares general Learn, private-user, approved-shared-case, or provider-purpose scope; contexts are not merged.
- Failures degrade explicitly with source/boundary notices and no fabricated case answer.

Contract area	Package rule
Ownership boundary	`app`, `intelligence`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Versioned context/advisory contract with strict validation and safe compatibility window. Responses carry source, scope, policy, safety and degraded-mode metadata.
Storage	Revision/supersedes/provenance/evidence fields and the private-note remediation under versioned migrations.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Data authorship/provenance rules

- Every assertion/revision records subject_ref, semantic_key, author_subject/person/role, authorship_class, provenance, verification_status, visibility, purpose, source_ref, derived_from, supersedes_id, promotion_event_id, policy version and immutable timestamps.
- Class 1 defaults IP-private; class 2 defaults IP-private/internal and visibly says reported by the intended parent, not confirmed by the GC.
- Class 3 is GC-authored and shared only per field/purpose; class 4 is GC-only private; class 5 requires an evidence-linked role-gated reviewer/provider rail.
- GC confirmation appends a class-3 revision referencing the class-2 assertion. Relationship answers can coexist; disagreement is not an overwrite.
- Unselected candidates, raw health assertions, ranking scores and private IP notes do not promote. Derived values inherit the weakest source status.
- Copying, promotion, generic proposal approval, IP assertion or AI inference can never create Natalyx-reviewed, provider-confirmed, medically cleared, legally cleared or insurance-verified truth.

4. Explicit PR sequence

1

Order	1
ID	P10-01
Title	Version the scoped intelligence context envelope
Repository	intelligence
Dependencies	P01-01, P04-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01, P04-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval

2

Order	2
ID	P10-02
Title	Enforce private/shared memory and evidence-bound proposal approval
Repository	app
Dependencies	P04-02, P10-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P10-01; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval

3

Order	3
ID	P10-03
Title	Harden Ask and Learn grounding and degraded-mode behavior
Repository	intelligence
Dependencies	P10-01, P11-02
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-01, P11-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval

4

Order	4
ID	P10-04
Title	Add intelligence safety evaluations and cross-service observability
Repository	intelligence
Dependencies	P10-02, P10-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-02, P10-03; Gate A0 active and verified; Luke review/final founder approval

P10-01 — Version the scoped intelligence context envelope

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P10-01
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01, P04-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Dependencies	P01-01, P04-02
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current case_context advisory stub

Objective

Define and validate versioned request/response envelopes that keep general Learn, private-user memory, approved shared-case facts, and provider-purpose context distinct.

Scope

- Explicit scope and audience
- Source/provenance/verification metadata
- Policy/model/version and redacted trace metadata

Explicit non-goals

- No durable app write
- No merged contexts

Expected files or modules

- `src/natalyx_intelligence/contracts/context.py`
- `src/natalyx_intelligence/case_context.py`
- `docs/agents/case-context.md`
- `tests/test_context_contract.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Versioned context/advisory contract with strict validation and safe compatibility window.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Cross-scope rejection
- unknown version
- private/shared fixtures
- redaction
- service auth

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P01-01, P04-02. Parallel opportunities: P10-04

Objective acceptance criteria

- □ Define and validate versioned request/response envelopes that keep general Learn, private-user memory, approved shared-case facts, and provider-purpose context distinct.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	FastAPI/Pydantic, AI safety, privacy
Higher-capability review	Mandatory before merge

P10-02 — Enforce private/shared memory and evidence-bound proposal approval

Field	Plan
Repository / base	`app` (`main`)
Project item	P10-02
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P04-02, P10-01; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Dependencies	P04-02, P10-01
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current facts/memory/proposals scaffolds, GC authorship audit

Objective

Store attributable private/shared memory revisions and require authorized evidence review before an intelligence proposal can affect case truth or provider material.

Scope

- Private and shared memory ledgers
- Affected-row inventory for source=node0_conversation_guide_note
- Immediate shared/intelligence exclusion plus idempotent person-owner quarantine/private remediation when rows exist
- Proposal preview/approve/reject
- Evidence/source and reviewer authority
- Audit/outbox

Explicit non-goals

- No automatic AI approval
- No guide notes default shared

Expected files or modules

- `app/case_intelligence.py`
- `app/storage/memory.py`
- `app/storage/proposals.py`
- `app/storage/facts.py`
- `tests/python/test_case_intelligence.py`

Change map

Area	Planned change
Schema/model	Revision/supersedes/provenance/evidence fields and the private-note remediation under versioned migrations.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Role and purpose authorization on every projection/write; ambiguous legacy ownership quarantines rather than broadens access.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Zero-row inventory evidence
- affected rows disappear from shared/Ask reads before remediation
- idempotent owner-safe migration
- ambiguous-owner quarantine
- no note content in logs
- private-to-shared denial
- wrong-role approval
- AI provenance remains proposal
- concurrent review
- provider projection

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P04-02, P10-01. Parallel opportunities: P10-03

Objective acceptance criteria

- □ Store attributable private/shared memory revisions and require authorized evidence review before an intelligence proposal can affect case truth or provider material.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.

- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	AI safety, authorization, data provenance
Higher-capability review	Mandatory before merge

P10-03 — Harden Ask and Learn grounding and degraded-mode behavior

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P10-03
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-01, P11-02; technical decisions/controls D-AI-01; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Dependencies	P10-01, P11-02
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current ask_natalyx.py and PR #124 Ask integration

Objective

Answer from the declared scope with approved sources/citations and explicit boundary/failure notices; never fabricate case context or silently fall across scope.

Scope

- Approved resource resolution
- Scope-specific prompt/context
- Timeout/unavailable/empty-source behavior
- Citation validation

Explicit non-goals

- No medical/legal advice
- No unapproved corpus fallback

Expected files or modules

- `src/natalyx_intelligence/ask_natalyx.py`
- `src/natalyx_intelligence/learn_resources.py`
- `tests/test_ask_natalyx.py`
- `docs/agents/ask-natalyx.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Responses carry source, scope, policy, safety and degraded-mode metadata.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Empty/unapproved corpus
- service/provider timeout
- case question without case scope
- citation mismatch
- safety escalation

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P10-01, P11-02. Parallel opportunities: P10-02

Objective acceptance criteria

- □ Answer from the declared scope with approved sources/citations and explicit boundary/failure notices; never fabricate case context or silently fall across scope.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	LLM orchestration, retrieval/citations, safety evaluation
Higher-capability review	Mandatory before merge

P10-04 — Add intelligence safety evaluations and cross-service observability

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P10-04
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-02, P10-03; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P10-02, P10-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current intelligence auth/telemetry tests

Objective

Add deterministic safety/grounding evaluations, correlation IDs, latency/failure metrics, and redacted traces across app-intelligence calls.

Scope

- Leakage/hallucination/fallback evals
- Correlation and contract version
- Safe logs and alarms

Explicit non-goals

- No prompt or participant payload in logs
- No PHI readiness claim

Expected files or modules

- `tests/evals/`
- `src/natalyx_intelligence/telemetry.py`
- `docs/runbooks/intelligence.md`
- `app/intelligence_auth.py (consumer follow-up PR if needed)`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- PII/PHI redaction fixtures
- correlation continuity
- failure classification
- model fallback
- eval threshold

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Metrics and redacted structured events; app-side consumer change must remain a separate coordinated PR if repository boundaries require it.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P10-02, P10-03. Parallel opportunities: P10-01

Objective acceptance criteria

- □ Add deterministic safety/grounding evaluations, correlation IDs, latency/failure metrics, and redacted traces across app-intelligence calls.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	AI evaluation, observability, privacy
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`intelligence`	advisory inference, scoped response/citation contract; no case truth

Merge order inside this package: `P10-01` → `P10-02` → `P10-03` → `P10-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Scope isolation; Proposal approval/rejection; Citation/source validity; Unavailable service; Prompt/context leakage; Trace redaction	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-DATA-01	Founder decision final; implementation evidence pending	Luke decides product/privacy outcome	Alan - implementation	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
D-AI-01	Founder scope/cost final; Alan later; PHI externally gated	Luke sets scope and cost; Alan decides implementation later	Alan - later before P10-01	P10-01, P10-02, P10-03, P12-01; Before P10-01	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.

D-DATA-01 - IP-GC field projection allowlist

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Use need-to-know sharing. Shared journey status and agreed relationship/outreach/logistics may cross roles; private IP notes, unconfirmed assertions and GC medical, reproductive, insurance, identity or wellbeing answers do not cross automatically. Provider packets require preview, consent and appropriate verification.
Founder constraints	Default-deny field-and-purpose allowlists; Source-labelled corrections; Named-purpose provider projection only
What remains for Alan	Define explicit field/purpose allowlists in code and projection contracts; Implement cross-role and provider negative tests
Evidence required	Cross-role denial matrix; Provider projection allowlist tests; Correction/source-label evidence
Approval rule	Luke approval is required for a broader sharing policy, not for compliant implementation detail.
External dependencies	None
Affected PRs / latest point	P04-02, P06-02, P06-03, P07-03, P07-04, P09-05, P10-01, P10-02; Before P04-02
Programme effect	Founder ambiguity is removed across WP04, WP06, WP07, WP09 and WP10; A1 still requires the technical denial evidence.
Independent recommendation	Default deny; expose only explicit purpose-limited projections. Class 1/2 and class 4 never automatically cross roles.

D-AI-01 - Intelligence context and approval authority matrix

Field	Recorded decision or requirement
Decision state	Final founder policy; PHI externally dependent
Founder outcome	AI is bounded and advisory: role-scoped education, retrieval, summaries, question drafts, next steps and provider preparation using approved Learn and authorized source-labelled context. It never diagnoses, ranks, approves, clears or writes durable truth; authorized human approval precedes memory, case or provider material.
Founder constraints	Deterministic or curated fallback; $10 per active case/month and $100 total beta/month ceilings with alerts and graceful fallback; No PHI before B2
What remains for Alan	Choose Bedrock model, prompts, retrieval, schemas, fallback, human-review rails and cost controls
Evidence required	Scope/authority negative tests; Fallback and malformed-response tests; Safety and cost evidence
Approval rule	Luke approves broader AI authority, higher cost or any PHI use; B2 external approval is mandatory for PHI.
External dependencies	B2 external security/privacy/compliance approval before PHI
Affected PRs / latest point	P10-01, P10-02, P10-03, P12-01; Before P10-01
Programme effect	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.
Independent recommendation	Generic case members cannot create reviewed/provider truth; intelligence remains advisory and scope explicit.

Implementation risk	Owner	Treatment
Private-to-shared leakage	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

Implementation risk	Owner	Treatment
Hallucinated truth	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Sensitive logs	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Contract version drift	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P10-01	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P10-02	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P10-02	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
main (behind origin/main)	9d824e8	dirty Learn registry/corpus plus uv.lock	P11-02/P10-03 as appropriate	Selective port from protected worktree	Do not pull/reset/stash; extract in isolated worktree later.

Package source inventory:

- Intelligence case_context advisory stub
- App proposal/fact/memory scaffolds
- Protected intelligence Learn corpus changes

9. Ownership recommendation

Alan implements service/app contracts; Luke approves truth/privacy policy. Higher-capability, security/privacy, and clinical/legal safety review required.

P10-01

PR	P10-01
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P10-02

PR	P10-02
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P10-03

PR	P10-03
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P10-04

PR	P10-04
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 11 — Learn release and design-system contract

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP11
Plan file	`11-learn-release-and-design-system-contract.md`
Proposed PRs	P11-01, P11-02, P11-03, P11-04

Plan	Value
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Learn artifact generation and catalog work is dirty/unreleased, while participant surfaces need governed recipes without making aesthetic purity a universal release gate.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Validated provenance-bearing Learn artifacts, approved-only serving, release-blocking DS accessibility/security/interaction contract, and app adoption of stable recipes/source cards.

Explicit non-goals.

- No invented final content
- No universal zero-hex rule
- No internal/operator DS merge
- No full legacy redesign

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-022	verified protected work	learn / dirty feature/learn-resource-rendering-contract@9c9a9c2	builder/validator/schema/templates/preview additions	A substantial artifact/rendering contract exists only as protected uncommitted work; no open Learn PR contains it.
E-023	verified protected work	intelligence / dirty main@9d824e8, behind origin/main	src/natalyx_intelligence/learn_resources.py; starting_natalyx_journey_v1.md	Catalog/resource edits are uncommitted and must not be overwritten or treated as released.
E-024	verified	design-system / main@43dab986; PR #7@cedfcc4; PR #8@4656426	components, tokens, Jinja attrs/parity guidance	Participant DS recipes and parity guidance exist; local brand assets are also dirty. Internal/operator DS is a separate repository and boundary.
E-029	verified preserved branch	app / feat/byos-journey-wave-2@5b63e3a7	exact three-commit stack on e853821	Wave 2 contains phase-1 cluster pages, metadata-only request/review, case anchoring and role/owner fixes. Controlled local evidence passed 535 tests with 9 skipped and 36/36 focused tests, but no branch checks, PostgreSQL, browser/accessibility or visual run exists.

3. Architecture and contracts

- Learn resources own authored artifacts/provenance; intelligence owns approved catalog delivery; app owns authorized presentation and context.
- Security, accessibility, responsive layout, and interaction defects can block release; token/style purity alone is backlog unless tied to concrete risk.
- Participant and internal/operator design systems remain separate.

Contract area	Package rule
Ownership boundary	`app`, `design-system`, `intelligence`, `learn`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.

Contract area	Package rule
API/service	Versioned artifact manifest consumed by intelligence. Catalog response includes manifest/version/source/approval fields. Read-only catalog contract with fail-closed approval checks.
Storage	No new storage; preserve current state read-only for this package.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Design-system release rubric

Class	Items
Release-blocking	XSS/escaping or unsafe raw slots; keyboard/focus/screen-reader failures; unusable responsive layout; misleading or nonfunctional interaction; motion without reduced-motion safety; contrast or error-state defects that block task completion
Desirable	token/recipe consistency; visual parity; removal of duplicate spacing/color literals; component naming/adoption
Deferred	zero raw hex values; zero inline geometry styles; full legacy strangler cleanup; broad participant redesign; internal/operator DS convergence

Style purity blocks a release only when tied to a concrete security, accessibility, interaction or material maintainability risk.

4. Explicit PR sequence

1

Order	1
ID	P11-01
Title	Publish validated Learn artifact and provenance contract
Repository	learn
Dependencies	P00-01
Gate	A2
Local preparation	Eligible for isolated, uncommitted, synthetic/no-invented-content preparation only
Merge readiness	Blocked pending PR dependencies P00-01; Gate A0 active and verified; Alan review

2

Order	2
ID	P11-02
Title	Serve approved Learn catalog versions from intelligence
Repository	intelligence
Dependencies	P11-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P11-01; Gate A0 active and verified; Alan review

3

Order	3
ID	P11-03
Title	Release participant design-system accessibility and interaction contract
Repository	design-system
Dependencies	P00-01
Gate	A1
Local preparation	Eligible for isolated, uncommitted, synthetic/no-invented-content preparation only
Merge readiness	Blocked pending PR dependencies P00-01; Gate A0 active and verified; Alan review

4

Order	4
ID	P11-04
Title	Adopt approved Learn and source-card recipes in app
Repository	app
Dependencies	P11-02, P11-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P11-02, P11-03; Gate A0 active and verified; Alan review

P11-01 — Publish validated Learn artifact and provenance contract

Field	Plan
Repository / base	`learn` (`main`)
Project item	P11-01
Gate / size	A2 / Large
Local preparation status	Eligible for isolated, uncommitted, synthetic/no-invented-content preparation only
Merge readiness	Blocked pending PR dependencies P00-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P00-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Protected feature/learn-resource-rendering-contract, feature/node0-beta-learn-resources

Objective

Turn the protected rendering-contract work into a small releaseable artifact schema/builder/validator with source provenance and no invented resources.

Scope

- Markdown/YAML source schema
- Article/PDF/context-packet artifacts
- Source/revision/audience/approval metadata
- Deterministic build

Explicit non-goals

- No final content beyond separately approved resources
- No app UI

Expected files or modules

- `schemas/`
- `src/natalyx_learn_resources/builder.py`
- `src/natalyx_learn_resources/validator.py`

- `tests/test_builder.py`
- `tests/test_validator.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Versioned artifact manifest consumed by intelligence.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Provenance and release procedure.

Tests and validation

Objective tests:

- Schema invalid cases
- deterministic output
- missing provenance
- unsafe links/HTML
- artifact checksum

Local validation:

- `uv run pytest`
- Run the builder/validator against approved fixture sources and compare deterministic checksums.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P00-01. Parallel opportunities: P11-03, P11-04

Objective acceptance criteria

- □ Turn the protected rendering-contract work into a small releaseable artifact schema/builder/validator with source provenance and no invented resources.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	content pipelines, schema validation, document generation
Higher-capability review	Risk-proportionate; not mandatory by programme default

P11-02 — Serve approved Learn catalog versions from intelligence

Field	Plan
Repository / base	`intelligence` (`main`)
Project item	P11-02
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P11-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P11-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Intelligence issue #14, protected dirty learn_resources.py

Objective

Load a pinned Learn artifact manifest, filter to approved audience/scope, and expose explicit empty/degraded behavior without hard-coded duplicate catalog truth.

Scope

- Manifest pin/checksum
- Approval/audience/cluster filters
- Source metadata
- Atomic catalog reload

Explicit non-goals

- No unapproved fallback catalog
- No app rendering

Expected files or modules

- `src/natalyx_intelligence/learn_resources.py`
- `src/natalyx_intelligence/catalog.py`
- `tests/pathfinder/test_corpus_loading.py`
- `docs/agents/learn-resources.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Catalog response includes manifest/version/source/approval fields.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Missing/bad manifest
- unapproved exclusion
- audience filter
- atomic reload
- stale version

Local validation:

- `uv run pytest`
- Run focused contract/migration/safety tests with redacted fixtures.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P11-01. Parallel opportunities: P11-03

Objective acceptance criteria

- □ Load a pinned Learn artifact manifest, filter to approved audience/scope, and expose explicit empty/degraded behavior without hard-coded duplicate catalog truth.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	content serving, FastAPI, provenance
Higher-capability review	Risk-proportionate; not mandatory by programme default

P11-03 — Release participant design-system accessibility and interaction contract

Field	Plan
Repository / base	`design-system` (`main`)
Project item	P11-03
Gate / size	A1 / Medium
Local preparation status	Eligible for isolated, uncommitted, synthetic/no-invented-content preparation only
Merge readiness	Blocked pending PR dependencies P00-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P00-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Design-system PRs #7 and #8, dirty local brand assets

Objective

Land the participant recipes/tokens/assets and parity guidance needed for secure escaping, accessible controls, responsive layouts, and stable interaction contracts.

Scope

- Jinja escaping/attrs contract
- 44px/keyboard/focus/reduced-motion requirements
- Responsive shell/recipe guidance
- Versioned vendoring

Explicit non-goals

- No universal raw-hex blocker
- No internal/operator DS
- No legacy app sweep

Expected files or modules

- `components/`
- `tokens/`
- `assets/`
- `docs/consumer-parity.md`
- `tests/`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	Participant design-system release only.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Template escaping
- raw-slot registry
- keyboard/focus
- responsive visual fixtures
- asset integrity

Local validation:

- `npm test`
- Run recipe escaping, accessibility and responsive fixture checks.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Release a versioned/pinned artifact or API contract first; consumers opt in explicitly and retain fail-closed/previous-version behavior during the compatibility window.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P00-01. Parallel opportunities: P11-01, P11-02

Objective acceptance criteria

- □ Land the participant recipes/tokens/assets and parity guidance needed for secure escaping, accessible controls, responsive layouts, and stable interaction contracts.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	design systems, accessibility, Jinja security
Higher-capability review	Risk-proportionate; not mandatory by programme default

P11-04 — Adopt approved Learn and source-card recipes in app

Field	Plan
Repository / base	`app` (`main`)
Project item	P11-04
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P11-02, P11-03; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P11-02, P11-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 Learn slot, PR #124 dependency note

Objective

Consume pinned Learn/catalog and participant-DS versions for source cards, reading views, suggestions, and honest empty/error states.

Scope

- Version-pinned catalog client
- Source/citation cards and reading shell
- No-resource/degraded states
- Cluster/context handoff

Explicit non-goals

- No final content invention
- No case truth write

Expected files or modules

- `app/learn_resources.py`
- `app/templates/learn_workspace.html`
- `app/templates/components/participant/`
- `tests/python/test_learn_workspace.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Read-only catalog contract with fail-closed approval checks.
UI/product surface	Governed participant recipes.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Approved resource
- missing/unapproved resource
- service timeout
- source links
- mobile/keyboard

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P11-02, P11-03. Parallel opportunities: P11-01

Objective acceptance criteria

- □ Consume pinned Learn/catalog and participant-DS versions for source cards, reading views, suggestions, and honest empty/error states.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`design-system`	participant recipes/tokens/accessibility/interaction contract
`intelligence`	advisory inference, scoped response/citation contract; no case truth
`learn`	authored Learn artifacts, provenance and release manifest

Merge order inside this package: `P11-01` → `P11-03` → `P11-02` → `P11-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Artifact schema/provenance; Approved-only catalog; Missing-resource empty state; Recipe escaping; Keyboard/reduced-motion/mobile	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect

Implementation risk	Owner	Treatment
Fake/unapproved content	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Cross-repo version drift	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Aesthetic review blocks recovery	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Internal DS leakage	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P11-03, P11-04	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#121	529d377e2337	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P11-03	Selective UI/test port, then supersede	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P11-03	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P11-03, P11-04	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
ds#7	cedfcc489b1f	Documents an escaping/attrs hardening follow-up relevant to participant recipe security.	P11-03	Retain for direct review/merge or fold into P11-03	Direct-reuse candidate; not yet merged.
ds#8	4656426e493e	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.	P11-03	Prefer direct review/merge if still correct	Direct-reuse candidate; local dirty asset delta separately protected.
feat/byos-journey-wave-2	5b63e3a7c364	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.	P11-03, P11-04	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.	Paused exact-head evidence; no direct merge
feature/learn-resource-rendering-contract	9c9a9c2	substantial uncommitted builder/schema/templates/resource work	P11-01	Selective port from protected worktree	Do not update/switch/clean; create a future isolated extraction worktree.
feature/node0-beta-learn-resources	e1fb2d6	uncommitted Node 0 Learn implementation	P11-01 after provenance/no-fake-content review	Selective port from protected worktree	Preserve until compared with rendering-contract branch.
main (behind origin/main)	9d824e8	dirty Learn registry/corpus plus uv.lock	P11-02/P10-03 as appropriate	Selective port from protected worktree	Do not pull/reset/stash; extract in isolated worktree later.
feature/ds-consumer-parity-guidance	4656426	dirty brand assets beyond PR #8	P11-03 after asset review	Selective port from protected worktree	Remote PR and local asset delta are separate evidence; do not overwrite.

Package source inventory:

- Dirty Learn rendering-contract branch
- Learn issue #3
- Intelligence issue #14 and dirty catalog work
- DS PRs #7/#8

9. Ownership recommendation

Luke is the proposed DRI and primary implementer across Learn, intelligence-catalog, participant-design-system and app-adoption PRs. Alan reviews technical contracts and failure behavior; accessibility and content/provenance review remain required.

P11-01

PR	P11-01
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P11-02

PR	P11-02
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P11-03

PR	P11-03
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P11-04

PR	P11-04
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.

- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 12 — Ask/Learn shell and mobile cutover

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP12
Plan file	`12-ask-learn-shell-and-mobile-cutover.md`
Proposed PRs	P12-01, P12-02, P12-03
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. Participant Ask/Learn entry points, case shell, legacy routes, and mobile interactions are not yet one coherent clickable journey with explicit context boundaries.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. A unified scope-labelled Ask/Learn shell, accessible mobile participant journey, and verified redirect-based legacy cutover with no broken or misleading controls.

Explicit non-goals.

- No final content authoring
- No staff/operator redesign
- No broad visual rebrand
- No removal before telemetry proves safe

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-006	verified	app / PR #122@e853821	PR comment Review round 2 addressed and changed workflow/tests	Updated PR #122 reports 499 tests plus return_to, TTL, CAS, self-invite, lens and Learn fail-closed fixes; green checks were observed. These are salvage candidates, not blanket merge approval.
E-012	verified	app / PR #122@e853821	app/storage/invitations.py and PR review-round comment	The updated case invitation work adds 14-day ceiling, CAS transitions, mismatch confirmation, self-invite refusal, return_to preservation and paired lens tests; it has no Node 0 workspace target.
E-024	verified	design-system / main@43dab986; PR #7@cedfcc4; PR #8@4656426	components, tokens, Jinja attrs/parity guidance	Participant DS recipes and parity guidance exist; local brand assets are also dirty. Internal/operator DS is a separate repository and boundary.

ID	Class	Repository / ref	File, construct or location	Finding
E-029	verified preserved branch	app / feat/byos-journey-wave-2 @5b63e3a7	exact three-commit stack on e853821	Wave 2 contains phase-1 cluster pages, metadata-only request/review, case anchoring and role/owner fixes. Controlled local evidence passed 535 tests with 9 skipped and 36/36 focused tests, but no branch checks, PostgreSQL, browser/accessibility or visual run exists.

3. Architecture and contracts

- The shell labels general, private, and shared-case context and never implies a source exists when it does not.
- Server owns state and permissions; client code enhances interaction only.
- Legacy routes redirect through allowlisted, telemetry-observed compatibility paths before deletion.

Contract area	Package rule
Ownership boundary	`app`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Pass versioned scoped envelopes only. Stable redirects with query sanitization.
Storage	No new storage; preserve current state read-only for this package.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

4. Explicit PR sequence

1

Order	1
ID	P12-01
Title	Unify Ask and Learn shell with explicit context scopes
Repository	app
Dependencies	P10-03, P11-04, P09-04
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-03, P11-04, P09-04; technical decisions/controls D-AI-01; Gate A0 active and verified; Alan review

2

Order	2
ID	P12-02
Title	Complete mobile participant interaction and accessibility cutover
Repository	app
Dependencies	P05-03, P06-03, P09-04, P09-06, P12-01
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P05-03, P06-03, P09-04, P09-06, P12-01; Gate A0 active and verified; Alan review

3

Order	3
ID	P12-03
Title	Retire legacy participant routes behind verified redirects
Repository	app
Dependencies	P12-02, P13-04
Gate	B1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P12-02, P13-04; Gate A0 active and verified; Alan review

P12-01 — Unify Ask and Learn shell with explicit context scopes

Field	Plan
Repository / base	`app` (`main`)
Project item	P12-01
Gate / size	A2 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P10-03, P11-04, P09-04; technical decisions/controls D-AI-01; Gate A0 active and verified; Alan review
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Dependencies	P10-03, P11-04, P09-04
Blocking decisions/checkpoints	D-AI-01
Source / salvage	Current Ask widget/Learn workspace, PR #124

Objective

Present general Learn, private personalization, and approved shared-case Ask as visibly distinct contexts with source cards and safe switching.

Scope

- Context scope badge/selector
- Suggested questions and citations
- Boundary and unavailable states
- Case-aware authorization

Explicit non-goals

- No context merge
- No case fact write

Expected files or modules

- `app/ask_natalyx.py`
- `app/templates/ask_natalyx*.html`
- `app/templates/learn_workspace.html`
- `app/static/js/learn_chat.js`
- `tests/python/test_ask_learn_shell.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Pass versioned scoped envelopes only.
UI/product surface	Unified but scope-explicit shell.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- General/private/shared views
- wrong-case denial
- logout/login
- empty/failure
- mobile/keyboard/screen-reader

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P10-03, P11-04, P09-04. Parallel opportunities: P12-03

Objective acceptance criteria

- □ Present general Learn, private personalization, and approved shared-case Ask as visibly distinct contexts with source cards and safe switching.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	privacy UX, FastAPI/Jinja, accessibility
Higher-capability review	Mandatory before merge

P12-02 — Complete mobile participant interaction and accessibility cutover

Field	Plan
Repository / base	`app` (`main`)
Project item	P12-02
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P05-03, P06-03, P09-04, P09-06, P12-01; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P05-03, P06-03, P09-04, P09-06, P12-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	PR #122 mobile journey, PR #124 Node 0, DS PR #8

Objective

Run a bounded participant-only mobile/accessibility pass across entry, auth, Node 0, GC claim, case, documents, Ask, and Learn using the released DS contract.

Scope

- Navigation/forms/cards/dialogs
- 320/375/768 widths
- Keyboard/focus/reduced motion
- Error/loading/empty states

Explicit non-goals

- No internal/operator surface
- No visual rebrand
- No zero-hex purity gate

Expected files or modules

- `app/templates/participant*/`
- `app/templates/node0/`
- `app/templates/case/`
- `app/static/participant*/`
- `tests/browser/participant_mobile.spec.*`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	Participant interaction defects and concrete DS adoption only.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- All required clickable journeys
- axe/manual keyboard
- reduced motion
- orientation/zoom
- slow dependency

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P05-03, P06-03, P09-04, P09-06, P12-01. Parallel opportunities: None inside this package beyond review/test preparation.

Objective acceptance criteria

- □ Run a bounded participant-only mobile/accessibility pass across entry, auth, Node 0, GC claim, case, documents, Ask, and Learn using the released DS contract.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	responsive UI, accessibility, browser testing
Higher-capability review	Risk-proportionate; not mandatory by programme default

P12-03 — Retire legacy participant routes behind verified redirects

Field	Plan
Repository / base	`app` (`main`)
Project item	P12-03
Gate / size	B1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P12-02, P13-04; Gate A0 active and verified; Alan review
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P12-02, P13-04
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current route inventory, PRs #112/#119 planning maps

Objective

Move legacy participant entry/case/Ask/Learn routes behind allowlisted redirects and telemetry, then remove only after a measured compatibility window.

Scope

- Route inventory and redirect map
- Bookmark/return_to compatibility
- Telemetry and rollback flag
- Later dead-code removal

Explicit non-goals

- No immediate destructive deletion
- No operator route change

Expected files or modules

- `app/routes/participant*.py`
- `app/main.py`
- `tests/python/route_inventory.json`
- `docs/runbooks/legacy-cutover.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Stable redirects with query sanitization.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Every legacy route
- query/open-redirect safety
- stale bookmarks
- rollback flag
- route parity

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate B1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Legacy hit counters and unknown-route alerts.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P12-02, P13-04. Parallel opportunities: P12-01

Objective acceptance criteria

- □ Move legacy participant entry/case/Ask/Learn routes behind allowlisted redirects and telemetry, then remove only after a measured compatibility window.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Luke
Primary implementer	Luke (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
PR-level ownership exception	None
Required expertise	Python/FastAPI, PostgreSQL, security-aware product engineering
Higher-capability review	Risk-proportionate; not mandatory by programme default

5. Cross-repository coordination

Repository	Authority after merge
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain

Merge order inside this package: `P12-01` → `P12-02` → `P12-03`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Clickable IP/GC flows; Context labels; No-JS/reduced-motion/keyboard; Mobile widths; Redirect and stale bookmark; Logout/login/resume	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-AI-01	Founder scope/cost final; Alan later; PHI externally gated	Luke sets scope and cost; Alan decides implementation later	Alan - later before P10-01	P10-01, P10-02, P10-03, P12-01; Before P10-01	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.

D-AI-01 - Intelligence context and approval authority matrix

Field	Recorded decision or requirement
Decision state	Final founder policy; PHI externally dependent
Founder outcome	AI is bounded and advisory: role-scoped education, retrieval, summaries, question drafts, next steps and provider preparation using approved Learn and authorized source-labelled context. It never diagnoses, ranks, approves, clears or writes durable truth; authorized human approval precedes memory, case or provider material.
Founder constraints	Deterministic or curated fallback; $10 per active case/month and $100 total beta/month ceilings with alerts and graceful fallback; No PHI before B2
What remains for Alan	Choose Bedrock model, prompts, retrieval, schemas, fallback, human-review rails and cost controls
Evidence required	Scope/authority negative tests; Fallback and malformed-response tests; Safety and cost evidence
Approval rule	Luke approves broader AI authority, higher cost or any PHI use; B2 external approval is mandatory for PHI.
External dependencies	B2 external security/privacy/compliance approval before PHI
Affected PRs / latest point	P10-01, P10-02, P10-03, P12-01; Before P10-01
Programme effect	P10/P12 synthetic planning may proceed; implementation completion waits for Alan's technical decision and PHI remains blocked through B2.
Independent recommendation	Generic case members cannot create reviewed/provider truth; intelligence remains advisory and scope explicit.

Implementation risk	Owner	Treatment
Context ambiguity	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Mobile-only dead ends	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Premature legacy removal	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Design drift	Luke	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
app#112	9605d88e31bd	Source-first 81-node/12-cluster surface map, participant-renderer boundaries, wave plan and cross-repo seams.	P12-03	Selective port, then supersede after extraction	Catalogued; content remains to be reconciled into replacement PR acceptance tests.
app#119	a8cd2f428535	Concise product-leverage framing and boundaries for the MVP flow.	P12-03	Retain as source or merge separately after plan decision	Catalogued; reconcile terminology into approved programme.
app#121	529d377e2337	Primary Node 0 participant model/UI, PDF/guide presentation and focused surface tests.	P12-02	Selective UI/test port, then supersede	Unextracted; #124 may contain later copies but equivalence must be proved by file/commit comparison.
app#122	e85382150853	Case projection/lens surfaces, safe return_to, invitation TTL/CAS/self-member/one-GC constraints, transactional membership, fail-closed Learn, SQL-limited timeline, responsive/a11y tests and Python CI workflow.	P12-02	Selective-port behavior and tests; supersede as programme architecture	Complete at e853821: authoritative behavior/tests and selectively portable implementation are classified; superseded architecture and evidence-only claims are named.
app#124	84a5a847fea7	Consolidated Node 0 questionnaire/guide/surfaces, GC onboarding, participant case shell, invitation delegation, DS assets and large focused test suites as extraction evidence.	P12-02	Fully supersede as a merge unit; preserve selective concepts, copy, assets and tests as evidence	Complete at 84a5a84: all consolidated-review findings are dispositioned and assigned; no mergeable/cherry-pick unit is approved.
ds#8	4656426e493e	Consumer parity and bounded app-owned layout guidance for Node 0/case/mobile adoption.	P12-02	Prefer direct review/merge if still correct	Direct-reuse candidate; local dirty asset delta separately protected.
feat/byos-journey-wave-2	5b63e3a7c364	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.	P12-02	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.	Paused exact-head evidence; no direct merge

Package source inventory:

- PR #122 responsive journey and lens behavior
- PR #124 Node 0 surfaces
- DS PR #8 parity guidance

9. Ownership recommendation

Luke is the proposed DRI and primary implementer for the participant shell and cutover. Alan reviews technical integration, authorization and failure behavior; accessibility review remains required where named.

P12-01

PR	P12-01
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	D-AI-01: Alan - later model/retrieval/schema/fallback mechanism
Reviewer	Alan
Final approver	Luke
Exception	None

P12-02

PR	P12-02
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

P12-03

PR	P12-03
WP DRI	Luke
Primary implementer	Luke (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Alan
Final approver	Luke
Exception	None

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.

Work package 13 — Deployment, cloud gates, and PHI readiness

Planning-only execution contract. The corrected work-package split remains proposed pending Alan's review. Local preparation and merge readiness are separate; the benched Project catalogue is not resynced. Project reference: MVP Product Flow Reshape.

Plan	Value
Work package	WP13
Plan file	`13-deployment-cloud-gates-and-phi-readiness.md`
Proposed PRs	P13-01, P13-02, P13-03, P13-04, P13-05, P13-06, P13-07
Inspection snapshot	2026-07-12 19:30 America/New_York

1. Purpose and outcome

Problem. The dev endpoint is presently 503, deployments do not prove application readiness, and test/deploy success cannot establish real-user or PHI readiness.

Why this package exists. It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

Completed outcome. Known-revision diagnostics, guarded rollback/evidence capture, automated synthetic A1/A2 journeys, and explicit B1/B2 operational/privacy/PHI go/no-go packets.

Explicit non-goals.

- No deployment in this planning task
- No real user before B1
- No PHI before B2
- No compliance claim from green CI

2. Verified current-state evidence

Classification is explicit: verified is observed code/PR/live behavior; verified limitation records an evidence gap; design authority is an approved planning source; design decision is proposed future behavior. None is silently treated as another.

ID	Class	Repository / ref	File, construct or location	Finding
E-001	verified	live dev / 2026-07-12 19:30 America/New_York	https://dev.app.natalyx.health/ and /health	Both requests returned HTTP 503 from awselb/2.0; the deployed participant app is not presently usable.
E-002	verified limitation	AWS dev / 2026-07-12 19:30 America/New_York	aws sts get-caller-identity --profile natalyx-dev	Local AWS SSO was expired, so current ECS desired/running counts and task revision were not independently verified. Gate A0 must refresh access and capture them.
E-007	verified	app / feature/byos-node0-consolidation@84a5a84	app/storage/schema.py:1-33,353-880	Schema ownership is a monolithic init_db/ensure_column routine with inline CREATE TABLE/ALTER TABLE and no ordered migration/version/checksum ledger.
E-008	verified	app / origin/main@fdf2bb4	app/main.py:295-313	Cloud startup validates auth/storage then calls storage.init_db in-process; readiness is not separated from migration execution.
E-026	verified	app / origin/main@fdf2bb4	app/main.py:145-148	/healthz returns only status ok when the process can serve; it does not report revision, database, migration, intelligence, storage, or readiness.
E-027	verified	app/infra / origin/main@fdf2bb4 / infra main@fec515f	app .github/workflows/aws-app-deploy.yml:60-190; infra CDK workflow	Deployment reads infra-owned SSM contracts and guards billable ECS updates, but successful image build/contract smoke does not prove a healthy service revision.
E-028	verified review	app / PR #124 comment 4948149089	Consolidated review — findings for the WP00 extraction ledger	Alan's completed review classifies #124 as high-value extraction material rather than a merge unit and identifies transaction, privacy, scoring, access, validation, invariant, exposure, retention, performance, CSS and CI findings.
E-030	independently reproduced defect	app / feat/byos-journey-wave-2@5b63e3a7	app/storage/schema.py orphan cleanup; document review CAS/work-item payload	A second init_db deletes canonical case-anchored reviewer work because participant_id is empty. Separately, a stale v1 work item can approve v2 after same-row resubmission because it is not bound to an immutable submission revision.
E-031	verified	app / feat/byos-journey-wave-2@5b63e3a7	case_members.py, case_documents.py, storage/documents.py, routes/agency.py	Membership roles still conflate participation, lens and access; document/reviewer loaders are broad or global; participant role/lens can be caller-supplied; operator actions lack a canonical organization-scoped staff capability; no RLS/non-owner runtime enforces denial.

3. Architecture and contracts

- Health, readiness, revision, dependency, migration, and safe configuration signals are distinct.
- Every cloud gate has prerequisites, evidence, owner, verifier, and failure/rollback response.
- B2 requires formal security/privacy/compliance evidence and human go/no-go; it is not a software test result.

Contract area	Package rule
Ownership boundary	`.github`, `app`, `infra`; repository-specific PRs remain separate.
Entities/state	No new runtime entity; evidence/coordination only.
Authorization	Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority.
API/service	Health/readiness endpoints with stable machine-readable reasons.
Storage	RLS policies and least-privilege grants under a reviewed migration.
Audit	Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads.
Failure	Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence.
Observability	Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging.
Compatibility	Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair.
Privacy/PHI	Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2.

Programme gate summary

Gate A0 — Infrastructure recovery and known-revision health

Field	Contract
Prerequisite PRs/gates	P01-01, P02-02, P13-01, P13-02
Prerequisite decisions	D-MIG-01, D-CI-01, D-OPS-01
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ GitHub Team is active for NatalyxPlatform; unittest and PostgreSQL checks are mandatory, routine bypass is restricted, and negative-bypass evidence is accepted.
- □ Alan's migration ADR records the supported low-maintenance framework, PostgreSQL authority, staged compatibility and the tested recovery path; P02-02/P02-03 evidence passes.
- □ CloudFormation/ECS/ALB state is captured from refreshed dev credentials and is stable.
- □ The running app and intelligence image SHAs/task definitions are known and map to reviewed commits.
- □ Liveness and readiness are both healthy; PostgreSQL, migrations and required private intelligence dependencies pass their bounded checks.
- □ A synthetic unauthenticated health request and authenticated minimal route succeed without PII/PHI.
- □ The fail-closed recovery runbook is exercised synthetically; the prior compatible image/schema pair is identified and the 30-minute planned-beta recovery target is evidenced.

Required evidence:

- GitHub Team/ruleset readback and negative-bypass result
- Migration ADR and consequential-migration recovery evidence
- AWS account/region and stack/service snapshots
- ALB status and target health
- image SHA/task definition
- readiness JSON/log correlation

- CI/PostgreSQL artifacts
- fail-closed synthetic recovery exercise

Failure and rollback response. Do not proceed to participant validation. Freeze live updates, retain evidence, restore the known compatible image/schema or forward-repair under the approved runbook, and keep the Project gate blocked.

Gate A1 — Synthetic core participant journey

Field	Contract
Prerequisite PRs/gates	P02-04, P03-01, P03-02, P03-04, P03-05, P04-01, P04-02, P04-03, P05-01, P05-02, P05-03, P11-03, P13-03
Prerequisite decisions	D-AUTH-02, D-DATA-01, D-DATA-02
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ A new synthetic IP signs up, is deterministically routed, enters Node 0 Free and no paid case is created.
- □ Candidate/outreach data remains IP-attributed and provisional; no score or answer becomes GC/provider truth.
- □ A synthetic GC begins a low-risk claim, binds identity/consent and sees only the allowed workspace projection.
- □ Cross-role negative tests prove class-2 IP assertions and class-4 GC-private answers do not leak.
- □ The admission proof is short-lived, claim/purpose/email-bound and cannot grant workspace or case membership; replay, mismatch, expiry and revocation fail closed.
- □ Both roles log out/in/resume and complete the mobile journey at tested widths.
- □ Known revision, audit events, correlation IDs and sanitized evidence are attached to the gate item.

Required evidence:

- A1 synthetic run artifact
- separate IP/GC subject IDs (synthetic only)
- authorization matrix results
- mobile screenshots/accessibility output
- audit event trace
- database invariant queries

Failure and rollback response. Stop before Complete, commercial or paid-case validation. Revert/disable the failing participant feature flag or deploy the known A0 revision; preserve synthetic data only for defect diagnosis, then purge under the runbook.

Gate A2 — Synthetic commercial, promotion, billing and document journey

Field	Contract
Prerequisite PRs/gates	P03-03, P06-01, P06-02, P06-03, P07-01, P07-02, P07-03, P07-04, P08-01, P08-02, P08-03, P08-04, P09-01, P09-02, P09-03, P09-04, P09-05, P09-06, P10-01, P10-02, P10-03, P10-04, P11-01, P11-02, P11-04, P12-01, P12-02, P13-04
Prerequisite decisions	D-COM-01, D-COM-02, D-COM-03, D-COM-04, D-PAY-01, D-COM-05, D-COM-06, D-COM-07, D-COM-08, D-DOC-01, D-DOC-02, D-PROV-01, D-AI-01
Responsible owner	Alan
Verifier	Luke

Acceptance matrix:

- □ Synthetic Complete entitlement does not itself create a case and records its commercial configuration version/credit.
- □ Synthetic configuration uses the founder-approved $495 Complete and $8,900 v1 journey schedules, including credits, refunds, loss policy and humane grace behavior; no real charge or live terms acceptance occurs.
- □ Only byos.workspace_promotion_authorized.v1 authorizes exactly one enrollment_pending case; callbacks/redirects/replays create none/duplicates.

- □ Recipient-bound paid-case invitation and consent grant correct membership; activation waits for approved evidence.
- □ Enrollment/activation/milestone/credit/refund ledgers reconcile, including duplicate and out-of-order provider events.
- □ Synthetic files follow upload/quarantine/scan/available/download/replacement/revocation/recovery and never become public.
- □ Provider packet contains only purpose-allowed consented sources; intelligence scopes and proposals do not upgrade truth.
- □ Financing is absent; scanning and AI stay within their $50/month and $100/month beta ceilings with alerts and fail-safe fallback.

Required evidence:

- A2 synthetic run artifact
- processor simulator event ledger
- promotion idempotency query
- invoice/refund explanations
- S3/KMS/IAM/quarantine assertions
- provider snapshot checksum
- Ask/proposal safety evals

Failure and rollback response. Disable commercial/document/provider features, retain free/safety content, prevent live money or real documents, reconcile synthetic ledger inconsistencies, and restore the A1-known revision or forward-repair as approved.

Gate B1 — Real-user and PII beta readiness

Field	Contract
Prerequisite PRs/gates	A2, P12-03, P13-05, P13-06, P13-07
Prerequisite decisions	D-PHI-01, D-PHI-02
Responsible owner	Luke
Verifier	External/TBD

Acceptance matrix:

- □ A written B1 data-classification boundary limits the invitation-only US cohort to at most five IP/GC pairs and permits only the approved nonclinical PII fields.
- □ Medical, reproductive-health, psychological, insurance-health and clinical-clearance data, real document bytes and real provider packets are expressly prohibited; AI receives only de-identified nonclinical context.
- □ The exact release uses a NOBYPASSRLS non-owner runtime, FORCE RLS on every in-scope table, verified transaction-local context and passing pool-reset/cross-org/cross-resource negative tests.
- □ Named beta users, support/on-call, access review, incident response, backup/restore, retention/deletion and complaint routes are ready.
- □ Alan's technical evidence is current; external privacy/legal advice is recorded; there is no unresolved critical finding and any high exception is written, time-limited and externally reviewed.
- □ A2 evidence is current for the exact candidate revision and rollback/feature-disable controls are operational.
- □ Consent/privacy/terms copy is approved and data-subject correction/deletion workflows are testable.

Required evidence:

- Signed B1 checklist and five-pair cohort roster
- data inventory/classification and prohibited-data tests
- runtime/migrator grants and RLS policy matrix
- cross-org/pool-reset results
- de-identification evidence
- access review
- incident tabletop

- backup/restore result
- vendor register
- external privacy/legal advice
- current A2 bundle no older than 60 days

Failure and rollback response. No real-user invitation. Continue synthetic-only, close control gaps, invalidate stale evidence and re-run the exact affected reviews/tests before a new go/no-go.

Gate B2 — PHI readiness and formal go/no-go

Field	Contract
Prerequisite PRs/gates	B1
Prerequisite decisions	D-PHI-01, D-PHI-02
Responsible owner	Luke
Verifier	External security/privacy/compliance reviewers

Acceptance matrix:

- □ Formal PHI data flows, risk assessment, policies, BAAs/vendor controls, least privilege, audit/log retention, encryption/key management, incident/breach response and backup/restore are approved.
- □ Application, infrastructure, document, provider and intelligence threat models have current evidence and no unresolved critical/high risk without accepted exception.
- □ Operational staff training, access provisioning/deprovisioning, monitoring, support and emergency stop are complete.
- □ The exact release passed A2 and B1 revalidation; a signed human go/no-go records scope, date, evidence versions and approvers.
- □ Qualified external security, privacy/compliance and legal/vendor reviewers are named; release evidence is no older than 30 days and underlying assessments are no older than 12 months.
- □ No critical exception is open; any high exception has joint Luke and relevant external-reviewer approval.
- □ The gate states explicitly that testing/deployment alone did not establish HIPAA/PHI readiness.
- □ PHI-capable upload, provider and intelligence paths remain disabled until this signed approval; B1 controls are necessary but not sufficient.

Required evidence:

- Signed go/no-go
- risk/security/privacy assessments
- BAA/vendor evidence
- access/log/incident/backup controls
- penetration/security testing
- training/operations evidence
- current technical gate bundles

Failure and rollback response. PHI remains prohibited. Keep B1 PII scope only if still independently valid; disable any PHI-capable upload/context path, remediate, and repeat formal review.

4. Explicit PR sequence

1

Order	1
ID	P13-01
Title	Expose known-revision health, readiness, and dependency diagnostics
Repository	app
Dependencies	P01-01
Gate	A0
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01; Luke review/final founder approval

2

Order	2
ID	P13-02
Title	Harden guarded deployment, rollback, and evidence capture
Repository	infra
Dependencies	P01-01, P13-01
Gate	A0
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01, P13-01; technical decisions/controls D-OPS-01; Luke review/final founder approval

3

Order	3
ID	P13-03
Title	Automate synthetic A1 core participant journey verification
Repository	app
Dependencies	P03-02, P04-03, P05-03, P11-03, P13-02
Gate	A1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P04-03, P05-03, P11-03, P13-02; Gate A0 active and verified; Luke review/final founder approval

4

Order	4
ID	P13-04
Title	Automate synthetic A2 commercial and document journey verification
Repository	app
Dependencies	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03
Gate	A2
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03; Gate A0 active and verified; Luke review/final founder approval

5

Order	5
ID	P13-05
Title	Enforce case RLS and a non-owner PostgreSQL runtime
Repository	app
Dependencies	P02-03, P03-05, P04-03, P08-04, P09-05, P10-02
Gate	B1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-05, P04-03, P08-04, P09-05, P10-02; Gate A0 active and verified; Luke review/final founder approval

6

Order	6
ID	P13-06
Title	Harden B1/B2 backups, audit, alarms, and cohort controls
Repository	infra
Dependencies	P09-02, P13-02, P13-05
Gate	B1
Local preparation	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-02, P13-02, P13-05; Gate A0 active and verified; Luke review/final founder approval

7

Order	7
ID	P13-07
Title	Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks
Repository	.github
Dependencies	P13-04, P13-05, P13-06
Gate	B1
Local preparation	Evidence-structure preparation only; technical and external evidence remains incomplete
Merge readiness	Blocked pending PR dependencies P13-04, P13-05, P13-06; Gate A0 active and verified; Luke review/final founder approval; qualified external attestations and B1 evidence

P13-01 — Expose known-revision health, readiness, and dependency diagnostics

Field	Plan
Repository / base	`app` (`main`)
Project item	P13-01
Gate / size	A0 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P01-01
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Current /healthz, direct 503 evidence

Objective

Separate liveness, readiness, build revision, migration, database, intelligence, storage, and safe-config status so Gate A0 can identify the actual failing dependency.

Scope

- Unauthenticated minimal liveness

- Safe readiness/dependency checks
- Revision/config identifiers without secrets
- Startup failure classification

Explicit non-goals

- No PHI/PII in diagnostics
- No dependency mutation

Expected files or modules

- `app/health.py`
- `app/main.py`
- `app/dev_diagnostics.py`
- `tests/python/test_health.py`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	Health/readiness endpoints with stable machine-readable reasons.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- DB/migration/intelligence/storage failures
- timeout
- redaction
- known SHA
- ALB contract

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Structured startup/readiness logs.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P01-01. Parallel opportunities: P13-03, P13-04, P13-05, P13-06, P13-07

Objective acceptance criteria

- □ Separate liveness, readiness, build revision, migration, database, intelligence, storage, and safe-config status so Gate A0 can identify the actual failing dependency.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	runtime operations, FastAPI, observability
Higher-capability review	Mandatory before merge

P13-02 — Harden guarded deployment, rollback, and evidence capture

Field	Plan
Repository / base	`infra` (`main`)
Project item	P13-02
Gate / size	A0 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P01-01, P13-01; technical decisions/controls D-OPS-01; Luke review/final founder approval
Technical-decision owner	D-OPS-01: Alan - immediate recovery/rollback/forward-repair mechanism
Dependencies	P01-01, P13-01
Blocking decisions/checkpoints	D-OPS-01
Source / salvage	Existing guarded app/infra workflows

Objective

Make app/intelligence deployment record the image SHA/task definition, wait for health, capture non-secret failure evidence, and roll back or stop safely under explicit guards.

Scope

- Known-revision deploy and service stabilization
- Least-privilege one-off migration task identity
- Migration/app compatibility preflight
- Rollback/forward-repair decision
- Evidence artifact

Explicit non-goals

- No automatic billable activation
- No planning-task deployment

Expected files or modules

- `.github/workflows/cdk.yml`
- `lib/stacks/platform-stack.ts`

- `lib/constructs/migration-task.ts`
- `scripts/verify-service-health.*`
- `docs/runbooks/deploy-rollback.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Guarded workflow/alarms/contracts; default desired counts remain safe.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- CDK/workflow contract
- unhealthy revision
- rollback compatibility
- billable flag false
- missing SSM output

Local validation:

- `npm test`
- `npm run build`
- `Synthesize the affected CDK stack with billable/live changes disabled.`

Cloud validation: run only through Gate A0 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Synthesize and review first; deploy only through the guarded dev workflow with billable/desired-count defaults unchanged, exact revision recorded, and explicit confirmation outside this planning task.

Rollback or forward repair

Return to the prior reviewed stack/task definition when compatible; otherwise keep resources fail-closed and forward-repair under the runbook. Never delete data-bearing resources as an automatic rollback.

Dependencies and parallel work

Hard dependencies: P01-01, P13-01. Parallel opportunities: P13-04, P13-05, P13-07

Objective acceptance criteria

- □ Make app/intelligence deployment record the image SHA/task definition, wait for health, capture non-secret failure evidence, and roll back or stop safely under explicit guards.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-OPS-01: Alan - immediate recovery/rollback/forward-repair mechanism
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	AWS ECS/ALB, GitHub Actions, rollback operations
Higher-capability review	Mandatory before merge

P13-03 — Automate synthetic A1 core participant journey verification

Field	Plan
Repository / base	`app` (`main`)
Project item	P13-03
Gate / size	A1 / Large
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P03-02, P04-03, P05-03, P11-03, P13-02; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P03-02, P04-03, P05-03, P11-03, P13-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Existing synthetic sandbox/scenario utilities

Objective

Run the entry, auth, Node 0 Free, low-risk GC claim, role separation, resume, and mobile journey against a known dev revision and publish a redacted A1 evidence bundle.

Scope

- A1 core IP/GC workspace journey
- Known SHA and evidence bundle
- Role/authorization negative cases
- Cleanup of synthetic records

Explicit non-goals

- No payment/promotion/document journey
- No real user
- No PHI
- No gate auto-approval

Expected files or modules

- `tests/cloud/a1/`
- `scripts/run_synthetic_a1.py`
- `.github/workflows/synthetic-cloud-gates.yml`
- `docs/runbooks/synthetic-gates.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Manual/guarded A1 synthetic runner with immutable artifact summary.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Happy path
- authorization denial
- retry/duplicate
- logout/login
- mobile
- dependency outage
- cleanup failure

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P03-02, P04-03, P05-03, P11-03, P13-02. Parallel opportunities: P13-01, P13-05, P13-06, P13-07

Objective acceptance criteria

- □ Run the entry, auth, Node 0 Free, low-risk GC claim, role separation, resume, and mobile journey against a known dev revision and publish a redacted A1 evidence bundle.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	browser automation, AWS dev operations, end-to-end testing
Higher-capability review	Mandatory before merge

P13-04 — Automate synthetic A2 commercial and document journey verification

Field	Plan
Repository / base	`app` (`main`)
Project item	P13-04
Gate / size	A2 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Dependencies	P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Payment document smoke paths, A1 harness

Objective

Extend the known-revision cloud harness with commitment, funding, exactly-once promotion, case invite, activation, billing, refund, document, handoff, Ask/Learn, and recovery evidence.

Scope

- A2 commercial/promotion/billing/document journey
- Duplicate/retry/refund/failure cases
- Known SHA and evidence bundle
- Synthetic cleanup

Explicit non-goals

- No real money
- No real user
- No PHI
- No gate auto-approval

Expected files or modules

- `tests/cloud/a2/`
- `scripts/run_synthetic_a2.py`
- `.github/workflows/synthetic-cloud-gates.yml`
- `docs/runbooks/synthetic-gates.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Guarded A2 synthetic runner; uses processor simulator and synthetic documents only.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Funding event exactly once
- promotion replay
- case invite
- milestone/refund
- upload/scan failure
- Ask scope
- rollback evidence

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate A2 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

Rollback or forward repair

Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

Dependencies and parallel work

Hard dependencies: P06-03, P07-04, P08-04, P09-05, P10-04, P12-02, P13-03. Parallel opportunities: P13-01, P13-02, P13-05, P13-06

Objective acceptance criteria

- □ Extend the known-revision cloud harness with commitment, funding, exactly-once promotion, case invite, activation, billing, refund, document, handoff, Ask/Learn, and recovery evidence.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	browser automation, payments testing, AWS dev operations, end-to-end testing
Higher-capability review	Mandatory before merge

P13-05 — Enforce case RLS and a non-owner PostgreSQL runtime

Field	Plan
Repository / base	`app` (`main`)
Project item	P13-05
Gate / size	B1 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P02-03, P03-05, P04-03, P08-04, P09-05, P10-02; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Dependencies	P02-03, P03-05, P04-03, P08-04, P09-05, P10-02
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	case-shell.md current app-layer-only/RLS gap, Wave 2 unguarded document/reviewer loaders

Objective

Move app runtime to a least-privilege non-DDL database role and enforce organization/resource access policies with RLS/FORCE RLS before any real-user PII gate can open.

Scope

- Runtime/migrator role separation with NOBYPASSRLS and runtime DML only
- RLS/FORCE RLS on workspace/case/participant/access/document/work-item/commercial/private-memory tables
- Transaction-local verified subject/org/resource/purpose context
- Organization-scoped staff capabilities
- Policy-negative and pool-reset tests

Explicit non-goals

- No PHI gate activation
- No trust in app guards alone
- No runtime DDL

Expected files or modules

- `migrations/versions/*_case_rls.py`
- `app/storage/connection.py`
- `app/storage/session_context.py`
- `tests/security/test_case_rls.py`
- `docs/security/rls.md`

Change map

Area	Planned change
Schema/model	RLS policies and least-privilege grants under a reviewed migration.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope.
Security/privacy	Database denies cross-case/ambient reads even if an app guard is missed.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- Unset context denies
- cross-org/case/owner/role denial
- staff queue organization scope
- connection-pool context reset
- migrator versus runtime grants
- background worker scope
- previous app compatibility during rollout

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate B1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

Rollback or forward repair

Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

Dependencies and parallel work

Hard dependencies: P02-03, P03-05, P04-03, P08-04, P09-05, P10-02. Parallel opportunities: P13-01, P13-02, P13-03, P13-04

Objective acceptance criteria

- □ Move app runtime to a least-privilege non-DDL database role and enforce organization/resource access policies with RLS/FORCE RLS before any real-user PII gate can open.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	PostgreSQL RLS, connection pooling, security architecture
Higher-capability review	Mandatory before merge

P13-06 — Harden B1/B2 backups, audit, alarms, and cohort controls

Field	Plan
Repository / base	`infra` (`main`)
Project item	P13-06
Gate / size	B1 / Exceptional
Local preparation status	Not ready; named PR, decision or programme-acceptance prerequisites remain
Merge readiness	Blocked pending PR dependencies P09-02, P13-02, P13-05; Gate A0 active and verified; Luke review/final founder approval
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval; D-PHI-02: Alan - evidence index; qualified external reviewers - attestations
Dependencies	P09-02, P13-02, P13-05
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Existing audit/backup/platform stack constructs

Objective

Add the infrastructure evidence and controls required for a named PII beta and later PHI review: backup/restore, data-event audit, least privilege, alarms, log retention, cohort disable, and emergency stop.

Scope

- RDS/S3 backup and restore evidence hooks
- CloudTrail S3 data events and access alarms
- Runtime/migrator/task-role least privilege
- Cohort disable and PHI gate kill switch
- Log retention/redaction contracts

Explicit non-goals

- No production/PHI activation
- No evidence claim without restore/tabletop

Expected files or modules

- `lib/stacks/platform-stack.ts`
- `lib/constructs/audit.ts`
- `lib/constructs/backup.ts`
- `test/security-controls.test.ts`
- `docs/runbooks/security-operations.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	Guarded, cost-aware controls with explicit dev defaults and documented operational owners.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Document the contract, version pins and safe partial-merge behavior.

Tests and validation

Objective tests:

- CDK policy assertions
- restore drill
- alarm path
- access-log redaction
- cohort/PHI gate disable
- role boundary

Local validation:

- `npm test`
- `npm run build`
- `Synthesize the affected CDK stack with billable/live changes disabled.`

Cloud validation: run only through Gate B1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Synthesize and review first; deploy only through the guarded dev workflow with billable/desired-count defaults unchanged, exact revision recorded, and explicit confirmation outside this planning task.

Rollback or forward repair

Return to the prior reviewed stack/task definition when compatible; otherwise keep resources fail-closed and forward-repair under the runbook. Never delete data-bearing resources as an automatic rollback.

Dependencies and parallel work

Hard dependencies: P09-02, P13-02, P13-05. Parallel opportunities: P13-01, P13-03, P13-04

Objective acceptance criteria

- □ Add the infrastructure evidence and controls required for a named PII beta and later PHI review: backup/restore, data-event audit, least privilege, alarms, log retention, cohort disable, and emergency stop.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Alan (proposed; pending Alan review)
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval; D-PHI-02: Alan - evidence index; qualified external reviewers - attestations
Reviewer	Luke
Final approver	Luke
PR-level ownership exception	None
Required expertise	AWS security, backup/restore, CloudTrail/CloudWatch, IAM
Higher-capability review	Mandatory before merge

P13-07 — Publish B1 and B2 privacy, PHI, and operational go-no-go runbooks

Field	Plan
Repository / base	`.github` (`main`)
Project item	P13-07
Gate / size	B1 / Large
Local preparation status	Evidence-structure preparation only; technical and external evidence remains incomplete
Merge readiness	Blocked pending PR dependencies P13-04, P13-05, P13-06; Gate A0 active and verified; Luke review/final founder approval; qualified external attestations and B1 evidence
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval; D-PHI-02: Alan - evidence index; qualified external reviewers - attestations
Dependencies	P13-04, P13-05, P13-06
Blocking decisions/checkpoints	None beyond dependencies and plan approval
Source / salvage	Natalyx compliance/company records by reference, all gate evidence

Objective

Create the programme evidence index and explicit human approval packets for PII beta (B1) and PHI readiness (B2), linking rather than duplicating controlled company/compliance records.

Scope

- Data classification and real-user boundary
- Access/logging/incident/backup/vendor/compliance evidence
- Named approvers and expiry/revalidation
- Rollback/stop response

Explicit non-goals

- No legal/compliance self-certification
- No PHI activation by merge

Expected files or modules

- `natalyx-repair-program/gates/B1-real-user-pii.md`
- `natalyx-repair-program/gates/B2-phi-go-no-go.md`
- `natalyx-repair-program/evidence-index.md`

Change map

Area	Planned change
Schema/model	None in this PR; any later schema change stays in its named migration PR.
API/routes	No public API/route contract change.
UI/product surface	No visible participant/operator UI change.
Infrastructure/workflow	No cloud/infrastructure resource or workflow change.
Migration/backfill	No migration or backfill.
Security/privacy	Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases.
Architecture/docs/config	Controlled go/no-go checklists and evidence links only.

Tests and validation

Objective tests:

- Evidence completeness review
- tabletop incident
- access review
- backup/restore evidence
- external security/privacy sign-off

Local validation:

- `git diff --check`
- `/Users/lukerhodes/.cache/codex-runtimes/codex-primary-runtime/dependencies/python/bin/python3 scripts/validate_programme.py`

Cloud validation: run only through Gate B1 on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

Diagnostics and observability

Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

Rollout

Merge only after document/sign-off checks; it changes no runtime. Re-sync the Project from the same canonical data.

Rollback or forward repair

Correct the canonical data and regenerate; do not rewrite implementation history.

Dependencies and parallel work

Hard dependencies: P13-04, P13-05, P13-06. Parallel opportunities: P13-01, P13-02, P13-03

Objective acceptance criteria

- □ Create the programme evidence index and explicit human approval packets for PII beta (B1) and PHI readiness (B2), linking rather than duplicating controlled company/compliance records.
- □ Every listed objective test passes locally and at the required gate where applicable.
- □ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- □ The planning record has matching dependency, local-preparation, merge-readiness, ownership, gate and disposition fields.
- □ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

Ownership and review

Role	Recommendation
Work-package DRI	Alan
Primary implementer	Joint (proposed; pending Alan review)
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval; D-PHI-02: Alan - evidence index; qualified external reviewers - attestations
Reviewer	Luke + External/TBD
Final approver	Luke
PR-level ownership exception	Alan assembles technical control evidence; Luke owns founder go/no-go; qualified external reviewers supply required attestations.
Required expertise	privacy/security governance, operations, compliance
Higher-capability review	Mandatory before merge

5. Cross-repository coordination

Repository	Authority after merge
`.github`	programme/governance evidence and Project control plane
`app`	routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain
`infra`	AWS resources, IAM/KMS/network/deployment/backup/audit contracts

Merge order inside this package: `P13-01` → `P13-02` → `P13-03` → `P13-05` → `P13-04` → `P13-06` → `P13-07`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

6. Test and acceptance matrix

Test dimension	Method	Objective result
Happy path	Known revision; ALB/ECS/dependency health; Rollback; Synthetic A1/A2; PII controls; PHI go/no-go evidence	Expected state/event/UI and named evidence are produced once.
Authorization failure	Wrong/no subject, role, membership, purpose or consent	403/404 as appropriate; no write or data leak; audit safe failure.
Role differences	Separate IP, GC, operator, provider/service identities	Only the approved lens/projection/actions appear.
Retries/duplicates	Repeat same canonical request/event	Original result returned; no duplicate membership/case/invoice/document/event.
Concurrency	Two writers/claimants/migrators/workers	One valid transition; loser receives deterministic stale/conflict response.
Stale state	Old revision/config/terms/token/artifact	Fail closed or approved compatibility behavior; never silent coercion.
Malformed input	Boundary, encoding, size, enum, redirect and identifier corpus	Validation before side effects; safe error.
Unavailable dependency	Database, intelligence, processor, S3/scanner or auth unavailable	Explicit degraded/blocked behavior; no hallucinated success.
Partial failure	Crash between external effect, DB transaction and outbox	Retry/forward repair converges from durable evidence.
Logout/login/resume	Session expires/restarts mid-flow	Return to authorized current state; no query/link authority.
Mobile/accessibility	320/375/768 widths, keyboard, focus, reduced motion	Core task completes with truthful state and no clipped controls.
Synthetic cloud	Known dev revision and synthetic identities/data	Gate artifact records SHA, dependency status, audit and cleanup.
Audit/observability	Inspect events/metrics/logs	Versions, actor/authority, correlation and reason present; no sensitive payload.

7. Risks and decisions

Founder outcomes, Alan's remaining technical ownership and external activation dependencies are recorded separately so a final policy is never mistaken for an active control.

ID	Status	Classification	Remaining owner	Affected / latest	Programme effect
D-AUTH-03	Resolved by reconciliation	Reconciliation settled	Alan - implementation	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12	No longer blocks the reconciled access/workspace PR sequence.
D-OPS-01	Founder constraints final; Alan immediate; operationally pending	Luke sets constraints; Alan decides immediately	Alan - immediate	P13-02; Before Gate A0 recovery action	P13-02 and Gate A0 remain blocked until the runbook and synthetic exercise pass; A1 must exercise the control.
D-PHI-01	Founder beta scope final; externally gated	Luke decides beta scope; external evidence required	Alan - B1/B2 evidence; external reviewers - approval	P09-01, P09-02, P09-05, P09-06, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.
D-PHI-02	Founder governance final; external reviewers pending	Luke decides governance; external evidence required	Alan - evidence; external reviewers - attestations; Luke - go/no-go	P13-06, P13-07; Before B1	B1 and B2 remain blocked until the corresponding evidence is current and the required humans sign off.

D-AUTH-03 - Identity, participation and access authorization contract

Field	Recorded decision or requirement
Decision state	Final
Founder outcome	Identity, participation, access grants, ownership and operation authorization remain separate; participation or lens alone never authorizes.
Founder constraints	Subject-derived authorization context; Application and RLS policy parity; Default-deny cross-resource access
What remains for Alan	Encode the settled contract in application services and RLS
Evidence required	Participation-without-grant and grant-without-capability denials; Application/RLS parity tests
Approval rule	No further founder decision unless the boundary changes.
External dependencies	None
Affected PRs / latest point	P03-02, P03-03, P03-05, P04-01, P04-02, P09-04, P13-05; Resolved 2026-07-12
Programme effect	No longer blocks the reconciled access/workspace PR sequence.
Independent recommendation	Identity binds principal to person; participation records domain relationship; grants carry resource-scoped capabilities; application and RLS consume a subject-derived context plus ownership and purpose. Participation/lens alone never authorizes.

D-OPS-01 - Known-good recovery revision and rollback sequence

Field	Recorded decision or requirement
Decision state	Final founder constraints
Founder outcome	Use fail-closed recovery: protect data integrity over uptime, restore only a reviewed proven-compatible application-image/schema pair, use safe read-only or unavailability when uncertain, and forward-repair when rollback is unsafe. Target 30-minute beta recovery; allow up to two hours emergency downtime and notify Luke when more than one hour is expected.
Founder constraints	Zero intended loss or corruption; No expensive 24/7 coverage for synthetic/early controlled beta without evidence; Real-user, destructive or uncertain-loss recovery requires explicit human approval
What remains for Alan	Choose exact compatibility, rollback/forward-repair and automation mechanisms; Name the evidence owner and write the runbook
Evidence required	Documented recovery procedure; Successful synthetic recovery exercise; Known-compatible image/schema evidence
Approval rule	Luke accepts A0 evidence and explicitly approves risky real-user/destructive recovery.
External dependencies	None
Affected PRs / latest point	P13-02; Before Gate A0 recovery action
Programme effect	P13-02 and Gate A0 remain blocked until the runbook and synthetic exercise pass; A1 must exercise the control.
Independent recommendation	Use a tested immutable image and compatible schema; do not deploy from dirty or draft branches.

D-PHI-01 - PII/PHI boundary, vendors and control set

Field	Recorded decision or requirement
Decision state	Final founder scope; externally dependent activation
Founder outcome	B1 is an invitation-only US beta of at most five IP/GC pairs using names, verified account/contact data, relationship/outreach/logistics, consent versions, nonclinical progress, feedback and processor references only. Medical, reproductive-health, psychological, insurance-health and clinical-clearance data are prohibited; no real document bytes or provider packets; AI receives de-identified nonclinical context only.
Founder constraints	RLS/non-owner runtime and cross-tenant negative evidence; Access review plus incident and restore exercises; No PHI before B2
What remains for Alan	Implement inventory, cohort controls, RLS, cross-tenant denials, de-identification and prohibited-data tests
Evidence required	Five-pair cohort configuration; RLS/cross-tenant/de-identification/prohibited-data tests; External privacy/legal approval
Approval rule	Luke accepts B1 only after required evidence and external privacy/legal advice; B2 requires qualified external approval.
External dependencies	External privacy/legal approval before B1; Qualified external security/privacy/compliance/legal approval before B2
Affected PRs / latest point	P09-01, P09-02, P09-05, P09-06, P13-05, P13-06, P13-07; Before B1 for PII scope; before B2 for PHI
Programme effect	B1 remains blocked until the limited PII control bundle is accepted; real documents/provider packets and all PHI remain blocked until B2.
Independent recommendation	B1 remains PII-only and content controls stay synthetic; B2 requires formal external security/privacy/compliance evidence before any PHI content path is enabled.

D-PHI-02 - B1/B2 approvers and evidence expiry

Field	Recorded decision or requirement
Decision state	Final founder governance; externally dependent activation
Founder outcome	Luke accepts B1 from Alan's technical evidence plus external privacy/legal advice, without separate independent technical sign-off. B1 evidence is at most 60 days old and reruns after material change or 180 days. B2 requires qualified external security, privacy/compliance and legal/vendor review with release evidence at most 30 days old and assessments at most 12 months old.
Founder constraints	No critical exceptions; High B1 exceptions are written, time-limited and externally reviewed; High B2 exceptions require joint Luke and relevant external approval
What remains for Alan	Maintain evidence recency, change-triggered reruns and exception records
Evidence required	B1/B2 evidence index with dates; Named reviewers; Exception register; Signed go/no-go
Approval rule	Luke is final product go/no-go within the specified external-review model.
External dependencies	Named privacy/legal advisers for B1; Named qualified security/privacy/compliance/legal reviewers for B2
Affected PRs / latest point	P13-06, P13-07; Before B1
Programme effect	B1 and B2 remain blocked until the corresponding evidence is current and the required humans sign off.
Independent recommendation	Luke is product go/no-go; independent qualified reviewers own their control attestations.

Implementation risk	Owner	Treatment
Unknown running revision	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Rollback incompatible with schema	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
Synthetic success overclaimed	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.
PHI enabled without organizational controls	Alan	Mitigate through explicit tests, rollout and forward-repair evidence before the associated gate.

8. Existing-work salvage

Source	Head	Valuable content	Destination	Recommendation	Extraction status/safety
feat/byos-journey-wave-2	5b63e3a7c364	Projection-driven phase-1 cluster presentation, exact route allowlisting, person-versus-case-side ownership intent, transactional submission/work-item and review/audit patterns, fail-closed item retention, approved Learn status selection and 36 focused tests.	P13-05	Preserve paused at the exact head; selectively port semantics/tests only after adaptation. Do not merge, cherry-pick, continue implementation or use its runtime schema directly.	Paused exact-head evidence; no direct merge

Package source inventory:

- Existing guarded AWS workflows and SSM contracts
- Current healthz
- Prior app/infra/intelligence AWS migration PR evidence

9. Ownership recommendation

Alan owns technical recovery/validation; Luke accepts gates and PHI boundary. External security, privacy, compliance, finance/payment, and clinical/legal review as applicable.

P13-01

PR	P13-01
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P13-02

PR	P13-02
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-OPS-01: Alan - immediate recovery/rollback/forward-repair mechanism
Reviewer	Luke
Final approver	Luke
Exception	None

P13-03

PR	P13-03
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P13-04

PR	P13-04
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	None; implement the settled founder constraints and approved contracts
Reviewer	Luke
Final approver	Luke
Exception	None

P13-05

PR	P13-05
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval
Reviewer	Luke
Final approver	Luke
Exception	None

P13-06

PR	P13-06
WP DRI	Alan
Primary implementer	Alan (proposed)
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval; D-PHI-02: Alan - evidence index; qualified external reviewers - attestations
Reviewer	Luke
Final approver	Luke
Exception	None

P13-07

PR	P13-07
WP DRI	Alan
Primary implementer	Joint (proposed)
Technical-decision owner	D-PHI-01: Alan - technical evidence; qualified external reviewers - control approval; D-PHI-02: Alan - evidence index; qualified external reviewers - attestations
Reviewer	Luke + External/TBD
Final approver	Luke
Exception	Alan assembles technical control evidence; Luke owns founder go/no-go; qualified external reviewers supply required attestations.

PR ownership inherits the proposed work-package split pending Alan's review. Named technical-decision ownership remains separate and never transfers implementation ownership across work packages; reciprocal and external review requirements remain intact.

10. Definition of done

- □ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and Project item complete.
- □ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- □ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- □ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- □ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- □ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- □ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.